

Exercise for chapter 1 to 3

Xiao Han (韩潇)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

xiao.han@nwpu.edu.cn

Chapter 1

$$PV = RT$$

1.2 The expression for the total derivative of V with respect to the dependent variables P and T is

$$dV = \left(\frac{\partial V}{\partial P} \right)_T dP + \left(\frac{\partial V}{\partial T} \right)_P dT$$

$$\ln V = \ln \frac{T}{P} + C$$

$$\frac{PV}{T} = e^C$$

$$R = e^C$$

Substitute the values of β_T and α obtained in Qualitative Problem 2 into this equation and integrate them to obtain the equation of state for an ideal gas.

$$\beta_T = -\frac{1}{P} \quad \alpha = \frac{1}{T}$$

$$\left(\frac{\partial V}{\partial P} \right)_T = -\frac{1}{P} V \quad \left(\frac{\partial V}{\partial T} \right)_P = \frac{1}{T} V$$

$$= -\frac{V}{P} \quad = \frac{V}{T}$$

$$dV = -\frac{V}{P} dP + \frac{V}{T} dT$$

$$\int \frac{dV}{V} = \int -\frac{dP}{P} + \int \frac{dT}{T}$$

$$\ln V = -\ln P + \ln T + C$$

Chapter 1

Solution to 1.2

According to Qualitative Problem 2,

$$\beta_T = \frac{1}{P}, \quad \alpha = \frac{1}{T}.$$

$$\text{So } \left(\frac{\partial V}{\partial P} \right)_T = -V\beta_T = -\frac{V}{P}, \quad \left(\frac{\partial V}{\partial T} \right)_P = V\alpha = \frac{V}{T}.$$

$$dV = \left(\frac{\partial V}{\partial P} \right)_T dP + \left(\frac{\partial V}{\partial T} \right)_P dT = -\frac{V}{P} dP + \frac{V}{T} dT.$$

Chapter 1

$$\text{Integrate } \frac{dV}{V} = -\frac{dP}{P} + \frac{dT}{T},$$

$$\ln V = -\ln P + \ln T + C = \ln \frac{T}{P} + C \text{ (C is constant).}$$

$$\text{So, } PV = e^C \cdot T.$$

Take $R = e^C$, so for an ideal gas, the equation of state is

$$PV = RT.$$

Chapter 1

1.3 The pressure temperature phase diagram in Figure 1.4 has no two-phase areas (only two-phase curves), but the temperature composition diagram in Figure 1.5 does have two-phase areas. Explain.

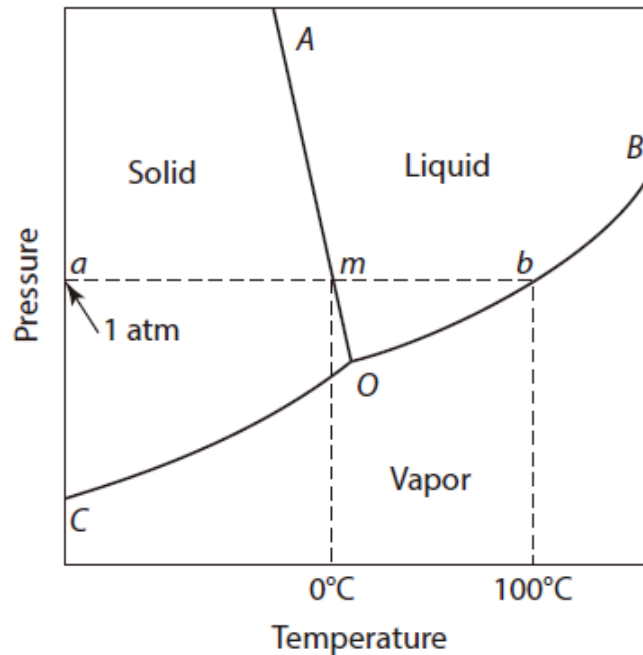


Figure 1.4 Schematic representation of part of the pressure–temperature equilibrium phase diagram for H_2O . The melting point is designated as m and the boiling point as b .

Chapter 1

1.3 The pressure temperature phase diagram in Figure 1.4 has no two-phase areas (only two-phase curves), but the temperature composition diagram in Figure 1.5 does have two-phase areas. Explain.

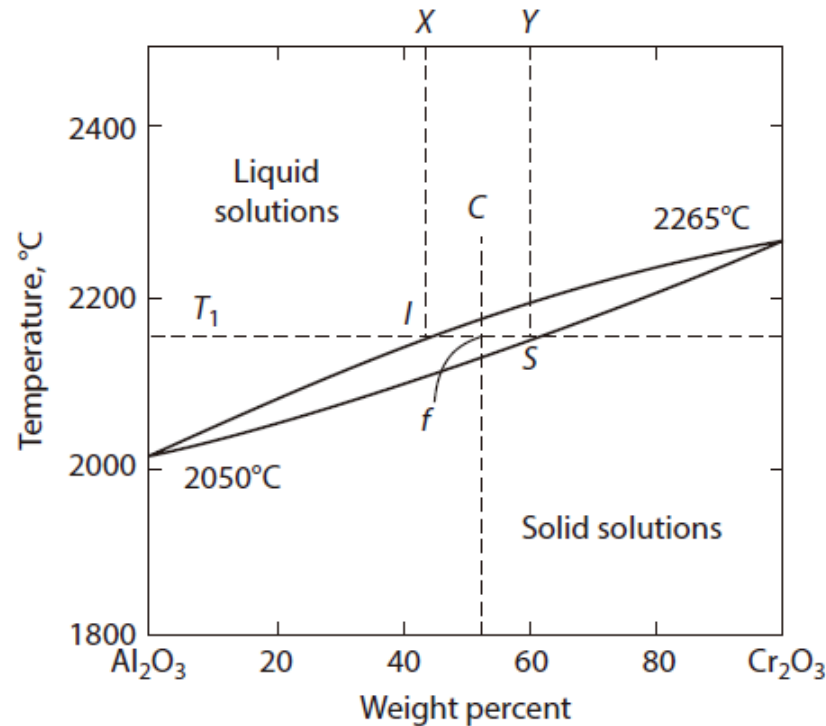


Figure 1.5 The temperature composition equilibrium phase diagram for the system Al_2O_3 - Cr_2O_3 at a constant pressure of 1 atm.

Chapter 1

Solution to 1.3

According to Phase Rule: $F=C-P+2$, where the degree of freedom F is a factor (such as temperature and pressure) that can be independently changed without changing the number of phases when the system is in equilibrium; C is the number of components in the system; P is the number of coexisting phases in the selected system.

- a) In Figure 1.4, it has only one component, when there are two coexisting phases, $F=1-2+2=1$, so it has one degree of freedom, that is, it has only two-phase curves.
- b) In Figure 1.5, it has two components, when there are two coexisting phases, $F=2-2+2=2$, so it has two degrees of freedom, that is it has two-phase areas.

Chapter 2

2.1 An monatomic ideal gas at 300 K has a volume of 15 liters at a pressure of 15 atm. Calculate:

a. The final volume of the system

b. The work done by the system

c. The heat entering or leaving the system

d. The change in the internal energy

e. The change in the enthalpy when the gas undergoes

i. A **reversible isothermal** expansion to a pressure of 10 atm

ii. A **reversible adiabatic** expansion to a pressure of 10 atm

The constant-volume molar heat capacity of the gas, c_v , has the value $1.5 R$.

$$T = 300$$

$$V = 15$$

$$P = 15$$

$$\frac{15 \times 15}{10}$$

$$U = 0. \quad Q = -W \int nRT \ln \frac{V_2}{V_1}$$

$$\Delta U = 0.$$

$$\Delta H = 0.$$

Chapter 2

Solution to 2.1

(i) Initial state : $p_1 = 15 \text{ atm}$, $V_1 = 15 \text{ liters}$, $T_1 = 300 \text{ K}$

Process: isothermal reversible

Final state: $p_2 = 10 \text{ atm}$, $V_2 = ?$, $T_2 = 300 \text{ K}$

a. Isothermal process, $p_1 V_1 = p_2 V_2$,

$$\text{So } V_2 = \frac{p_1 V_1}{p_2} = \frac{15 \text{ atm} \cdot 15 \text{ liters}}{10 \text{ atm}} = 22.5 \text{ liters}$$

b. $p_1 V_1 = p_2 V_2 = nRT$,

$$\begin{aligned} w &= \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln V \Big|_{V_1}^{V_2} = nRT \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{V_2}{V_1} \\ &= 15 \text{ atm} \times 15 \text{ liters} \ln \frac{22.5 \text{ liters}}{15 \text{ liters}} \end{aligned}$$

Chapter 2

Solution to 2.1

b. $w = 15 \text{ atm} \times 15 \text{ liters} \ln \frac{22.5 \text{ liters}}{15 \text{ liters}} =$

$$15 \times 101325 \text{ Pa} \times 15 \times 10^{-3} \text{ m}^3 \ln \frac{22.5}{15} = 9244 \text{ J}$$

For ideal gas, the internal energy is the function only of temperature. So:

c. $Q = w = 9244 \text{ J};$

d. $\Delta U = 0;$

e. $\Delta H = 0.$

Chapter 2

(ii) Initial state : $p_1 = 15 \text{ atm}$, $V_1 = 15 \text{ liters}$, $T_1 = 300 \text{ K}$

Process: Adiabatic reversible

Final state: $p_2 = 10 \text{ atm}$, $V_2 = ?$, $T_2 = ?$

$$\frac{p_1}{p_2} = \left(\frac{V_2}{V_1} \right)^\gamma$$

a. $p_1 V_1^\gamma = p_2 V_2^\gamma$, for a monatomic ideal gas, $\gamma = \frac{5}{3}$.

$$\text{So } V_2 = \left(\frac{p_1 V_1^\gamma}{p_2} \right)^{\frac{1}{\gamma}} = \left(\frac{15 \text{ atm} 15^{5/3} \text{ liters}}{10 \text{ atm}} \right)^{\frac{3}{5}} = 19.13 \text{ liters}$$

Chapter 2

$$\text{b. } w = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \left(\frac{p_1 V_1^\gamma}{V^\gamma} \right) dV = \frac{p_1 V_1^\gamma}{-\gamma+1} V^{-\gamma+1} \Big|_{V_1}^{V_2}$$

$$= \frac{15 \text{ atm} 15^{5/3} \text{ l}^5}{2/3} (19.13^{-2/3} - 15^{-2/3}) \text{ l}^5$$

$$= -1.5 \times 15 \times 101325 \times 15^{5/3} \times 10^{-3} \times (19.13^{-2/3} - 15^{-2/3}) \text{ J} = 5119 \text{ J}$$

$$w = -\Delta U = -n c_v \Delta T = -\frac{p_1 V_1}{R T_1} 1.5 R \left(\frac{p_2 V_2}{n R} - T_1 \right)$$

$$= -9.14 \times 1.5 \times 8.3145 \times (255.07 - 300) = 5121.65 \text{ J}$$

Chapter 2

c. For adiabatic process, $Q = 0$

d. $\Delta U = -w = -5121.96 \text{ J}$

e. $\Delta H = \Delta U + \Delta(PV)$

$$= -5121.96 + (10 \times 19.13 - 15 \times 15) \times 101325 \times 10^{-3}$$

$$= -8536.61 \text{ J}$$

Chapter 2

2.3 The initial state of a quantity of monatomic ideal gas is $P=1$ atm, $V=1$ liter, and $T=373$ K. The gas is isothermally expanded to a volume of 2 liters and is then cooled at constant pressure to the volume V . This volume is such that a reversible adiabatic compression to a pressure of 1 atm returns the system to its initial state. All of the changes of state are conducted reversibly. Calculate the value of V and the total work done on or by the gas.

$P_1=1$ $V_1=1$ $T_1=373$ $U=Q+w.$

$$P_2=\frac{1}{2} \quad V_2=2 \quad T_2=373 \quad \textcircled{1} w=nRT \ln \frac{V_2}{V_1}$$

$$P_3=P_2 \quad V_3= \quad T_3=\frac{373}{2} \quad \textcircled{2} w=\int P dv = -\frac{1}{2} R$$

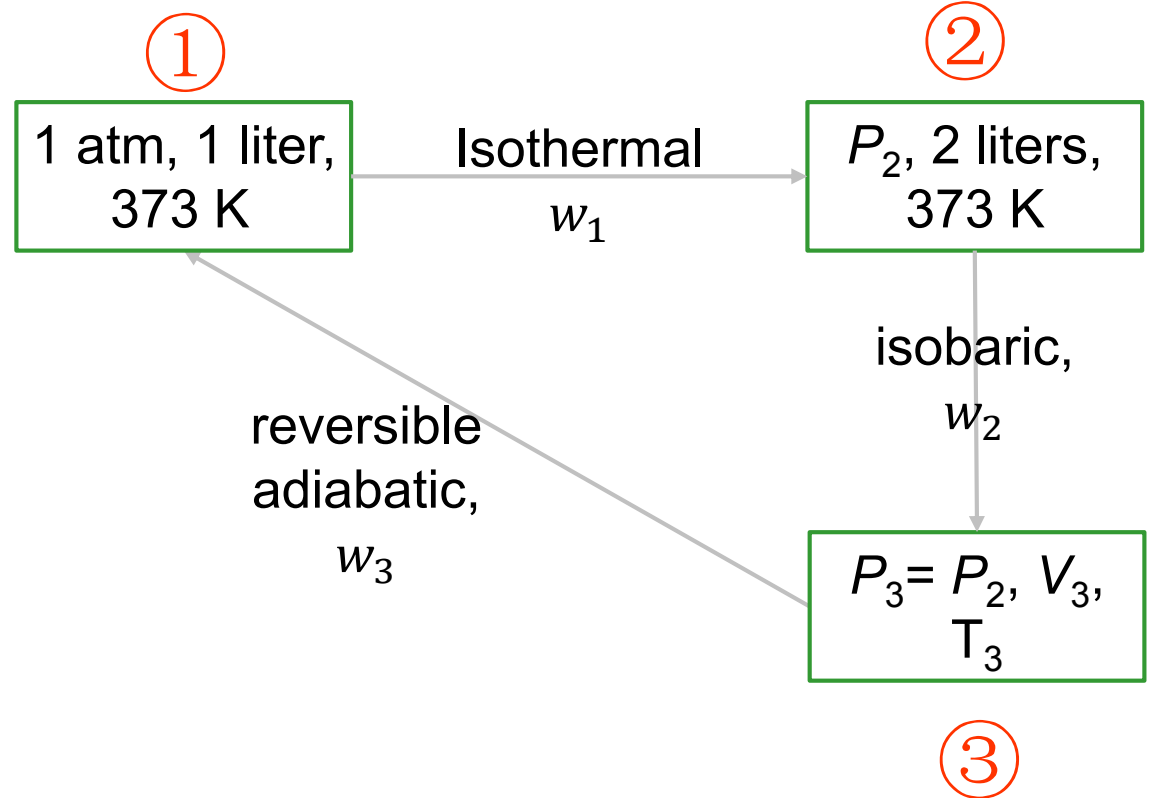
绝热可逆.

Chapter 2

Solution to 2.3

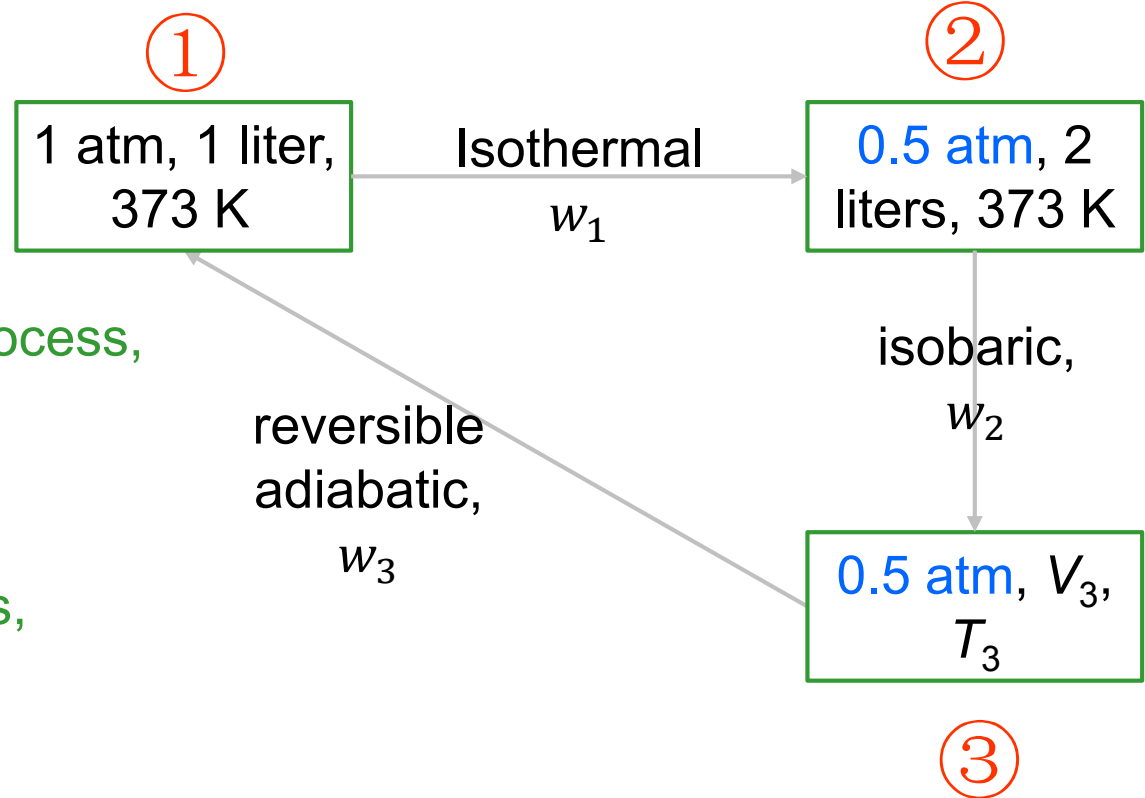
The process is shown in the figure on the right.

$$\begin{aligned} P_3 &= P_2 = \frac{P_1 V_1}{V_2} \\ &= \frac{1 \text{ atm} \times 1 \text{ liter}}{2 \text{ liters}} \\ &= 0.5 \text{ atm} \end{aligned}$$



Chapter 2

Solution to 2.3



For reversible adiabatic process,

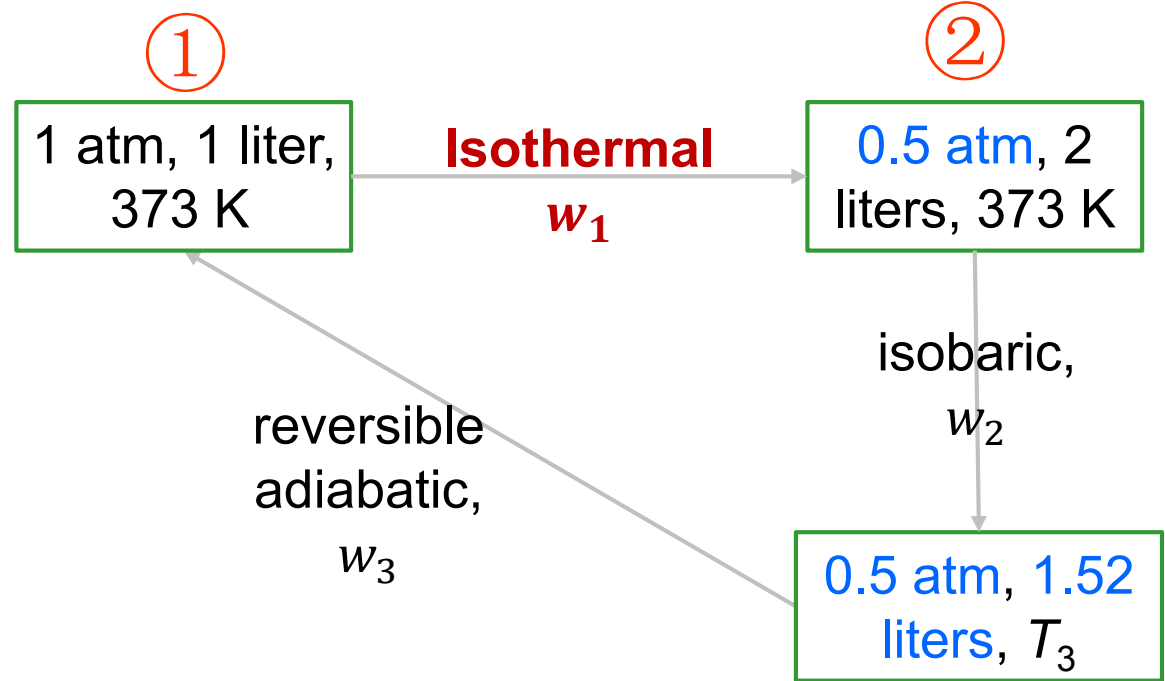
$$P_3 V_3^\gamma = P_1 V_1^\gamma,$$

for an monatomic ideal gas,

$$\gamma = \frac{5}{3}$$

$$V_3 = \left(\frac{P_1 V_1^\gamma}{P_3} \right)^{\frac{1}{\gamma}} = \left(\frac{1 \text{ atm} 1^\gamma \text{ liter}^\gamma}{0.5 \text{ atm}} \right)^{\frac{1}{\gamma}} = 1.52 \text{ liters}$$

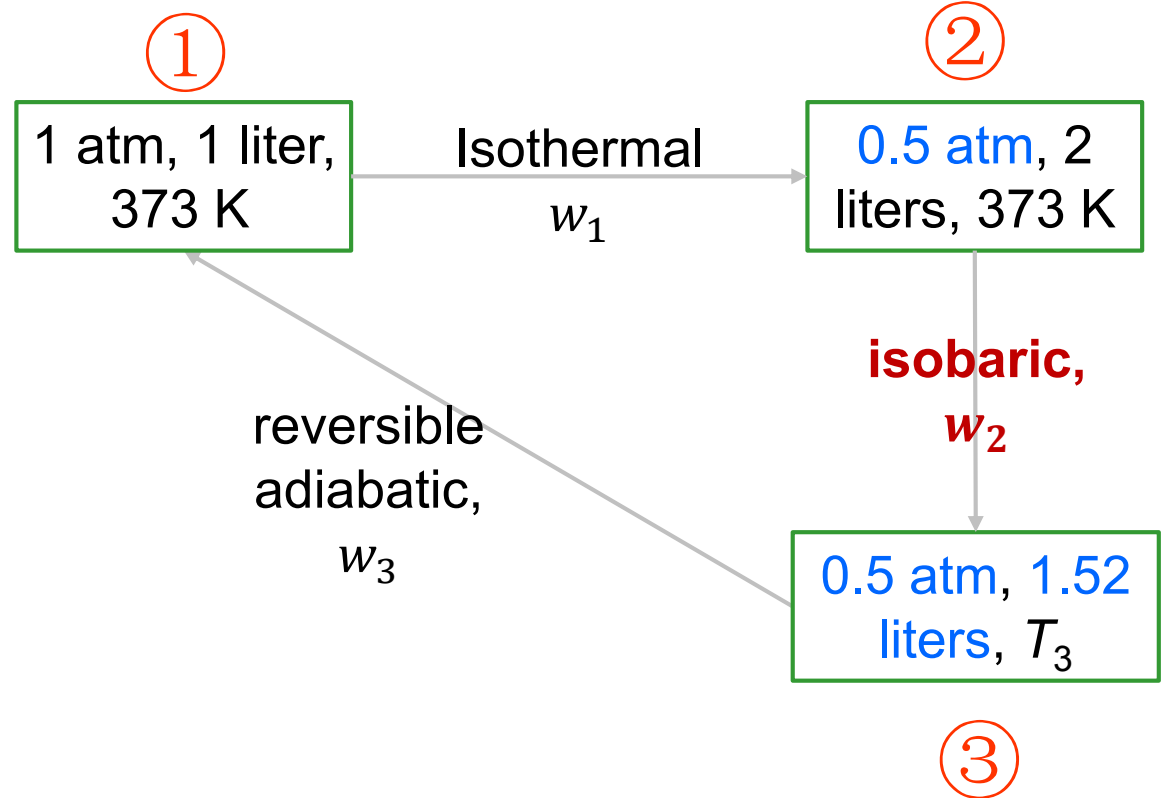
Chapter 2



$$w_1 = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln V \Big|_{V_1}^{V_2} = nRT \ln \frac{V_2}{V_1} = P_1 V_1 \ln \frac{V_2}{V_1}$$

$$= 1 \text{ atm} \cdot 1 \text{ liter} \ln \frac{2}{1} = 1 \times 101325 \times 1 \times 10^{-3} \ln 2 \text{ J} = 70.233 \text{ J}$$

Chapter 2

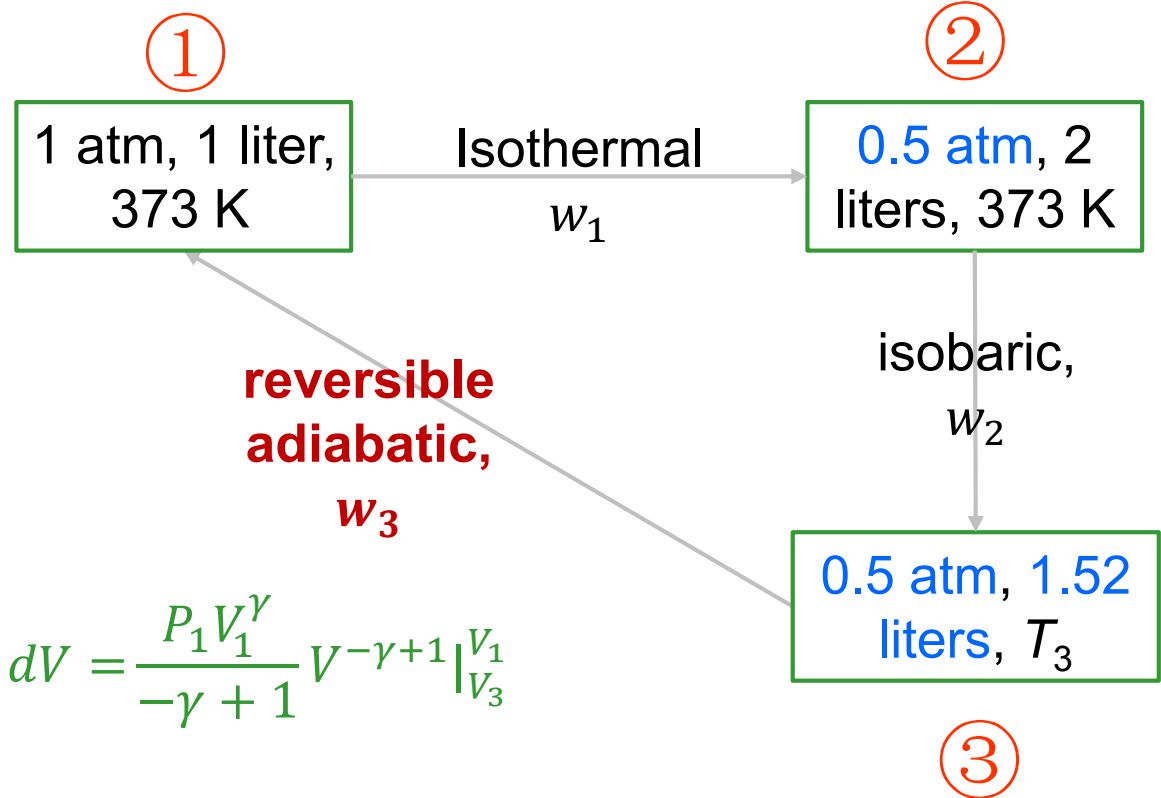


$$w_2 = P_2 \Delta V = P_2 (V_3 - V_2)$$

$$= 0.5 \text{ atm} \times (1.52 - 2) \text{ liters} = 0.5 \times 101325 \times (1.52 - 2) \times 10^{-3} \text{ J}$$

$$= -24.318 \text{ J}$$

Chapter 2



$$w_3 = \int_{V_3}^{V_1} P dV = \int_{V_3}^{V_1} \left(\frac{P_1 V_1^\gamma}{V^\gamma} \right) dV = \frac{P_1 V_1^\gamma}{-\gamma + 1} V^{-\gamma+1} \Big|_{V_3}^{V_1}$$

$$= 1 \times 101325 \times (-1.5) \times (1 - 1.52^{-2/3}) \times 10^{-3} \text{ J} = -37.019 \text{ J}$$

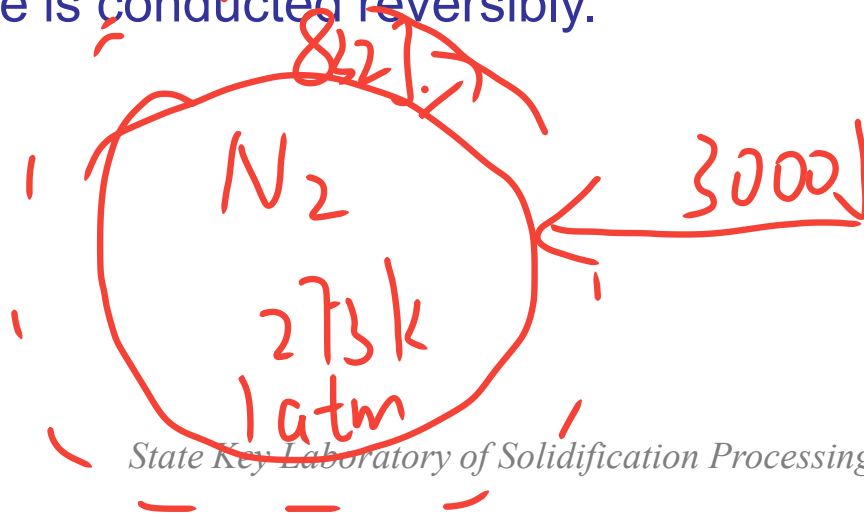
$$w = w_1 + w_2 + w_3 = (70.233 - 24.318 - 37.019) \text{ J} = 8.896 \text{ J}$$

Chapter 2

2.5 One mole of N_2 gas is contained at 273 K and a pressure of 1 atm. The addition of 3000 J of heat to the gas at constant pressure causes 832 J of work to be done during the expansion. Calculate

- The final state of the gas
- The values of ΔU and ΔH for the change of state
- The values of c_v and c_p for N_2

Assume that nitrogen behaves as an ideal gas, and that the change of state is conducted reversibly.



$$\begin{aligned} PV &= nRT & T_L &= \frac{PV_2}{R} \\ V &= \frac{RT}{P} & \Delta U &= Q - W \\ W &= P(V_2 - V_1) & \Delta H &= 3000 \text{ J} \\ V_2 &= \frac{W}{P} + V_1 \end{aligned}$$

Solution to 2.5

Initial state: 1 mole N_2 , $P_1 = 1 \text{ atm}$, $V_1 = ??$, $T_1 = 273 \text{ K}$,

Process: isobaric reversible, $\Delta H = 3000 \text{ J}$. $W = 832 \text{ J}$.

Final state: 1 mole N_2 , $P_2 = 1 \text{ atm}$, $V_2 = ??$, $T_2 = ??$,

a. Constant pressure, $P_2 = P_1 = 1 \text{ atm}$, according to $P_1 V_1 = nRT_1$,

$$\text{so } V_1 = \frac{nRT_1}{P_1} = \frac{1 \times 8.3144 \times 273}{1 \times 101325} \text{ m}^3 = 0.0224 \text{ m}^3 = 22.4 \text{ liters}$$

$$w = P_1 \Delta V = P(V_2 - V_1),$$

$$V_2 = \frac{w}{P_1} + V_1 = \left(\frac{832}{1 \times 101325} \times 10^3 + 22.4 \right) \text{ liters} = 30.6 \text{ liters}$$

$$T_2 = \frac{P_2 V_2}{nR} = \frac{1 \times 101325 \times 30.6 \times 10^{-3}}{1 \times 8.3144} \text{ K} = 373 \text{ K}$$

Chapter 2

b. $\Delta H = q_p = 3000 \text{ J}$

$$\Delta U = q_p - w = (3000 - 832) \text{ J} = 2168 \text{ J}$$

c. $c_v = \frac{\Delta U}{n\Delta T} = \frac{2168}{373-273} \frac{\text{J}}{\text{mol} \cdot \text{K}} = 21.68 \text{ J/mol} \cdot \text{K}$

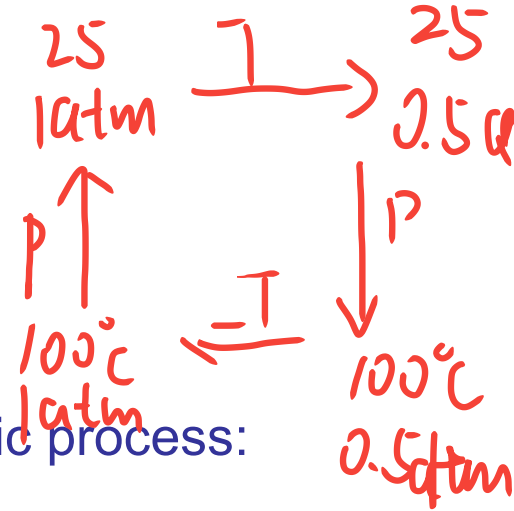
$$c_p = \frac{\Delta H}{n\Delta T} = \frac{3000}{373-273} \text{ J/mol} \cdot \text{K} = 30 \text{ J/mol} \cdot \text{K}$$

$$C_v = \frac{\Delta U}{\Delta T}$$
$$C_p = \frac{\Delta H}{\Delta T}$$

Chapter 2

2.7 One mole of a monatomic ideal gas at 25 °C and 1 atm undergoes the following reversibly conducted cycle:

- An isothermal expansion to 0.5 atm, followed by
- An isobaric expansion to 100 °C, followed by
- An isothermal compression to 1 atm, followed by
- An isobaric compression to 25 °C



The system then undergoes the following reversible cyclic process:

- An isobaric expansion to 100 °C, followed by
- A decrease in pressure at constant volume to the pressure P atm, followed by
- An isobaric compression at P atm to 24.5 liters, followed by
- An increase in pressure at constant volume to 1 atm

$$P_1 \propto \frac{TV}{298} = R \quad w_1 = nRT \ln \frac{P_1}{P_2}$$

$$213 \quad 298 \text{ K} \quad w_2 = \int P dv$$

Calculate the value of P which makes the work done on the gas during the first cycle equal to the work done by the gas during the second cycle.



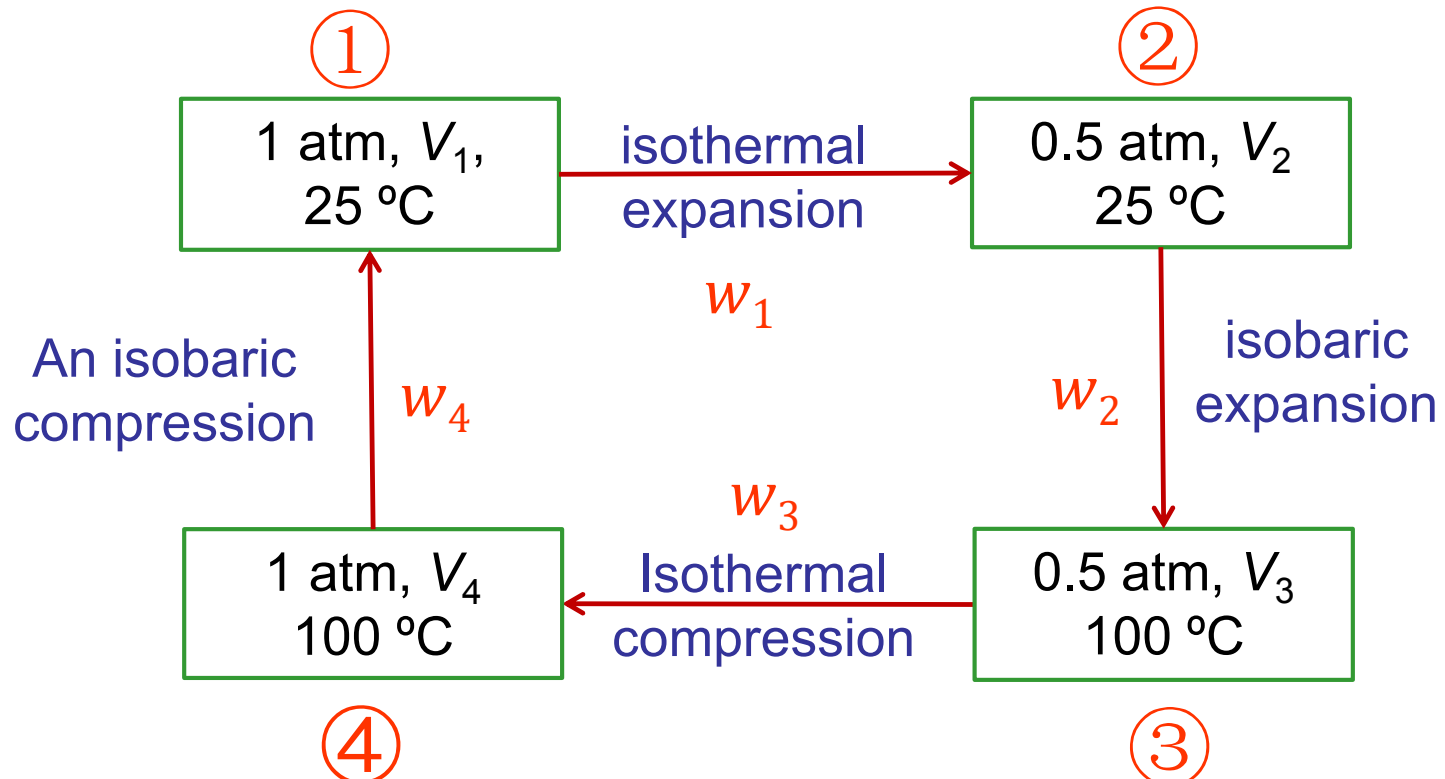
Chapter 2

Solution to 2.7

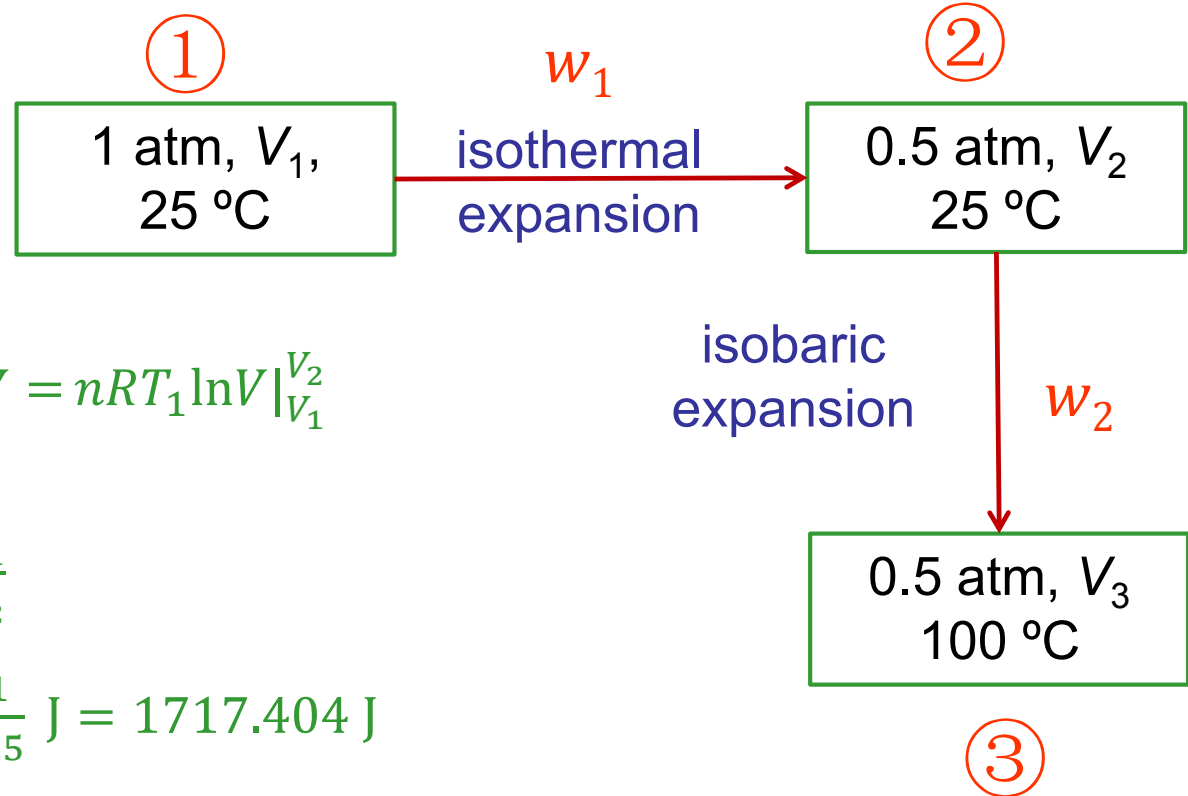
One mole of an monatomic ideal gas

The first reversible
cyclic process

- An isothermal expansion to 0.5 atm,
- An isobaric expansion to 100 °C,
- An isothermal compression to 1 atm,
- An isobaric compression to 25 °C



Chapter 2



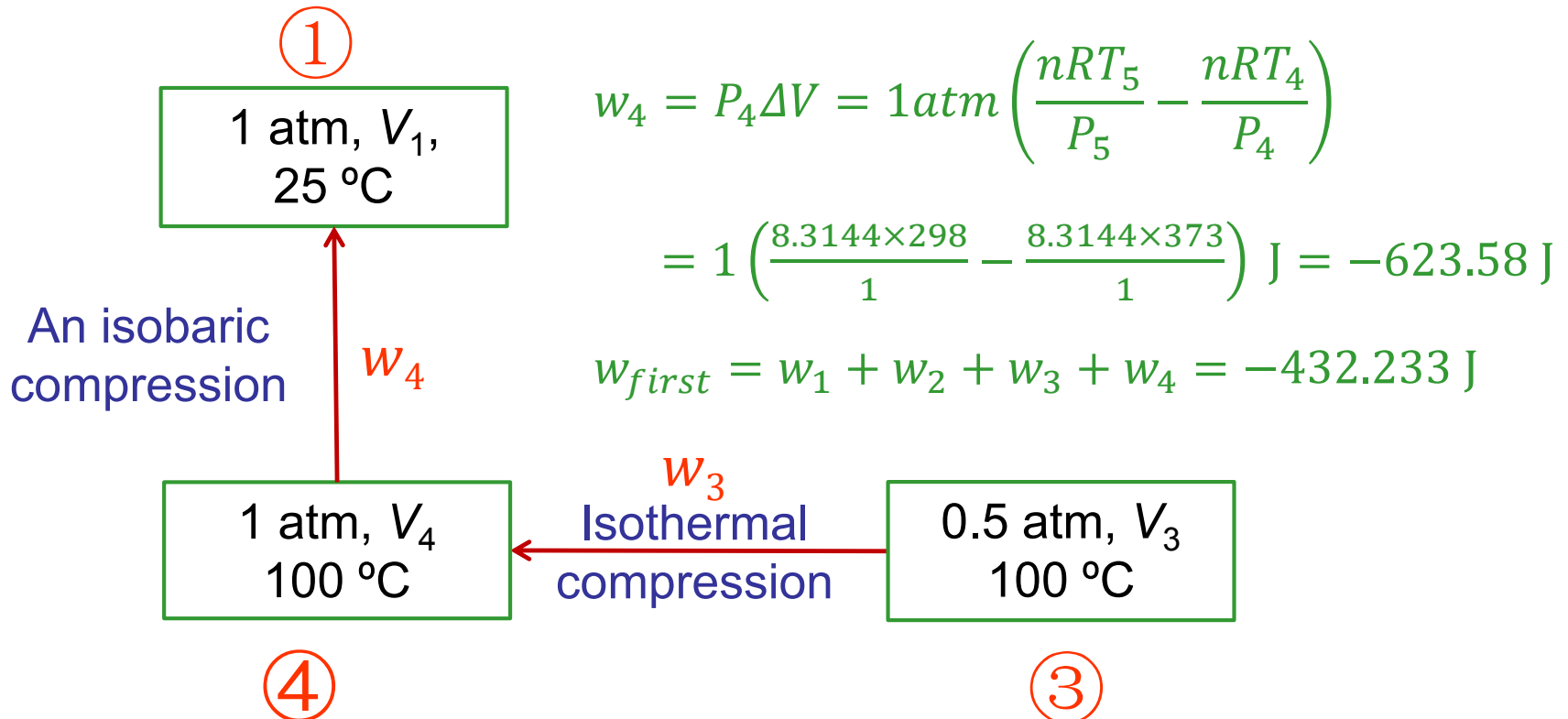
$$\begin{aligned}w_1 &= \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{nRT_1}{V} dV = nRT_1 \ln V \Big|_{V_1}^{V_2} \\&= nRT_1 \ln \frac{V_2}{V_1} = nRT_1 \ln \frac{P_1}{P_2} \\&= 1 \times 8.3144 \times 298 \ln \frac{1}{0.5} \text{ J} = 1717.404 \text{ J}\end{aligned}$$

$$\begin{aligned}w_2 &= P_2 \Delta V = 0.5 \text{ atm} \left(\frac{nRT_3}{P_3} - \frac{nRT_2}{P_2} \right) \\&= 0.5 \left(\frac{8.3144 \times 373}{0.5} - \frac{8.3144 \times 298}{0.5} \right) \text{ J} = 623.58 \text{ J}\end{aligned}$$

Chapter 2

$$w_3 = \int_{V_3}^{V_4} P dV = \int_{V_3}^{V_4} \frac{nRT_3}{V} dV = nRT_3 \ln V \Big|_{V_3}^{V_4} = nRT_3 \ln \frac{V_4}{V_3} = nRT_3 \ln \frac{P_3}{P_4}$$

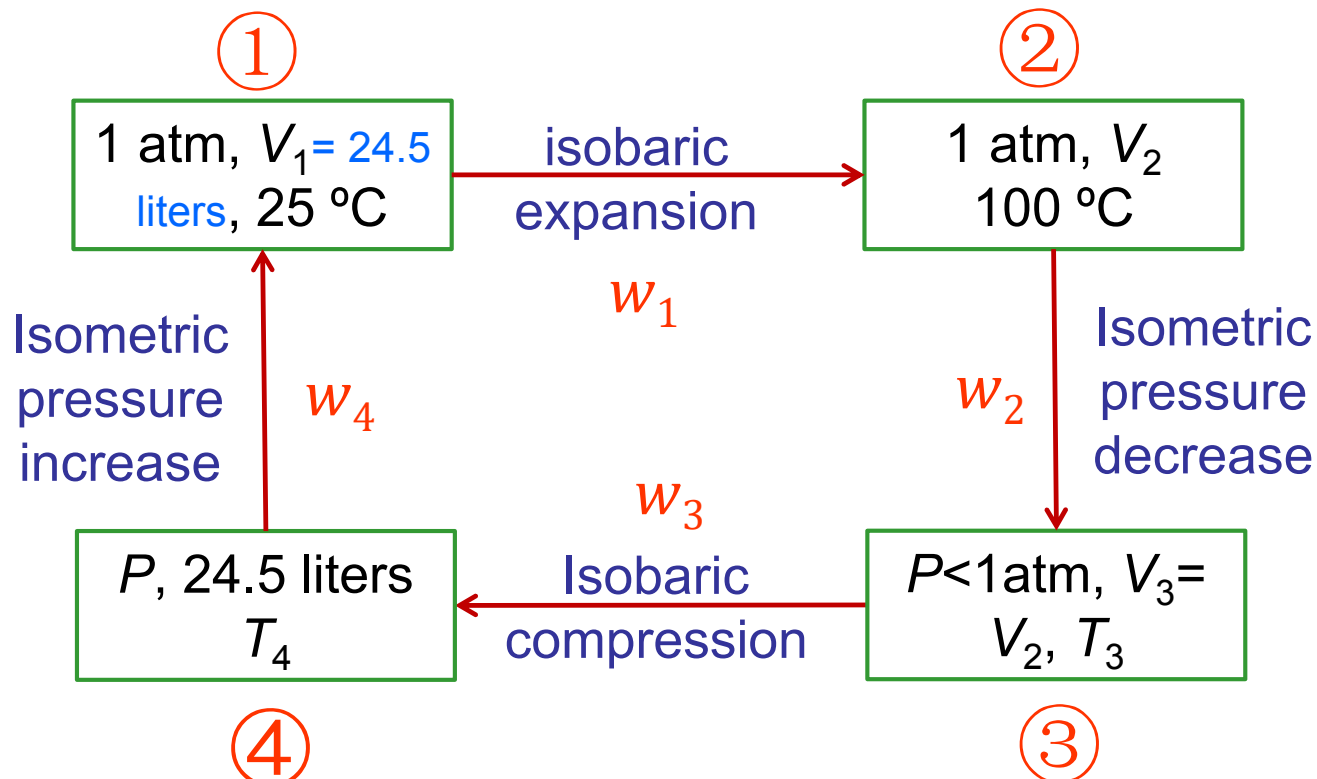
$$= 1 \times 8.3144 \times 373 \ln \frac{0.5}{1} \text{ J} = -2149.637 \text{ J}$$



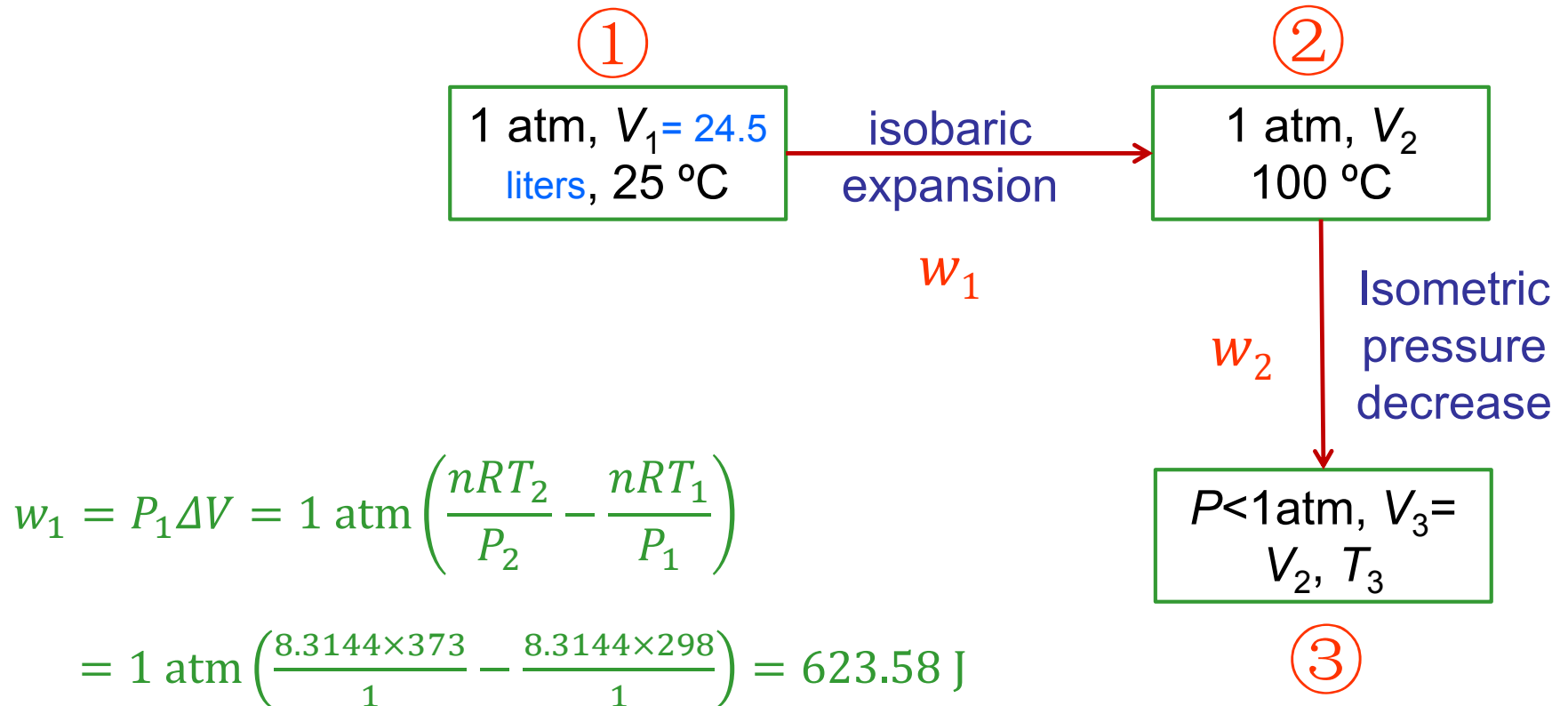
Chapter 2

The second reversible cyclic process

- An isobaric expansion to 100 °C,
- A decrease in pressure at constant volume to the pressure P atm,
- An isobaric compression at P atm to 24.5 liters
- An increase in pressure at constant volume to 1 atm



Chapter 2



$$= P \times 101325 \left(24.5 \times 10^{-3} - \frac{8.3144 \times 373}{101325} \right) \text{ J} = -618.8087 P \text{ J}$$

①

$$w_4 = 0$$
 w_4

$$= (623.5875 - 618.8087P) \text{ J}$$

 w_3
$$P < 1 \text{ atm}, V_3 = V_2, T_3$$

④

③

Chapter 2

Because the work done on the gas during the first cycle equal to the work done by the gas during the second cycle,

$$W_{\text{second}} = -W_{\text{first}}$$

That is, $(623.5875 - 618.8087P) \text{ J} = 432.233 \text{ J}$

So $P = (623.5875 - 432.233) \div 618.8087 = 0.3 \text{ atm}$

Chapter 3

3.1 The initial state of 1 mole of a monatomic ideal gas is $P = 10 \text{ atm}$ and $T = 300 \text{ K}$. Calculate the change in the entropy of the gas for

a. An isothermal decrease in the pressure to 5 atm

b. A reversible adiabatic expansion to a pressure of 5 atm

c. A constant-volume decrease in the pressure to 5 atm

Handwritten solutions for the three cases:

a. $PV = nRT$
 $V = \frac{R300}{10}$
 $S = \int \frac{q}{T} d$

b. $\Delta S = 0$

c. $V = \frac{R300}{10}$
 $\Delta S = \int \frac{q}{T} dU$

Diagram showing the initial state (10 atm, $\frac{R300}{10}$, $T = 300 \text{ K}$) and the final state (5 atm, $\frac{R300}{10}$, $T = 150$).

Equations for the final state:

$$\frac{PV}{AT} = \frac{PV}{T} \quad \frac{10}{300} \frac{5}{150}$$

$$\int \frac{nC_v dT}{T} = nR \ln \frac{1}{2}$$

Chapter 3

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: $P_1 = 10 \text{ atm}$, $T_1 = 300 \text{ K}$, $V_1 = ??$

Process: An isothermal decrease in the pressure

Final state: $P_2 = 5 \text{ atm}$, $T_2 = 300 \text{ K}$, $V_2 = ??$

a. Isothermal process, $\Delta U = 0$,

$$\text{so, } q = w = \int_{V_1}^{V_2} P dV = nRT \ln \frac{V_2}{V_1}$$

$$\Delta S = \frac{q}{T} = nR \ln \frac{V_2}{V_1} = nR \ln \frac{P_1}{P_2} = 1 \times 8.3144 \times \ln \frac{10}{5} \text{ J/K} = 5.76 \text{ J/K}$$

Chapter 3

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: $P_1 = 10 \text{ atm}$, $T_1 = 300 \text{ K}$, $V_1 = ??$

Process: Reversible adiabatic expansion

Final state: $P_2 = 5 \text{ atm}$, $T_2 = ??$, $V_2 = ??$

b. Reversible adiabatic process, $q = 0$, $\Delta S = 0$.

Chapter 3

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: $P_1 = 10 \text{ atm}$, $T_1 = 300 \text{ K}$, $V_1 = ??$

Process: Isometric decrease in the pressure

Final state: $P_2 = 5 \text{ atm}$, $T_2 = ??$, $V_2 = ??$

c. Constant-volume process, $\Delta V = 0$, so $w = 0$,

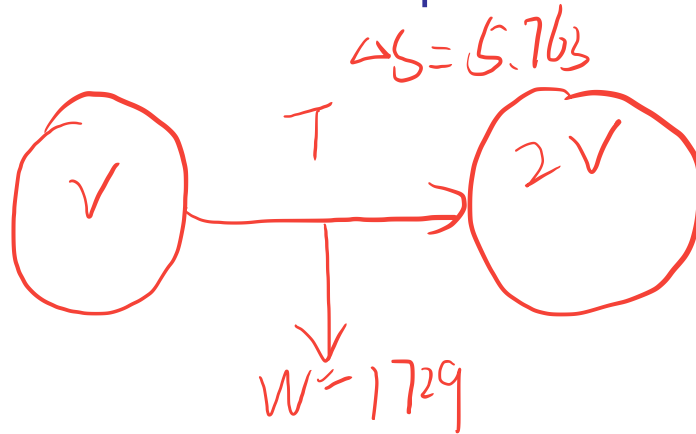
$$q = \Delta U = \int n c_v dT$$

$$\Delta S = \int_{T_1}^{T_2} \frac{n c_v}{T} dT = n c_v \ln \frac{T_2}{T_1} = n c_v \ln \frac{P_2}{P_1}$$

$$= 1 \times 1.5 \times 8.3144 \ln \frac{5}{10} \text{ J/K} = -8.64 \text{ J/K}$$

Chapter 3

3.3 One mole of a monatomic ideal gas undergoes a reversible expansion at constant pressure, during which the entropy of the gas increases by 14.41 J/K and the gas absorbs 6236 J of thermal energy. Calculate the initial and final temperatures of the gas. One mole of a second monatomic ideal gas undergoes a reversible isothermal expansion, during which it doubles its volume, performs 1729 J of work, and increases its entropy by 5.763 J/K. Calculate the temperature at which the expansion was conducted.



$$\Delta S = \frac{W}{T} = \frac{1729}{5.763} \quad T = 300 \text{ K}$$

Chapter 3

Solution to 3.3

(i) Reversible expansion at constant pressure,

Substance: One mole of a monatomic ideal gas

Process: The entropy of the gas increases by 14.41 J/K

The gas absorbs 6236 J of thermal energy

$$\Delta S = nc_p \ln \frac{T_2}{T_1} = 1 \times 2.5 \times 8.3144 \times \ln \frac{T_2}{T_1} \text{ J/K} = 14.41 \text{ J/K}$$

Solve it,
$$\frac{T_2}{T_1} = 2 \quad (1)$$

$$q_p = nc_p(T_2 - T_1) = 1 \times 2.5 \times 8.3144 \times (T_2 - T_1) \text{ J} = 6236 \text{ J}$$

Solve it,
$$(T_2 - T_1) = 300 \text{ K} \quad (2)$$

Combining (1) and (2), $T_1 = 300 \text{ K}$, $T_2 = 600 \text{ K}$.

Chapter 3

Solution to 3.3

(ii) Reversible isothermal expansion,

Substance: One mole of a monatomic ideal gas

Process:

it doubles its volume

performs 1729 J of work

increases its entropy by 5.763 J/K

$$\Delta U = 0,$$

So

$$w = q = T\Delta S$$

$$T = \frac{w}{\Delta S} = \frac{1729}{5.763} \text{ K} = 300\text{K}$$

Chapter 3

3.5 One mole of copper at a uniform temperature of 0 °C is placed in thermal contact with a second mole of copper which, initially, is at a uniform temperature of 100 °C. Calculate the temperature of the 2-mole system, which is contained in an adiabatic enclosure, when thermal equilibrium is attained.

Why is the common uniform temperature not exactly 50 °C?

How much thermal energy is transferred, and how much entropy is produced by the transfer?

The constant-pressure molar heat capacity of solid copper varies with temperature as $c_p = 22.64 + 6.28 \times 10^{-3} T$ J/mole·K

Chapter 3

Solution to 3.5

According to the item, $T_1 = 273.15\text{K}$, $T_2 = 373.15\text{K}$,

$$q = \int nc_p dT = \int (22.64 + 6.28 \times 10^{-3}T) dT$$

$$\begin{aligned} q_{total} &= \int_{T_1}^{T_e} nc_p dT + \int_{T_2}^{T_e} nc_p dT \\ &= [22.64T + 3.14 \times 10^{-3}T^2]_{T_1}^{T_e} + [22.64T + 3.14 \times 10^{-3}T^2]_{T_2}^{T_e} \\ &= 0 \text{ (adiabatic enclosure system)} \end{aligned}$$

Solve this, $T_e = 323.46 \text{ K}$

Chapter 3

$$q_{trans} = \int_{T_1}^{T_e} n c_p dT = [22.64T + 3.14 \times 10^{-3} T^2]_{T_1}^{T_e} = 1236.92 \text{ J}$$

$$\Delta S_{irr} = \int_{T_1}^{T_e} \frac{n c_p}{T} dT + \int_{T_2}^{T_e} \frac{n c_p}{T} dT$$

$$= [22.64 \ln T + 6.28 \times 10^{-3} T]_{T_1}^{T_e} + [22.64 \ln T + 6.28 \times 10^{-3} T]_{T_2}^{T_e}$$

$$= 0.619 \text{ J/K}$$

Chapter 3

Solution to 3.5

The final temperature is $T_e = 323.46$ K, which is greater than 323 K because the heat capacity increases with increasing temperature.

Thus the decrease in temperature caused by withdrawing heat q from hot copper is less than the increase in temperature caused by adding heat q to cold copper.

Review of chapter 1 to 3

Xiao Han (韩潇)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

xiao.han@nwpu.edu.cn

Chapter 1

- ✓ **Heat** is identified as the process in which energy is transferred from one region to another down a temperature gradient.
- ✓ **Work** is identified as the process in which energy is transferred from one region to another not due to a temperature gradient.

Chapter 1

- ✓ The **system** is that part of the universe we wish to investigate in detail.
- ✓ The **surroundings** is that part of the universe outside the system which may interact with it by exchanging energy or matter.
- ✓ The **simple thermodynamic systems**, is the case in which the surroundings interacts with the system only via pressure and temperature changes. The composition remains constant in simple systems

Chapter 1

- ✓ *Isolated systems*: In these systems, no work is done on or by the system. In addition, energy or matter may not enter or leave it. Thus, the energy of these systems remains constant, as does the overall composition. Isolated systems are therefore unaffected by changes in the surroundings.
- ✓ *Closed systems*: These are systems which may receive (or give off) energy from (or to) the surroundings. The boundaries are called diathermal; that is, they allow thermal energy to transfer through them into or out of the system.
- ✓ *Open systems*: These are systems which can exchange both energy and matter with the surroundings. Neither the energy nor the composition of these systems need remain constant.

Chapter 1

The boundaries or walls of the system are classified as follows:

- ✓ *Adiabatic*: No thermal energy can pass through.
- ✓ *Diathermal*: Thermal energy can pass through.
- ✓ *Permeable*: Matter can pass through.
- ✓ *Impermeable*: Matter cannot pass through.
- ✓ *Semipermeable*: Some components are able to pass through, while others are not.

Chapter 1

- ✓ If it was possible to know the masses, velocities, positions, and all modes of motion of all of the constituent particles in a system, this mass of knowledge would serve to describe the *microscopic state* of the system, which, in turn, would determine all of the properties of the system.
- ✓ In the absence of such detailed knowledge as is required to determine the microscopic state of the system, thermodynamics begins with a consideration of the properties of the system which, when determined, define the *macroscopic state* of the system; i.e., when all of the properties are fixed then the macroscopic state of the system is fixed.

Chapter 1

- ✓ *independent variables*
- ✓ *dependent variables*
- ✓ *thermodynamic variables*
- ✓ *Extrinsic properties*
- ✓ *equation of state*
- ✓ The isothermal compressibility with dimensions of P^{-1} .
- ✓ The coefficient of thermal expansion with dimensions of T^{-1} .
- ✓ Absolute zero of temperature
- ✓ the ideal gas temperature scale

$$\beta_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

Chapter 1

$$1 \text{ atm} = 101325 \text{ newtons/meter}^2$$

$$1 \text{ liter} \cdot \text{atm} = 101.325 \text{ newton} \cdot \text{meters} = 101.325 \text{ joules}$$

$$R = 0.082057 \text{ liter} \cdot \text{atm/degree} \cdot \text{mole} = 8.314 \text{ joules/degree} \cdot \text{mole}$$

- ✓ Extensive properties have values which depend on the size of the system, and the values of intensive properties are independent of the size of the system.
- ✓ Volume is an extensive property, and temperature and pressure are intensive properties.

Chapter 1

- ✓ A **phase** is defined as being a finite volume in the physical system within which the properties are uniformly constant, i.e., do not experience any abrupt change in passing from one point in the volume to another. Within any of the one phase areas in the phase diagram, the system is said to be **homogeneous**.
- ✓ The system is **heterogeneous** when it contains two or more phases, e.g., coexisting ice and liquid water (on the line AO) is a heterogeneous system comprising two phases, and the phase boundary between the ice and the liquid water is that very thin region across which the density changes abruptly from the value for homogeneous ice to the higher value for liquid water.

Chapter 1

- ✓ **The First Law** not only states that the energy of the universe is conserved but it also posits that the various forms of energy (e.g., thermal, electrical, magnetic, and mechanical) can be converted into other forms of energy. When first delineated, it was the fact that thermal energy (heat) could be converted to mechanical work that was of special interest (heat engines). This law also defines an important extensive thermodynamic state function called the internal energy, U , of the system under investigation.

Chapter 1

- ✓ Although the Second Law is the one that often gets the most attention in popular discussions of science. This law allows us to make important predictions of the direction in which a system will evolve with time during spontaneous processes, if other important caveats are taken into account.
- ✓ The Second Law introduces another important extensive thermodynamic state function called entropy, S . A short version of the Second Law is that the entropy of the universe never decreases.

Chapter 1

- ✓ The Third Law states that when a system which is in complete internal equilibrium approaches the absolute zero in temperature, all of the aspects of its entropy approach zero.
- ✓ Sometimes, the Third Law is stated as follows: a system can never be taken to the absolute zero in temperature. This is also called the unattainability principle .

Chapter 2

- ✓ *a mechanical equivalent of heat*
- ✓ In describing work, it is conventional to assign a negative value to work done on a body and a positive value to work done by a body
- ✓ In describing heat changes it is conventional to assign a negative value to heat which flows out of a body (an **exothermic** process) and a positive value to heat which flows into a body (an **endothermic** process).

$$\Delta U = U_B - U_A = q - w$$

$$dU = \delta q - \delta w$$

The first law of thermodynamics

- ✓ The use of the symbol “ d ” indicates a differential element of a state function or state property, the integral of which is independent of the path, and use of the symbol “ δ ” indicates a differential element of some quantity which is not a state function.

Chapter 2

$$U = U(V, T) \quad dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- ✓ In this manner, we can examine processes in which the volume V is maintained constant (**isochore** or **isometric** processes), or the pressure P is maintained constant (**isobaric** processes), or the temperature T is maintained constant (**isothermal** processes). We can also examine **adiabatic** processes in which $q=0$.

Isometric process $dU = \delta q_V \quad \Delta U = q_V$

Isobaric process $H = U + PV$

$$(U_2 + P_2 V_2) - (U_1 + P_1 V_1) = H_2 - H_1 = q_P$$

Chapter 2

Heat capacity $C = \frac{q}{\Delta T} \quad C = \frac{\delta q}{\delta T}$

$$C_V = \left(\frac{\delta q}{dT} \right)_V \quad \text{The heat capacity at constant volume}$$

$$C_P = \left(\frac{\delta q}{dT} \right)_P \quad \text{The heat capacity at constant pressure}$$

$$C_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{dU'}{dT} \right)_V \quad dU' = C_V dT$$

$$C_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH'}{dT} \right)_P \quad dH' = C_P dT$$

Chapter 2

- ✓ The **specific heat** of the system is the heat capacity per gram at constant P , and the **molar heat capacity** is the heat capacity per mole at constant pressure or at constant volume. Thus, for a system containing n moles:

$$C_V = nc_V$$

$$C_P = nc_P$$

Chapter 2

$$\frac{PdV}{dT} \text{ or } P\left(\frac{\partial V}{\partial T}\right)_P \quad \rightarrow \quad c_P - c_V = P\left(\frac{\partial V}{\partial T}\right)_P$$

$$H = U + PV$$

$$c_P = \left(\frac{\delta q}{dT}\right)_P = \left(\frac{dH}{dT}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P$$

$$c_V = \left(\frac{\delta q}{dT}\right)_V = \left(\frac{dU}{dT}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V$$

Chapter 2

$$U = U(V, T)$$

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V$$

$$\begin{aligned} c_P - c_V &= \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V \\ &= \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right] \end{aligned}$$

Chapter 2

$$P \left(\frac{\partial V}{\partial T} \right)_P$$

- ✓ The work done by the system per degree rise in temperature in expanding against the constant external pressure P acting on the system.

$$\left(\frac{\partial V}{\partial T} \right)_P \left(\frac{\partial U}{\partial V} \right)_T$$

- ✓ The work done per degree rise in temperature in expanding against the internal cohesive forces acting between the constituent particles of the substance.

Chapter 2

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$



Divided by dT

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \frac{dV}{dT}$$



$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$

Chapter 2

Prove:
$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

Suppose: $U = U(T, V), \quad V = V(T, P)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP$$

Substituting the expression of dV , we have:

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T \left[\left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP \right]$$

Chapter 2

Rearranging to separate dP from dT , we have:

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \\ &= \left(\frac{\partial U}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \end{aligned}$$

Because U is a function of T and P . $U = U(T, P)$

$$dU = \left(\frac{\partial U}{\partial P} \right)_T dP + \left(\frac{\partial U}{\partial T} \right)_P dT$$

According to the two expression of dU , we have:

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$

Chapter 2

✓ For an ideal gas

$$c_P - c_V = R$$

$\left(\frac{\partial U}{\partial V}\right)_T = 0$	$\left(\frac{\partial U}{\partial P}\right)_T = 0$	$\left(\frac{\partial H}{\partial P}\right)_T = 0$	$\left(\frac{\partial H}{\partial V}\right)_T = 0$
--	--	--	--

✓ For a gas

The throttle process is an **isenthalpic** process.

Chapter 2

✓ For a phase-transformation

$$H_S(T_1) - H_L(T_1) = H_S(T_2) - H_L(T_2) + \int_{T_2}^{T_1} (c_P^S - c_P^L) dT$$

$$\left(\frac{\partial \Delta H}{\partial T} \right)_P = \Delta c_P$$

✓ Reversible adiabatic processes

$$dU = c_V dT$$

$$\delta w = P dV$$

$$C_V dT = -P dV$$

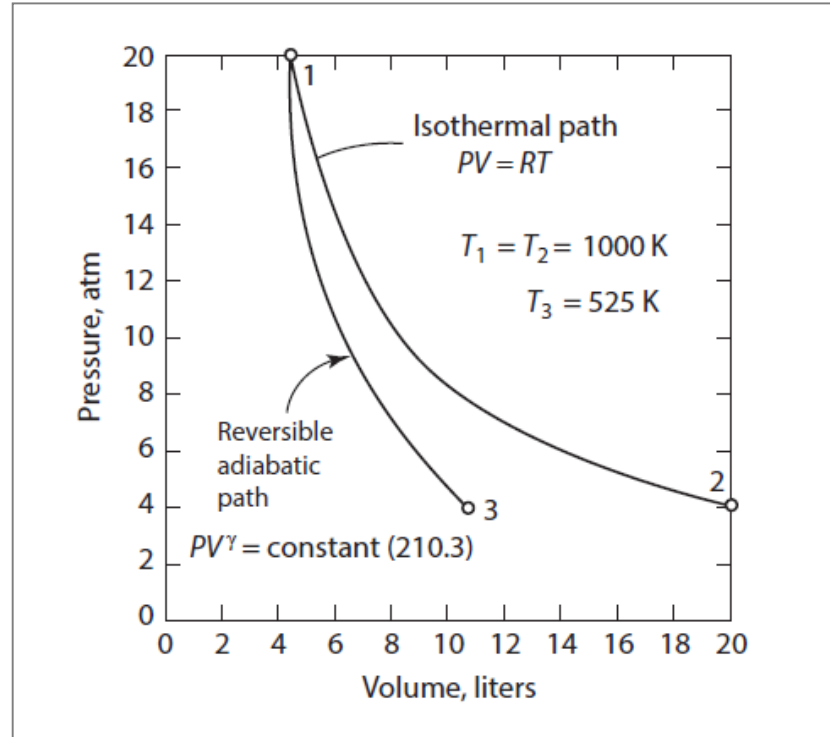
$$P_2 V_2^\gamma = P_1 V_1^\gamma = \text{constant}$$

$$\left(\frac{T_2}{T_1} \right) = \left(\frac{V_1}{V_2} \right)^{\gamma-1} \quad \gamma = \frac{5}{3}$$

Chapter 2

✓ Reversible isothermal process

$$w = q = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$



Chapter 2

$$dU = \delta q - \sum \delta w_i$$

✓ Magnetic Work

$$\delta w' = -V \mu_0 \mathfrak{N} d\mathbf{M}$$

$$\delta w' = -V \mu_0 \int_{M_1}^{M_2} \mathfrak{N} d\mathbf{M}$$

✓ Electrical Work

$$\delta w' = -V \mathbf{E} d\mathbf{D}$$

$$w' = -V \int_{D_1}^{D_2} \mathbf{E} d\mathbf{D}$$

✓ Surface work

$$\delta w' = -\gamma dA$$

$$\delta w' = -\sigma dA$$

Chapter 3

- ✓ A process which involves the spontaneous movement of a system from a non-equilibrium state to an equilibrium state is called **a natural or spontaneous process**.
- ✓ As such a process cannot be reversed without the application of an external agency (a process which would leave a permanent change in the external agency), such a process is said to be irreversible. (The terms natural, spontaneous, and irreversible are synonymous in this context.)

Chapter 3

- ✓ If it is considered that an irreversible process is one in which the energy of the system undergoing the process is degraded, then the possibility that the extent of degradation can differ from one process to another suggests that a *quantitative* measure of the extent of degradation, or degree of irreversibility, exists.
- ✓ As the degree of irreversibility of a process is variable, it should be possible for the process to be conducted in such a manner that the degree of irreversibility is minimized. The ultimate of this minimization is a process in which the degree of irreversibility is zero and in which no degradation occurs.

Chapter 3

- ✓ This limit, toward which the behavior of real systems can be made to approach, is the **reversible process**. If a process is reversible, then the concept of spontaneity is no longer applicable.
- ✓ Spontaneity occurred as a result of the system moving, of its own accord, from a non-equilibrium state to an equilibrium state.

Chapter 3

- ✓ If the spontaneity is removed, it is apparent that at all times during the process, the system is at equilibrium. Thus **a reversible process** is the one during which the system is never away from equilibrium, and a reversible process which takes the system from the state A to the state B is one in which the process path passes through a continuum of equilibrium states. Such a path is, of course, **imaginary**.
- ✓ But it is possible to conduct an actual process in such a manner that it is virtually reversible. Such an actual process (**a quasi-static process**) is one which proceeds under the influence of an infinitesimally small driving force such that, during the process, the system is never more than an infinitesimal distance from equilibrium. If, at any point along the path, the external influence is removed, then the process ceases, or, if the direction of the external influence is reversed, then the direction of the process is reversed.

Chapter 3

✓ Three points have emerged from the discussion so far:

1. Entropy is a thermodynamic state variable (function).
2. Entropy is not created when a system undergoes a reversible process; entropy is transferred from one part of the system/surroundings to another part.
3. The total entropy of the universe increases when an irreversible process occurs.
4. For all processes, we can write $\Delta S_{\text{system}} = q/T + \Delta S_{\text{irr}}$ and $q \leq q_{\text{rev}}$.
5. For all processes, the entropy of the universe increases or stays the same. The total entropy of the universe never decreases.

Chapter 3

- ✓ A **heat engine** is a device which converts heat into work, and it is interesting to note that the first steam engine, which was built in 1769, was in operation for a considerable number of years before the reverse process, i.e., the conversion of work into heat, was investigated.
- ✓ In the operation of a heat engine, some of the energy that has been transferred from a high-temperature heat reservoir is converted into work, with the remainder of the energy being transferred to a low-temperature heat reservoir.

Chapter 3

- ✓ The efficiency of a heat engine is given by

$$\text{Efficiency} = \frac{\text{work obtained}}{\text{heat input}} = \frac{w}{q_2}$$

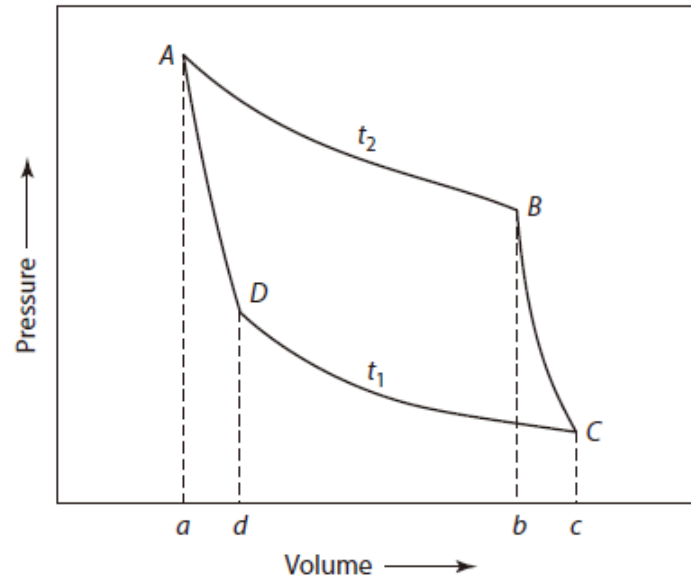


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

$$\text{Efficiency} = \frac{w}{q_2} = \frac{q_2 - q_1}{q_2} = 1 - \frac{q_1}{q_2}$$

Chapter 3

- ❑ The principle of Thomsen states that it is impossible, by means of a cyclic process, to take heat from a reservoir and convert it to work without, in the same operation, transferring heat to a cold reservoir.
- ❑ The principle of Clausius states that it is impossible to transfer heat from a cold to a hot reservoir without, in the same process, converting a certain amount of work to heat.

Chapter 3

- ✓ The efficiency of a Carnot cycle is:

$$\text{Efficiency} = \frac{q_2 - q_1}{q_2} = \frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2}$$

- ✓ This defines an absolute thermodynamic scale of temperature which is independent of the working substance. It is seen that the zero of this temperature scale is that temperature of the cold reservoir which makes the Carnot cycle 100% efficient.

Chapter 3

$$dS = \frac{\delta q_{rev}}{T}$$

- ✓ The Second Law of Thermodynamics can thus be stated as
 - The incremental change in entropy dS of a system into which (or out of which) thermal energy has been reversibly added (or expelled) is given as $dS = \delta q_{rev}/T$, and the function S is a state function of the system.
 - The entropy of a system in an adiabatic enclosure can never decrease.
 - The entropy increases during an irreversible process.
 - The entropy remains constant during a reversible process.
 - The entropy remains constant if the system was initially in equilibrium.

Chapter 3

The Second Law: $dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}}$

The First Law: $\delta q = dU_{\text{system}} + \delta w$

$$\left. \begin{array}{l} dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}} \\ \delta q = dU_{\text{system}} + \delta w \end{array} \right\} dS_{\text{system}} = \frac{dU_{\text{system}} + \delta w}{T} + dS_{\text{irr}}$$

✓ If the temperature remains constant throughout the process (and equal to the temperature of the reservoir supplying heat to the system), then integration from state A to state B gives

$$\delta w = TdS_{\text{system}} - dU_{\text{system}} - TdS_{\text{irr}}$$

$$\delta w \leq TdS_{\text{system}} - dU_{\text{system}}$$

$$w_{\text{max}} = q_{\text{rev}} = T(S_B - S_A) - (U_B - U_A) \quad \leftarrow w \leq T(S_B - S_A) - (U_B - U_A)$$

Chapter 3

- ✓ For an incremental change in the state of a closed system, the First Law of Thermodynamics gives:

$$dU = \delta q - \delta w$$

- ✓ If the process occurs reversibly, the Second Law of Thermodynamics gives:

$$dS = \frac{\delta q}{T} \text{ or } \delta q = TdS \qquad \delta w = PdV$$

- ✓ Combination of the two laws gives the equation

$$dU = TdS - PdV$$

Chapter 3

- ✓ In an *isolated* system of constant internal energy, U , and constant volume, V , equilibrium is attained when the entropy of the system is a maximum, consistent with the fixed values of U and V .
- ✓ This *spontaneous* process is, by definition, irreversible, and during the movement of the system from its initial non-equilibrium state to its final equilibrium state the entropy of the system increases. The attainment of the equilibrium state coincides with the entropy reaching a *maximum value*, and hence entropy can be used as a criterion for determination of the equilibrium state.

Thermodynamics of Materials & Phase transformations

The End



Exercise for Chapter 4 and 5

Xiao Han (韩潇)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

xiao.han@nwpu.edu.cn

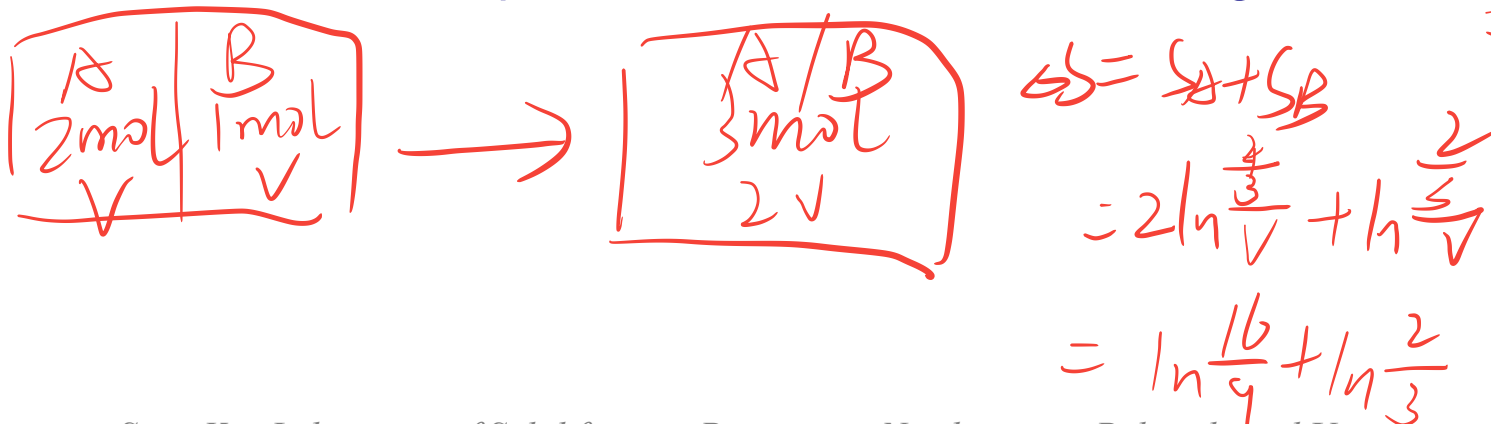
Chapter 4

4.1. A rigid container is divided into two compartments of equal volume by a partition.

One compartment contains 1 mole of ideal gas *A* at 1 atm, and the other contains 1 mole of ideal gas *B* at 1 atm. Calculate the increase in entropy which occurs when the partition between the two compartments is removed.

If the first compartment had contained 2 moles of ideal gas *A*, what would have been the increase in entropy when the partition was removed?

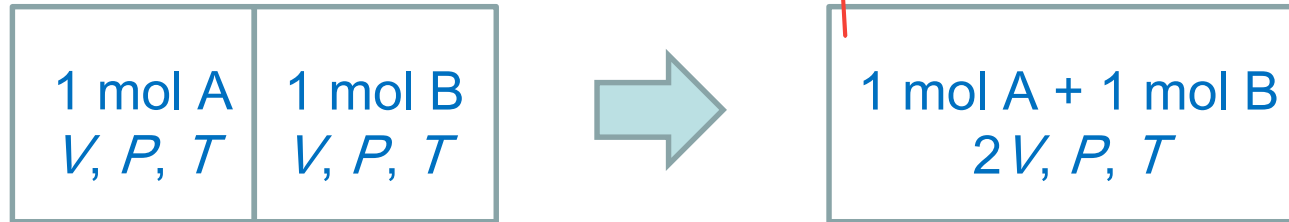
Calculate the corresponding increases in entropy in each of the preceding two situations if both compartments had contained ideal gas *A*.


$$\begin{aligned} \Delta S &= S_A + S_B \\ &= 2 \ln \frac{4}{3} + 1 \ln \frac{2}{3} \\ &= \ln \frac{16}{9} + \ln \frac{2}{3} \end{aligned}$$

Chapter 4

Solution to 4.1

(1)

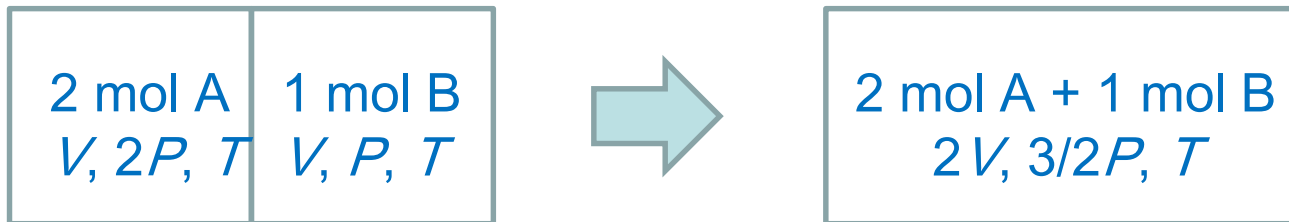


The volume occupied by 1 mol of A doubles $\therefore \Delta S_A = R \ln 2$

The volume occupied by 1 mol of B doubles $\therefore \Delta S_B = R \ln 2$

$$\therefore \Delta S_{\text{Total}} = \Delta S_A + \Delta S_B = 2R \ln 2$$

(2)



The volume occupied by 2 mol of A doubles $\therefore \Delta S_A = 2R \ln 2$

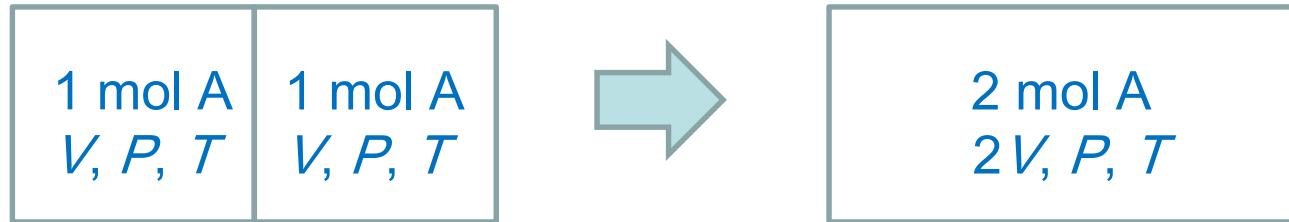
The volume occupied by 1 mol of B doubles $\therefore \Delta S_B = R \ln 2$

$$\therefore \Delta S_{\text{Total}} = \Delta S_A + \Delta S_B = 3R \ln 2$$

Chapter 4

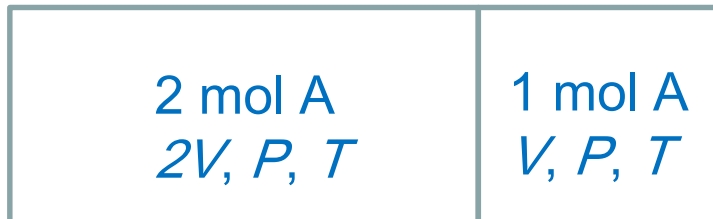
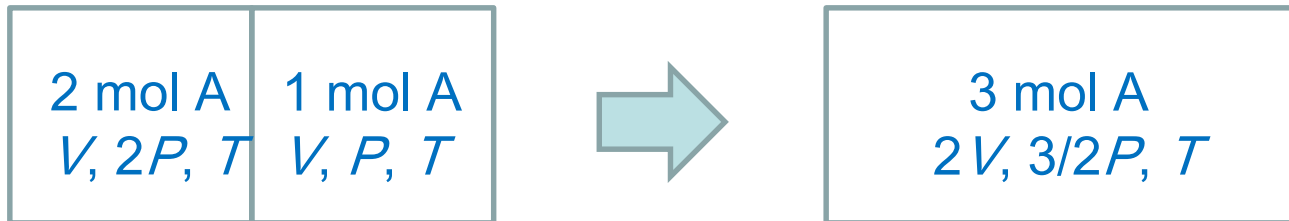
Solution to 4.1

(3)



$\Delta S = 0$ because the state of the gas doesn't change.

(4)



Double the volume of the 2 mol of A $\Delta S = 2R \ln 2$

Remove the partition $\Delta S = 0$

Decrease the volume of the 3 mol of A by the factor of $2/3$ $\Delta S = 3R \ln(2/3)$

$$\therefore \Delta S_{\text{Total}} = 2R \ln 2 + 3R \ln(2/3) = R \ln(32/27)$$

Chapter 4

4.2 Show that, when n atoms of A and n atoms of B form a randomly mixed solution, the fraction of the total number of distinguishable complexions which occur in the most probable distribution decreases with the increasing value of n .

Chapter 4

Solution to 4.2

The total number of distinguishable complexions is

$$\Omega_{\text{total}} = \frac{(2n)!}{n!n!} \quad (1)$$

the number of complexions which is the most probable distribution is

$$\Omega_{\text{max}} = \left[\frac{n!}{(0.5n)!(0.5n)!} \right] \left[\frac{n!}{(0.5n)!(0.5n)!} \right] \quad (2)$$

So the ratio of (2) to (1) is the fraction of the total number of distinguishable complexions which occur in the most probable distribution

$$f(n) = \frac{\Omega_{\text{max}}}{\Omega_{\text{total}}} = \frac{\left[\frac{n!}{(0.5n)!(0.5n)!} \right] \left[\frac{n!}{(0.5n)!(0.5n)!} \right]}{\frac{(2n)!}{n!n!}} = \frac{(n!)^4}{((0.5n)!)^4 (2n)!}$$

which decreases with the increasing value of n .

Chapter 4

4.4 On the assumption that copper–nickel alloys are random mixtures of copper and nickel atoms, calculate the mass of copper which, when mixed with 100 g of nickel, causes an increase in entropy of 15 J/K. The gram atomic weights of Cu and Ni are, respectively, 63.55 and 58.69.

Chapter 4

Solution to 4.4

Assume $n_{\text{Cu}} = \frac{m_{\text{Cu}}}{63.55\text{g/mol}} = n \text{ mol}$ $n_{\text{Ni}} = \frac{100\text{g}}{58.69\text{g/mol}} = 1.70 \text{ mol}$

$$X_{\text{Cu}} = \frac{n_{\text{Cu}}}{n_{\text{Cu}} + n_{\text{Ni}}} = \frac{n}{n + 1.70} \quad X_{\text{Ni}} = \frac{n_{\text{Ni}}}{n_{\text{Cu}} + n_{\text{Ni}}} = \frac{1.70}{n + 1.70} = 79.17\%$$

$$\begin{aligned} \Delta S_{\text{conf}} &= -k_{\text{B}}(n_{\text{Cu}} + n_{\text{Ni}})(X_{\text{Cu}} \ln X_{\text{Cu}} + X_{\text{Ni}} \ln X_{\text{Ni}}) \\ &= -(n_{\text{Cu}} + 1.70) \times 8.3144 \times \left(\frac{n}{n + 1.70} \ln \frac{n}{n + 1.70} + \frac{1.70}{n + 1.70} \ln \frac{1.70}{n + 1.70} \right) \\ &= 15 \text{ J/K} \end{aligned}$$

Solve it, $n_{\text{Cu}}=1.025 \text{ mol}$, $m_{\text{Cu}}=65.14\text{g}$.

Chapter 4

4.5 We saw in Sec. 4.6 (Eq. 4.20) that $A = -nk_{\text{B}}T \ln Z$. We will see in chapter 5 that: $(\partial A / \partial T)_V = -S$. Perform the differentiation of A with respect to temperature and obtain the relationship for the entropy of an isolated system in terms of its partition function Z ; further obtain the relationship for the internal energy of an isolated system in terms of its partition function Z

Chapter 4

Solution to 4.5 $A = -nk_{\text{B}}T \ln Z$

$$\left(\frac{\partial A}{\partial T} \right)_V = -S = -nk_{\text{B}} \ln Z - nk_{\text{B}}T \frac{\partial \ln Z}{\partial T}$$

Thus: $S = nk_{\text{B}} \ln Z + nk_{\text{B}}T \frac{\partial \ln Z}{\partial T}$

Note that if we multiply both sides by T we obtain:

$$TS = nk_{\text{B}}T \ln Z + nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T}$$

$$-nk_{\text{B}}T \ln Z = nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T} - TS$$

Thus: $U = nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T}$

Chapter 4

4.6 A weak magnetic field is applied to a system of up and down spins. The up spins have a slightly lower energy state than down spins. This is because the up spins are favored by the weak magnetic field. Thus,

$$\varepsilon^{\uparrow} < \varepsilon^{\downarrow}$$

a. Determine the partition function for this system under the influence of the weak magnetic field.

b. Determine the ratio of $\frac{n^{\uparrow}}{n^{\downarrow}}$ for very high temperatures and very low temperatures.

Chapter 4

Solution to 4.6

a.
$$Z = \sum \exp\left(\frac{-\varepsilon_i}{k_B T}\right) = \exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)$$

$$\frac{n^\uparrow}{n} = \frac{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)} \quad \frac{n^\downarrow}{n} = \frac{\exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)}$$

Thus:
$$\frac{n^\uparrow}{n^\downarrow} = \frac{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)} = \exp\left(\frac{\varepsilon^\downarrow - \varepsilon^\uparrow}{k_B T}\right)$$

Chapter 4

Solution to 4.6

b.

$$\frac{n^{\uparrow}}{n^{\downarrow}} = \frac{\exp\left(\frac{-\varepsilon^{\uparrow}}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^{\downarrow}}{k_B T}\right)} = \exp\left(\frac{\varepsilon^{\downarrow} - \varepsilon^{\uparrow}}{k_B T}\right)$$

$T \rightarrow \infty$, $n \rightarrow 1$, the numbers of up and down of are equal

The numerator of the exponential is positive: Thus when T decreases, the exponential becomes large.

As $T \rightarrow 0$, $n \rightarrow n^{\uparrow}$ and $n^{\downarrow} \rightarrow 0$

Chapter 5

5.1 Show that

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{c_p}{TV\alpha}$$

Solution:

$$dU = TdS - PdV \Rightarrow dS = \frac{dU}{T} + \frac{PdV}{T} \Rightarrow \left(\frac{\partial S}{\partial V}\right)_P = \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_P + \frac{P}{T}$$

$$U = H - PV \Rightarrow \left(\frac{\partial U}{\partial V}\right)_P = \left(\frac{\partial H}{\partial V}\right)_P - P$$

$$\left(\frac{\partial H}{\partial V}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \frac{c_P}{V\alpha}$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{1}{T}\left(\frac{\partial H}{\partial V}\right)_P$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{c_P}{V\alpha T}$$

Chapter 5

5.2 Show that

$$\left(\frac{\partial S}{\partial P}\right)_V = \frac{c_p \beta_T}{T \alpha} - V \alpha$$

Solution:

$$dU = TdS - PdV \Rightarrow \left(\frac{\partial S}{\partial P}\right)_V = \frac{1}{T} \left(\frac{\partial U}{\partial P}\right)_V$$
$$\left(\frac{\partial U}{\partial P}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V \quad \left(\frac{\partial U}{\partial T}\right)_V = c_V \quad \left(\frac{\partial T}{\partial P}\right)_V = \frac{-\left(\frac{\partial V}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_P} = \frac{\beta_T}{\alpha}$$

$$c_P - c_V = \frac{VT\alpha^2}{\beta_T} + \left(\frac{\partial S}{\partial P}\right)_V = \frac{c_V \beta_T}{T \alpha} \Rightarrow \left(\frac{\partial S}{\partial P}\right)_V = \frac{c_p \beta_T}{T \alpha} - V \alpha$$

Chapter 5

5.3 Show that

$$\left(\frac{\partial A}{\partial P}\right)_V = -\frac{S\beta_T}{\alpha}$$

Solution:

$$\left(\frac{\partial A}{\partial P}\right)_V = \left(\frac{\partial A}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V$$

$$\left(\frac{\partial T}{\partial P}\right)_V = \frac{\beta_T}{\alpha}$$

$$dA = \left(\frac{\partial A}{\partial T}\right)_V dT + \left(\frac{\partial A}{\partial V}\right)_T dV$$

$$dA = -SdT - PdV$$

$$\left(\frac{\partial A}{\partial T}\right)_V = -S$$

$$\left(\frac{\partial T}{\partial P}\right)_V = \frac{\beta_T}{\alpha}$$



$$\left(\frac{\partial A}{\partial P}\right)_V = -\frac{S\beta_T}{\alpha}$$

Chapter 5

5.4 Show that

$$\left(\frac{\partial A}{\partial V}\right)_P = -\left(\frac{S}{V\alpha} + P\right)$$

Solution:

$$\left(\frac{\partial A}{\partial V}\right)_P = \left(\frac{\partial A}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \left(\frac{\partial A}{\partial T}\right)_P \frac{1}{\alpha V}$$

$$dA = -SdT - PdV \quad \Rightarrow \quad \left(\frac{\partial A}{\partial T}\right)_P = -S - P\left(\frac{\partial V}{\partial T}\right)_P$$



$$\left(\frac{\partial A}{\partial V}\right)_P = (-S - P\alpha V) \frac{1}{\alpha V} = -\left(\frac{S}{\alpha V} + P\right)$$

Chapter 5

5.5 Show that

$$\left(\frac{\partial H}{\partial S}\right)_V = T \left(1 + \frac{V\alpha}{c_v\beta_T}\right)$$

Solution:

$$\left(\frac{\partial H}{\partial S}\right)_V = \left(\frac{\partial H}{\partial T}\right)_V \left(\frac{\partial T}{\partial S}\right)_V \quad \left(\frac{\partial T}{\partial S}\right)_V = \frac{T}{c_v}$$

$$H = U + PV \quad \Rightarrow \quad \left(\frac{\partial H}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V + V \left(\frac{\partial P}{\partial T}\right)_V = c_v + \frac{V\alpha}{\beta_T}$$



$$\left(\frac{\partial H}{\partial S}\right)_V = \frac{T}{c_v} \left[c_v + \frac{V\alpha}{\beta_T} \right] = T \left[1 + \frac{V\alpha}{c_v\beta_T} \right]$$

Chapter 5

5.6 Show that

$$\left(\frac{\partial H}{\partial V}\right)_S = -\frac{c_P}{c_v \beta_T}$$

Solution:

$$dH = TdS + VdP$$

Maxwell relations



$$\left(\frac{\partial H}{\partial V}\right)_S = V \left(\frac{\partial P}{\partial V}\right)_S$$

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$



$$\text{From problem 5.1: } \left(\frac{\partial S}{\partial V}\right)_P = \frac{c_P}{TV\alpha} \quad \left(\frac{\partial P}{\partial V}\right)_S = -\frac{\left(\frac{\partial S}{\partial V}\right)_P}{\left(\frac{\partial S}{\partial P}\right)_V} = -\frac{c_P}{Vc_v\beta_T}$$

$$\text{From problem 5.2: } \left(\frac{\partial S}{\partial P}\right)_V = \frac{c_v\beta_T}{T\alpha}$$

Chapter 5

5.7 Show that

$$\left(\frac{\partial c_P}{\partial P}\right)_T = -TV\left(\alpha^2 + \frac{d\alpha}{dT}\right)$$

Solution: $c_P = \left(\frac{\partial H}{\partial T}\right)_P \rightarrow \left(\frac{\partial c_P}{\partial P}\right)_T = \left(\frac{\partial}{\partial P}\left(\frac{\partial H}{\partial T}\right)_P\right)_T = \left(\frac{\partial}{\partial T}\left(\frac{\partial H}{\partial P}\right)_T\right)_P$

$$dH = TdS + VdP \rightarrow \left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial S}{\partial P}\right)_T + V \quad \& \quad \left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

$$\begin{aligned}\left(\frac{\partial c_P}{\partial P}\right)_T &= \left(\frac{\partial}{\partial T}\left[-T\left(\frac{\partial V}{\partial T}\right)_P + V\right]\right)_P = \left(\frac{\partial}{\partial T}[-T\alpha V + V]\right)_P \\ &= -\alpha V - TV\frac{d\alpha}{dT} - T\alpha^2 V + \alpha V = -TV\left(\alpha^2 + \frac{d\alpha}{dT}\right)\end{aligned}$$

Chapter 5

5.8 Show that

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{T\alpha V}{c_P}$$

Solution:

$$\left(\frac{\partial T}{\partial P}\right)_S = -\frac{\left(\frac{\partial S}{\partial P}\right)_T}{\left(\frac{\partial S}{\partial T}\right)_P} = -\frac{-\alpha VT}{c_P}$$

$$\left(\frac{\partial S}{\partial T}\right)_P = \frac{c_P}{T}$$

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P = -\alpha V$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{\alpha VT}{c_P}$$

Chapter 5

5.9 Show that

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{c_P}{c_v V \beta_T}$$

Solution:

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{\left(\frac{\partial S}{\partial V}\right)_P}{\left(\frac{\partial S}{\partial P}\right)_V}$$

$$\left(\frac{\partial S}{\partial P}\right)_V = \left(\frac{\partial S}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V = -\frac{c_v}{T} \frac{\left(\frac{\partial V}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_P} = \frac{c_v}{T} \frac{\beta_T V}{\alpha V} = \frac{c_v \beta_T}{T \alpha}$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \left(\frac{\partial S}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \frac{c_P}{T} \frac{1}{\alpha V}$$



$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{c_P}{c_v V \beta_T}$$

Chapter 5

5.10 Show that

$$\left(\frac{\partial^2 G}{\partial P^2}\right)_T = -\frac{1}{\left(\frac{\partial^2 A}{\partial V^2}\right)_T}$$

Solution:

$$dG = -SdT + VdP \Rightarrow \left(\frac{\partial G}{\partial P}\right)_T = V \Rightarrow \left(\frac{\partial^2 G}{\partial P^2}\right)_T = \left(\frac{\partial V}{\partial P}\right)_T$$

$$dA = -SdT - PdV \Rightarrow \left(\frac{\partial A}{\partial V}\right)_T = -P$$

$$\left(\frac{\partial^2 A}{\partial V^2}\right)_T = -\left(\frac{\partial P}{\partial V}\right)_T = -\frac{1}{\left(\frac{\partial^2 G}{\partial P^2}\right)_T}$$

Chapter 5

5.11 Joule and Thomson showed experimentally that when a steady stream of nonideal gas is passed through a thermally insulated tube, in which is inserted a throttle valve, the temperature of the gas changes and the state of the gas is changed from P_1, T_1 to P_2, T_2 . Show that this process is isenthalpic. The change in T is described in terms of the Joule–Thomson coefficient, μ_{J-T} , as follows:

$$\mu_{J-T} = \left(\frac{\partial T}{\partial P} \right)_H$$

Show that

$$\mu_{J-T} = -\frac{V}{c_P} (1 - \alpha T)$$

and show that the Joule–Thomson coefficient for an ideal gas is zero.

Chapter 5

Solution to 5.11

- a. The process is adiabatic. Therefore $q = 0$ and work done by the gas, $w = P_2 V_2 - P_1 V_1$. Thus $\Delta H = q - w + (P_2 V_2 - P_1 V_1) = 0$. **isenthalpic**

b.

$$\mu_{J-T} = \left(\frac{\partial T}{\partial P} \right)_H \quad \left(\frac{\partial T}{\partial P} \right)_H = - \frac{\left(\frac{\partial H}{\partial P} \right)_T}{\left(\frac{\partial H}{\partial T} \right)_P} = - \frac{1}{c_P} \left(\frac{\partial H}{\partial P} \right)_T$$
$$dH = TdS + VdP$$

$$\left(\frac{\partial H}{\partial P} \right)_T = T \left(\frac{\partial S}{\partial P} \right)_T + V = -T \left(\frac{\partial V}{\partial T} \right)_P + V = -T\alpha V + V$$

$$\mu_{J-T} = - \frac{1}{c_P} (-T\alpha V + V) = \frac{V}{c_P} (T\alpha - 1)$$

Chapter 5

c.

$$\mu_{J-T} = -\frac{1}{c_P}(-T\alpha V + V) = \frac{V}{c_P}(T\alpha - 1)$$

For an ideal gas

$$\alpha = \frac{1}{T}, \quad \text{So} \quad \alpha T = 1 \quad \text{and} \quad \mu_{J-T} = 0$$

Chapter 5

5.12 Determine the values of $\Delta U'$, $\Delta H'$, $\Delta S'$, $\Delta A'$, and $\Delta G'$ for the following processes (in (c), (d), and (e), show that an absolute value of the entropy is required):

- a. The four processes in Problem 4.1
- b. The expansion of 1 mole of an ideal gas at the pressure P and the temperature T into a vacuum to double its volume
- c. The adiabatic expansion of 1 mole of an ideal gas from P_1 , T_1 to P_2 , T_2
- d. The expansion of 1 mole of an ideal gas at constant pressure from V_1 , T_1 to V_2 , T_2
- e. The expansion of 1 mole of an ideal gas at constant volume from P_1 , T_1 to P_2 , T_2

Chapter 5

Solution to 5.12

a. The four processes in Problem 4.1

A rigid container is divided into two compartments of equal volume by a partition. One compartment contains 1 mole of ideal gas A at 1 atm, and the other contains 1 mole of ideal gas B at 1 atm.

(1) when the partition between the two compartments is removed.

(2) If the first compartment had contained 2 moles of ideal gas A, when the partition was removed.

(3, 4) In each of the preceding two situations if both compartments had contained ideal gas A.

Chapter 5

Solution to 5.12

a. 1. $\Delta U = \Delta H = 0, \Delta S = R \ln 4, \Delta A = \Delta G = -T \Delta S = -RT \ln 4$

2. $\Delta U = \Delta H = 0, \Delta S = R \ln 8, \Delta A = \Delta G = -RT \ln 8$

3. $\Delta U = \Delta H = \Delta S = \Delta A = \Delta G = 0$

4. $\Delta U = \Delta H = 0, \Delta S = R \ln \left(\frac{32}{27} \right), \Delta A = \Delta G = -RT \ln \left(\frac{32}{27} \right)$

Chapter 5

Solution to 5.12

b. The expansion of 1 mole of an ideal gas at the pressure P and the temperature T into a vacuum to double its volume

b. $\Delta U = \Delta H = 0, \Delta S = R \ln(V_2 / V_1) = R \ln 2, \Delta A = \Delta G = -RT \ln 2$

c. The adiabatic expansion of 1 mole of an ideal gas from P_1 , T_1 to P_2 , T_2

c. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = 0, \Delta A = \Delta U - S(T_2 - T_1),$
 $\Delta G = \Delta H - S(T_2 - T_1)$

Chapter 5

Solution to 5.12

d. The expansion of 1 mole of an ideal gas at constant pressure from V_1, T_1 to V_2, T_2

d. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = c_p \ln(T_2 / T_1),$
 $\Delta A = \Delta U - S_1 (T_2 - T_1) - T_2 \Delta S, \Delta G = \Delta H - S_1 (T_2 - T_1) - T_2 \Delta S$

e. The expansion of 1 mole of an ideal gas at constant volume from P_1, T_1 to P_2, T_2

e. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = c_v \ln(T_2 / T_1),$
 $\Delta A = \Delta U - S_1 (T_2 - T_1) - T_2 \Delta S, \Delta G = \Delta H - S_1 (T_2 - T_1) - T_2 \Delta S$

Chapter 5

5.13 Show that

$$1. TdS = c_V dT + \frac{T\alpha}{\beta_T} dV$$

$$2. TdS = c_P dT - TV\alpha dP$$

$$3. TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$

Chapter 5

Solution 5.13

a.

$$S = S(T, V)$$

$$\therefore dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

under constant volume

$$TdS = \delta q_v = dU = c_v dT$$

$$\therefore \left(\frac{\partial S}{\partial T} \right)_V = \frac{c_v}{T}$$

according to the Maxwell relation

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial V}{\partial T} \right)_P \left(\frac{\partial P}{\partial V} \right)_T$$

$$\therefore TdS = c_v dT + \frac{T\alpha}{\beta_T} dV$$

Chapter 5

Solution 5.13

b.

$$S = S(T, P)$$

$$\therefore dS = \left(\frac{\partial S}{\partial T} \right)_P dT + \left(\frac{\partial S}{\partial P} \right)_T dP$$

under constant pressure

$$TdS = \delta q_p = dH = c_p dT$$

$$\therefore \left(\frac{\partial S}{\partial T} \right)_P = \frac{c_p}{T}$$

according to the Maxwell relation

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P = -\alpha V$$

$$\therefore TdS = c_p dT - TV\alpha dP$$

Chapter 5

Solution 5.13

C.

$$S = S(P, V)$$

$$\therefore dS = \left(\frac{\partial S}{\partial V} \right)_P dV + \left(\frac{\partial S}{\partial P} \right)_V dP$$

$$\left(\frac{\partial S}{\partial V} \right)_P = \frac{\left(\frac{\partial S}{\partial T} \right)_P}{\left(\frac{\partial V}{\partial T} \right)_P} = \frac{\frac{c_P}{T}}{\alpha V} = \frac{c_P}{T \alpha V}$$

$$\left(\frac{\partial S}{\partial P} \right)_V = \left(\frac{\partial S}{\partial T} \right)_V \left(\frac{\partial T}{\partial P} \right)_V = \frac{c_V \beta_T}{T \alpha}$$

$$\therefore TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$

Chapter 5

5.14. Starting from the following equation valid for one mole of a closed system, obtain Maxwell relationships.

$$dH = TdS + VdP - \mu_0 VMd\aleph$$

Solution:

$$\left(\frac{\partial T}{\partial P}\right)_{S,\aleph} = \left(\frac{\partial V}{\partial S}\right)_{P,\aleph}$$

$$\left(\frac{\partial V}{\partial \aleph}\right)_{S,P} = -\mu_0 V \left(\frac{\partial M}{\partial P}\right)_{S,\aleph}$$

$$\left(\frac{\partial T}{\partial \aleph}\right)_{S,P} = -\mu_0 V \left(\frac{\partial M}{\partial S}\right)_{P,\aleph}$$

Chapter 5

5.15. Starting from the following equation valid for one mole of a closed system, obtain Maxwell relationships.

$$dA = -SdT - PdV + \mu_0 V H dM$$

Solution:

$$\left(\frac{\partial S}{\partial V} \right)_{T,M} = \left(\frac{\partial P}{\partial T} \right)_{V,M}$$

$$\left(\frac{\partial S}{\partial M} \right)_{T,V} = -\mu_0 V \left(\frac{\partial H}{\partial T} \right)_{M,V}$$

$$\left(\frac{\partial P}{\partial M} \right)_{V,T} = -\mu_0 V \left(\frac{\partial H}{\partial V} \right)_{M,T}$$

Chapter 5

5.16 The cycle shown in Figure 5.8 consists of two isotherms (AB and CD) and two isobars (BC and DA). Use the TdS equations to draw the T-S diagram for this cycle.

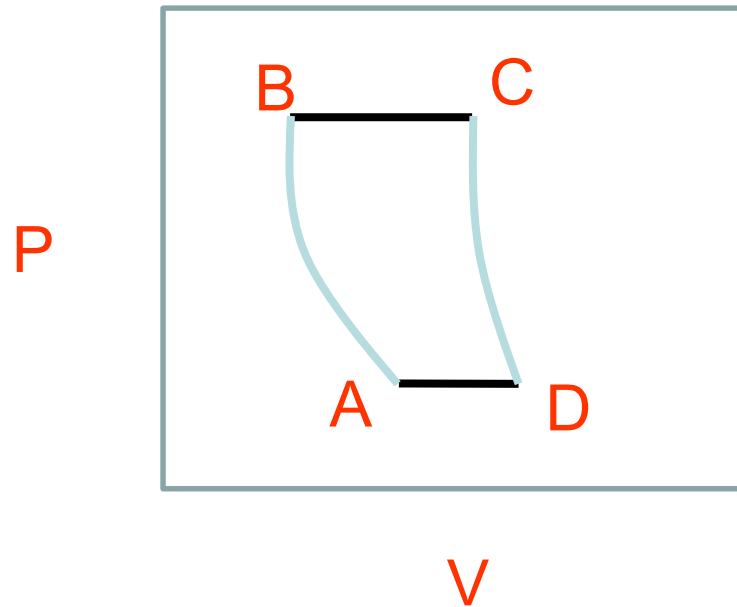


Figure 5.8 Pressure– volume diagram of two isothermals (AB and CD) and two isobaric changes (BC and DA) making up a cycle.

Chapter 5

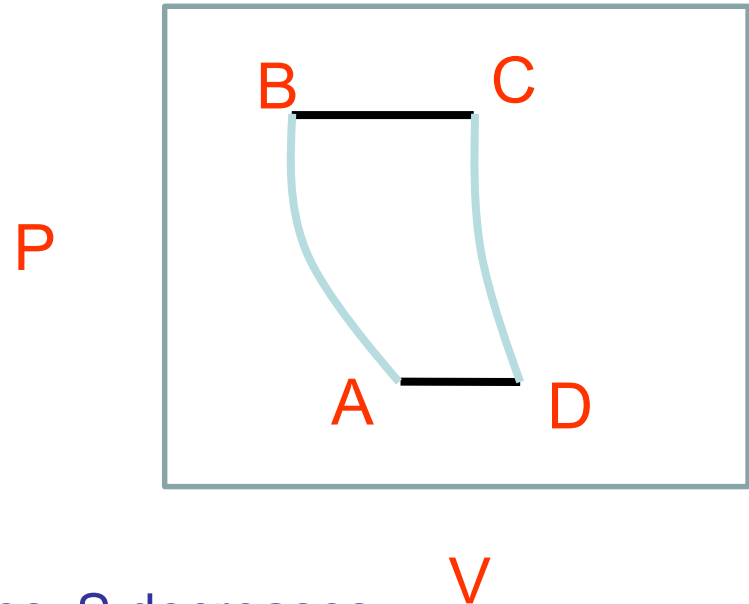
Solution 5.16

Use the following TdS equations

$$1. TdS = c_V dT + \frac{T\alpha}{\beta_T} dV$$

$$2. TdS = c_P dT - TV\alpha dP$$

$$3. TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$



A→B from 1 at constant T, if V decreases, S decreases

B→C from 3, at constant P, V increases then S increases.

From 2, at constant P if S increases T increases

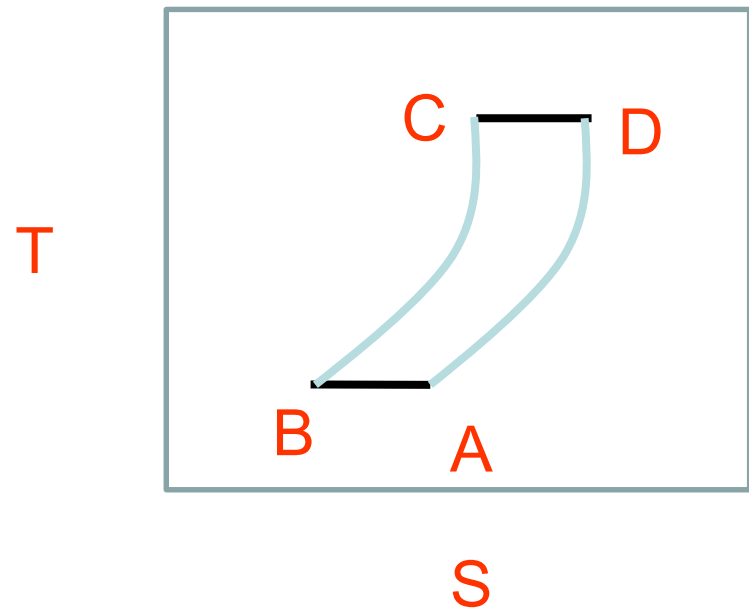
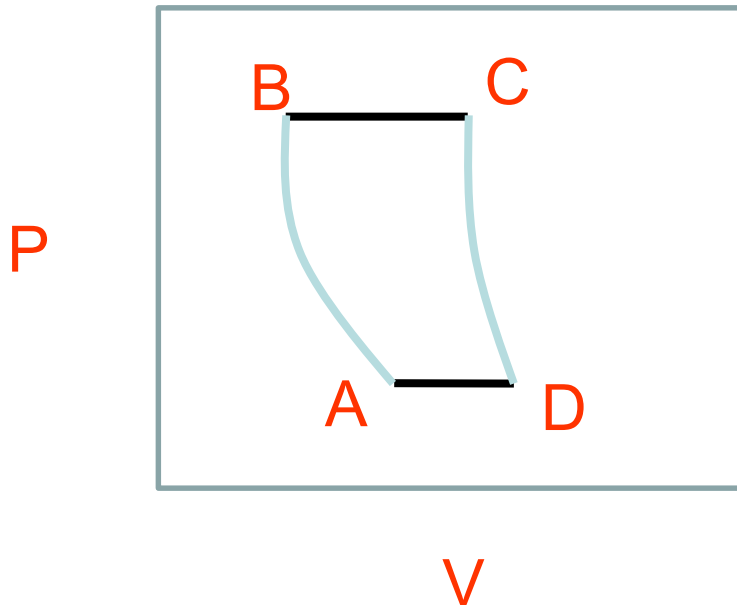
C→D from 2, at constant T if P decreases then S increases

D→A from 3, at constant P, if V decreases S decreases

From 2, at constant P if S decreases, T decreases.

Chapter 5

Solution 5.16



Chapter 5

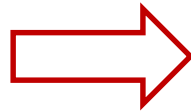
5.17. Show that: $\frac{\beta_T}{\beta_S} = \frac{c_P}{c_V} = \gamma$ $\beta_T \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$ $\beta_S \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_S$

Hint: use the TdS equations.

Solution: Start with the First and Second TdS equations:

$$\begin{aligned} 1. \quad TdS &= C_V dT + \frac{T\alpha}{\beta_T} dV \\ 2. \quad TdS &= C_P dT - TV\alpha dP \end{aligned}$$

Set $dS=0$



$$\begin{aligned} 1. \quad C_V dT &= -\frac{T\alpha}{\beta_T} dV \\ 2. \quad C_P dT &= TV\alpha dP \end{aligned}$$



$$\frac{C_P}{C_V} = -\frac{TV\alpha}{\frac{T\alpha}{\beta_T}} \left(\frac{\partial P}{\partial V} \right)_S = (V\beta_T) \frac{1}{V\beta_S} = \frac{\beta_T}{\beta_S}$$

Thermodynamics of Materials & Phase transformations

Review for Chapter 4 and 5

Chapter 4

1. A single macrostate of a system, which is determined when the independent variables of the system are fixed, contains a very large number of microstates, each of which is characterized by the manner in which

- The thermal energy of the system is distributed among the particles
- The particles are distributed in the configurations available to them

2. Although the occurrence of a system in any one of its microstates is equally probable (for the case of the closed system), greatly differing numbers of microstates occur in differing distributions. The distribution which contains the largest number of microstates is the most probable distribution.

Chapter 4

3. In real systems (with a very large number of particles) the number of microstates in the most probable distribution is significantly larger than the sum of all of the other microstates occurring in all of the other distributions. This most probable distribution is the equilibrium thermodynamic state of the system.

4. The relationship between the number of microstates available to the system, Ω , and the entropy of the system is given by the Boltzmann equation:

$$S = k_B \ln \Omega$$

in which k_B is Boltzmann's constant

Chapter 4

5. If a situation arises which allows an increase in the number of microstates available to the system (at constant energy), then spontaneous redistribution of the energy among the particles (or particles over the available configurations) occurs until the newly available most probable distribution occurs. The Boltzmann equation shows that an increase in the number of microstates made available to the system causes an increase in the entropy of the system.

6. The total entropy of a system is the sum of each of the aspects of entropy:

- Thermal entropy, S'_{th}
- Configurational entropy, S'_{conf}
- Any other aspect of entropy, such as spin entropy, electric dipole entropy, and so on

Chapter 4

7. The thermal entropy arises from the number of ways in which the thermal energy available to the system can be shared among the constituent particles, Ω_{th} .
8. The configurational entropy arises from the number of ways in which the particles can be distributed over the configurations available to them, Ω_{conf} .
9. The spin entropy arises from the number of ways in which the spins can be distributed over the sites available to them, Ω_{spin} .
10. Since any of the thermal distributions can be combined with any of the configurational distributions, the total number of microstates available to the system is the product $\Omega_{\text{th}} \Omega_{\text{conf}}$ ($\Omega_{\text{th}} \Omega_{\text{conf}} \Omega_{\text{spin}}$ if spins are present), and hence, from the logarithmic form of the Boltzmann equation, the total entropy of the system is the sum of the individual entropies of the system.

Chapter 4

✓ Partition function

$$P = \sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} = e^{-\beta \varepsilon_0} + e^{-\beta \varepsilon_1} + e^{-\beta \varepsilon_2} + \dots + e^{-\beta \varepsilon_r} \quad i = 0, 1, 2, \dots, r$$

$$n_i = \frac{n}{P} e^{-\beta \varepsilon_i} \quad i = 0, 1, 2, \dots, r \quad \beta = \frac{1}{k_B T}$$

- ✓ From the point of view of microstates, an irreversible process is one which takes the system from **a less probable state** to **the most probable state**, and from the point of view of macrostates, an irreversible process takes the system from a **nonequilibrium state** to the **equilibrium state**.

Chapter 5

1. The Helmholtz free energy, A , is defined by $A \equiv U - TS$. In a closed system held at constant T and V , the Helmholtz free energy can only
 - Decrease, for a spontaneous process
 - Remain constant if the system is in equilibrium

Equilibrium at constant T and V is attained when the Helmholtz free energy achieves its minimum value.

2. The Gibbs free energy, G , is defined by $G \equiv H - TS$. In a closed system at constant pressure and temperature, the Gibbs free energy is a minimum at equilibrium. During an isothermal, isobaric process during which no form of work other than P - V work is performed (i.e., $w=0$), G can only
 - Decrease for a spontaneous process
 - Remain constant if the system is in equilibrium

Equilibrium is attained at constant T and P when the Gibbs free energy reaches its minimum value.

Chapter 5

3. For a change of state at constant pressure, $\Delta H = q_p$.
4. The chemical potential, μ_i , of the i th component can be expressed as

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_j, j \neq i} = \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_j, j \neq i} = \left(\frac{\partial H}{\partial n_i} \right)_{S, P, n_j, j \neq i} = \left(\frac{\partial A}{\partial n_i} \right)_{S, V, n_j, j \neq i}$$

5. Internal energy varies with S , V , and composition as

$$dU = TdS - PdV + \sum_{i=1}^{i=k} \mu_i dn_i$$

Enthalpy varies with S , P , and composition as

$$dH = TdS + VdP + \sum_{i=1}^{i=k} \mu_i dn_i$$

Chapter 5

5. Helmholtz free energy varies with T , V , and composition as

$$dA = -SdT - PdV + \sum_{i=1}^{i=k} \mu_i dn_i$$

and Gibbs free energy varies with T , P , and composition as

$$dG = -SdT + VdP + \sum_{i=1}^{i=k} \mu_i dn_i$$

6. For one mole of a closed system with magnetic terms, the following holds.

$$dG = -SdT + VdP - V\mu_0 M d\mathfrak{H}$$

Chapter 5

7. The Maxwell relations for one mole of a closed simple system are

$$\begin{aligned}\left(\frac{\partial T}{\partial V}\right)_S &= -\left(\frac{\partial P}{\partial S}\right)_V & \left(\frac{\partial S}{\partial V}\right)_T &= \left(\frac{\partial P}{\partial T}\right)_V \\ \left(\frac{\partial T}{\partial P}\right)_S &= \left(\frac{\partial V}{\partial S}\right)_P & \left(\frac{\partial S}{\partial P}\right)_T &= -\left(\frac{\partial V}{\partial T}\right)_P\end{aligned}$$

8. With x , y , and z as state functions, the upstairs-downstairs-inside-out formula is

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

Chapter 5

9. The Gibbs-Helmholtz equations are

$$\frac{d(G/T)}{dT} = -\frac{H}{T^2}$$

$$\frac{d(A/T)}{dT} = -\frac{U}{T^2}$$

Chapter 5

$$H \equiv U + PV$$

$$G \equiv H - TS$$

$$A \equiv U - TS$$

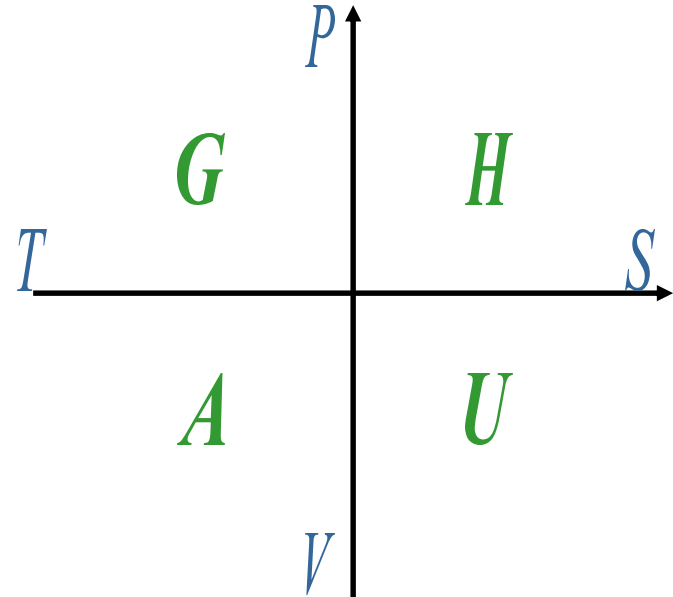
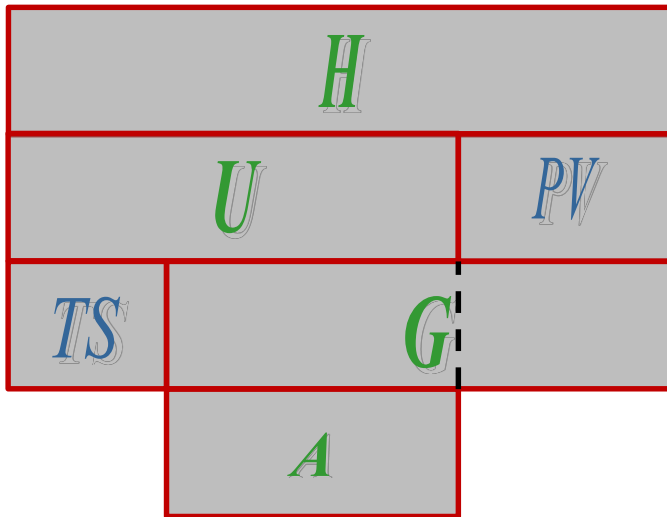
$$G \equiv A + PV$$

$$dU = TdS - PdV$$

$$dH = TdS + VdP$$

$$dA = -SdT - PdV$$

$$dG = -SdT + VdP$$



Thermodynamics of Materials & Phase transformations

The End



Exercise for Chapter 6 and 7

Xiao Han(韩潇)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

xiao.han@nwpu.edu.cn

Summary of chapter 6

1. Knowledge of the heat capacities and the entropies of substances and the heats of formation of compounds allows the enthalpy and entropy changes to be evaluated for any process—that is, for phase changes and chemical reactions.
2. The heat capacity of solids can be modeled by the statistical thermodynamic models of Einstein and Debye. The Debye model is best at low temperatures as it displays the experimentally observed T^3 dependence at low temperatures for non conductors.
3. Heat capacities can also be fit to experimental data via an equation of the form

$$c_P = a + bT + \frac{c}{T^2}$$

Summary of chapter 6

4. Since enthalpy does not have an absolute value, it is conventional to assign the value of zero to the enthalpy of all elements in their stable states of existence at 298 K and to consider changes in enthalpy with respect to this reference state.
5. The entropy of all substances which are in complete internal equilibrium is zero at 0 K. This is the Nernst–Planck–Simon statement of the Third Law of Thermodynamics.
6. Both the enthalpy and entropy are dependent on pressure and temperature; however, the pressure dependence of the enthalpy and entropy of condensed phases is normally small enough to be ignored, especially when the pressure of interest is in the range 0–1 atm.

Summary of chapter 6

7. The variation of enthalpies of transformation with temperature and pressure can be calculated by the application of the First Law of Thermodynamics to the changes in enthalpy with temperature and pressure, as well as the temperature dependence of the heat capacities and expansion coefficients.
8. The enthalpies and entropies of transformation at the melting point and boiling point of metals can be estimated by the rules of Richard and Trouton.

Summary of chapter 6

9. The determination of ΔH_T and ΔS_T for any change of state at any temperature and pressure allows the all-important change in the Gibbs free energy for the change of state to be calculated as

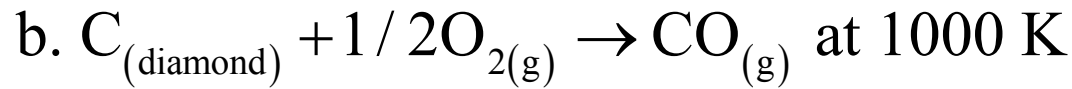
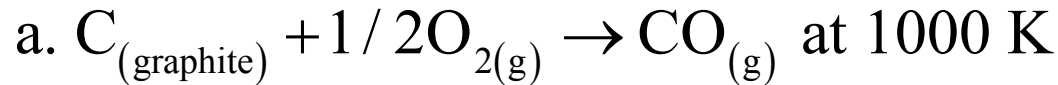
$$\Delta G_T = \Delta H_T - T \Delta S_T$$

10. Since consideration of the Gibbs free energy in any isothermal, isobaric process provides the criterion for equilibrium, the equilibrium state of a system can be determined from knowledge of the thermochemical properties of the system.

Chapter 6

放热

6.2 Which of the following two reactions is the more exothermic?



$$C_{P,C_{(\text{graphite})}} = 0.11 + 38.94 \times 10^{-3} T - 1.48 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-1100 \text{ K})$$

$$C_{P,C_{(\text{diamond})}} = 9.12 + 13.22 \times 10^{-3} T - 6.19 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-1200 \text{ K})$$

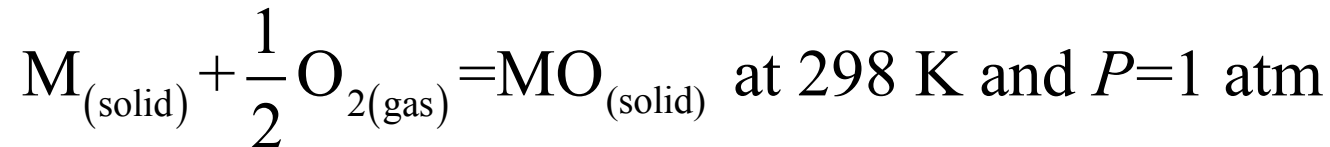
$$C_{P,O_{2(g)}} = 29.96 + 4.18 \times 10^{-3} T - 1.67 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-3000 \text{ K})$$

$$C_{P,CO_{(g)}} = 28.41 + 4.10 \times 10^{-3} T - 0.46 \times 10^5 T^{-2} \quad \text{J/mole} \cdot \text{K} \quad (298-2500 \text{ K})$$

$$\Delta H_{298 \text{ K}}^0 (\text{graphite} \rightarrow \text{diamond}) = 1900 \quad \text{J}$$

6.4 Enthalpy as a function of temperature and composition

- ✓ Since H does not have an absolute value (only changes in H can be measured), it is convenient to introduce a convention which will allow the comparison of different enthalpy–temperature diagrams. **This convention assigns the value of zero to the enthalpy of *elements in their stable states at 298 K (25°C) and $P=1$ atm.***
- ✓ The enthalpy of a compound at 298 K and $P=1$ atm is simply the enthalpy of formation of one mole of the compound from its elements at 298 K and $P=1$ atm.



$$\Delta H_{298} = H_{\text{MO(s)298}} - H_{\text{M(s)298}} - \frac{1}{2} H_{\text{O2(gas)298}}$$

Chapter 6

Solution to 6.2

$$C_{P,C_{(\text{graphite})}} = 0.11 + 38.94 \times 10^{-3} T - 1.48 \times 10^5 T^{-2} - 17.38 \times 10^6 T^2$$

J/mole · K (298-1100 K)

$$\begin{aligned} H_{(\text{graphite}, 1000)} &= 0.11(1000 - 298) + 38.94 / 2 \times 10^{-3} (1000^2 - 298^2) \\ &\quad + 1.48 \times 10^5 (1 / 1000 - 1 / 298) - 17.38 \times 10^6 / 3 (1000^3 - 298^3) \\ &= 11830 \text{ J/mole} \end{aligned}$$

Chapter 6

$$C_{P,C(\text{diamond})} = 9.12 + 13.22 \times 10^{-3} T - 6.19 \times 10^5 T^{-2}$$

J/mole · K (298-1200 K)

$$\Delta H_{298 \text{ K}}^0 (\text{graphite} \rightarrow \text{diamond}) = 1900 \quad \text{J/mole}$$

$$\begin{aligned} H_{(\text{diamond}, 1000)} &= 1900 + 9.12(1000 - 298) + 13.22 / 2 \times 10^{-3} (1000^2 - 298^2) \\ &\quad + 6.19 \times 10^5 (1 / 1000 - 1 / 298) \\ &= 12867 \quad \text{J/mole} \end{aligned}$$

For diamond $\rightarrow$ graphite $\Delta H_{1000} = 11830 - 12867 = -1037 \text{ J/mole}$

Thus the oxidation of diamond at 1000 K is 1037 J/mole more exothermic than the oxidation of graphite.

Chapter 6

6.3 Calculate the change in enthalpy and the change in entropy at 1000 K for the reaction $\text{CaO}_{(s)} + \text{TiO}_{2(s)} \rightarrow \text{CaTiO}_{3(s)}$

Chapter 6

Table A.2 The Constant-Pressure Molar Heat Capacities of Various Substances ($c_p = a + bT + cT^{-2}$ J/mole·K)

Substance	a	b×10 ³	c×10 ⁻⁵	Range, K	Remarks
Ag	21.30	8.54	1.51	298–1234 (T_m)	
Ag(l)	30.50	—	—	1234–1600	
Al(s)	20.67	12.38	—	298–933 (T_m)	
Al(l)	31.76	—	—	933–1600	
Al ₂ O ₃	106.6	17.78	-28.53	298–2325 (T_m)	
Ba(s)	-473.2	1587.0	128.2	298–648	
Ba(g)	-5.69	80.33	—	648–1003	
BaO	53.30	4.35	-8.30	298–2286 (T_m)	
BaTiO ₃	121.46	8.54	-19.16	298–1800	
C(graphite)	0.11	38.94	-1.48	298–1100	-17.38 × 10 ⁻⁸ T ²
C ₂ (graphite)	24.43	0.44	-31.63	1100–4000	
C(diamond)	9.12	13.22	-6.19	298–1200	
CO	28.41	4.10	-0.46	298–2500	
CO ₂	44.14	9.04	-8.54	298–2500	
Ca(s)	25.37	-7.26	—	298–716	23.72 × 10 ⁻⁶ T ²
Ca(g)	-0.36	41.25	—	716–1115	
CaO	49.62	4.51	-6.95	298–1177	
CaTiO ₃	127.49	5.69	-27.99	298–1530	
Cr(s)	24.43	9.87	-3.68	298–2130 (T_m)	
Cr ₂ O ₃	119.37	9.30	-15.65	298–1800	
Cu(s)	22.64	6.28	—	298–1356 (T_m)	
Fe(s)	37.12	6.17	—	298–1183/1664–1809	
Fe(l)	24.47	8.45	—	1187–1664	
Fe(g)	41.8	—	—	1809–1873	
H ₂ O(g)	30.00	10.71	0.33	298–2500	
O ₂ (g)	29.96	4.18	-1.67	298–3000	
2MgO·2Al ₂ O ₃ ·5SiO ₂	626.34	91.21	-200.83	298–1738 (T_m)	
N ₂	27.87	4.27	—	298–2500	
Si ₃ N ₄	70.54	98.74	—	298–900	
SiO ₂ (α-quartz)	43.89	1.00	-6.02	298–847	
Ti	22.09	10.46	—	298–1155	
TiO ₂ (rutile)	75.19	1.17	-18.20	298–1800	
Zr(s)	21.97	11.63	—	298–1136	
Zr(l)	23.22	4.64	—	1136–2128	
ZrO ₂ (s)	69.62	7.53	-14.06	298–1478	
ZrO ₂ (g)	74.48	—	—	1478–2950 (T_m)	

Table A.3 The Standard Molar Heats of Formation and Molar Entropies of Various Substances at 298 K

Substance	ΔH_{298}° , J	S_{298}° , J/K
Al ₂ O ₃	-1,675,700	50.9
Ba	—	62.4
BaO	-548,100	72.1
BaTiO ₃	-1,653,100	107.9
C ₂ (graphite)	—	5.73
C ₂ (diamond)	1,900	2.43
CH ₄	-74,800	186.3
CO	-110,500	197.5
CO ₂	-393,500	213.7
Ca	—	41.6
CaO	-634,900	38.1
CaTiO ₃	-1,660,600	93.7
3CaO·Al ₂ O ₃ ·3SiO ₂	-6,646,300	241.4
CaO·Al ₂ O ₃ ·SiO ₂	-3,293,200	144.8
CaO·Al ₂ O ₃ ·2SiO ₂	-4,223,700	202.5
2CaO·Al ₂ O ₃ ·SiO ₂	-3,989,400	198.3
Cr ₂ O ₃	-1,134,700	81.2
H ₂ O(g)	-241,800	232.9
N ₂	—	191.5
O ₂	—	205.1
SiO ₂ (α-quartz)	-910,900	41.5
Si ₃ N ₄	-744,800	113.0
Ti	—	30.7
TiO	-543,000	34.7
Ti ₂ O ₃	-1,521,000	77.2
Ti ₃ O ₅	-2,459,000	129.4
TiO ₂	-944,000	50.6
Zr	—	39.0
ZrO ₂	-1,100,800	50.4

Table A.5 Molar Heats of Melting and Transformation

Substance	Trans.	ΔH_{trans} , J	T_{trans} , K
Ag	s → l	11,090	1,234
Al	s → l	10,700	934
Al ₂ O ₃	s → l	107,500	2,324
Au	s → l	12,600	1,338
Ba	α → β	630	648
Ba	β → l	7,650	1,003
Cu	s → l	12,970	1,356
Ca	α → β	900	716
CaF ₂	s → l	31,200	1,691
Fe	α → γ	670	1,187
Fe	γ → δ	840	1,664
Fe	δ → l	13,770	1,809
H ₂ O	s → l	6,008	273
K ₂ O·B ₂ O ₃	s → l	62,800	1,220
MgF ₃	s → l	58,160	1,563
Na ₂ O·B ₂ O ₃	s → l	67,000	1,240
Pb	s → l	4,810	600
PbO	s → l	27,480	1,158
Si	s → l	50,200	1,685
V	s → l	22,840	2,193
Zr	α → β	3,900	1,136
ZrO ₂	α → β	5,900	1,478

Chapter 6

Solution to 6.3

(1) Change in enthalpy



In the range 298 – 716K,

$$\Delta c_p = 9.27 + 9.68 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$



In the range 716 – 1000 K,

$$\Delta c_p = 35 - 38.83 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\Delta H_{1000}$$

$$= -634900 + 9.27(716 - 298) + 4.84 \times 10^{-3}(716^2 - 298^2) + 6.115$$

$$\times 10^5 \left(\frac{1}{716} - \frac{1}{298} \right) + 900 + 35(1000 - 716) - 19.415 \times 10^{-3}$$

$$(1000^2 - 716^2) + 6.115 \times 10^5 \left(\frac{1}{1000} - \frac{1}{716} \right)$$

$$= -630836 \text{ J}$$

Chapter 6

Solution to 6.3

For $\text{Ti} + \text{O}_2 = \text{TiO}_2$, $\Delta H_{298} = -944000 \text{ J}$

In the range $298 - 1000\text{K}$,

$$\Delta c_p = 23.14 - 13.47 \times 10^{-3}T - 16.53 \times 10^5 T^{-2}$$

$$\Delta H_{1000}$$

$$= -944000 + 23.14(1000 - 298) - 6.735 \times 10^{-3}(1000^2 - 298^2)$$

$$+ 16.53 \times 10^5 \left(\frac{1}{1000} - \frac{1}{298} \right)$$

$$= -937787 \text{ J}$$

Chapter 6

Solution to 6.3



In the range 298 – 716 K,

$$\Delta c_p = 35.09 - 3.78 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$



In the range 716 – 1000 K, for $\text{Ca}(\beta) + \text{Ti} + \frac{3}{2}\text{O}_2 = \text{CaTiO}_3$,

$$\Delta c_p = 60.82 - 52.29 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

$$\begin{aligned} \Delta H_{1000} &= -1660600 + 35.09(716 - 298) - 1.89 \times 10^{-3}(716^2 - 298^2) + 25.485 \\ &\times 10^5 \left(\frac{1}{716} - \frac{1}{298} \right) + 900 + 60.82(1000 - 716) - 26.145 \times 10^{-3}(1000^2 \\ &- 716^2) + 25.485 \times 10^5 \left(\frac{1}{1000} - \frac{1}{716} \right) \\ &= -1649106 \text{ J} \end{aligned}$$

Chapter 6

Solution to 6.3

For $\text{CaO(s)} + \text{TiO}_2\text{(s)} = \text{CaTiO}_3\text{(s)}$

$$\Delta H_{1000} = -1649106 + 630836 + 937787 = -80483 \text{ J}$$

Chapter 6

Solution to 6.3

(2) Change in entropy

$$\text{For } \text{Ca}(\alpha) + \frac{1}{2} \text{O}_2 = \text{CaO}, \quad \Delta S_{298} = 38.1 - 41.6 - 205.1 \times 0.5 = -106.05 \text{ J/K}$$

$$\text{In the range } 298 - 716\text{K}, \quad \Delta c_p = 9.27 + 9.68 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\text{in the range } 716 - 1000 \text{ K}, \quad \Delta c_p = 35 - 38.83 \times 10^{-3}T - 6.115 \times 10^5 T^{-2}$$

$$\text{At } 716 \text{ K}, \quad \text{Ca}(\alpha) \rightarrow \text{Ca}(\beta), \quad \Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$$

$$\Delta S_{1000}$$

$$= -106.05 + 9.27 \ln \left(\frac{716}{298} \right) + 9.68 \times 10^{-3} (716 - 298) + 3.0525$$

$$\times 10^5 \left(\frac{1}{716^2} - \frac{1}{298^2} \right) + \frac{900}{716} + 35 \ln \left(\frac{1000}{716} \right) - 38.83 \times 10^{-3} (1000 - 716)$$

$$+ 3.0525 \times 10^5 \left(\frac{1}{1000^2} - \frac{1}{716^2} \right) = -97.6 \text{ J/K}$$

Chapter 6

Solution to 6.3

$$\text{For Ti} + \text{O}_2 = \text{TiO}_2, \quad \Delta S_{298} = 50.6 - 205.1 - 30.7 = -185.2 \text{ J/K}$$

In the range 298 – 1000K,

$$\Delta c_p = 23.14 - 13.47 \times 10^{-3}T - 16.53 \times 10^5 T^{-2}$$

$$\begin{aligned} \Delta S_{1000} &= -185.2 + 23.14 \ln\left(\frac{1000}{298}\right) - 13.47 \times 10^{-3}(1000 - 298) + 8.265 \\ &\quad \times 10^5 \left(\frac{1}{1000^2} - \frac{1}{298^2}\right) \\ &= -175.1 \text{ J/K} \end{aligned}$$

Chapter 6

For $\text{Ca}(\alpha) + \text{Ti} + \frac{3}{2} \text{O}_2 = \text{CaTiO}_3$,

$$\Delta S_{298} = 93.7 - 41.6 - 30.7 - \frac{3}{2} \times 205.1 = -286.25 \text{ J/K}$$

In the range 298 – 716 K,

$$\Delta c_p = 35.09 - 3.78 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

At 716 K, $\text{Ca}(\alpha) \rightarrow \text{Ca}(\beta)$, $\Delta H_{\alpha \rightarrow \beta} = 900 \text{ J}$

In the range 716 – 1000 K,

$$\Delta c_p = 60.82 - 52.29 \times 10^{-3}T - 25.485 \times 10^5 T^{-2} \text{ J/K}$$

ΔS_{1000}

$$\begin{aligned} &= -286.25 + 35.09 \ln\left(\frac{716}{298}\right) - 3.78 \times 10^{-3}(716 - 298) + 12.743 \\ &\times 10^5 \left(\frac{1}{716^2} - \frac{1}{298^2} \right) + \frac{900}{716} + 60.82 \ln\left(\frac{1000}{716}\right) - 52.29 \times 10^{-3}(1000 - 716) \\ &+ 12.743 \times 10^5 \left(\frac{1}{1000^2} - \frac{1}{716^2} \right) = -265.9 \text{ J/K} \end{aligned}$$

Chapter 6

Solution to 6.3

For $\text{CaO(s)} + \text{TiO}_2 = \text{CaTiO}_3\text{(s)}$

$$\Delta S_{1000} = -265.9 + 97.6 + 175.1 = 6.8 \text{ J/K}$$

6.6 Experimental verification of the third law

Assigning a value of zero to S_0 allows the absolute value of the entropy of any material to be determined as

$$S_T = \int_0^T \frac{c_P}{T} dT \text{ J/K}$$

and molar entropies are normally tabulated at 298 K, where:

$$S_{298} = \int_0^{298} \frac{c_P}{T} dT \text{ J/K}$$

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The variations, with temperature, of the molar entropies of several elements and compounds are shown in Figure 6.12. With the constant-pressure molar heat capacity of the solid expressed in the form

$$c_{P(s)} = a + bT + cT^{-2}$$

6.6 Experimental verification of the third law

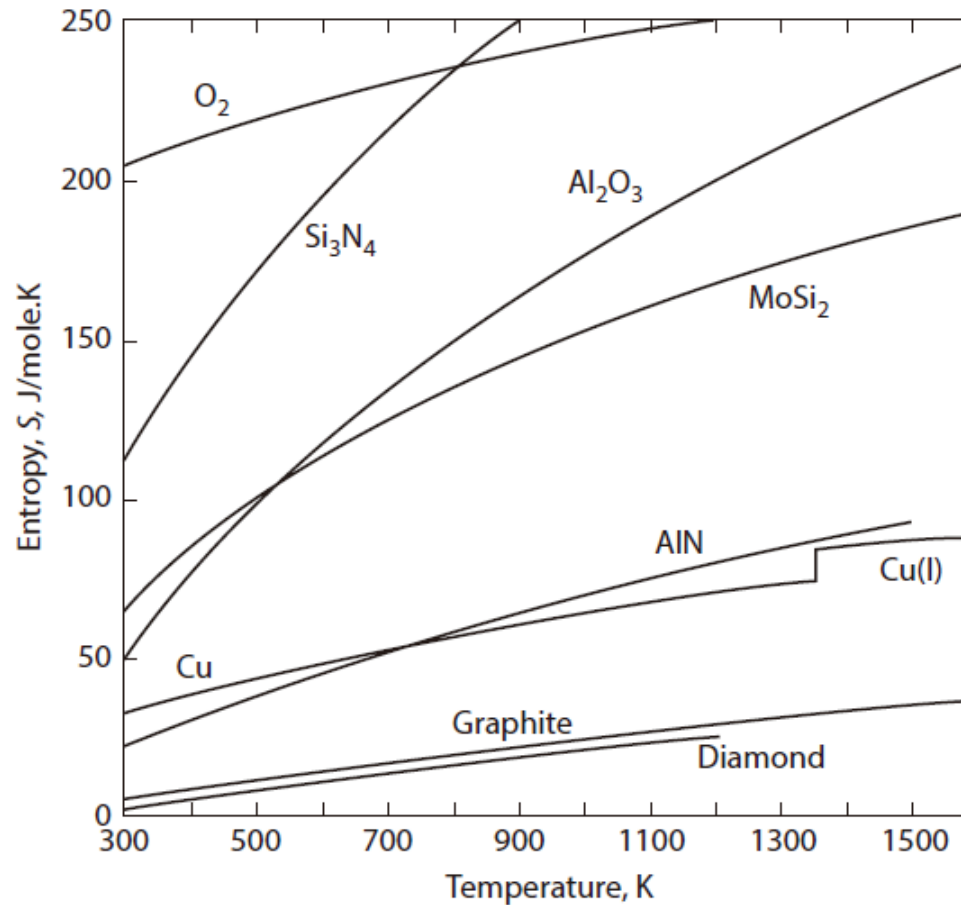


Figure 6.12 The variation, with temperature, of the molar entropies of several elements and compounds.

Chapter 6

6.4 Copper exists in the state $T=298\text{ K}$, $P=1\text{ atm}$. Calculate the temperature to which the copper must be raised at 1 atm pressure to cause the same increase in molar enthalpy as is caused by increasing its pressure to 1000 atm at 298 K .

The molar volume of Cu at 298 K is 7.09 cm^3 , and the volumetric coefficient of thermal expansion is $0.501 \times 10^{-4}\text{ K}^{-1}$. These values can be taken as being independent of pressure in the range $1\text{--}1000\text{ atm}$.

$$\Delta H = \int_{P_1}^{P_2} V(1 - \alpha T) dp.$$

6.7 The influence of pressure on enthalpy and entropy

For one mole of a closed system of fixed composition undergoing a change of pressure at constant temperature,

$$dH = \left(\frac{\partial H}{\partial P} \right)_T dP$$

$$dH = TdS + VdP \quad \Rightarrow \quad \frac{dH}{dP} = T \frac{dS}{dP} + V \quad \Rightarrow \quad \left(\frac{\partial H}{\partial P} \right)_T = T \left(\frac{\partial S}{\partial P} \right)_T + V$$

A Maxwell relation gives: $\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$

$$\left(\frac{\partial H}{\partial P} \right)_T = -T \left(\frac{\partial V}{\partial T} \right)_P + V$$

6.7 The influence of pressure on enthalpy and entropy

$$\left(\frac{\partial H}{\partial P}\right)_T = -T\left(\frac{\partial V}{\partial T}\right)_P + V$$

The isobaric coefficient of thermal expansion, α , is defined as

$$\left(\frac{\partial V}{\partial T}\right)_P = \alpha V \quad \left(\frac{\partial H}{\partial P}\right)_T = -TV\alpha + V = V(1 - \alpha T)$$

The change in molar enthalpy caused by the change in state from (P_1, T) to (P_2, T) is thus

$$\Delta H = H(P_2, T) - H(P_1, T) = \int_{P_1}^{P_2} V(1 - \alpha T) dP$$

For an ideal gas, $\alpha = 1/T$, and thus, this equation shows once again that the enthalpy of an ideal gas is independent of pressure.

Chapter 6

Solution to 6.4

$$P_1=1 \text{ atm}, P_2=1000 \text{ atm}, T=298\text{K}$$

$$\begin{aligned}\Delta H &= V(1-\alpha T)(P_2 - P_1) \\ &= 7.09 \times 10^{-3} (1 - 0.493 \times 10^{-3} \times 298)(1000 - 1) \times 101.3 \\ &= 612 J\end{aligned}$$

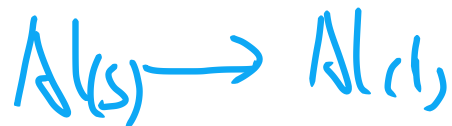
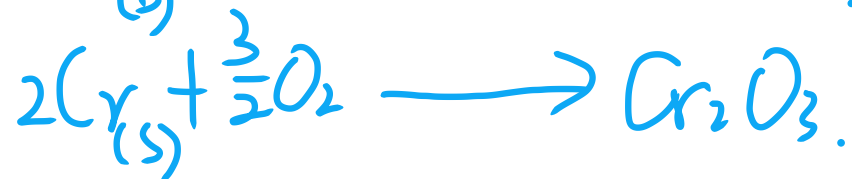
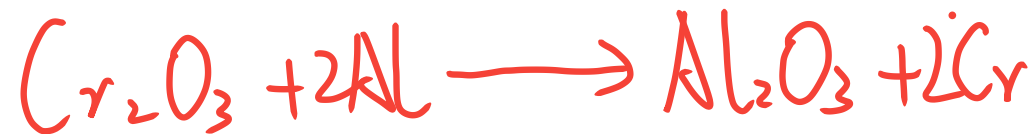
$$T_1=298\text{K}, T_2=T, P=1 \text{ atm}$$

$$\Delta H = \int_{T_1}^T c_p dT = 22.64(T - 298) - 6.28 \times 10^{-3} (T^2 - 298^2)$$

$$\text{So } T=327\text{K}$$

Chapter 6

6.6 An adiabatic vessel contains 1000 g of liquid aluminum at 700°C. Calculate the mass of Cr_2O_3 at room temperature, which, when added to the liquid aluminum (with which it reacts to form Cr and Al_2O_3), raises the temperature of the resulting mixture of Al_2O_3 , Cr_2O_3 , and Cr to 1600 K.



Chapter 6

Solution to 6.6

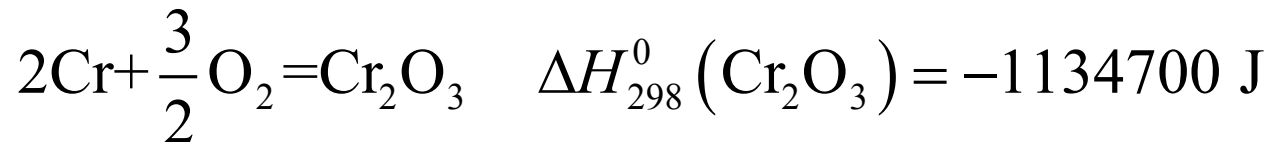
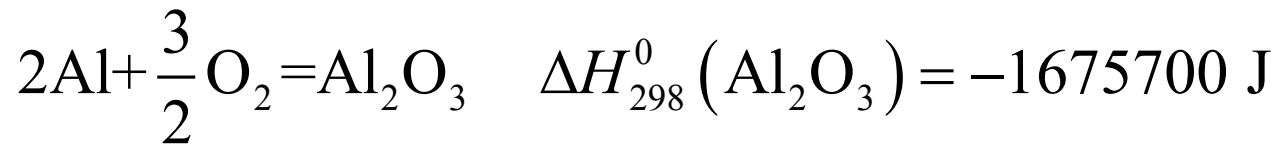
For Cr_2O_3 $c_p = 119.37 + 9.30 \times 10^{-3} T - 15.65 \times 10^5 T^{-2}$ at 298K-1800K

For Al_2O_3 $c_p = 106.6 + 17.78 \times 10^{-3} T - 28.53 \times 10^5 T^{-2}$ at 298K-2325K

For Cr(s) $c_p = 24.43 + 9.87 \times 10^{-3} T - 3.68 \times 10^5 T^{-2}$ at 298K-2130K

For Al(s) $c_p = 20.67 + 12.38 \times 10^{-3} T$ at 298K-933K

For Al(l) $c_p = 31.76$ at 933K-1600K



Chapter 6

Solution to 6.6

According to the meaning of the questions, all Al have transferred to Al_2O_3 . So in the final state, there are $500/27$ moles Al_2O_3 and $1000/27$ moles Cr and (x) moles Cr_2O_3

Process 1: the temperature of the liquid Al decrease from 973K to 298K

$$\begin{aligned}\Delta H_1 &= 1000 / 27 \left(\int_{973}^{933} 31.76 dT + \Delta H(l \rightarrow s) + \int_{933}^{298} 20.67 + 12.38 \times 10^{-3} T dT \right) \\ &= 1000 / 27 (-1270.4 - 10700 - 17964) = -1108669.635\text{J} \quad \checkmark\end{aligned}$$

Process 2: reaction at 298K $2\text{Al} + \text{Cr}_2\text{O}_3 = 2\text{Cr} + \text{Al}_2\text{O}_3$

$$\Delta H_m^{298} = \Delta H_{298}^0(\text{Al}_2\text{O}_3) - \Delta H_{298}^0(\text{Cr}_2\text{O}_3) = -541000\text{J}$$

$$\Delta H_2 = \frac{500}{27} \Delta H_m^{298} = -10018218.52\text{J} \quad \checkmark$$

Chapter 6

Process 3: the temperature of the mixture raise from 298K to 1600K

$$\begin{aligned}\Delta H_3 &= 1000 / 27 \int_{298}^{1600} c_P(\text{Cr}) dT + x \int_{298}^{1600} c_P(\text{Cr}_2\text{O}_3) dT + 500 / 27 \int_{298}^{1600} c_P(\text{Al}_2\text{O}_3) dT \\&= 1000 / 27 \int_{298}^{1600} 24.43 + 9.87 \times 10^{-3} T - 3.68 \times 10^5 T^{-2} dT \\&\quad + x \int_{298}^{1600} 119.37 + 9.30 \times 10^{-3} T - 15.65 \times 10^5 T^{-2} dT \\&\quad + 500 / 27 \int_{298}^{1600} 106.6 + 17.78 \times 10^{-3} T - 28.53 \times 10^5 T^{-2} dT \\&= 1000 / 27 \left(24.43T + 4.935 \times 10^{-3} T^2 + 3.68 \times 10^5 T^{-1} \right)_{298}^{1600} \\&\quad + x \left(119.37T + 4.65 \times 10^{-3} T^2 + 15.65 \times 10^5 T^{-1} \right)_{298}^{1600} \\&\quad + 500 / 27 \left(106.6T + 8.89 \times 10^{-3} T^2 + 28.53 \times 10^5 T^{-1} \right)_{298}^{1600} \\&= 1000 / 27 \times 42998.3 + 162637.248x + 500 / 27 \times 152971.43 \\&= 4425334.404 + 162637.248x\end{aligned}$$

Chapter 6

According to the meanings of the question, we have:

$$\Delta H_1 + \Delta H_2 + \Delta H_3 = 0$$

So $x=41.2055$ mole, $n(\text{Cr}_2\text{O}_3)=x+500/27=59.72$ mole,

$m(\text{Cr}_2\text{O}_3)=n(\text{Cr}_2\text{O}_3)M(\text{Cr}_2\text{O}_3)=9078$ g

Chapter 6

6.8 Calculate the value of ΔG at 800 K for the reaction



What percentage error occurs if it is assumed that Δc_p for the reaction is zero?

Chapter 6

Solution to 6.8



Substance	$\Delta H_{298}^0(\text{J})$	$S_{298}^0(\text{J/K})$	a	b	c
Si_3N_4	-744800	113.0	70.54	98.74×10^{-3}	-
SiO_2	-910900	41.5	43.89	1.00×10^{-3}	-6.02×10^5
O_2	-	205.1	29.96	4.18×10^{-3}	-1.67×10^5
N_2	-	191.5	27.87	4.27×10^{-3}	-

$$c_p = a + bT + cT^{-2} \quad (\text{J/mole} \cdot \text{K})$$

Chapter 6



$$\begin{aligned}\Delta H_{298} &= 3\Delta H_{\text{SiO}_2} + 2\Delta H_{\text{N}_2} - (\Delta H_{\text{Si}_3\text{N}_4} + 3\Delta H_{\text{O}_2}) \\ &= 3 \times (-910900) - (-744800) \\ &= -1987900 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{298} &= 3S_{\text{SiO}_2} + 2S_{\text{N}_2} - (S_{\text{Si}_3\text{N}_4} + 3S_{\text{O}_2}) \\ &= 3 \times 41.5 + 2 \times 191.5 - (113.0 + 3 \times 205.1) \\ &= -220.8 \text{ J/K}\end{aligned}$$

$$\begin{aligned}\Delta c_p &= 3 \times 43.89 + 3 \times 1.0 \times 10^{-3}T - 3 \times 6.02 \times 10^5 T^{-2} + 2 \times 27.87 + 2 \\ &\times 4.27 \times 10^{-3}T - 70.54 - 98.74 \times 10^{-3}T - 3 \times 29.96 - 3 \times 4.18 \\ &\times 10^{-3}T + 3 \times 1.67 \times 10 T^{-2} \\ &= 26.99 - 0.09974T - 1305000.0T^{-2}\end{aligned}$$

Chapter 6

$$\begin{aligned}\Delta H_{800} &= \Delta H_{298} + \int_{298}^{800} \Delta c_p dT \\ &= -1987900 + \int_{298}^{800} (26.99 - 0.09974T - 1305000.0T^{-2}) dT \\ &= -2004587.1 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{800} &= \Delta S_{298} + \int_{298}^{800} \frac{\Delta c_p}{T} dT \\ &= -220.8 + \int_{298}^{800} \frac{(26.99 - 0.09974T - 1305000.0T^{-2})}{T} dT \\ &= -250.5 \text{ J/K}\end{aligned}$$

$$\begin{aligned}\Delta G_{800} &= \Delta H_{800} - 800\Delta S_{800} \\ &= -2004587.1 - 800 \times (-250.5) \\ &= -1804.2 \text{ kJ}\end{aligned}$$

Chapter 6

If $\Delta c_p = 0$, then

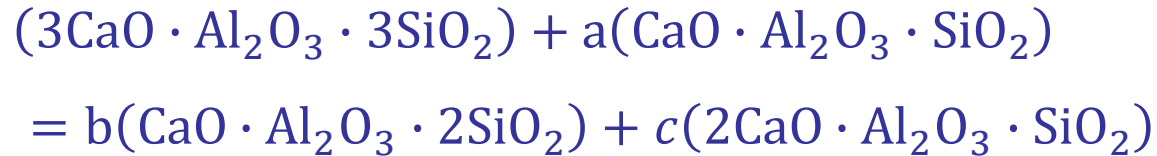
$$\begin{aligned}\Delta G_{800} &= \Delta H_{298} - 800\Delta S_{298} \\ &= -1987900 - 800 \times (-220.8) \\ &= -1811.3 \text{ kJ}\end{aligned}$$

$$Error\% = \frac{1811.3 - 1804.2}{1804.2} \times 100\% = 0.39\%$$

The error is only about 0.4%

Chapter 6

6.9 Determine the stoichiometric coefficients for the reaction



and calculate ΔH_{298} , ΔS_{298} , and ΔG_{298} for the reaction.

Substance	$\Delta H_{298}^0(\text{J})$	$S_{298}^0(\text{J/K})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	– 6,646,300	241.4
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	– 3,293,200	144.8
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	– 4,223,700	202.5
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	– 3,989,400	198.3

Chapter 6

Solution to 6.9

$$(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) + a(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2) \\ = b(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + c(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

According to the conservation of mass

$$3 + a = b + 2c$$

$$1 + a = b + c$$

$$3 + a = 2b + c$$

$$a = 3$$

$$b = c = 2$$

So

$$\Delta H_{298}$$

$$= -\Delta H_{298}(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) - 3\Delta H_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$+ 2\Delta H_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + 2\Delta H_{298}(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$= 6646300 + 3 \times 3293200 - 2 \times 4223700 - 2 \times 3989400$$

$$= 99700 \text{ J}$$

Substance	$\Delta H_{298}^0(\text{J})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	- 6,646,300
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	- 3,293,200
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	- 4,223,700
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	- 3,989,400

Chapter 6

$$\Delta S_{298}$$

$$= -S_{298}(3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2) - 3S_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$+ 2S_{298}(\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2) + 2S_{298}(2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2)$$

$$= -241.4 - 3 \times 144.8 + 2 \times 202.5 + 2 \times 198.3$$

$$= 125.8 \text{ J/K}$$

$$\Delta G_{298} = \Delta H_{298} - T\Delta S_{298} = 62211.6 \text{ J}$$

Substance	$S_{298}^0(\text{J/K})$
$3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2$	241.4
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	144.8
$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	202.5
$2\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	198.3

Chapter 6

6.11 *A function $\Omega \equiv -A + \mu N$, has been defined, where A is the Helmholtz free energy, μ is the chemical potential and N is the number of particles in the system. Obtain an expression for $d\Omega$ and thus determine the independent variables for this function.

Chapter 6

Solution to 6.11

$$d\Omega = -dA + \mu dN + Nd\mu$$

$$dA = -SdT - PdV + \mu dN$$

$$d\Omega = SdT + PdV - \mu dN + \mu dN + Nd\mu$$

$$d\Omega = SdT + PdV + Nd\mu$$

$$\Omega = \Omega(T, V, \mu)$$

Chapter 6

6.12*(a) Given that the Helmholtz energy can be written as

$$A = -N_{\text{system}} k_B T \ln Z$$

Obtain simplified expressions for both A and S of an Einstein solid.

(b) Show that the entropy approaches 0 as the temperature approaches 0 in such a solid.

Chapter 6

Solution to 6.12

$$\varepsilon_i = \left(i + \frac{1}{2}\right) h\nu$$

From the text we saw that Z for the Einstein solid can be written as:

$$Z = \sum \exp\left(-\frac{\varepsilon_i}{kT}\right) = \exp\left[-\frac{1}{2} \frac{h\nu}{k_B T}\right] \left[\frac{1}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} \right]$$

$$A = -N_{\text{system}} k_B T \ln Z$$

$$A = \frac{1}{2} N_{\text{system}} h\nu + N_{\text{system}} k_B T \ln \left[1 - \exp\left(\frac{-h\nu}{k_B T}\right) \right]$$

$$-\left(\frac{\partial A}{\partial T}\right)_V = S = N_{\text{system}} k_B \ln Z + N_{\text{system}} k_B T \left(\frac{\partial \ln Z}{\partial T}\right)_V$$

Chapter 6

$$S = N_{\text{system}} k_B \ln Z + N_{\text{system}} k_B T \left(\frac{\partial \ln Z}{\partial T} \right)_V$$

$$Z = \exp \left[-\frac{1}{2} \frac{h\nu}{k_B T} \right] \left[\frac{1}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} \right]$$

$$S = N_{\text{system}} k_B \left(\frac{h\nu}{k_B T} \right) \frac{\exp \left(\frac{-h\nu}{k_B T} \right)}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} - N_{\text{system}} k_B \ln \left[1 - \exp \left(\frac{-h\nu}{k_B T} \right) \right]$$

Since $N_{\text{system}} = 3n$

$$S = 3n k_B \left(\frac{h\nu}{k_B T} \right) \frac{\exp \left(\frac{-h\nu}{k_B T} \right)}{1 - \exp \left(\frac{-h\nu}{k_B T} \right)} - 3n k_B \ln \left[1 - \exp \left(\frac{-h\nu}{k_B T} \right) \right]$$

Chapter 6

$$S = 3nk \left(\frac{h\nu}{k_B T} \right) \frac{\exp\left(\frac{-h\nu}{k_B T}\right)}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} - 3nk_B \ln \left[1 - \exp\left(\frac{-h\nu}{k_B T}\right) \right]$$

$$\frac{\left(\frac{h\nu}{k_B T} \right) \exp\left(\frac{-h\nu}{k_B T}\right)}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)} \rightarrow 0 \text{ as } T \rightarrow 0$$

$$1 - \exp\left(\frac{-h\nu}{k_B T}\right) \rightarrow 1 \text{ as } T \rightarrow 0; \ln 1 = 0$$

So $S \rightarrow 0$ as $T \rightarrow 0$

Chapter 6

6.13* Obtain an expression for the internal energy of an Einstein solid as the temperature gets very large, differentiate it with respect to T , and show that the Dulong and Petit value for the heat capacity is attained.

Chapter 6

Solution to 6.13

$$U' = \frac{3}{2}nhv + \frac{3nhv}{\left(\exp\left(\frac{hv}{k_B T}\right) - 1\right)}$$

$$\text{as } T \rightarrow \infty \quad \exp\left(\frac{hv}{k_B T}\right) \rightarrow 1 + \frac{hv}{k_B T}$$

$$U' = \frac{3}{2}nhv + \frac{3nhv}{\left(\frac{hv}{k_B T}\right)} = \frac{3}{2}nhv + 3nk_B T$$

$$\text{Thus } C_V = \frac{\partial U'}{\partial T} = 3nk_B = 3R \quad \text{for one mole substance}$$

Summary of Chapter 7

1. Knowledge of the dependencies, on temperature and pressure, of the changes in molar enthalpy and molar entropy caused by phase changes in a system allows determination of the corresponding change in the molar Gibbs free energy of the system.
2. Since a closed one-component system has only two independent variables, the dependence of G can be examined most simply by choosing the state variables T and P as the independent variables (these are the natural independent variables when G is the dependent variable). The phases in which the material can exist can thus be represented in a three-dimensional diagram using the state function G , the independent state variables P , and T as coordinates.

Summary of Chapter 7

3. If sections of these three-dimensional diagrams are taken at constant P , one obtains G versus T plots for the phases. The phase that is in equilibrium at a given temperature is the one with the lowest G . Crossings of the curve represent temperatures at which two (or three) phases are in equilibrium. The slope of the G versus T plot is the negative of the entropy, and the curvature is related to the heat capacity of the phases delineated by the curves.

4. Sections of the three-dimensional diagrams may be taken at constant temperature. Such plots of P versus T show which phase is in equilibrium as a function of pressure. Their slopes are equal to the molar volume of the phases, and their curvatures are related to the isothermal compressibility of the phase being plotted.

Summary of Chapter 7

5. In the three-dimensional diagrams, the various states in which the material can exist occur as surfaces. In any state, which is determined by the values of P and T , the stable phase is that which has the lowest Gibbs free energy. The surfaces in the diagram intersect with one another along curves, and these curves represent the variations of P with T required for equilibrium between the two phases.

Summary of Chapter 7

6. The intersection of the surfaces for the solid and liquid phases gives the variation of the equilibrium melting temperature with pressure. The intersection of the surfaces for the liquid and vapor phases gives the variation of the boiling temperature with pressure. The normal melting and boiling points of the material occur on these intersections at $P = 1$ atm. Three surfaces intersect at a point in the diagram, and the values of P and T at which this intersection occurs are those of the invariant triple point at which an equilibrium occurs among three phases. In a one-component system, no more than three phases can coexist in equilibrium with one another.

Summary of Chapter 7

7. The three-dimensional G - T - P diagram illustrates the differences between stable, metastable, and unstable states and, hence, shows the difference between reversible and irreversible process paths. At any value of P and T , the stable phase is that which has the lowest Gibbs free energy, and phases which have higher values of G at the same values of P and T are metastable with respect to the phase of lowest value of G . Phases with a value of G at any combination of P and T which do not lie on a surface in the diagram are unstable. A reversible process path involving a change in P and/or T lies on a phase surface, and the state of a phase is changed reversibly only when, during the change, the state of the system does not leave the surface of the phase. If the process path leaves the phase surface, then the change of state, which necessarily passes through non-equilibrium states, is irreversible.

Summary of Chapter 7

8. Since the perspective representation, in two dimensions, of a three-dimensional diagram is difficult, it is normal practice to present the phase diagram for a one component system as the basal plane of the G - T - P diagram (i.e., a P - T diagram), onto which are projected the lines along which two surfaces intersect (equilibrium between two phases) and the points at which three surfaces intersect (equilibrium among three phases). Such a diagram contains areas in which a single phase is stable, which are separated by curves along which two phases exist at equilibrium, and points at the intersection of three curves at which three phases coexist in equilibrium. The curves for equilibrium between a condensed phase and the vapor phase are called *vapor pressure curves*, and they are exponential in form. In view of the fact that saturated vapor pressures can vary over several orders of magnitude, pressure–temperature phase diagrams can often be presented in more useful form as plots of $\ln P$ versus $1/T$ than as plots of P versus T .

Summary of Chapter 7

9. The development of phase diagrams for one-component systems demonstrates the use of the Gibbs free energy as a criterion for equilibrium when T and P are chosen as the independent state variables.

10. The equilibrium Gibbs phase rule for a one-component simple system is where F is the number of degrees of freedom and the ϕ number of phases allowed to be present in equilibrium.

$$F = 3 - \phi$$

11. Thermodynamic equilibrium is affected by the application of external fields, such as a magnetic or electrical field.

Chapter 7

7.1 Using the vapor pressure–temperature relationships for

$\text{CaF}_2(\alpha)$, $\text{CaF}_2(\beta)$, and liquid CaF_2 , calculate

a. The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\alpha) - \text{CaF}_2(\beta) - \text{CaF}_2(v)$ and $\text{CaF}_2(\beta) - \text{CaF}_2(l) - \text{CaF}_2(v)$

b. The normal boiling temperature of CaF_2

c. The molar latent enthalpy of the transformation $\text{CaF}_2(\alpha) \rightarrow \text{CaF}_2(\beta)$

d. The molar latent enthalpy of melting of $\text{CaF}_2(\beta)$

Chapter 7

Solution to 7.1

The saturated vapor pressures of the three state of CaF_2 are:

$$\text{CaF}_2(\alpha): \ln p \text{ (atm)} = -54350/T - 4.525 \ln T + 56.57$$

$$\text{CaF}_2(\beta): \ln p \text{ (atm)} = -53780/T - 4.525 \ln T + 56.08$$

$$\text{CaF}_2(l): \ln p \text{ (atm)} = -50200/T - 4.525 \ln T + 53.96$$

a. The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\alpha) - \text{CaF}_2(\beta) - \text{CaF}_2(v)$:

$$-54350/T - 4.525 \ln T + 56.57 = -53780/T - 4.525 \ln T + 56.08 \Rightarrow$$

$$T = 1163.26 \text{ K}$$

$$p = 2.54 \times 10^{-10} \text{ Pa}$$

Chapter 7

$$\text{CaF}_2(\beta): \ln p \text{ (atm)} = -53780/T - 4.525 \ln T + 56.08$$

$$\text{CaF}_2(l): \ln p \text{ (atm)} = -50200/T - 4.525 \ln T + 53.96$$

The temperatures and pressures of the triple points for the equilibria

$\text{CaF}_2(\beta) - \text{CaF}_2(l) - \text{CaF}_2(v)$:

$$-\frac{53780}{T} - 4.525 \ln T + 56.08 = -\frac{50200}{T} - 4.525 \ln T + 53.96 \Rightarrow$$

$$T = 1688.67 \text{ K}$$

$$p = 8.31 \times 10^{-5} \text{ atm}$$

Chapter 7

b. The normal boiling temperature of CaF_2

When $p=1$ atm, we have

$$0 = \frac{50200}{T_b} - 4.525 \ln T_b + 53.96$$

$$\Rightarrow T_b = 2775.71 \text{ K}$$

$$\text{CaF}_{2(l)}: \ln p (\text{atm}) = -50200/T - 4.525 \ln T + 53.96$$

Chapter 7

c. The molar latent enthalpy of the transformation $\text{CaF}_2(\alpha) \rightarrow \text{CaF}_2(\beta)$

According to the Clausius-Clapeyron equation $d \ln p = \frac{\Delta H}{RT^2} dT$, we have:

$$\frac{d \left(-\frac{54350}{T} - 4.525 \ln T + 56.57 \right)}{dT} = \frac{\Delta H_{(\alpha \rightarrow v)}}{RT^2}$$

$$\frac{d \left(-\frac{53780}{T} - 4.525 \ln T + 56.08 \right)}{dT} = \frac{\Delta H_{(\beta \rightarrow v)}}{RT^2}$$

$$\frac{d \left(-\frac{50200}{T} - 4.525 \ln T + 53.96 \right)}{dT} = \frac{\Delta H_{(l \rightarrow v)}}{RT^2}$$

Chapter 7

$$\Delta H_{(\alpha \rightarrow v)} = 54350R - 4.525RT$$

$$\Delta H_{(\beta \rightarrow v)} = 53780R - 4.525RT$$

$$\Delta H_{(l \rightarrow v)} = 50200R - 4.525RT$$

So,

$$\Delta H_{(\alpha \rightarrow \beta)} = \Delta H_{(\alpha \rightarrow v)} - \Delta H_{(\beta \rightarrow v)} = 570R = 4739.265 \text{ J}$$

d. The molar latent enthalpy of melting of $\text{CaF}_2(\beta)$

We can also get,

$$\Delta H_{(\beta \rightarrow l)} = \Delta H_{(\beta \rightarrow v)} - \Delta H_{(l \rightarrow v)} = 3580R = 29765.91 \text{ J}$$

7.5 Equilibrium between the vapor phase and a condensed phase

For condensed phase–vapor equilibria, the Clapeyron equation can be written as

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{T\Delta V} \xrightarrow{\Delta V = V^{\text{vapor}}} \left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{TV^{\text{vapor}}}$$

If it is further assumed that the vapor in equilibrium with the condensed phase behaves ideally (i.e., $PV=RT$), then

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{P\Delta H}{RT^2}$$

7.5 Equilibrium between the vapor phase and a condensed phase

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{P\Delta H}{RT^2} \quad \Rightarrow \quad \frac{dP}{P} = \frac{\Delta H}{RT^2} dT$$



$$d \ln P = \frac{\Delta H}{RT^2} dT$$

This is Clausius–Clapeyron equation

7.5 Equilibrium between the vapor phase and a condensed phase

$$d \ln P = \frac{\Delta H}{RT^2} dT$$

If ΔH is independent of temperature — that is, if $c_p(\text{vapor})=c_p(\text{condensed phase})$ — integration this equation gives

$$\ln P = \frac{\Delta H}{RT} + \text{constant}$$

where the initial pressure is assumed to be 1 atm.

If we use the boiling point values of the pressure and temperature (P_0 and T_b), we can write

$$\ln \frac{P}{P_0} = -\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right)$$

Chapter 7

7.3 One mole of SiCl_4 vapor is contained at 1 atm pressure and 350 K in a rigid container of fixed volume. The temperature of the container and its contents is cooled to 280 K. At what temperature does condensation of the SiCl_4 vapor begin, and what fraction of the vapor has condensed when the temperature is 280 K?

Chapter 7

Solution to 7.3

The saturated vapor pressures of the SiCl_4 is

$$\ln p = -3620 / T + 10.96 \Rightarrow \ln V = 3620 / T - 10.96 + \ln RT$$

At first, there is one mole of SiCl_4 vapor contained at 1 atm pressure and 350 K in a rigid container of fixed volume. So $V=2910.75 \text{ J/atm}$

$$\ln 2910.75 = 3620 / T - 10.96 + \ln 8.3145T \Rightarrow T = 328.3701 \text{ K}$$

Chapter 7

When $T=280\text{K}$, the system reaches equilibrium. so

$$\ln p = -3620 / 280 + 10.96 \Rightarrow P = 0.1396 \text{ atm}$$

$$PV = nRT \Rightarrow n_{350} = \frac{V}{350R}, n_{280} = \frac{0.1396V}{280R}$$

$$\eta = \frac{\frac{V}{350R} - \frac{0.1396V}{280R}}{\frac{V}{350R}} = 82.55\%$$

Chapter 7

7.6 Below the triple point ($-56.2\text{ }^{\circ}\text{C}$), the vapor pressure of solid CO_2 is given as

$$\ln P(\text{atm}) = -\frac{3116}{T} + 16.01$$

The molar latent enthalpy of melting of CO_2 is 8330 J . Calculate the vapor pressure exerted by liquid CO_2 at $25\text{ }^{\circ}\text{C}$ and explain why solid CO_2 is referred to as “dry ice”

Chapter 7

Solution to 7.6

– 56.2 °C

$$\ln P(\text{atm}) = -\frac{3116}{T} + 16.01$$

$$d \ln P = \frac{\Delta H}{RT^2} dT$$

The triple point pressure is

$$\ln p = -3116 / 216.8 + 16.01 \Rightarrow p = 5.14 \text{ atm}$$

$$\frac{d \ln p}{dT} = \frac{3116}{T^2} = \frac{\Delta H_{(s \rightarrow v)}}{RT^2}$$

$$\Delta H(s \rightarrow v) = 8.3144 \times 3116 = 25907 \text{ J}$$

$$\Delta H(s \rightarrow l) = 8330 \text{ J}$$

$$\Delta H(l \rightarrow v) = 25907 - 8330 = 17577 \text{ J}$$

Chapter 7

For liquid CO₂

$$\ln p_{\text{CO}_2(l)} = \frac{-\Delta H_{(l \rightarrow v)}}{RT} + \text{constant}$$

At the triple point

$$\ln 5.14 = \frac{-17577}{216.8R} + \text{constant} \Rightarrow \text{constant} = 11.39$$

$$\ln p_{\text{CO}_2(l)} = \frac{-\Delta H_{(l \rightarrow v)}}{RT} + 11.39$$

At 298K

$$p_{\text{CO}_2(l)} = 73.3 \text{ atm}$$

The triple-point pressure is 5.14 atm, and, as the 1 atm isobar does not pass through the liquid-phase field, liquid CO₂ is not stable at atmospheric pressure at 25 °C.

Chapter 7

7.7 The molar volumes of solid and liquid lead (铅) at the normal melting temperature of lead are, respectively, 18.92 and 19.47 cm³. Calculate the pressure which must be applied to lead in order to increase its melting temperature by 20 °C.

Chapter 7

Solution to 7.7

For Pb at 600K

$$\Delta H(s \rightarrow l) = 4810 \text{ J}$$

$$\left(\frac{dp}{dT}\right)_{eq} = \frac{\Delta H}{T\Delta V} = \frac{4810}{T(19.47 - 18.92) \times 10^{-6}} = \frac{8.745 \times 10^9}{T}$$

Increase its melting temperature by 20 °C

$$P - 1 = \int_{600}^{620} \frac{8.745 \times 10^9}{T} dT = 8.745 \times 10^9 \ln T_{600}^{620} = 2829 \text{ atm}$$

So $P=2830 \text{ atm}$

Chapter 7

7.9 Measurements of the saturated vapor pressure of liquid NdCl_5 give 0.3045 atm at 478 K and 0.9310 atm at 520 K. Calculate the normal boiling temperature of NdCl_5 .

Chapter 7

Solution to 7.9

The vapor pressure equation is of the form

0.3045 atm at 478 K
0.9310 atm at 520 K

$$\ln p = \frac{A}{T} + B$$

$$\ln(0.3045) = \frac{A}{478} + B$$

$$\ln(0.9310) = \frac{A}{520} + B$$

So, $A = -6614$ and $B = 12.56$

At $P = 1 \text{ atm}$, $T_b = 523 \text{ K}$

Chapter 7

7.10* Three allotropes α , β and γ of a certain element are in equilibrium at its triple point (Figure 7.19). It is known that

$$V_m^\gamma > V_m^\alpha \text{ and } S_m^\gamma < S_m^\beta$$

Determine which regions of the diagram are α , β and γ . Explain your reasoning.

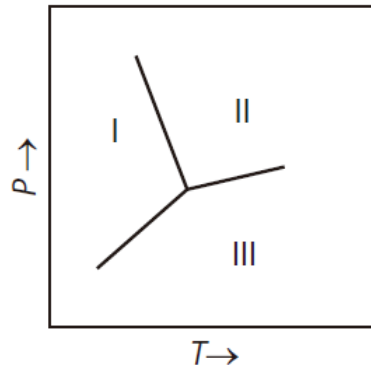


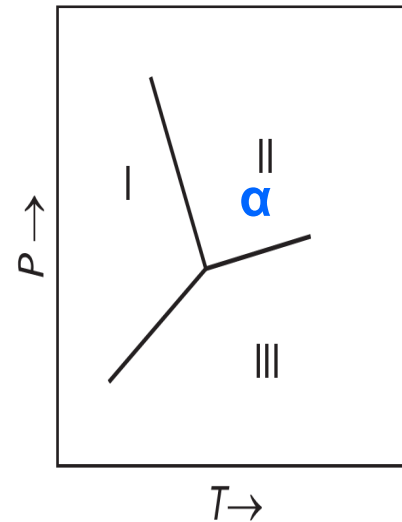
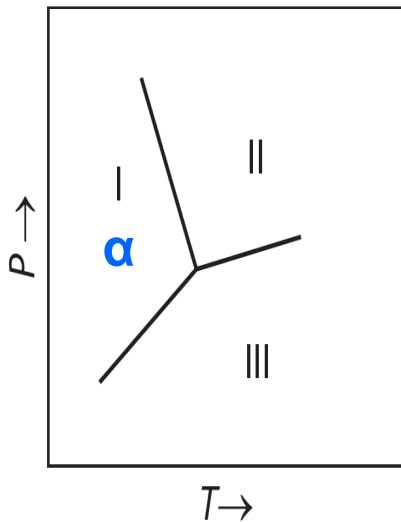
Figure 7.19 Schematic pressure–temperature phase diagram for a one-component system in the vicinity of its triple point. All phases are solids.

Chapter 7

Solution to 7.10

If the pressure is increased, equilibrium of the system favors the lowest molar volume phase.

Therefore, α is in either region I or II of Figure 7.19, since $V_m^\gamma > V_m^\alpha$.



Chapter 7

Assumed I= α , if II= β , then
III= γ .

$$\left(\frac{dP}{dT}\right)_{\alpha\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\alpha}{V^\gamma - V^\alpha} > 0$$

$$\therefore V_m^\gamma > V_m^\alpha$$

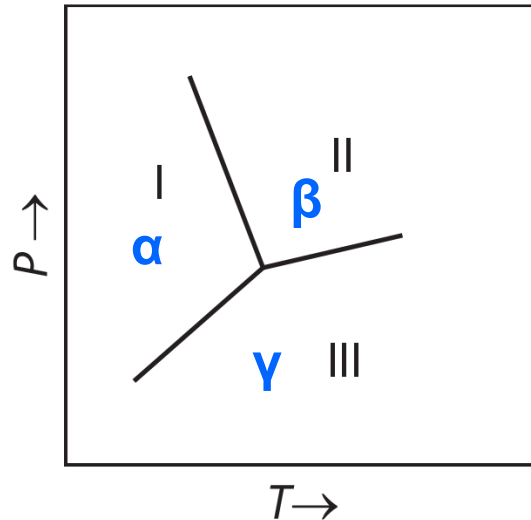
$$\therefore S^\gamma > S^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\beta\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\beta}{V^\gamma - V^\beta} > 0$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma > V_m^\alpha$$



$$\left(\frac{dP}{dT}\right)_{\alpha\beta} = \frac{\Delta S}{\Delta V} = \frac{S^\beta - S^\alpha}{V^\beta - V^\alpha} < 0$$

$$\therefore V_m^\beta > V_m^\alpha,$$

$$\therefore S^\alpha > S^\beta$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore S^\alpha > S^\beta > S^\gamma$$

$$\therefore S^\alpha > S^\gamma$$

It is impossible since $S^\gamma > S^\alpha$.

$$\therefore \beta = \text{III}, \quad \gamma = \text{II}.$$

Chapter 7

But, at the triple point, increasing the pressure moves equilibrium to the lowest molar volume phase.

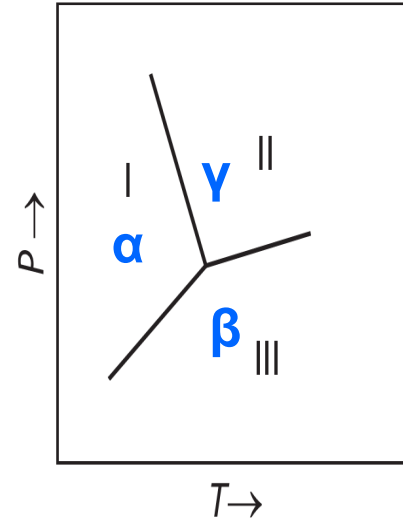
$$\because V_m^\gamma > V_m^\alpha$$

$$\therefore \gamma \neq \text{II}$$

$$\therefore \alpha \neq \text{I}$$

$$\therefore \alpha = \text{II}$$

Thus, region II is the α phase field.



Chapter 7

If I=β, then III=γ.

$$\left(\frac{dP}{dT}\right)_{\beta\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\beta}{V^\gamma - V^\beta} > 0$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma$$

$$\therefore V_m^\beta > V_m^\gamma > V_m^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\alpha\gamma} = \frac{\Delta S}{\Delta V} = \frac{S^\gamma - S^\alpha}{V^\gamma - V^\alpha} > 0$$

$$\therefore V_m^\gamma > V_m^\alpha$$

$$\therefore S^\gamma > S^\alpha$$

$$\therefore S^\beta > S^\gamma$$

$$\therefore S^\beta > S^\gamma > S^\alpha$$

$$\left(\frac{dP}{dT}\right)_{\beta\alpha} = \frac{\Delta S}{\Delta V} = \frac{S^\alpha - S^\beta}{V^\alpha - V^\beta} < 0$$

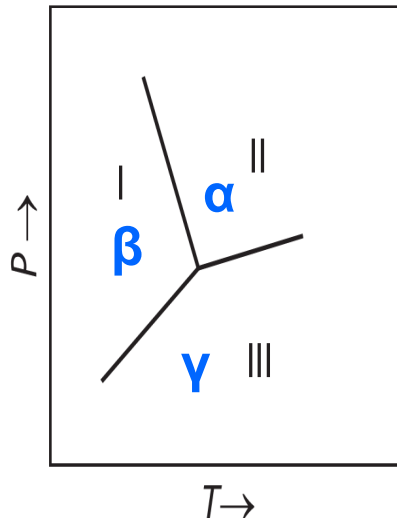
$$\therefore S^\beta > S^\alpha$$

$$\therefore V_m^\beta < V_m^\alpha$$

It is impossible since $V_m^\beta > V_m^\gamma > V_m^\alpha$.

$$\therefore \beta \neq I$$

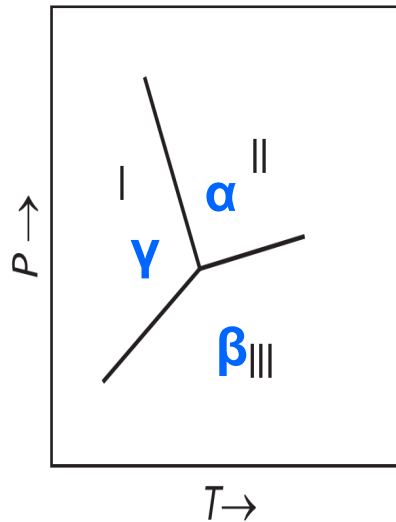
$$\therefore \beta = III, \gamma = I.$$



Chapter 7

Solution to 7.10

Therefore



$$V_m^\gamma > V_m^\alpha \text{ and } S_m^\gamma < S_m^\beta$$

Thermodynamics & Phase transformations

The End



Exercise for chapter 8

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

Chapter 8

8.1. Demonstrate the law of corresponding states by writing the van der Waals equation in terms of the reduced variables. Calculate the compressibility factor for a van der Waals fluid at its critical point and compare the result with the values obtained for real gases at their critical points listed in Table 8.1. Calculate the value of $(\partial U/\partial V)_T$ for a van der Waals fluid.

Chapter 8

Solution to 8.1

For the mole

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$\left(P_R P_{cr} + \frac{a}{V_R^2 V_{cr}^2}\right)(V_R V_{cr} - b) = RT_R T_{cr}$$

$$\left(P_R \frac{R}{27b^2} + \frac{a}{V_R^2 9b^2}\right)(V_R 3b - b) = RT_R \frac{8a}{27bR}$$

$$\frac{ab}{27b^2} \left(P_R + \frac{3}{V_R^2}\right)(3V_R - 1) = \frac{8a}{27b} T_R$$

$$\left(P_R + \frac{3}{V_R^2}\right)(3V_R - 1) = 8T_R$$

Chapter 8

Solution to 8.1

$$\frac{P_{cr} V_{cr}}{RT_{cr}} = \frac{a}{27b^2} \frac{3b}{R} \frac{27bR}{8a} = \frac{3}{8} = 0.375$$

$$dU = TdS - PdV$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial r}\right)_T - P = T\left(\frac{\partial P}{\partial T}\right)_r - P$$

For a van der Waals gas

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

So $\left(\frac{\partial P}{\partial T}\right)_r = \frac{R}{V - b}$ and $\left(\frac{\partial V}{\partial r}\right)_T = \frac{RT}{V - b} - P = \frac{a}{V^2}$

Chapter 8

8.2 n moles of an ideal gas A and $(1 - n)$ moles of an ideal gas B, each at 1 atm pressure, are mixed at total constant pressure. What ratio of A to B in the mixture maximizes the decrease in the Gibbs free energy of the system? If the decrease in the Gibbs free energy is ΔG^M , to what value must the pressure be increased in order to increase the Gibbs free energy of the gas mixture by $1/2 \Delta G^M$?

Chapter 8

Solution to 8.2

Mix n moles of A and $(1-n)$ moles of B at constant $P=1$ atm

$$\Delta G^{Mix} = RT[n \ln n + (1 - n) \ln(1 - n)]$$

$$\begin{aligned} \frac{d\Delta G^{Mix}}{dn} &= RT[\ln n + 1 - \ln(1 - n) - 1] = RT \ln \frac{n}{1 - n} \\ &= 0 \quad \text{when } n = 0.5 \end{aligned}$$

Chapter 8

Solution to 8.2

So Maximum decrease occurs when $n_A = n_B = 0.5$

Increasing the pressure from 1 atm to P atm

increases the free energy by

$$\Delta G = RT \ln P$$

So
$$-\frac{1}{2} \Delta G^{Mix} = RT \ln P = \frac{1}{2} RT \ln 2$$

So
$$P = \sqrt{2} = 1.414 \text{ atm}$$

Chapter 8

8.3 You are responsible for the purchase of oxygen gas which, before use, will be stored at a pressure of 200 atm at 300 K in a cylindrical vessel of diameter 0.2 m and height 2 m. Would you prefer that the gas behaved ideally or as a van der Waals fluid? The van der Waals constants for oxygen are $a = 1.36 \text{ liters}^2 \text{ atm mole}^{-2}$ and $b = 0.0318 \text{ liters/mole}$.

Chapter 8

Solution to 8.3

The volume of the tank is

$$\pi r^2 h = 3.142 \times 0.1^2 \times 2 = 0.06284 m^3 = 62.84 l$$

Thus if the gas is ideal, the tank contains

$$n = \frac{PV'}{RT} = \frac{200 \times 62.84}{0.08206 \times 300} = 511 \text{ moles}$$

If the gas is a van der Waals gas

$$\left(P + \frac{an^2}{V^2} \right) (V' - nb) = nRT$$

Chapter 8

Solution to 8.3

$$\left(200 + \frac{1.36n^2}{62.84^2}\right)(62.84 - 0.0318n) = n \times 0.08206 \times 300$$

Which gives $n=565$ moles

It is thus less expensive to fill the tank if the gas behavior is ideal

Chapter 8

8.4 The virial equation of state for n-butane at 460 K is $Z = 1 + A/V + B/V^2$, in which $A = -265 \text{ cm}^3/\text{g} \cdot \text{mole}$ and $B = 30250 \text{ cm}^6/\text{g} \cdot \text{mole}^2$. Calculate the work required to reversibly compress 1 mole of n-butane from 50 to 100 atm at 460 K.

Chapter 8

Solution to 8.4

$$Z = 1 - \frac{265}{V} + \frac{30250}{V^2}$$

So

$$P(atm) = 82.057 \times 460 \left[\frac{1}{V} - \frac{265}{V^2} + \frac{30250}{V^3} \right]$$

At $P_1 = 50 atm$, this give $V_1 = 394.6 cm^3$

And at $P_2 = 100 atm$, $V_2 = 176.9 cm^3$

Chapter 8

Solution to 8.4

So

$$w = \int_1^2 P dV = RT \int_1^2 \left[\frac{1}{V} - \frac{265}{V^2} + \frac{30250}{V^3} \right] dV$$
$$= RT \left[\ln \frac{V_2}{V_1} + 265 \left(\frac{1}{V_2} - \frac{1}{V_1} \right) - \frac{30250}{2} \left(\frac{1}{V_2^2} - \frac{1}{V_1^2} \right) \right]$$
$$= 8.3144 \times 460 \left[\ln \frac{176.9}{394.6} + 265 \left(\frac{1}{176.9} - \frac{1}{394.6} \right) - \frac{30250}{2} \left(\frac{1}{176.9^2} - \frac{1}{394.6^2} \right) \right]$$
$$= -1384J$$

Chapter 8

- 8.5** For sulfur dioxide, $T_{cr} = 430.7K$ and $P_{cr} = 77.8atm$. Calculate
- The critical van der Waals constants for the gas.
 - The critical volume of van der Waals SO_2 .
 - The pressure exerted by 1 mole of SO_2 occupying a volume of 500 cm^3 at 500 K. Compare this with the pressure which would be exerted by an ideal gas occupying the same molar volume at the same temperature

Chapter 8

Solution to 8.5

$$T_{cr} = \frac{8a}{27bR}$$

$$P_{cr} = \frac{a}{27b^2}$$

$$\frac{T_{cr}}{P_{cr}} = \frac{8b}{R}$$

So

$$b = \frac{430.7 \times 0.08206}{77.8 \times 8} = 0.0567 \text{ l/mole}$$

$$a = 27b^2P_{cr} = 27 \times 0.0567^2 \times 77.8 = 6.77 \frac{\text{atm} \cdot \text{l}^2}{\text{mole}^2}$$

Chapter 8

Solution to 8.5

$$V_{cr} = 3b = 0.170 \text{ l/mole}$$

$$P(\text{van der Waals}) = \frac{RT}{V - b} - \frac{a}{V^2} = \frac{0.08206 \times 500}{0.5 - 0.0567} - \frac{6.77}{0.5^2} \\ = 65.5 \text{ atm}$$

$$P(\text{ideal gas}) = \frac{RT}{V} = \frac{0.08206 \times 500}{0.5} = 82.1 \text{ atm}$$

Chapter 8

8.6 One hundred moles of hydrogen gas at 298 K are reversibly and isothermally compressed from 30 to 10 liters. The van der Waals constants for hydrogen are $a = 0.2461 \text{ liter}^2 \cdot \text{atm} \cdot \text{mole}^{-2}$ and $b = 0.02668 \text{ liter/mole}$, and in the range of pressure 0–1500 atm, the virial equation for hydrogen is $PV = RT(1 + 6.4 \times 10^{-4} P)$. Calculate the work that must be done on the system to effect the required change in volume and compare this with the values that would be calculated assuming that (a) hydrogen behaves as a van der Waals fluid and (b) hydrogen behaves as an ideal gas.

Chapter 8

Solution to 8.6

$$n = 100 \text{ moles}, T = 298\text{K}, V_1 = 30\text{l}, V_2 = 10\text{l}$$

$$PV = nRT(1 + 6.4 \times 10^{-4}P)$$

$$w = \int P dV$$

$$V = nRT \left(\frac{1}{P} + 6.4 \times 10^{-4} \right)$$

$$dV = nRT d \left(\frac{1}{P} \right) = -nRT \frac{dP}{P^2}$$

So

$$w = nRT \ln \frac{P_1}{P_2}$$

Chapter 8

Solution to 8.6

$$PV - 6.4 \times 10^{-4} RT P n = nRT$$

$$P = \frac{nRT}{V - n6.4 \times 10^{-4} RT}$$

So
$$P_1 = \frac{100 \times 0.08206 \times 298}{30 - 100 \times 6.4 \times 10^{-4} \times 0.08206 \times 298} = 86 \text{ atm}$$

$$P_2 = \frac{100 \times 0.08206 \times 298}{10 - 100 \times 6.4 \times 10^{-4} \times 0.08206 \times 298} = 290 \text{ atm}$$

So
$$w = 100 \times 8.3144 \times 298 \ln \frac{86}{290} = -301 \text{ kJ}$$

Chapter 8

Solution to 8.6

Van der Waals:

$$P = \frac{nRT}{V - bn} - \frac{an^2}{V^2}$$

$$a = 0.2461 \frac{\text{atm l}^2}{\text{mole}}, b = 0.02668 \text{ l/mole}$$

$$\begin{aligned} w &= \int P dV = \int \left(\frac{nRT}{V - nb} - \frac{an^2}{V^2} \right) dV \\ &= nRT \ln \left(\frac{V_2 - nb}{V_1 - nb} \right) - an^2 \left(\frac{1}{V_1} - \frac{1}{V_2} \right) \\ &= -309 \text{ kJ} \end{aligned}$$

Ideal gas

$$w = nRT \ln \frac{V_2}{V_1} = 100 \times 8.3144 \times 298 \ln \frac{10}{30} = -272 \text{ kJ}$$

Chapter 8

8.7 Using the virial equation of state for hydrogen at 298 K given in Problem 8.6, calculate

- a. The fugacity of hydrogen at 500 atm and 298 K
- b. The pressure at which the fugacity is twice the pressure
- c. The change in the Gibbs free energy caused by a compression of 1 mole of hydrogen at 298 K from 1 to 500 atm What is the magnitude of the contribution to (c) caused by the nonideality of hydrogen?

Chapter 8

Solution to 8.7

a)
$$\frac{PY}{RT} = Z = 1 + 6.4 \times 10^{-4} P$$

So
$$Z - 1 = 6.4 \times 10^{-4} P$$

$$\ln \frac{f}{P} = \int_0^P \frac{Z - 1}{P} dP = 6.4 \times 10^{-4} (500 - 0) = 0.3194$$

So
$$\frac{f}{P} = 1.376, f = 1.376 \times 500 = 688 \text{ atm}$$

Chapter 8

Solution to 8.7

b)

$$\frac{f}{P} = 2$$

So

$$\ln \frac{f}{P} = 0.693 = 6.4 \times 10^{-4}(P - 0)$$

$$\frac{f}{P} = 2 \text{ at } P = 1083 \text{ atm}$$

Chapter 8

Solution to 8.7

c)

$$\Delta G = \int V dP$$

$$V = RT \left(\frac{1}{P} + 6.4 \times 10^{-4} \right)$$

$$\begin{aligned} \Delta G &= RT \ln \frac{P_2}{P_1} + RT 6.4 \times 10^{-4} (P_2 - P_1) \\ &= 8.3144298 \ln \frac{500}{1} + 8.3144298 6.4 \times 10^{-4} (500 - 1) \\ &= 15400 + 790 = 16190J \end{aligned}$$

And the contribution due to the non-ideal behavior is 790J.

Chapter 8

8.8 Show that the truncated Kamerlingh Onnes virial equation

$$\frac{PV}{RT} = 1 + \frac{B'(T)}{T}$$

reduces to $P(V - b') = RT$, and find the expression for b'

Chapter 8

Solution to 8.8

$$\frac{PV}{\left(1 + \frac{B'(T)}{V}\right)} = RT$$

If $V \gg B'(T)$

then using the expansion

$$\frac{1}{1+x} = 1 - x \quad \text{For} \quad x \ll 1$$

$$PV \left(1 - \frac{B'(T)}{V}\right) = P(V - B'(T)) RT$$

So $b' = B'(T)$

Chapter 8

8.9 Figure 8.8 shows an isothermal variation, with pressure, of the volume of a van der Waals fluid at a temperature below its critical temperature. Redraw this figure and below it sketch the Helmholtz free energy versus volume plot. Be sure to indicate important points on the A versus V plot.

Chapter 8

Solution to 8.9

$$dA_T = -PdV$$

$$\left(\frac{\partial A}{\partial V}\right)_T = -P$$

so the slope should always be < 0

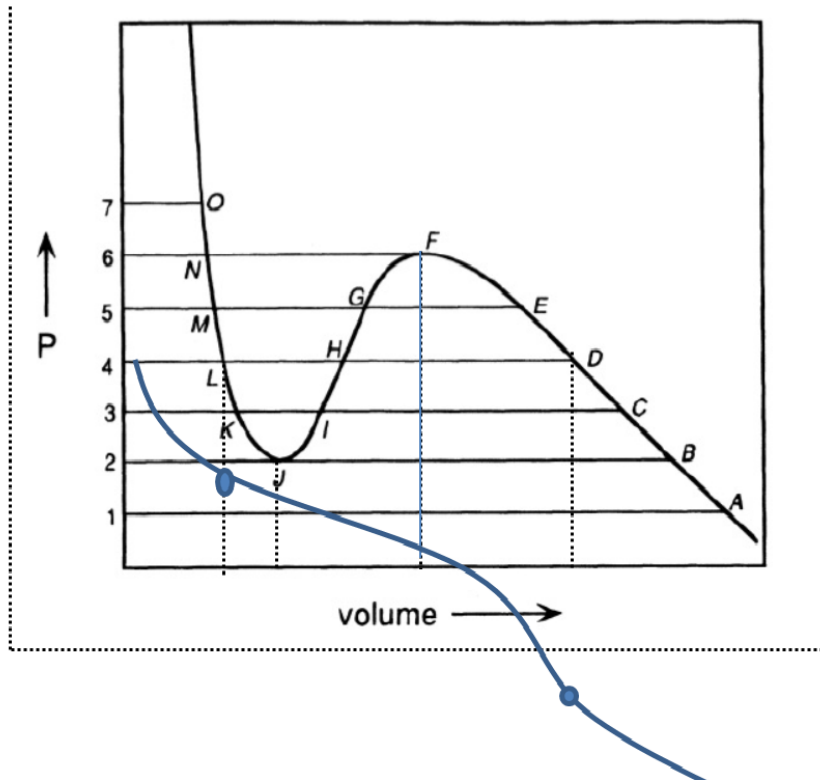
$$\left(\frac{\partial^2 A}{\partial V^2}\right)_T = \left(\frac{\partial P}{\partial V}\right)_T$$

and there are two extrema in P vs V

Chapter 8

Solution to 8.9

Thus there are two points where A vs. V changes its curvature . Also the common tangent delineates the molar volumes of the equilibrium phases.



Chapter 8

8.10 Obtain a simplified expression for the total derivative of the pressure for a van der Waals gas.

Chapter 8

Solution to 8.10

$$dP = \left(\frac{\partial P}{\partial T} \right)_V dT + \left(\frac{\partial P}{\partial V} \right)_T dV$$

$$P = \frac{RT}{(V - b)} - \frac{a}{V^2} \quad \text{for a van der Waals' gas}$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V - b} \quad \text{for a van der Waals' gas}$$

$$\left(\frac{\partial P}{\partial V} \right)_T = -\frac{RT}{(V - b)^2} + \frac{2a}{V^3} \quad \text{for a van der Waals' gas}$$

Chapter 8

Solution to 8.10

Thus

$$dP = \frac{R}{V-b} dT - \left(\frac{RT}{(V-b)^2} + \frac{2a}{V^3} \right) dV$$

which reduces to

$$dP = \frac{R}{V} dT - \frac{RT}{V^2} dV \quad \text{if } a = 0 \text{ and } b = 0$$

Chapter 8

8.11 Obtain a simplified expression for the total derivative of the pressure for a gas that obeys the truncated Kamerlingh Onnes virial equation:

$$\frac{PV}{RT} = 1 + \frac{B'(T)}{T}$$

Chapter 8

Solution to 8.11

$$P = \left(1 + \frac{B'T}{V}\right) \frac{RT}{V}$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V} + \frac{B'(T)R}{V^2} + \frac{RT}{V^2} \left(\frac{\partial B}{\partial T}\right)$$

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{2B'T}{V^3} - \frac{RT}{V^2}$$

$$dP = \left[\frac{R}{V} + \frac{B'(T)R}{V^2} + \frac{RT}{V^2} \left(\frac{\partial B'(T)}{\partial T}\right) \right] dT - \left[\frac{RT}{V^2} + \frac{2B'(T)RT}{V^3} \right] dV$$

which reduces to $dP = \frac{P}{V} dT - \frac{RT}{V^2} dV$ if $B'(T) = 0$

Chapter 8

8.12 Derive an expression for Z_{cr} of a van der Waals fluid

Chapter 8

Solution to 8.12

The equation of state for the van der Waals fluid is

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

Where P is the pressure in P_a

R is the gas constant (8.314 J/K)

V is the molar volume of the gas (m^3)

a and b are empirical constants that vary from gas to gas.

To calculate Z_{cr} we must obtain: P_{cr} , V_{cr} and T_{cr}

Chapter 8

Solution to 8.12

At the critical pressure and volume, the critical isotherm has a horizontal inflection:

$$\left(\frac{\partial P}{\partial V}\right)_{T_C} = \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_C} = 0$$

$$\left(\frac{\partial P}{\partial V}\right)_{T_C} = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3} = 0$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_{T_C} = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4} = 0$$

Solving we get:

$$RT_{cr} = \frac{8a}{27b} \quad V_{cr} = 3b \quad P_{cr} = \frac{a}{27b^2}$$

$$Z_{cr} = \frac{P_{cr}V_{cr}}{RT_{cr}} = \frac{3}{8}$$

Chapter 8

8.13 The ideal entropy of mixing of a binary solution is given as

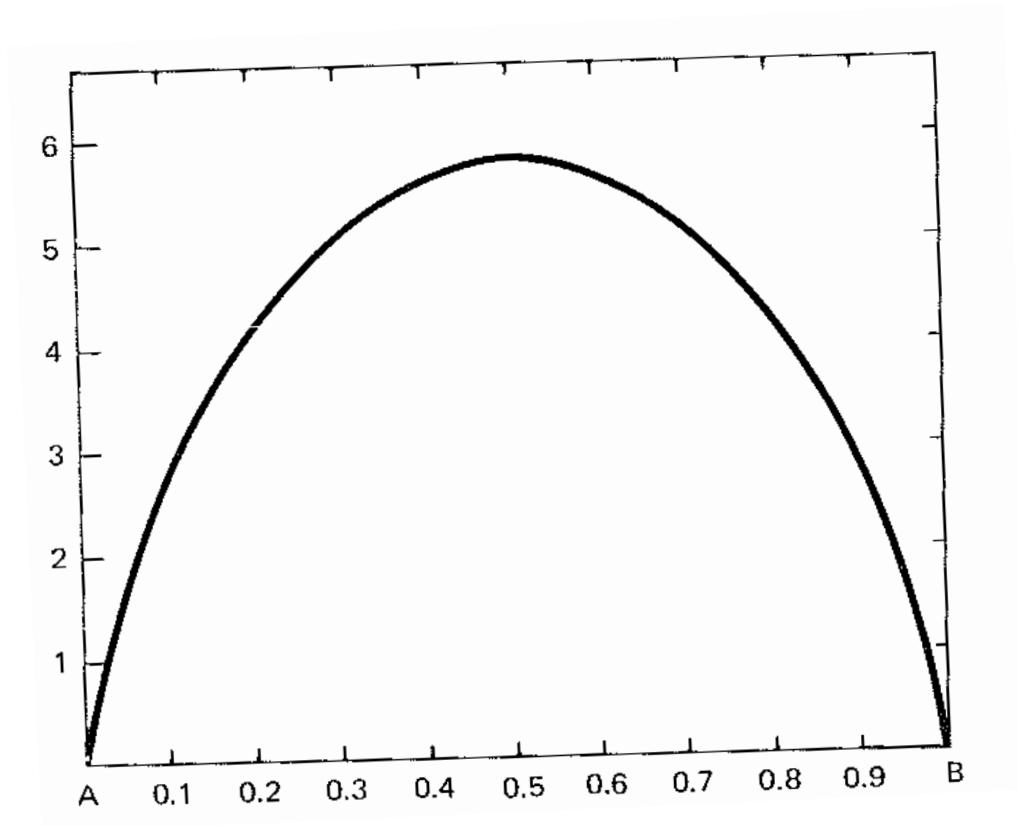
$$\Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B)$$

- Sketch the entropy of mixing versus X_B .
- Calculate (using the expression for the ideal entropy of mixing) the ideal entropy of mixing at $X_B = 0.25$.
- Calculate the slope of this curve at $X_B = 0.25$.
- Calculate the partial molar entropy of component B for the composition $X_B = 0.25$.
- Calculate the partial molar entropy of component A for the composition $X_B = 0.25$.
- Show that your answers in (d) and (e) are consistent with your answer in (c).

Chapter 8

Solution to 8.13

a)



Chapter 8

Solution to 8.13

b) $\Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B) = 4.56 J/mol$

c) $\Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B)$

$$\frac{d\Delta S_{mix}}{dX_B} = -R[\ln X_B - \ln X_A] = R \ln 3 = 9.13 J/mol$$

d) $\overline{S}_B = -R \ln X_B = R \ln 4 = 11.52 J/mol$

e) $\overline{S}_A = -R \ln X_A = R \ln \frac{3}{4} = 2.39 J/mol$

f) $\frac{d\Delta S_{mix}}{dX_B} = R \ln 3 = 9.13 J/mol$

$$\Delta S_{mix} = \overline{S}_B - \overline{S}_A = R \ln 3 = 9.13 J/mol$$

Thermodynamics of Materials & Phase transformations

The End



Exercise for chapter 9

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

Chapter 9

9.1. One mole of solid Cr_2O_3 at 2500 K is dissolved in a large volume of a liquid Raoultian solution of Al_2O_3 and Cr_2O_3 , in which $X_{Cr_2O_3} = 0.2$. and which is also at 2500 K. Calculate the changes in enthalpy and entropy caused by the addition. The normal melting temperature of Cr_2O_3 is 2538 K, and it can be assumed that $\Delta S_{m,Cr_2O_3} = \Delta S_{m,Al_2O_3}$.

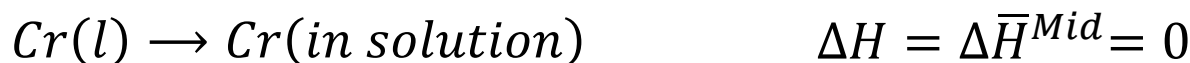
Chapter 9

Solution to 9.1

$$\Delta S_{\text{melting}} = \frac{\Delta H_{\text{melting}}}{T_m} \quad \Delta H_{m, \text{Al}_2\text{O}_3} = 107500 \text{ J}$$

$$\Delta S_{m, \text{Al}_2\text{O}_3} = \frac{107500}{2324} = 46.25 \text{ J/K} = \Delta S_{m, \text{Cr}_2\text{O}_3}$$

So $\Delta H_{m, \text{Cr}_2\text{O}_3} = T_m \Delta S_m = 2538 \times 46.25 = 117400 \text{ J}$



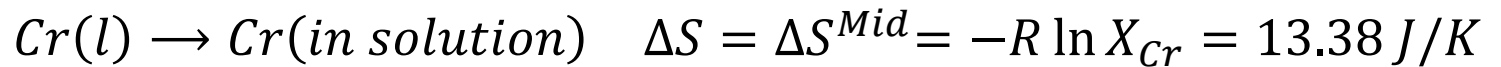
So $\Delta H_{\text{Total}} = 117400 \text{ J}$

Table A.5 Molar Heats of Melting and Transformation

Substance	Trans.	$\Delta H_{\text{trans}}, \text{ J}$	$T_{\text{trans}}, \text{ K}$
Ag	$s \rightarrow l$	11,090	1,234
Al	$s \rightarrow l$	10,700	934
Al₂O₃	$s \rightarrow l$	107,500	2,324
Au	$s \rightarrow l$	12,600	1,338
Ba	$\alpha \rightarrow \beta$	630	648
Ba	$\beta \rightarrow l$	7,650	1,003
Cu	$s \rightarrow l$	12,970	1,356
Ca	$\alpha \rightarrow \beta$	900	716
CaF ₂	$s \rightarrow l$	31,200	1,691

Chapter 9

Solution to 9.1



So $\Delta S_{Total} = 46.25 + 13.38 = 59.63 \text{ J/K}$

Chapter 9

9.2 When 1 mole of argon gas is bubbled through a large volume of an Fe–Mn melt of $X_{Mn} = 0.5$ at 1863 K, the evaporation of Mn into the Ar causes the mass of the melt to decrease by 1.5 g. The gas leaves the melt at a pressure of 1 atm. Calculate the activity coefficient of Mn in the liquid alloy.

Chapter 9

Solution to 9.2

Table A.4 The Saturated Vapor Pressures of Various Substances

$\ln p \text{ (atm)} = -A/T + B \ln T + C$				
Substance	A	B	C	Range, K
CaF _{2(α)}	54,350	-4.525	56.57	298–1430
CaF _{2(β)}	53,780	-4.525	56.08	1430–1691 (<i>T_m</i>)
CaF _{2(l)}	50,200	-4.525	53.96	1691–2783 (<i>T_b</i>)
Fe _(l)	45,390	-1.27	23.93	1809 (<i>T_m</i>)–3330 (<i>T_b</i>)
Hg _(l)	7,611	-0.795	17.168	298–630 (<i>T_b</i>)
Mn _(l)	33,440	-3.02	37.68	1517 (<i>T_m</i>)–2348 (<i>T_b</i>)
SiCl _{4(l)}	3,620	—	10.96	273–333 (<i>T_b</i>)
Zn _(l)	15,250	-1.255	21.79	693 (<i>T_m</i>)–1177 (<i>T_b</i>)

The saturated vapor pressure of liquid M_n at 1863K is

$$P_{Mn}^{\theta} = \exp \left(\frac{-33440}{1863} - 3.02 \ln 1863 + 37.68 \right) = 0.0493 \text{ atm}$$

The number of mole of M_n removed is $\frac{1.50}{54.94} = 0.0273$

The partial pressure of M_n in the gas mixture is

$$P_{Mn} = \frac{n_{Mn}}{n_{Mn} + n_{Ar}} = \frac{0.0273}{1.0273} = 0.0266$$

Chapter 9

Solution to 9.2

So

$$a_{Mn} = \frac{P_{Mn}}{P_{Mn}^{\ominus}} = \frac{0.0266}{0.0493} = 0.539$$

$$a_i = \frac{p_i}{p_i^{\circ}}$$

$$\gamma_{Mn} = \frac{a_{Mn}}{X_{Mn}} = \frac{0.539}{0.5} = 1.08$$

Chapter 9

9.3 The variation, with composition, of G^{XS} for liquid Fe–Mn alloys at 1863 K is as follows

X_{Mn}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
G^{XS}, J	395	703	925	1054	1100	1054	925	703	395

- Does the system exhibit regular solution behavior?
- Calculate $\bar{G}_{Fe}^{XS}$ and $\bar{G}_{Mn}^{XS}$ at $X_{Mn} = 0.6$.
- Calculate ΔG^M at $X_{Mn} = 0.4$.
- Calculate the partial pressures of Mn and Fe exerted by the alloy of $X_{Mn} = 0.2$.

Chapter 9

Solution to 9.3

a) Calculation of $\alpha = \frac{G^{XS}}{X_{Fe}X_{Mn}}$ for each composition gives an average value of 4396 J, therefore with respect to G^{XS} , the system is regular.

b)
$$\bar{G}_{Fe}^{XS} = \alpha X_{Mn}^2 = 4396 \times 0.6^2 = 1583 \text{ J}$$

$$\bar{G}_{Mn}^{XS} = \alpha X_{Fe}^2 = 4396 \times 0.4^2 = 703 \text{ J}$$

c)
$$\Delta G^M = 8.3144 \times 1868(0.4 \ln 0.4 + 0.6 \ln 0.6) + 4396 \times 0.4 \times 0.6 = -9370 \text{ J}$$

Chapter

Solution to 9.3

d) At $X_{Mn} = 0.2$

$$\gamma_{Mn} = \exp\left(\frac{4396 \times 0.8^2}{8.3144 \times 1863}\right) = 1.2$$

$$\gamma_A = \exp\left(\frac{\alpha X_B^2}{RT}\right) \quad (9.64)$$

$$\gamma_B = \exp\left(\frac{\alpha X_A^2}{RT}\right) \quad (9.65)$$

So $a_{Mn} = 1.2 \times 0.2 = 0.24$ $\gamma_i = \frac{a_i}{X_i}$

$$P_{Mn}^\ominus = \exp\left(\frac{-33440}{1863} - 3.02 \ln 1863 + 37.68\right) = 0.0493 \text{ atm}$$

So $P_{Mn} = a_{Mn} P_{Mn}^\ominus = 0.24 \times 0.0493 = 0.0018 \text{ atm}$

Table A.4 The Saturated Vapor Pressures of Various Substances

$$\ln p \text{ (atm)} = -A/T + B \ln T + C$$

Substance	A	B	C	Range, K
CaF _{2(α)}	54,350	-4.525	56.57	298–1430
CaF _{2(β)}	53,780	-4.525	56.08	1430–1691 (<i>T_m</i>)
CaF _{2(l)}	50,200	-4.525	53.96	1691–2783 (<i>T_b</i>)
Fe _(l)	45,390	-1.27	23.93	1809 (<i>T_m</i>)–3330 (<i>T_b</i>)
Hg _(l)	7,611	-0.795	17.168	298–630 (<i>T_b</i>)
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SiCl _{4(l)}	3,620	—	10.96	273–333 (<i>T_b</i>)
Zn _(l)	15,250	-1.255	21.79	693 (<i>T_m</i>)–1177 (<i>T_b</i>)

Chapter 9

Solution to 9.3

$$\gamma_{Fe} = \exp\left(\frac{4396 \times 0.2^2}{8.3144 \times 1863}\right) = 1.011$$

So $a_{Fe} = 1.011 \times 0.8 = 0.809$

$$P_{Fe}^{\ominus} = \exp\left(\frac{-45390}{1863} - 1.27 \ln 1863 + 23.93\right) = 4.55 \times 10^{-5} \text{ atm}$$

So $P_{Fe} = a_{Fe} P_{Fe}^{\ominus} = 0.809 \times 4.55 \times 10^{-5} = 3.68 \times 10^{-5} \text{ atm}$

Chapter 9

9.4 Calculate the heat required to form a liquid solution at 1356 K, starting with 1 mole of Cu and 1 mole of Ag at 298 K. At 1356 K, the molar heat of mixing of liquid Cu and liquid Ag is given by $\Delta H^M = -20590X_{Cu}X_{Ag}$

Chapter 9

Solution to 9.4

Heat 1 mole of Cu from 298K to 1356K and melt it

$$\begin{aligned}\Delta H &= \int_{298}^{1356} C_{p,Cu} dT + \Delta H_{m,Cu} \\ &= 22.64(1356 - 298) + \frac{6.28 \times 10^{-3}}{2}(1356^2 - 298^2) + 12970 \\ &= 42420J\end{aligned}$$

Chapter 9

Solution to 9.4

Heat 1 mole of Ag from 298K to 1356K and melt it

$$\begin{aligned}\Delta H &= \int_{298}^{1234} C_{p,Ag(s)} dT + \int_{1234}^{1356} C_{p,Ag(l)} dT + \Delta H_{m,Ag} \\ &= 21.3 \times (1234 - 298) + \frac{8.54 \times 10^{-3}}{2} (1234^2 - 298^2) \\ &\quad - 1.51 \times 10^5 \left(\frac{1}{1234} - \frac{1}{298} \right) + 30.5(1356 - 1234) + 11090 = 41260J\end{aligned}$$

In 2 moles of equimolar liquid solution $\Delta H^M = -20590X_{Cu}X_{Ag}$

$$\Delta H = 2(-20590 \times 0.5^2) = -10360J$$

So total requirement = $42420 + 41260 - 10360 = 73380J$

Chapter 9

9.5 Melts in the system Pb–Sn exhibit regular solution behavior. At 473°C, $a_{Pb} = 0.055$ in a liquid solution of $X_{Pb} = 0.1$. Calculate the value of α for the system and calculate the activity of Sn in the liquid solution of $X_{Sn} = 0.5$ at 500°C .

Chapter 9

Solution to 9.5

$$\gamma_{Pb} = \frac{a_{Pb}}{X_{Pb}} = \frac{0.055}{0.1} = 0.55$$

$$RT \ln \gamma_A = \Delta \bar{H}_A^M = \alpha X_B^2$$

So

$$\alpha = \frac{8.3144 \times 746 \times \ln 0.55}{0.9^2} = -4578 \text{ J}$$

At 500°C (773K) and $X_{Sn} = 0.5$

$$RT \ln \gamma_A = \Delta \bar{H}_A^M = \alpha X_B^2$$

$$\ln \gamma_{Sn} = \frac{-4578 \times 0.5^2}{8.3144 \times 773} = -0.178$$

So

$$a_{Sn} = 0.5 \times \exp(-0.178) = 0.418$$

$$\gamma_i = \frac{a_i}{X_i}$$

Chapter 9

9.6

- a. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K assuming the solution is ideal .
- b. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K that is a regular solution with $\Delta H_{mixing} = 16628X_A X_B$.

Chapter 9

Solution to 9.6

a) For ideal and regular solutions $\Delta G^M (X_B = 0.5) = \Delta \bar{G}_A^M = \Delta \bar{G}_B^M$

$$\Delta G^{M,id} = RT(X_A \ln X_A + X_B \ln X_B)$$

$$\Delta G^{M,id}(X_B = 0.5) = -RT\Delta S^M (X_B = 0.5)$$

$$= -8314 \left(\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \ln \frac{1}{2} \right) = -5763 \text{ J}$$

$$\Delta \bar{G}_B = RT \ln a_B = -5763 \text{ J}$$

$$a_B(X_B = 0.5) = \exp \left(-\frac{5763}{8314} \right) = 0.5$$

Chapter 9

Solution to 9.6

b)

$$\Delta G^M = \alpha X_A X_B - T \Delta S^{M, \text{ideal}} \quad \Delta S^M = -R (X_A \ln X_A + X_B \ln X_B)$$

$$\begin{aligned} \Delta G^{M, \text{reg}}(0.5) &= 16626 \times \frac{1}{4} + 8314 \left(\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \ln \frac{1}{2} \right) \\ &= -1609 \text{ J} \end{aligned}$$

$$\Delta \bar{G}_B = RT \ln a_B = -1609 \text{ J}$$

$$a_B(X_B = 0.5) = \exp \left(-\frac{1609}{8314} \right) = 0.824$$

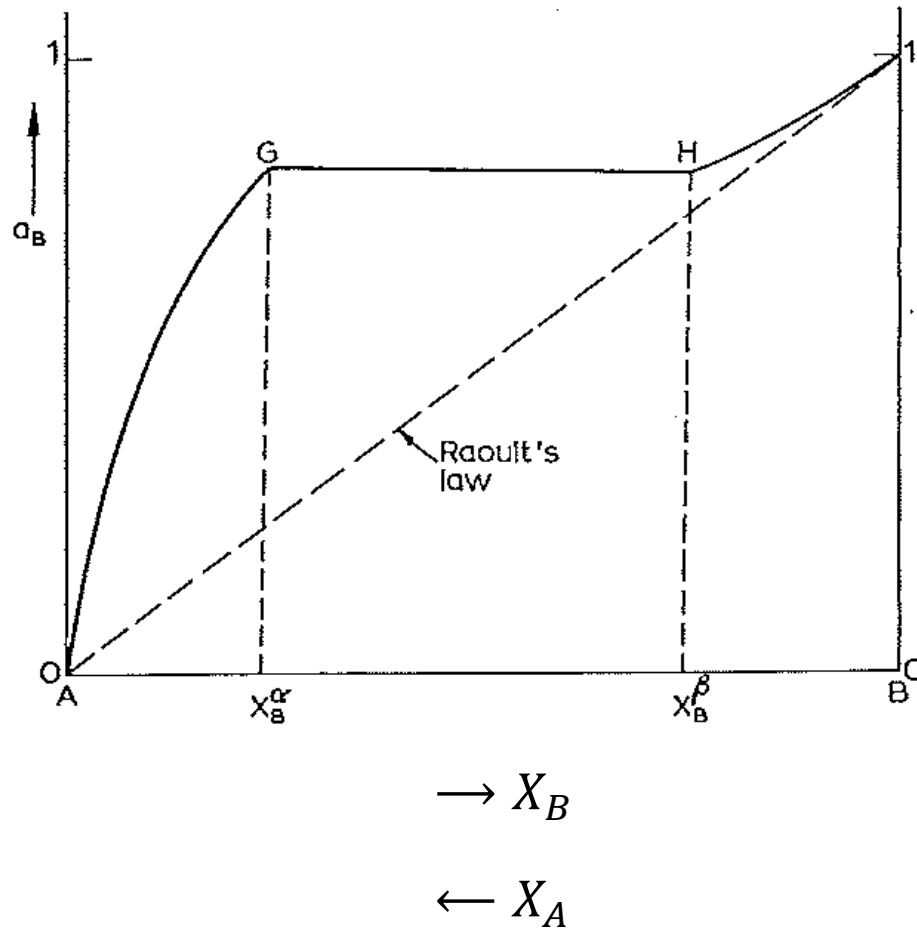
Chapter 9

9.7 A regular solution exhibits a miscibility gap. Sketch the activity of X_B vs. X_B at a temperature within the miscibility gap.

Denote regions where Henry's law is obeyed and where Raoult's law is obeyed if applicable .

Chapter 9

Solution to 9.7



Chapter 9

9.8 Tin obeys Henry's law in dilute liquid solutions of Sn and Cd, and the Henrian activity coefficient of Sn, γ_{Sn}° , varies with temperature as

$$\ln \gamma_{Sn}^\circ = -\frac{840}{T} + 1.58$$

Calculate the change in temperature when 1 mole of liquid Sn and 99 moles of liquid Cd are mixed in an adiabatic enclosure. The molar constant-pressure heat capacity of the alloy formed is 29.5 J/K

Chapter 9

Solution to 9.8

$$\frac{\partial(R \ln \gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta \bar{H}_i^M$$

$$\Delta \bar{H}_{Sn}^M = \frac{R \partial \ln \gamma_{Sn}^\circ}{\partial\left(\frac{1}{T}\right)} = -8.3144 \times 840 = -6984 J$$

$$\Delta H^M = X_A \Delta \bar{H}_A^M + X_B \Delta \bar{H}_B^M$$

$$\begin{aligned} \Delta H^M(\text{per mole}) &= X_{Sn} \Delta \bar{H}_{Sn}^M + X_{Cd} \Delta \bar{H}_{Cd}^M \quad \Delta \bar{H}_{Cd}^M = 0 \\ &= 0.01 \times (-6984) = -69.84 J = C_P \Delta T \\ &= 29.5 \Delta T \end{aligned}$$

So

$$\Delta T = \frac{69.84}{29.5} = 2.37 K$$

Chapter 9

9.9 Use the Gibbs–Duhem equation to show that, if the activity coefficients of the components of a binary solution can be expressed as

$$\ln \gamma_A = \alpha_1 X_B + \frac{1}{2} \alpha_2 X_B^2 + \frac{1}{3} \alpha_3 X_B^3 + \dots$$

and

$$\ln \gamma_B = \beta_1 X_A + \frac{1}{2} \beta_2 X_A^2 + \frac{1}{3} \beta_3 X_A^3 + \dots$$

over the entire range of composition, then $\alpha_1 = \beta_1 = 0$, and that, if the variation can be represented by the quadratic terms alone, then $\alpha_2 = \beta_2$.

Chapter 9

Solution to 9.9

$$\ln \gamma_A = \alpha_1 X_B + \frac{1}{2} \alpha_2 X_B^2 + \frac{1}{3} \alpha_3 X_B^3 + \dots$$

So $X_A d\ln \gamma_A = \alpha_1 X_A dX_B + \alpha_2 X_A X_B dX_B + \alpha_3 X_A X_B^2 dX_B + \dots$

$$\ln \gamma_B = \beta_1 X_A + \frac{1}{2} \beta_2 X_A^2 + \frac{1}{3} \beta_3 X_A^3 + \dots$$

So $X_B d\ln \gamma_B = \beta_1 X_B dX_A + \beta_2 X_B X_A dX_A + \beta_3 X_B X_A^2 dX_A + \dots$

For the Gibbs–Duhem equation

$$X_A d\ln \gamma_A = -X_B d\ln \gamma_B$$

Chapter 9

Solution to 9.9

So
$$\alpha_1 X_A dX_B + \alpha_2 X_A X_B dX_B = \beta_1 X_B dX_A + \beta_2 X_B X_A dX_A$$

If the power series are fluctuated at the quadratic terms.

Equating of the coefficients in terms containing X_A (or X_B) requires that $\alpha_1 = \beta_1 = 0$ and equating of the coefficients in the quadratic terms requires that $\alpha_2 = \beta_2$.

Chapter 9

9.10 The activity coefficient of Zn in liquid Zn–Cd alloys at 435°C can be represented as

$$\ln \gamma_{\text{Zn}} = 0.875X_{\text{Cd}}^2 - 0.30X_{\text{Cd}}^3$$

Derive the corresponding expression for the dependence of $\ln \gamma_{\text{Cd}}$ on composition and calculate the activity of cadmium in the alloy of $X_{\text{Cd}} = 0.5$ at 435°C.

Chapter 9

Solution to 9.10

$$\ln \gamma_{Zn} = 0.875X_{Cd}^2 - 0.30X_{Cd}^3$$

$$d \ln \gamma_{Zn} = 1.750X_{Cd}dX_{Cd} - 0.9X_{Cd}^2dX_{Cd}$$

$$-\frac{X_{Zn}}{X_{Cd}}d \ln \gamma_{Zn} = -1.75 X_{Zn}dX_{Cd} + 0.9X_{Zn}X_{Cd}dX_{Cd}$$

$$= 1.75X_{Zn}dX_{Zn} - 0.9X_{Zn}(1 - X_{Zn})dX_{Zn}$$

$$= 0.85X_{Zn}dX_{Zn} + 0.9X_{Zn}^2dX_{Zn}$$

$$X_A d \ln \gamma_A = -X_B d \ln \gamma_B$$

So $\ln \gamma_{Cd} = 0.425X_{Zn}^2 + 0.30X_{Zn}^3$

So At $X_{Cd} = 0.5$, $\gamma_{Cd} = 1.155$ and $a_{Cd} = 0.577$

Chapter 9

9.11 The molar excess Gibbs free energy of formation of solid solutions in the system Au–Ni can be represented by

$$G^{XS} = X_{Ni}X_{Au}(24140X_{Au} + 38280X_{Ni} - 14230X_{Au}X_{Ni}) \left(1 - \frac{T}{2660}\right) J$$

Calculate the activities of Au and Ni in the alloy of $X_{Au} = 0.5$ at 1100 K.

Chapter 9

Solution to 9.11

Let $X_{Au} = x$

$$G^{XS} = X_{Ni}X_{Au}(24140X_{Au} + 38280X_{Ni} - 14230X_{Au}X_{Ni})\left(1 - \frac{T}{2660}\right) J$$

$$\begin{aligned} G^{XS} &= x(1-x)[24140x + 38280(1-x) - 14230x(1-x)]\left(1 - \frac{1100}{2660}\right) J \\ &= [38280x - 66650x^2 + 42600x^3 - 14230x^4] \times 0.586 \end{aligned} \quad (1)$$

$$\frac{\partial G^{XS}}{\partial x} = [38280 - 133300x + 127800x^2 - 56920x^3] \times 0.586$$

$$(1-x)\frac{\partial G^{XS}}{\partial x} = [38280 - 171580x + 261100x^2 - 184720x^3 + 56920x^4] \times 0.586$$

Chapter 9

Solution to 9.11

$$\bar{G}_{Au}^{XS} = G^{XS} + (1-x) \frac{\partial G^{XS}}{\partial x} \quad \Delta \bar{G}_B^M = \Delta G^M + X_A \frac{d\Delta G^M}{dX_B}$$
$$= [38280 - 133300x + 194450x^2 - 142120x^3 + 42690x^4] \times 0.586$$

So $\bar{G}_{Au}^{XS} = 3105 \text{ J} = 8.3144 \times 1100 \ln \gamma_{Au}$ $\bar{G}_A^{XS} = RT \ln \gamma_A$

$$\gamma_{Au} = 1.39$$

$$a_{Au} = 0.5 \times 1.39 = 0.695 \quad \gamma_i = \frac{a_i}{X_i}$$

From (1), At $X_{Au} = 0.5$, $G^{XS} = 4051 \text{ J} = 8.3144 \times 1100 \times [0.5 \ln a_{Ni} + 0.5 \ln a_{Au}]$

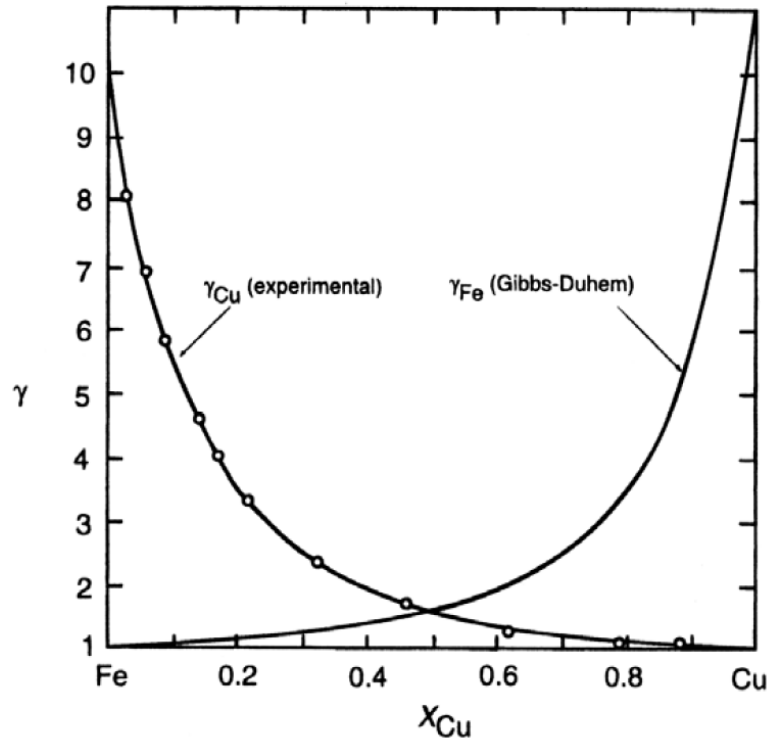
So $a_{Ni} = 0.85$

Chapter 9

9.12 The activities of liquid iron–copper solutions at 1550°C as a function of composition are shown in Figure 9.9. Sketch the activity coefficients as a function of composition at this temperature

Chapter 9

Solution to 9.12



Chapter 9

9.13 Use the equation $\Delta G^{XS} = b(T - T_c)\eta^2 + c\eta^4$ to derive the following equation:

For $\eta = \eta(T): \eta_{eq} = \left(1 - \frac{T}{T_c}\right)^{1/2}$ Where $b > 0$

Hint: Use the Third Law of Thermodynamics to set $\eta = 1$ at $T = 0$.

Chapter 9

Solution to 9.13

$$\Delta G^{XS} = b(T - T_C)\eta^2 + c\eta^4$$

$$\frac{\partial \Delta G^{XS}}{\partial \eta} = 0 \quad \text{for equilibrium}$$

$$\frac{\partial \Delta G^{XS}}{\partial \eta} = 2b(T - T_C)\eta + 4c\eta^3$$

Thus $\eta = 0$ and $\eta^2 = -\frac{b(T-T_C)}{2c}$ are extrema

Chapter 9

Solution to 9.13

By taking the second derivative $\eta = 0$ can be shown to be a maximum when $T < T_c$ and the other two extrema are minima when $T > T_c$.

$$\eta^2 = -\frac{b(T-T_c)}{2C}$$

At $T = 0$, $\eta = 1$ (by the third law)

Thus: $T_c = \frac{2C}{b}$ which leads to $\eta^2 = \left(1 - \frac{T}{T_c}\right)$

Chapter 9

9.14 It has been found that in a certain solution, the activity $a_A = X_A$ over a certain range of composition. Determine the relationship between a_B and X_B over the same range of composition.

Chapter 9

Solution to 9.14

$$a_A = X_A$$

$$d(\ln a_A) = d(\ln X_A)$$

$$X_A d(\ln a_A) + X_B d(\ln X_B) = 0 \quad \text{Gibbs-Duhem}$$

Thus

$$d(\ln a_2) = -\frac{X_A}{X_B} d(\ln a_A) = -\frac{X_A}{X_B} d(\ln X_A) = \frac{dX_B}{X_B}$$

Integrating

$$\ln a_2 = \ln X_2 + \ln \gamma \quad \text{Where } \gamma \text{ is a constant}$$

Thus

$$a_B = \gamma X_B = \gamma_B X_B$$

This is Herry's law

Chapter 9

9.15 All regular solutions with positive heats of mixing have the same value of the activity of its components (A or B) at the critical point of the miscibility gap.

a. Calculate this activity.

Chapter 9

Solution to 9.15

a)

$$a_B = \gamma_B X_B \qquad \gamma_A = \exp\left(\frac{\alpha X_B^2}{RT}\right) \qquad (9.64)$$

$$a_B = \exp\left(\frac{a_0 X_B^2}{RT}\right) X_B$$

$$T_C = \frac{a_0}{2R}$$

$$a_B = \exp\left(\frac{a_0 \left(\frac{1}{2}\right)^2}{R \frac{a_0}{2R}}\right) \left(\frac{1}{2}\right) = \exp(0.5) \left(\frac{1}{2}\right) = 0.824$$

Chapter 9

9.16 At a certain temperature, T , the A–B system exhibits regular solution behavior. The activity coefficient of A is given by

$$\ln \gamma_A = -b(1 - X_A)^2$$

where b is a constant at the given T .

Compute the corresponding equation for the variation of γ_B with composition at the same temperature. Be sure to state the justification for the steps of your solution.

Chapter 9

Solution to 9.16

$$\ln \gamma_A = -b(1 - X_A)^2$$

$$X_A d\bar{G}_A + X_B d\bar{G}_B = 0$$

$$d\bar{G}_A = RT d \ln a_A = RT d \ln(\gamma_A X_A)$$

$$X_A d \ln \gamma_A + X_B d \ln \gamma_B = 0$$

$$d \ln \gamma_B = -\frac{X_A}{X_B} d \ln \gamma_A = -\frac{X_A}{X_B} (-2b)(1 - X_A)(-dX_A)$$

$$= -2b \frac{X_A}{X_B} X_B dX_A = -2b X_A dX_A$$

So

$$\int_{\gamma}^{\gamma_B} d \ln \gamma_B = \ln \gamma_B = -bX_A^2 = -b(1 - X_B)^2$$

Thermodynamics of Materials & Phase transformations

The End



Exercise for chapter 10

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

Chapter 10

10.1 CaF_2 and MgF_2 are mutually insoluble in the solid state and form a simple binary eutectic system. Calculate the composition and temperature of the eutectic melt assuming that the liquid solutions are Raoultian. The actual eutectic occurs at $X_{\text{CaF}_2} = 0.45$ and $T = 1243 \text{ K}$.

Chapter 10

Solution to 10.1

Let $x = X_{CaF_2}$

For the CaF_2 liquid line

$$\Delta G_M^\circ = 31200 - 18.45T = -8.3144T \ln x$$

For the MgF_2 liquid line

$$\Delta G_M^\circ = 58160 - 37.86T = -8.3144T \ln(1 - x)$$

Simultaneous solution gives the point of intersection of the liquid lines at

$$T = 1317K$$

$$X_{CaF_2} = 0.53$$

The actual eutectic is

$$T = 1243K$$

$$X_{CaF_2} = 0.45$$

$$\Delta G_{m(A)}^\circ = -RT \ln a_A$$

if the liquid solutions are Raoultian,

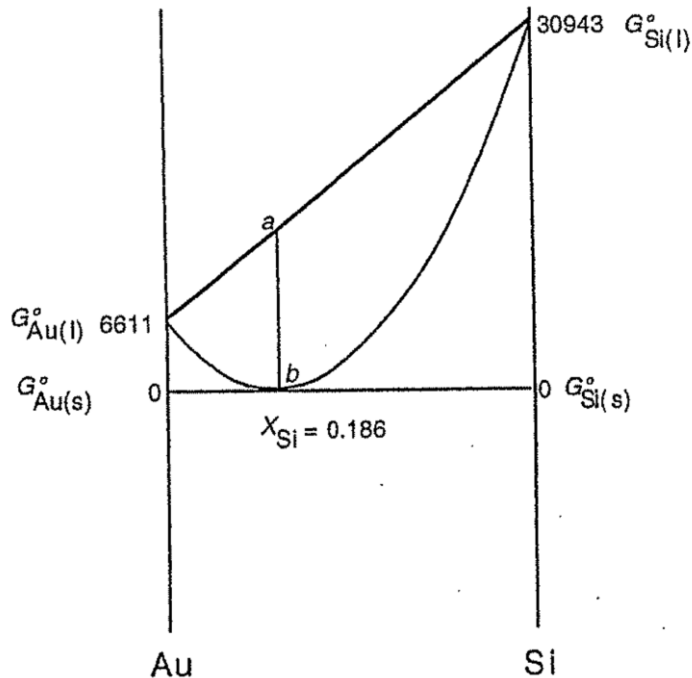
$$\Delta G_{m(A)}^\circ = -RT \ln X_A$$

Chapter 10

10.2 Gold and silicon are mutually insoluble in the solid state and form a eutectic system with a eutectic temperature of 636 K and a eutectic composition of $X_{Si} = 0.186$. Calculate the Gibbs free energy of the eutectic melt relative to (a) unmixed liquid Au and liquid Si, and (b) unmixed solid Au and solid Si.

Chapter 10

Solution to 10.2



At 636K

$$\Delta G_{m(A)}^{\circ} = \Delta H_{m(A)}^{\circ} \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right)$$

$$\Delta G_{m,Au}^{\circ} = 12600 \left[\frac{1338 - 636}{1338} \right] = 6611 \text{ J}$$

and

$$\Delta G_{m,Si}^{\circ} = 50200 \left[\frac{1685 - 636}{1685} \right] = 30943 \text{ J}$$

a).

$$\begin{aligned} \Delta G^M &= ab = -(0.186 \times 30943 + 0.814 \times 6611) \text{ J} \\ &= -11140 \text{ J} \end{aligned}$$

b). $\Delta G^M = 0$

Chapter 10

10.3 Al_2O_3 , which melts at 2324 K, and Cr_2O_3 , which melts at 2538 K, form complete ranges of solid and liquid solutions. Assuming that $\Delta S_{m,Al_2O_3}^\circ = \Delta S_{m,Cr_2O_3}^\circ$ and that the solid and liquid solutions in the system $Al_2O_3 - Cr_2O_3$ behave ideally, calculate

- The temperature at which equilibrium melting begins when an alloy of $X_{Al_2O_3} = 0.5$ is heated
- The composition of the melt which first forms
- The temperature at which equilibrium melting is complete
- The composition of the last-formed solid

Chapter 10

Solution to 10.3

$$\frac{\Delta G_{m,Al_2O_3}^\circ}{RT} = \frac{12929}{T} - 5.563$$

$$\frac{\Delta G_{m,Cr_2O_3}^\circ}{RT} = \frac{14119}{T} - 5.563$$

$$X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)}{\exp\left(-\frac{\Delta G_{m(A)}^\circ}{RT}\right) - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)}$$

a) The solidus temperature T , at $X_{Al_2O_3} = 0.5$ is given by

$$\frac{1 - \exp\left(-\frac{14119}{T} + 5.563\right)}{\exp\left(-\frac{12929}{T} + 5.563\right) - \exp\left(-\frac{14119}{T} + 5.563\right)} = 0.5$$

Which has the solution $T = 2418 \text{ K}$

Chapter 10

Solution to 10.3

b) The corresponding liquidus composition is

$$X_{Al_2O_3}(liquidus) = 0.5 \left[\exp \left(-\frac{12929}{2418} + 5.563 \right) \right] = 0.621$$

c) The liquidus temperature T , at $X_{Al_2O_3} = 0.5$ is given by

$$\frac{1 - \exp \left(-\frac{14119}{T} + 5.563 \right)}{\exp \left(-\frac{12929}{T} + 5.563 \right) - \exp \left(-\frac{14119}{T} + 5.563 \right)} \times \exp \left(-\frac{12929}{T} + 5.563 \right) = 0.5$$

$$X_{A(l)} = \frac{\left[1 - \exp \left(-\frac{\Delta G_{m(B)}^\circ}{RT} \right) \right] \exp \left(-\frac{\Delta G_{m(A)}^\circ}{RT} \right)}{\exp \left(-\frac{\Delta G_{m(A)}^\circ}{RT} \right) - \exp \left(-\frac{\Delta G_{m(B)}^\circ}{RT} \right)}$$

Which has the solution $T = 2444 \text{ K}$

Chapter 10

Solution to 10.3

d) The corresponding solidus composition is

$$X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G_m^\circ(B)}{RT}\right)}{\exp\left(-\frac{\Delta G_m^\circ(A)}{RT}\right) - \exp\left(-\frac{\Delta G_m^\circ(B)}{RT}\right)}$$

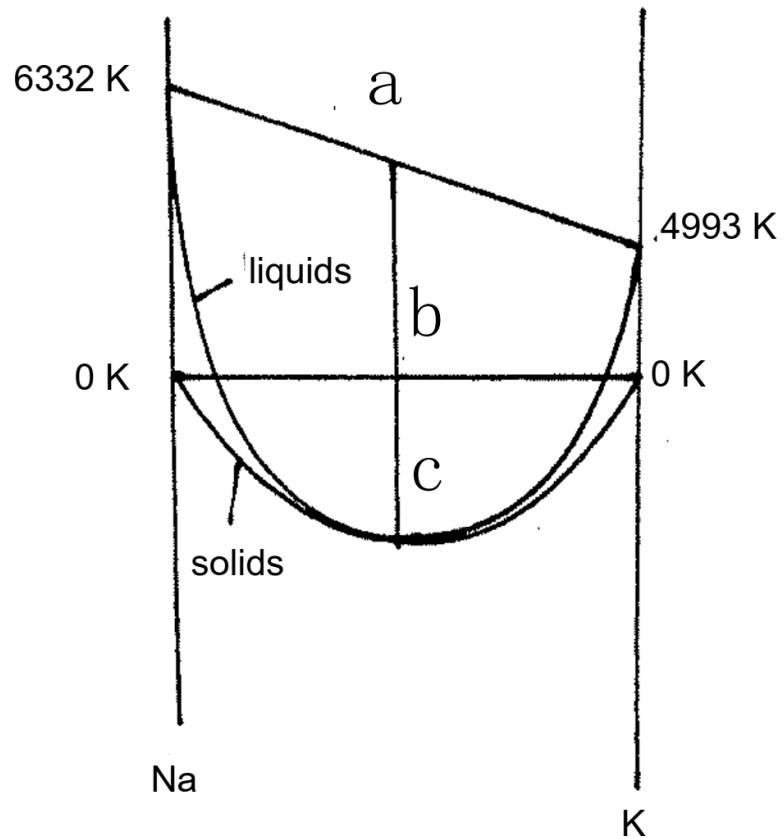
$$X_{Al_2O_3}(\text{solidus}) = \frac{1 - \exp\left(-\frac{14119}{2444} + 5.563\right)}{\exp\left(-\frac{12929}{2444} + 5.563\right) - \exp\left(-\frac{14119}{2444} + 5.563\right)}$$
$$= 0.38$$

Chapter 10

10.4 $Na_2O.B_2O_3$ and $K_2O.B_2O_3$ form complete ranges of solid and liquid solutions and the solidus and liquidus show a common minimum at the equimolar composition and $T = 1123\text{ K}$. Calculate the molar Gibbs free energy of formation of the equimolar solid solution from solid $Na_2O.B_2O_3$ at 1123 K, assuming that the liquid solutions are ideal.

Chapter 10

Solution to 10.4



Chapter 10

Solution to 10.4

$$\Delta G_{m(A)}^{\circ} = \Delta H_{m(A)}^{\circ} \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right)$$

At

$$T = 1123 \text{ K}$$

$$\Delta G_{m,Na_2O.B_2O_3}^{\circ} = 67000 \left(\frac{1240 - 1123}{1240} \right) = 6322 \text{ J}$$

$$\Delta G_{m,K_2O.B_2O_3}^{\circ} = 62800 \left(\frac{1220 - 1123}{1220} \right) = 4993 \text{ J}$$

$$\Delta G^M(\text{flow liquids}) = 8.3144 \times 1123 [0.5 \ln 0.5 + 0.5 \ln 0.5] = -6472 \text{ J}$$

= ac

$$\Delta G_s^M = RT(X_A \ln X_A + X_B \ln X_B)$$

$$ab = 0.5 \times 6322 + 0.5 \times 4993 = 5658 \text{ J}$$

So

$$\Delta G^M = bc = -6472 + 5658 = -814 \text{ J}$$

Chapter 10

10.5 SiO_2 , which melts at 1723°C , and TiO_2 , which melts at 1842°C , are immiscible in the solid state, and the $SiO_2 - TiO_2$ binary system contains a monotectic equilibrium at 1794°C , at which temperature virtually pure TiO_2 is in equilibrium with two liquids containing mole fractions of SiO_2 of 0.04 and 0.76. If, for the purpose of simple calculation, it is assumed that the compositions of the two liquids are $X_{SiO_2} = 0.24$ and $X_{SiO_2} = 0.76$ and that the liquid solutions are regular in their behavior, what is the value of α_l and at what temperature does the liquid immiscibility gap disappear?

Chapter 10

Solution to 10.5

At the composition of the double tangent to the ΔG^M curve

$$\Delta G^M = RT(X_A \ln X_A + X_B \ln X_B) + \alpha X_A X_B$$
$$\frac{d\Delta G^M}{dx} = RT \ln \left(\frac{x}{1-x} \right) + \alpha_l(1-2x) = 0$$

At $T = 2067 \text{ K}$

$$8.3144 \times 2067 \ln \left(\frac{0.24}{1-0.24} \right) + \alpha_l(1-2 \times 0.24) = 0$$

Which gives $\alpha_l = 38096 \text{ J}$

$$T_{cr} = \frac{\alpha}{2R} \quad T_{Cr} = \frac{\alpha_l}{2R} = \frac{38096}{2 \times 8.3144} = 2291 \text{ K}$$

Chapter 10

10.6 The binary system Ge–Si contains complete solid and liquid solutions. The melting temperatures are $T_{m, Si} = 1685\text{ K}$, and $T_{m, Ge} = 1210\text{ K}$, and $\Delta H_{m, Si}^{\circ} = 50200\text{ J}$. At 1200°C , the liquidus and solidus compositions are, respectively, $X_{Si} = 0.32$ and $X_{Si} = 0.665$. Calculate the value of $\Delta H_{m, Ge}^{\circ}$, assuming that

- The liquid solutions are ideal.
- The solid solutions are ideal.

Which assumption gives the better estimate? The actual value of $\Delta H_{m, Ge}^{\circ}$ at $T_{m, Ge}$ is $36,900\text{ J}$.

Chapter 10

Solution to 10.6

At $T = 1473 \text{ K}$

$$\Delta G_{m(A)}^{\circ} = \Delta H_{m(A)}^{\circ} \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right) \quad X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G_{m(B)}^{\circ}}{RT}\right)}{\exp\left(-\frac{\Delta G_{m(A)}^{\circ}}{RT}\right) - \exp\left(-\frac{\Delta G_{m(B)}^{\circ}}{RT}\right)}$$

$$\Delta G_{m, Si}^{\circ} = 50200 \left(\frac{1685 - 1473}{1685} \right) = 6316 \text{ J}$$

$$\frac{\Delta G_{m, Si}^{\circ}}{RT} = \frac{6316}{8.3144 \times 1473} = 0.5157$$

$$X_{Ge}(\text{solidus}) = \frac{1 - \exp(-0.5157)}{\exp\left(-\frac{\Delta G_{m, Ge}^{\circ}}{8.3144 \times 1473}\right) - \exp(-0.5157)} = 0.335$$

Which gives $\Delta G_{m, Ge}^{\circ} = -7197 \text{ J}$ $\Delta H_{m, Ge}^{\circ} = \frac{-7197}{\left(\frac{1210 - 1473}{1210}\right)} = 33111 \text{ J}$

Chapter 10

Solution to 10.6

$$X_{A(l)} = \frac{\left[1 - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)\right] \exp\left(-\frac{\Delta G_{m(A)}^\circ}{RT}\right)}{\exp\left(-\frac{\Delta G_{m(A)}^\circ}{RT}\right) - \exp\left(-\frac{\Delta G_{m(B)}^\circ}{RT}\right)}$$

$$X_{Ge}(\text{liquidus}) = \frac{[1 - \exp(-0.5157)] \exp\left(-\frac{\Delta G_{m,Ge}^\circ}{8.3144 \times 1473}\right)}{\exp\left(-\frac{\Delta G_{m,Ge}^\circ}{8.3144 \times 1473}\right) - \exp(-0.5157)} = 0.68$$

Which gives

$$\Delta G_{m,Ge}^\circ = -4679 \text{ J}$$

$$\Delta H_{m,Ge}^\circ = \frac{-4679}{\left(\frac{1210 - 1473}{1210}\right)} = 21527 \text{ J}$$

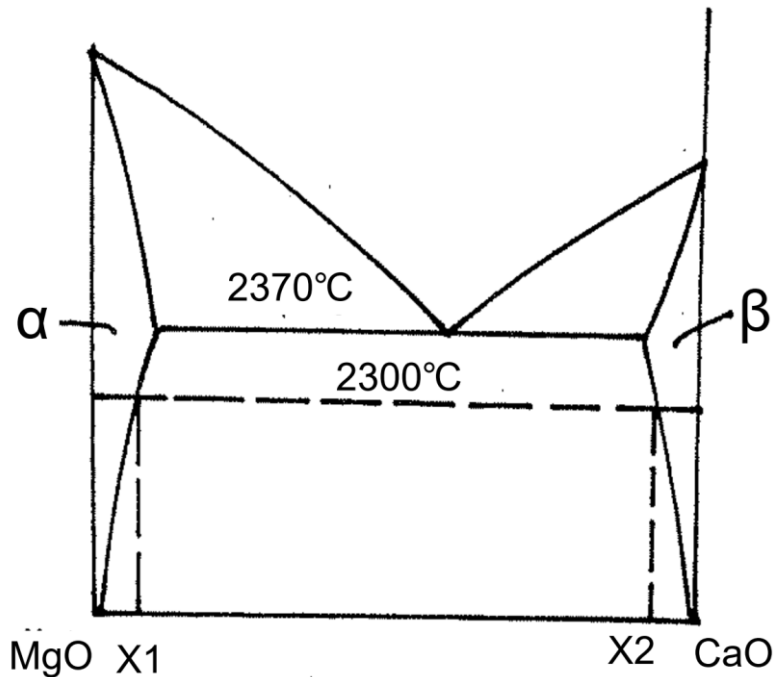
So The estimate from the solidus(33100 J) is closer to the actant value of 36900 J

Chapter 10

10.7 CaO and MgO form a simple eutectic system with limited ranges of solid solubility. The eutectic temperature is 2370°C. Assuming that the solutes in the two solid solutions obey Henry's law with $\gamma_{\text{CaO}}^{\circ}$ in MgO = 12.88 and $\gamma_{\text{MgO}}^{\circ}$ in CaO = 6.23 at 2300°C, calculate the solubility of CaO in MgO and the solubility of MgO in CaO at 2300°C.

Chapter 10

Solution to 10.7



$$a_{CaO}(\text{in } \beta \text{ at } X_2) = a_{CaO}(\text{in } \alpha \text{ at } X_1)$$

$$1 - X_1 = 6.23(1 - X_2) \quad (i)$$

$$a_{MgO}(\text{in } \alpha \text{ at } X_1) = a_{MgO}(\text{in } \beta \text{ at } X_2)$$

$$12.88X_1 = X_2 \quad (ii)$$

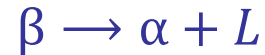
Solution of (i) and (ii) gives

$$X_1 = X_{CaO} = 0.066$$

$$X_2 = X_{MgO} = 0.85$$

Chapter 10

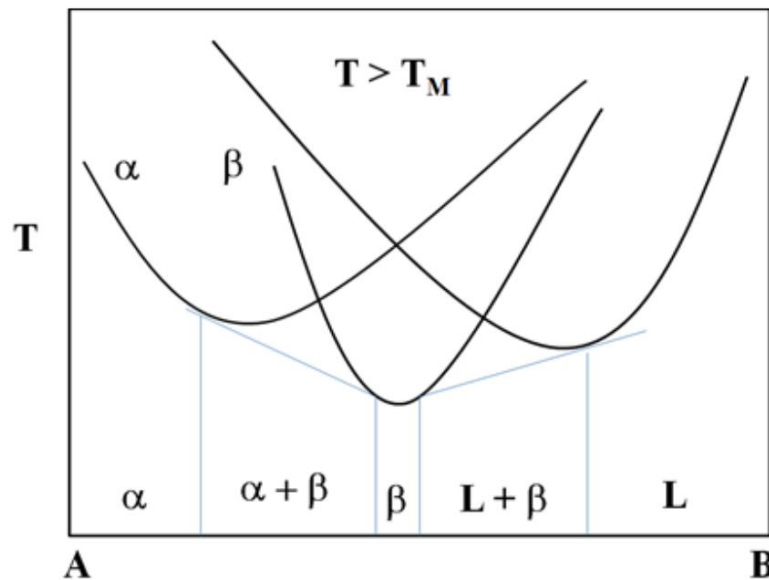
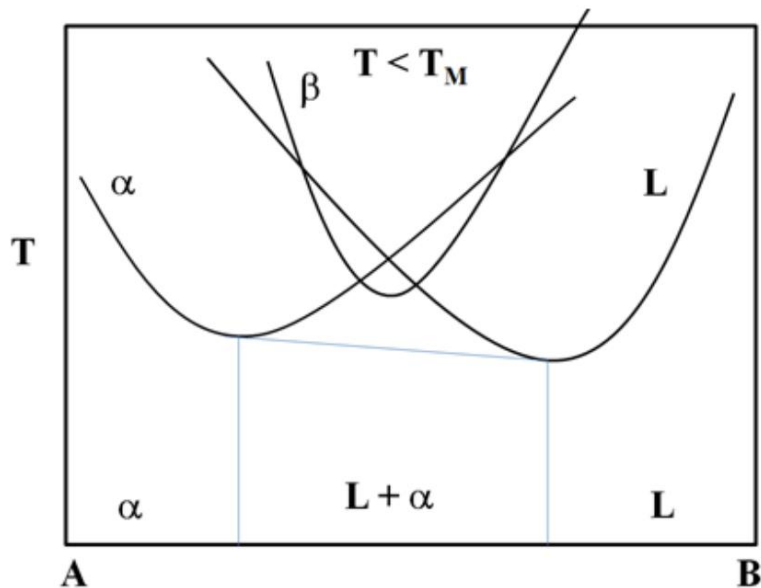
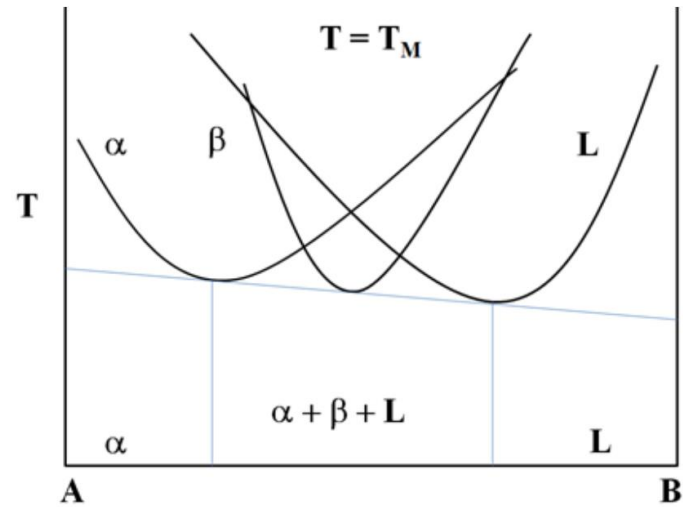
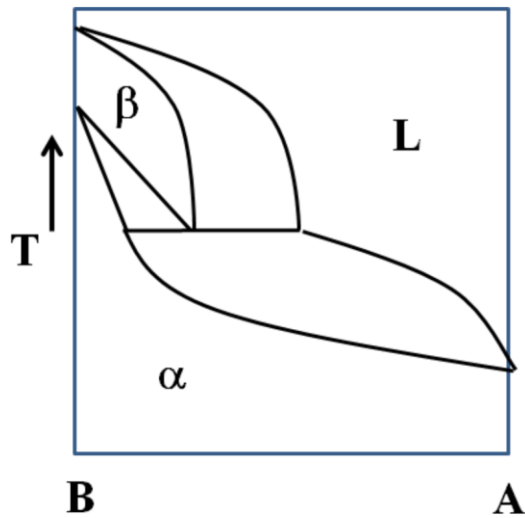
10.8 A metatectic binary phase diagram displays the following invariant transformation on cooling:



Sketch such a phase diagram and then draw free energy curves of mixing just below, at, and just above the invariant temperature.

Chapter 10

Solution to 10.8



Chapter 10

10.9 The free energy of mixing of a regular solution is given by

$$\Delta G_{mix} = \alpha X_A X_B + RT(X_A \ln X_A + X_B \ln X_B)$$

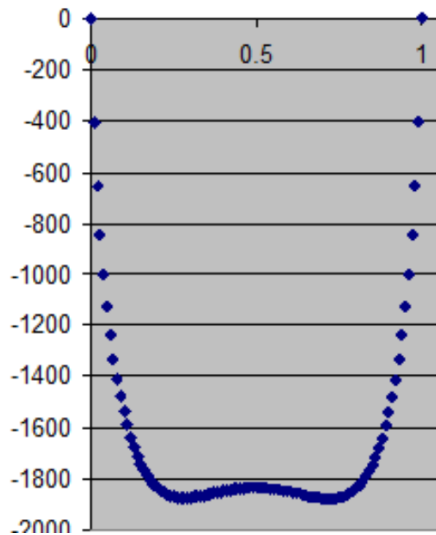
$$\alpha = 24943 \text{ J/mole}$$

- Plot ΔG_{mix} versus X_B at 1400, 1500, and 1600 K.
- Plot $\partial \Delta G_{mix} / \partial X_B$ versus X_B at the same temperatures as (a).
- Determine the critical temperature for this alloy. Show your work.

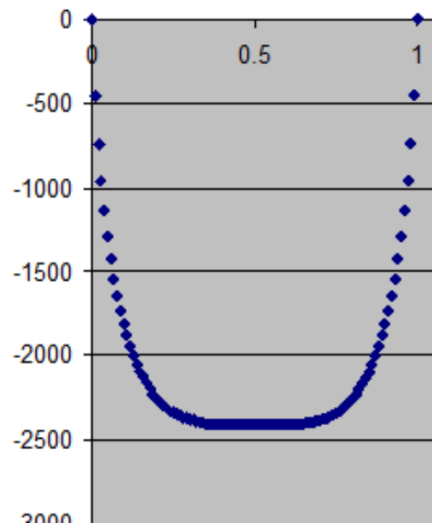
Chapter 10

Solution to 10.9

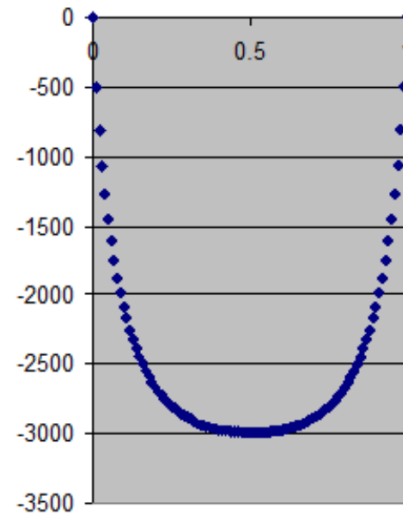
a)



1400 K



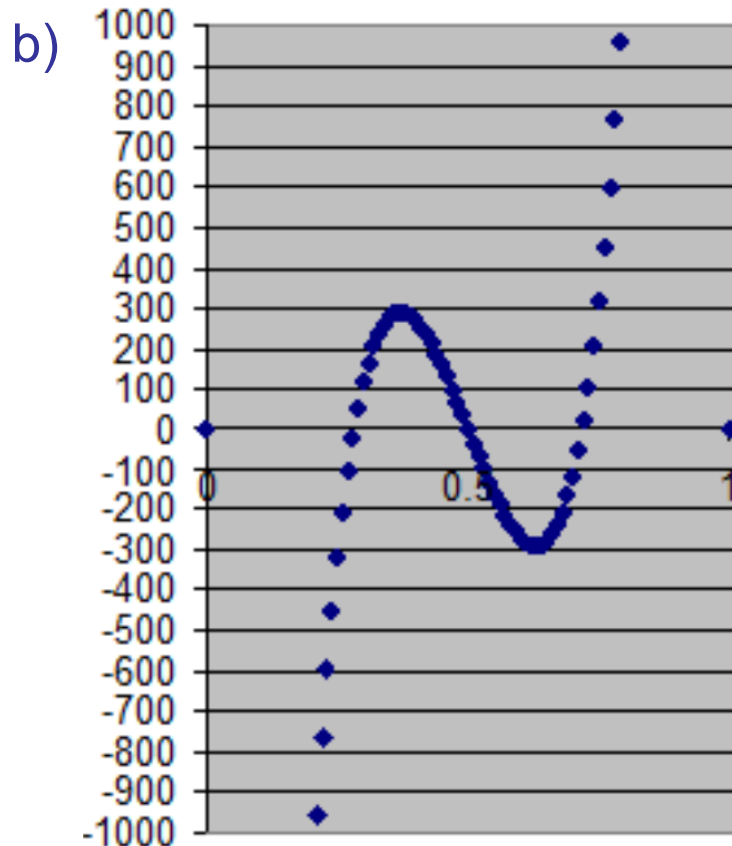
1500 K



1600 K

Chapter 10

Solution to 10.9

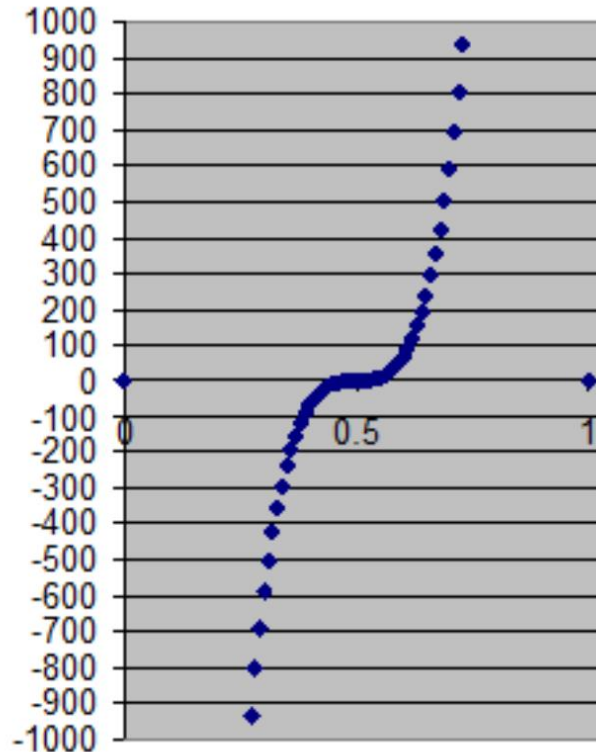


Plot of the first derivative for 1400K. The first and third zeros are the minima of free energy plot: they give the compositions of the two phases in equilibrium. The maximum in this plot and the minimum give us the spinodals.

Chapter 10

Solution to 10.9

b)

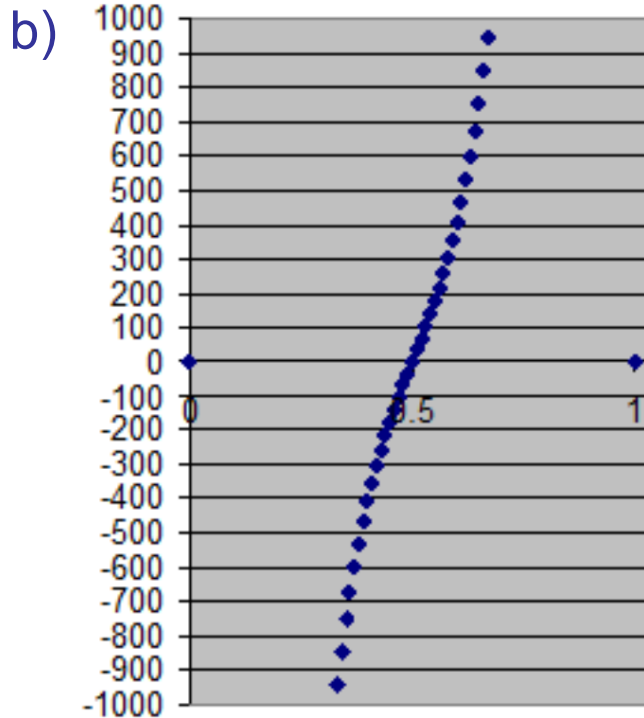


Plot of the first derivative for 1500K.

Only one zero for this plot at 0.5. In fact when it passes through zero the curve has zero slope. This demonstrates that this is the critical temperature

Chapter 10

Solution to 10.9



Plot of the first derivative for 1600K. The zero value is the minimum in free energy vs. X_B plot (0.5)

c) From the answers above we estimate T_C to 1500K. Or, since a regular solution we calculate it to be $T_C = \frac{a_0}{2R} = 1500K$

Chapter 10

10.10 A certain solid solution of A and B has a ΔG_{mix}^{XS} as follows:

$$\Delta G_{mix}^{XS} = \Delta G_{mix} - \Delta G_{mix}^{id} = X_A X_B (a_1 X_A + a_2 X_B) + RT(X_A \ln X_A + X_B \ln X_B)$$

Where:

$$a_1 = 12,500 \text{ J/mole}$$

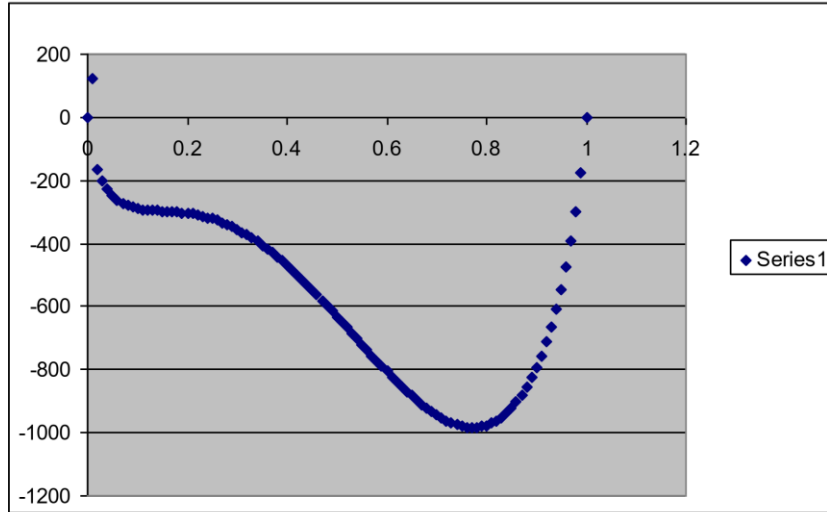
$$a_2 = 5,500 \text{ J/mole}$$

- Plot ΔG_{mix}^{XS} at $T = 500$ and 700 K.
- Sketch the T versus X_B phase diagram of this alloy.
- Determine the critical temperature and composition of this alloy.

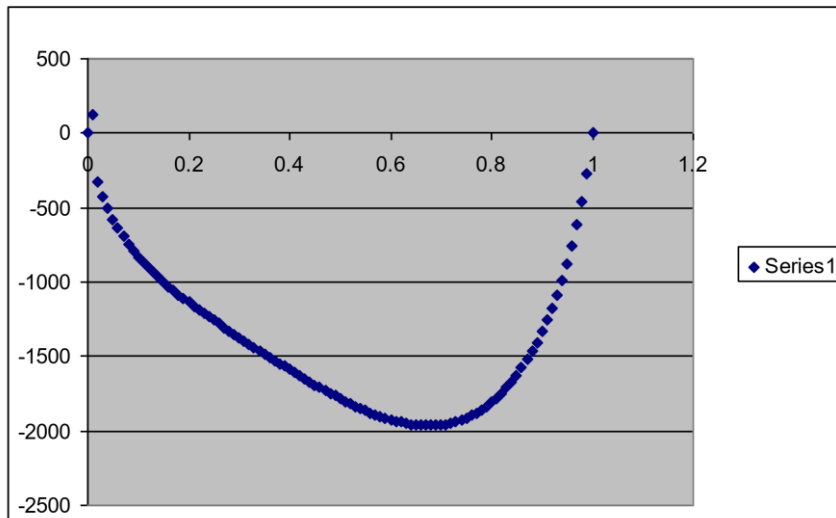
Chapter 10

Solution to 10.10

a)



Plot of ΔG_{mix}^{XS} vs. X_B at 500 K, showing that a miscibility gap will exist in the T vs. X_B phase diagram with T_C being less than $X_B = 0.5$.



Plot of ΔG_{mix}^{XS} vs. X_B at 700K, showing that 700 K is above T_C .

Chapter 10

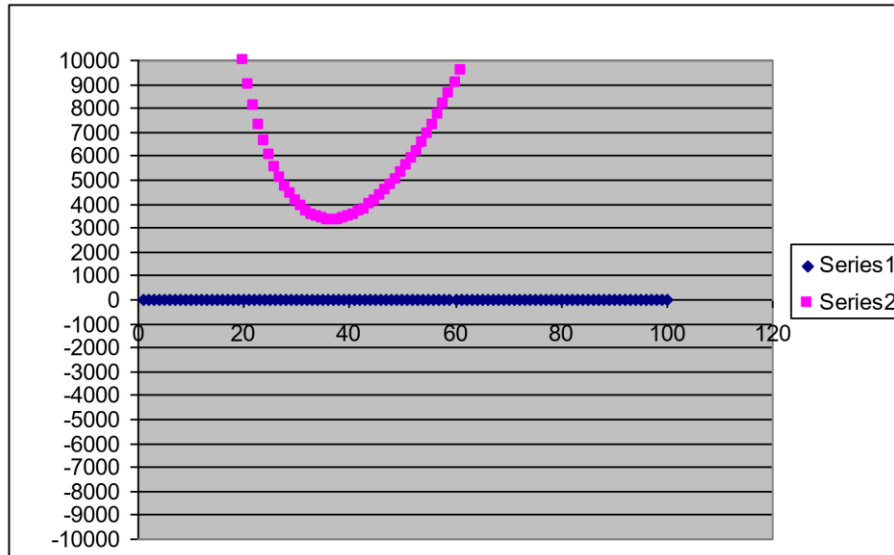
Solution to 10.10

b) Plot will be asymmetric with T_C less than $X_B = 0.5$.

Chapter 10

Solution to 10.10

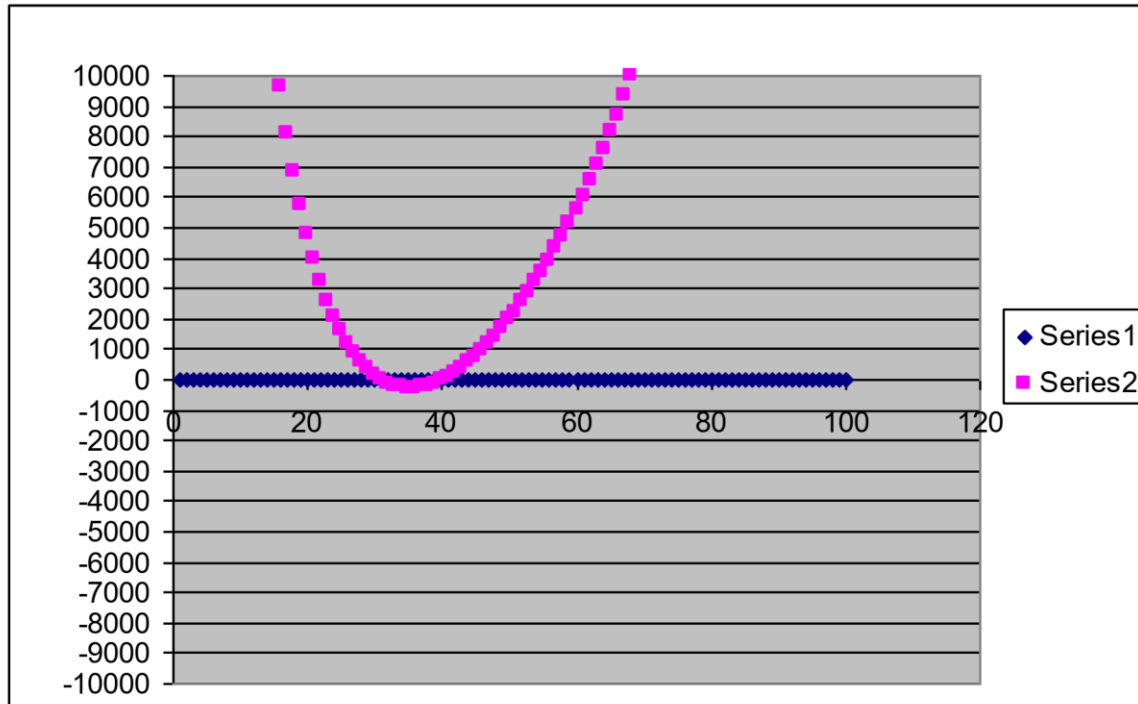
c) A graphical way of doing this is to plot $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial X_B^2}$ vs. X_B and see what T has the curve just touching the 0 ordinate. Below is the $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial X_B^2}$ vs. X_B for 700 K .



This is above T_C . The 600K plot shows two crossings of the $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial X_B^2} = 0$. Hence $T < T_C$.

Chapter 10

Solution to 10.10



By trying various values of T , T_C can be found to be 607.5 K and the composition of the critical point is $X_B = 0.35$.

Chapter 10

10.11 The phase diagram of an alloy that has a eutectoid transformation is shown in Figure 10.33.

- Sketch the Gibbs free energy curves for this alloy at $T = T'$.
- Show that the α/γ solvus must enter the α/β two-phase field (as shown by the arrow)

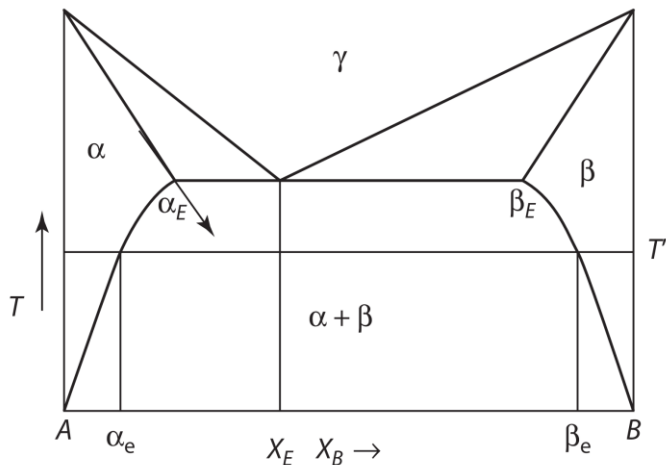
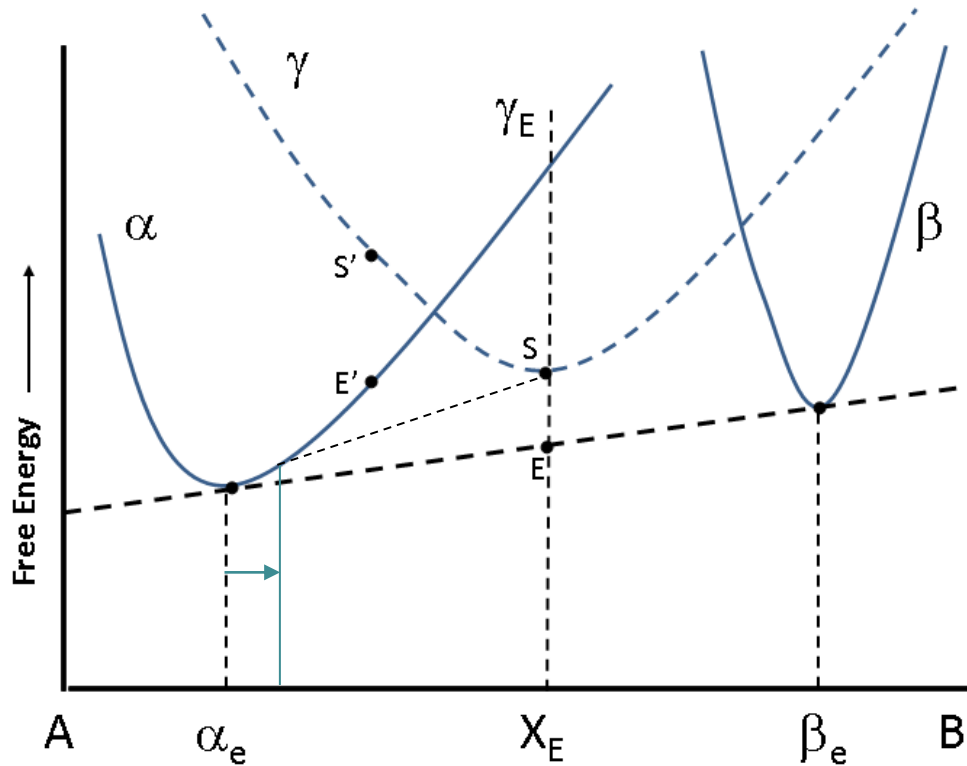


Figure 10.33: A eutectoid phase diagram showing the extension of the α/γ equilibrium curve into the α/β solvus.

Chapter 10

Solution to 10.11

a)



b) The common tangent of γ and α below T_E intersects the α free energy curve to the right of α_e .

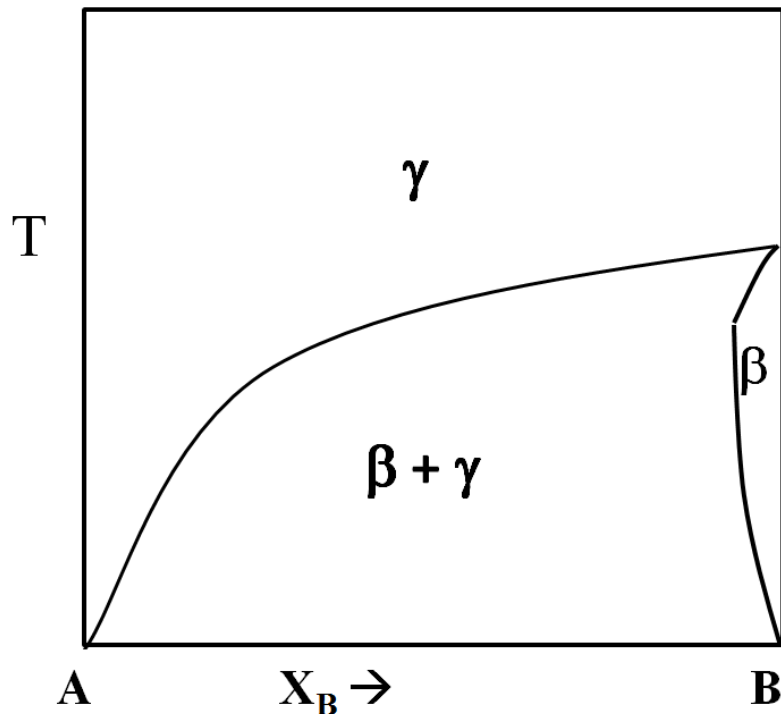
Chapter 10

10.12 A eutectoid phase diagram is shown in Figure 10.33. Draw the phase diagram that would result if for some reason it were impossible to form the α phase.

Chapter 10

Solution to 10.12

A possible diagram is shown below. Note that the γ/β curve is the same as the one in Fig.10.28a until the temperature where the eutectoid temperature was. is reached. It then is extrapolated to 0 K. The 0 K phases present are γ and β .



Thermodynamics of Materials & Phase transformations

The End



Exercise for chapter 11

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

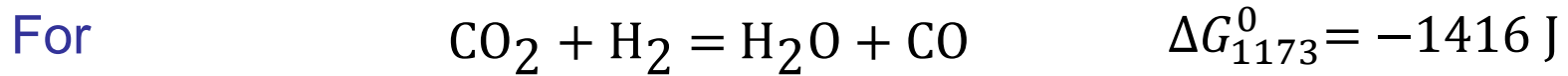
<http://teacher.nwpu.edu.cn/kehua>

Chapter 11

11.1 A gas mixture of 50% CO , 25% CO₂, and 25% H₂ (by volume) is fed to a furnace at 900°C. Determine the composition of the equilibrium CO–CO₂–H₂–H₂O gas if the total pressure of the gas in the furnace is 1 atm.

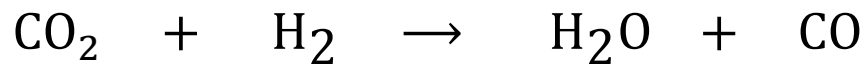
Chapter 11

Solution to 11.1



$$\Delta G^\circ = -RT \ln K_p$$

So $K_p = \exp\left(\frac{1416}{8.3144 \times 1173}\right) = 1.156$



$$n_T = 0.25 - x + 0.25 - x + x + 0.5 + x = 1$$

Chapter 11

Solution to 11.1

with $P_T = 1 \text{ atm}$

$$K_P = \frac{P_{H_2O} \cdot P_{CO}}{P_{CO_2} \cdot P_{H_2}}$$

$$1.156 = \frac{x(0.5 + x)}{(0.25 - x)^2}$$

$$x = 0.0676$$

So

$$P_{CO_2} = P_{H_2} = 0.25 - 0.0676 = 0.182 \text{ atm}$$

$$P_{CO} = 0.5 + 0.0676 = 0.568 \text{ atm}$$

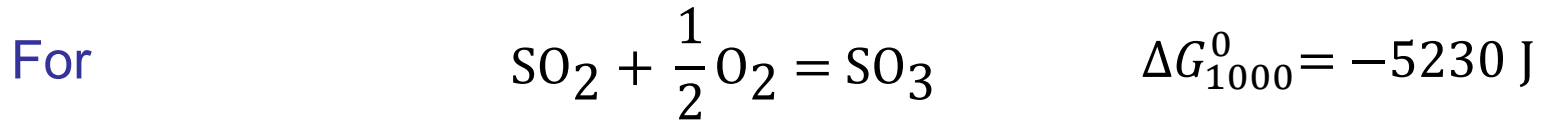
$$P_{H_2O} = 0.0676 \text{ atm}$$

Chapter 11

11. 2 How much heat is evolved when 1 mole of SO_2 and $\frac{1}{2}$ mole of O_2 , each at 1 atm pressure, react to form the equilibrium SO_3 – SO_2 – O_2 mixture at 1000 K and 1 atm pressure?

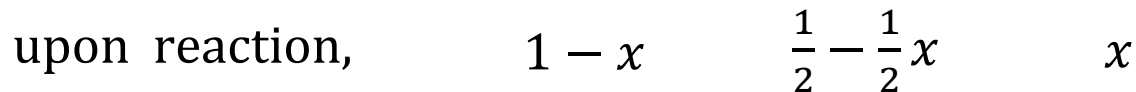
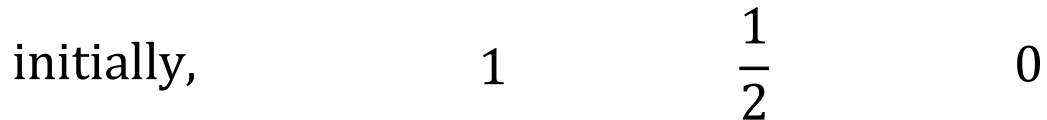
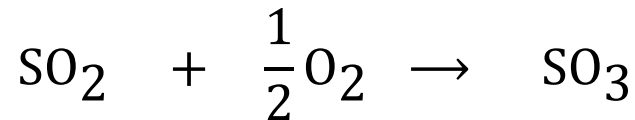
Chapter 11

Solution to 11.2



$$\Delta G^\circ = -RT \ln K_p$$

So $K_{P,1000} = \exp\left(\frac{5230}{8.3144 \times 1000}\right) = 1.876$



$$n_T = 1 - x + \frac{1}{2} - \frac{1}{2}x + x = \frac{1}{2}(3 - x)$$

Chapter 11

Solution to 11. 2

$$\text{So } P_{SO_2} = \frac{2(1-x)}{3-x} P, \quad P_{O_2} = \frac{1-x}{3-x} P, \quad P_{SO_3} = \frac{2x}{3-x} P$$

$$K_P = \frac{P_{SO_3}}{P_{SO_2} \cdot P_{O_2}^{1/2}} = 1.876 = \frac{x \cdot (3-x)^{1/2}}{(1-x)^{1/2} \cdot (1-x)}$$

$$x = 0.463$$

For formation of 1 mole of SO_3 , $\Delta H = \Delta H^0 = -94600 \text{ J}$

$$\begin{aligned} \text{For formation of 0.463 mole of } SO_3, \quad \Delta H &= 0.463 \times (-94600) \\ &= -43800 \text{ J} \end{aligned}$$

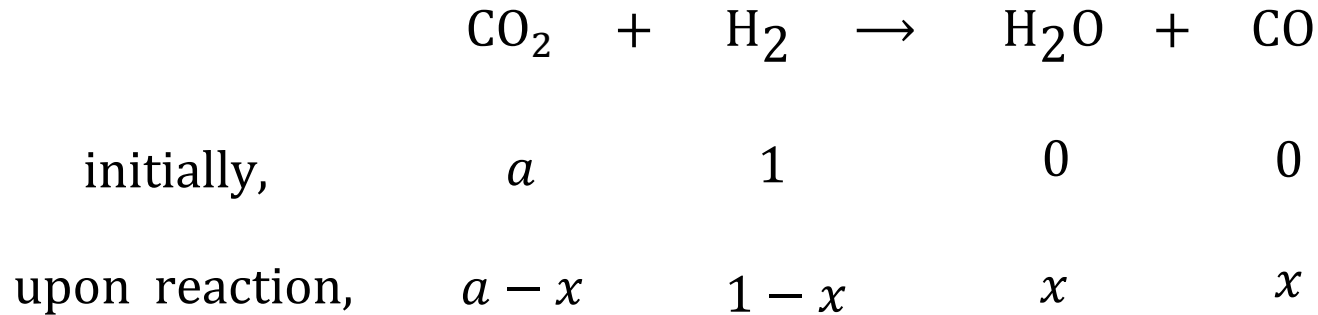
So 43800 J are evolved.

Chapter 11

11. 3 A $\text{CO}_2\text{--CO--H}_2\text{O--H}_2$ gas mixture at a total pressure of 1 atm exerts a partial pressure of oxygen of 10^{-7} atm at 1600°C . In what ratio were the CO_2 and H_2 mixed to produce the gas with this oxygen pressure?

Chapter 11

Solution to 11.3



For $\text{H}_2 + \frac{1}{2}\text{O}_2 = \text{H}_2\text{O} \quad \Delta G_{1873}^0 = -142892 \text{ J}$

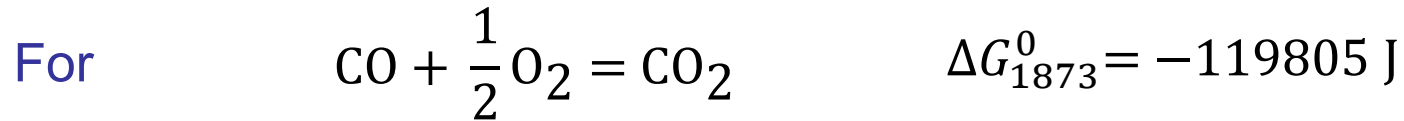
$$\Delta G^\circ = -RT \ln K_p$$

So
$$K_P = \exp\left(\frac{142892}{8.3144 \times 1873}\right) = 9659 = \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2} \cdot P_{\text{O}_2}^{1/2}} = \frac{x}{(1-x)10^{-3.5}}$$

$$x = 0.753$$

Chapter 11

Solution to 11.3



So $K_P = 2193 = \frac{P_{\text{CO}_2}}{P_{\text{CO}} \cdot P_{\text{O}_2}^{1/2}} = \frac{a - x}{x \cdot 10^{-3.5}} = \frac{a - 0.753}{0.753 \cdot 10^{-3.5}}$

$$a = 1.276$$

So $\frac{\text{CO}_2}{\text{H}_2} = a = 1.276$

Chapter 11

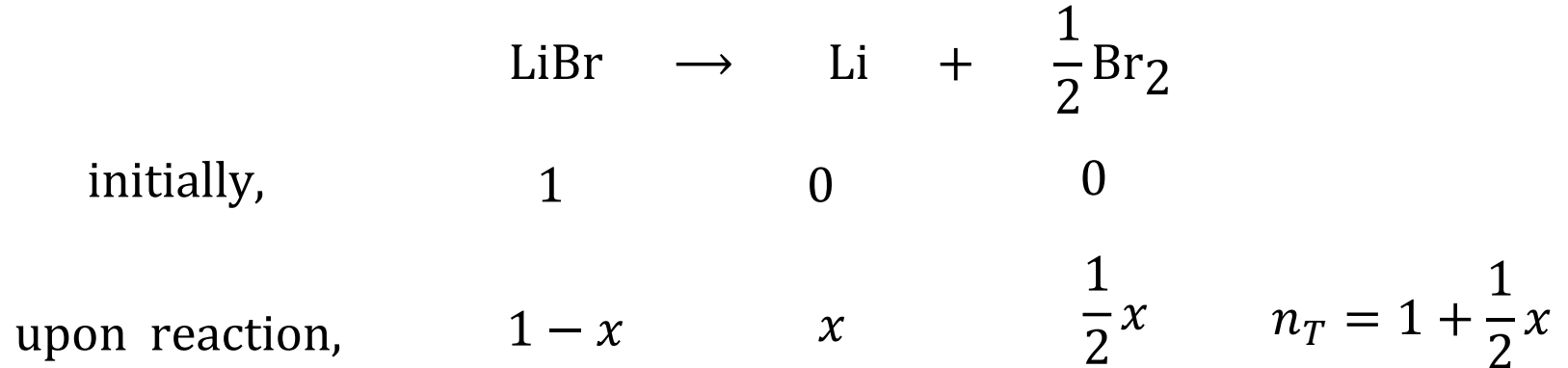
11.4 Lithium bromide vapor dissociates according to



At what temperature does the partial pressure of Li reach the value of 10^{-5} atm when the gas is heated at a constant total pressure of 1 atm?

Chapter 11

Solution to 11.4



So with $P_T = 1 \text{ atm}$, $P_{\text{Li}} = \frac{x}{1 + \frac{1}{2}x} = 10^{-5}$

$$x = 10^{-5}$$

$$P_{\text{LiBr}} = \frac{1-x}{1 + \frac{1}{2}x} \approx 1 \text{ atm}$$

$$P_{\text{Br}_2} = 0.5 \times 10^{-5} \text{ atm}$$

Chapter 11

Solution to 11.4

So

$$K_p = \frac{P_{Li} \cdot P_{Br_2}^{1/2}}{P_{LiBr}} = (0.5 \times 10^{-5})^{1/2} \times 10^{-5}$$

$$= 2.236 \times 10^{-8}$$

$$\Delta G^\circ = -RT \ln K_p$$

$$\Delta G^\circ = 333900 - 42.09 T = -8.3144 T \ln 2.236 \times 10^{-8}$$

$$T = 1771 \text{ K}$$

Chapter 11

11. 5 When SO_3 is decomposed at the constant pressure P and $T = 1000 \text{ K}$, the partial pressure of O_2 in the equilibrium gas is 0.05 atm . What is the pressure P ? If the pressure of this equilibrated gas is increased to 1 atm , to what value must the temperature be decreased to produce a gas mixture in which $P_{\text{O}_2} = 0.05 \text{ atm}$?

Chapter 11

Solution to 11.5

For $\text{SO}_3 = \text{SO}_2 + \frac{1}{2}\text{O}_2 \quad \Delta G_{1000}^{\circ} = 5236 \text{ J}$

So $K_P = 0.533 \quad \Delta G^{\circ} = -RT \ln K_P$

$$P_{\text{SO}_2} = 2P_{\text{O}_2}$$

$$P_{\text{SO}_3} = P_T - 3P_{\text{O}_2}$$

So
$$K_P = 0.533 = \frac{P_{\text{SO}_2} \cdot P_{\text{O}_2}^{1/2}}{P_{\text{SO}_3}} = \frac{2 \times 0.05 \times (0.05)^{1/2}}{(P_T - 0.15)}$$

$$P_T = 0.192 \text{ atm}$$

Chapter 11

Solution to 11.5

$$\text{with } P_T = 1 \text{ atm} \quad K_P = \frac{P_{SO_2} \cdot P_{O_2}^{1/2}}{P_{SO_3}} = \frac{2 \times 0.05 \times (0.05)^{1/2}}{1 - 0.15}$$
$$= 0.0263$$

$$\Delta G^\circ = -RT \ln K_p$$

$$\Delta G^\circ = 94600 - 89.37 T = -8.3144 T \ln 0.0263$$

$$T = 792 \text{ K}$$

Chapter 11

11.6 For the dissociation of nitrogen according to



$$\Delta G^0 = 945,000 - 114.9T \text{ J}$$

- a. Calculate the equilibrium partial pressure of N in nitrogen gas at 3000 K and a total pressure of 1 atm.
- b. The total pressure of the gas, at 3000 K, at which the partial pressure of N_2 is 90% of the total pressure.

Chapter 11

Solution to 11.6

a. For $N_2 = 2N$ $\Delta G_{3000}^0 = 600300 \text{ J}$

So $K_P = 3.53 \times 10^{-11} = \frac{P_N^2}{(1 - P_N)}$ $\Delta G^\circ = -RT \ln K_p$

$$P_N = 5.94 \times 10^{-6} \text{ atm}$$

b. For $P_T = P_{N_2} + P_N$

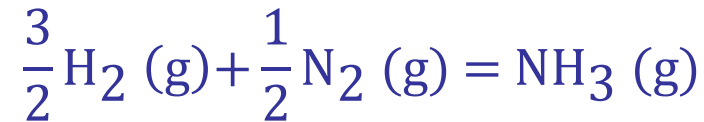
$$= 0.9P_T + 0.1P_T$$

So $K_P = 3.53 \times 10^{-11} = \frac{(0.1P_T)^2}{0.9P_T} = \frac{0.01}{0.9} P_T$

$$P_T = 3.18 \times 10^{-9} \text{ atm}$$

Chapter 11

11.7 Ammonia gas is heated to 300°C. At what total pressure is the mole fraction of N₂ in the equilibrium gas mixture equal to 0.2 ? Calculate the standard enthalpy change and the standard entropy change for the reaction



at 300°C.

Chapter 11

Solution to 11.7

For $2\text{NH}_3 = \text{N}_2 + 3\text{H}_2$ $\Delta G_{573}^0 = -25021 \text{ J}$

$$\Delta G^\circ = -RT \ln K_p$$

So $K_p = 190.1$

As $P_{\text{H}_2} = 3P_{\text{N}_2}$ $x_{\text{H}_2} = 3x_{\text{N}_2}$

So $x_{\text{N}_2} = 0.2$, $x_{\text{H}_2} = 0.6$, $x_{\text{NH}_3} = 1 - 0.2 - 0.6 = 0.2$

So
$$K_p = 190.1 = \frac{P_{\text{N}_2} \cdot P_{\text{H}_2}^3}{P_{\text{NH}_3}^2} = \frac{x_{\text{N}_2} \cdot x_{\text{H}_2}^3 \cdot P_T^2}{x_{\text{NH}_3}^2} = \frac{0.2 \times 0.6^3 \times P_T^2}{0.2^2}$$

$$P_T = 13.3 \text{ atm}$$

Chapter 11

Solution to 11.7

$$\Delta G^0 = -87030 + 25.8 T \ln T + 31.7T$$



So
$$\frac{\partial \Delta G^0}{\partial T} = -\Delta S^0 = 25.8 \ln T + 25.8 + 31.7$$

So
$$\Delta S_{573}^0 = -221.4 \text{ J/K} \quad \Delta H_{573}^0 = \Delta G_{573}^0 + 573 \cdot \Delta S_{573}^0$$
$$= -25021 + 221.4 \times 573$$

For
$$\frac{3}{2}\text{H}_2 (\text{g}) + \frac{1}{2}\text{N}_2 (\text{g}) = \text{NH}_3 (\text{g}) \quad = -101841 \text{ J}$$

So
$$\Delta H_{573}^0 = -50900 \text{ J} \quad \text{and} \quad \Delta S_{573}^0 = -110.7 \text{ J/K}$$

Chapter 11

11.8 By establishing the equilibrium



at 500 K in a mixture of PCl_5 and PCl_3 , a gas is obtained at 1 atm total pressure in which the partial pressure of Cl_2 is 0.1 atm. In what ratio were PCl_5 and PCl_3 mixed to obtain this equilibrium gas?

Chapter 11

Solution to 11.8

For $\text{PCl}_3 + \text{Cl}_2 = \text{PCl}_5$ $\Delta G_{500}^0 = -2672 \text{ J}$

$$\Delta G^\circ = -RT \ln K_p$$

So $K_p = 1.901$



initially, $a \quad 0 \quad 1$

upon reaction, $a + x \quad x \quad 1 - x \quad n_T = a + 1 + x$

with $P_T = 1 \text{ atm}$ $P_{\text{Cl}_2} = \frac{x}{a + 1 + x} = 0.1$

Chapter 11

Solution to 11.8

$$K_p = 1.901 = \frac{P_{PCl_5}}{P_{PCl_3} \cdot P_{Cl_2}} = \frac{(1-x)(a+1+x)}{x \cdot (a+x)}$$

So $a = 2.692, \quad x = 0.410$

$$\frac{PCl_3}{PCl_5} = a = 2.692$$

Chapter 11

11.9 Air and hydrogen are mixed in the ratio 1/4 and are heated to 1200 K. Calculate the partial pressures of H_2 and O_2 in the equilibrium gas at 1 atm pressure and at 10 atm pressure. Air contains 21 volume percent O_2 .

Chapter 11

Solution to 11.9

$$n_{H_2} = 20 \qquad n_{air} = 80$$

So $n_{O_2} = 16.8 \qquad n_{H_2} = 63.2$

For $H_2 + \frac{1}{2}O_2 = H_2O \qquad \Delta G_{1200}^0 = -180480 \text{ J}$

$$\Delta G^0 = -RT \ln K_p$$

So
$$K_P = 7.178 \times 10^7 = \frac{P_{H_2O}}{P_{H_2} \cdot P_{O_2}^{1/2}}$$

$$\frac{n_{H_2}}{n_{O_2}} = \frac{20}{16.8} = 1.1905 = \frac{2n_{H_2} + 2n_{H_2O}}{2n_{O_2} + n_{H_2O}}$$

Chapter 11

Solution to 11.9

$$\frac{n_{O_2}}{n_{H_2}} = \frac{16.8}{63.2} = 0.2658 = \frac{2n_{O_2} + 2n_{H_2O}}{126.4}$$

So

$$2n_{O_2} + n_{H_2O} = 0.2658 \times 126.4 = 33.6$$

$$2n_{H_2} + 2n_{H_2O} = 33.6 \times 1.1905 = 40$$

So

$$n_{H_2O} = 20 - n_{H_2} = 33.6 - 2n_{O_2} \quad n_{O_2} = 6.8 + \frac{1}{2}n_{H_2}$$

$$n_T = 20 - n_{H_2} + n_{H_2} + 6.08 + \frac{1}{2}n_{H_2} + 63.2$$

$$= 90 + \frac{1}{2}n_{H_2}$$

Chapter 11

Solution to 11.9

$$K_P = 7.178 \times 10^7 = \frac{n_{H_2O} \cdot n_T^{1/2}}{n_{H_2} \cdot n_{O_2}^{1/2} \cdot P^{1/2}}$$

$$\text{with } P_T = 1 \text{ atm, } 7.178 \times 10^7 = \frac{(20 - n_{H_2}) \cdot \left(90 + \frac{1}{2}n_{H_2}\right)^{1/2}}{n_{H_2} \cdot \left(6.8 + \frac{1}{2}n_{H_2}\right)^{1/2}}$$

So $n_{H_2} = 9.43 \times 10^{-7} \text{ moles}$ $n_{O_2} = 6.8 \text{ moles}$ $n_T = 90 \text{ moles}$

So $P_{H_2} = \frac{n_{H_2}}{n_T} \cdot P_T = \frac{9.43 \times 10^{-7}}{90} \times 1 = 1.05 \times 10^{-8} \text{ atm}$

$$P_{O_2} = \frac{6.8}{90} \times 1 = 0.0756 \text{ atm}$$

Chapter 11

Solution to 11.9

$$\text{with } P_T = 10 \text{ atm,} \quad 7.178 \times 10^7 = \frac{(20 - n_{H_2}) \cdot \left(90 + \frac{1}{2}n_{H_2}\right)^{1/2}}{n_{H_2} \cdot \left(6.8 + \frac{1}{2}n_{H_2}\right)^{1/2} \cdot \sqrt{10}}$$

$$\text{So} \quad n_{H_2} = 2.98 \times 10^{-7} \text{ moles} \quad n_{O_2} = 6.8 \text{ moles}$$

$$\text{So} \quad P_{H_2} = \frac{n_{H_2}}{n_T} \cdot P_T = \frac{2.98 \times 10^{-7}}{90} \times 10 = 3.31 \times 10^{-8} \text{ atm}$$

$$P_{O_2} = \frac{6.8}{90} \times 10 = 0.756 \text{ atm}$$

Chapter 11

11.10 One mole of each of hydrogen, iodine vapor, and HI gas are allowed to react at 1500 K and $P = 1$ atm. Calculate the mole fractions of H_2 , I_2 , and HI in the equilibrium mixture. The temperature is then changed to that value at which P_{HI} in the equilibrated gas is five times P_{H_2} . What is this temperature?

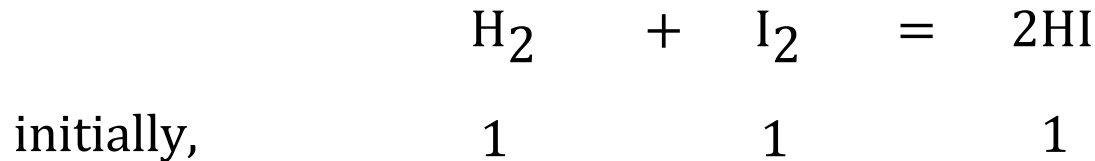
Chapter 11

Solution to 11.10

For $\text{H}_2 + \text{I}_2 = 2\text{HI}$ $\Delta G_{1500}^0 = -34845 \text{ J}$

$$\Delta G^0 = -RT \ln K_p$$

So $K_p = 16.345 = \frac{P_{\text{HI}}^2}{P_{\text{H}_2} \cdot P_{\text{I}_2}}$



upon reaction,	$1 - x$		$1 - x$		$1 + 2x$	$n_T = 3$
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So $K_p = \frac{(1 + 2x)^2}{(1 - x)^2} = 16.345$ $x = 0.504$

Chapter 11

Solution to 11.10

So $x_{H_2} = x_{I_2} = \frac{1 - 0.504}{3} = 0.165,$

$$x_{HI} = \frac{1 + 2 \times 0.504}{3} = 0.669$$

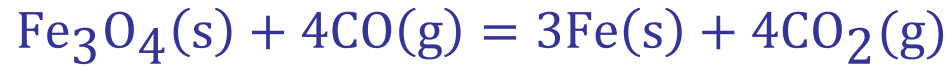
For $P_{HI} = 5P_{H_2}$ $K_P = \frac{(5P_{H_2})^2}{P_{HI}^2} = 25$

So $\Delta G^0 = -8370 - 17.65T = -8.3144 T \ln 25$

$$T = 918 \text{ K}$$

Chapter 11

11.11 Consider the reduction of magnetite by CO to form pure iron :



Calculate the partial pressure of CO and CO₂ in terms of the equilibrium constant K_P for the reaction.

Chapter 11

Solution to 11.11

Assume the activity of the solids to be unity and that the gases behave as ideal gases.

For
$$K_P = \frac{a_{Fe}^3 \cdot a_{CO_2}^4}{a_{Fe_3O_4} \cdot a_{CO}^4} = \frac{P_{CO_2}^4}{P_{CO}^4}$$

So
$$K_P^{1/4} = \frac{P_{CO_2}}{P_{CO}}$$

Also the total pressure is 1 atmosphere : $P_{CO_2} + P_{CO} = 1$

Solving the two equations simultaneously:

$$P_{CO} = \frac{1}{\left(K_P^{1/4} + 1\right)} \qquad P_{CO_2} = \frac{K_P^{1/4}}{\left(K_P^{1/4} + 1\right)}$$

Chapter 11

11.12 It is known that at 300 K, the value of K_p for a certain reaction is 10^{12} . For the reaction, ΔH° is 100 kJ/mole.

a. Determine if this reaction is favorable at 800 K and estimate K_p (800K).

Explain.

b. The actual value of K_p (800) is 35. Explain any discrepancy from your estimate.

Chapter 11

Solution to 11.12

a. First, estimate K_p at 800 K.

$$\ln \left(\frac{K_p(800\text{K})}{K_p(300\text{K})} \right) = \frac{\Delta H^0}{R} \left(\frac{1}{300} - \frac{1}{800} \right)$$

$$\ln \left(\frac{K_p(800\text{K})}{K_p(300\text{K})} \right) = \frac{-100000}{8.3144} \left(\frac{1}{300} - \frac{1}{800} \right) = -25.06$$

$$\frac{K_p(800\text{K})}{K_p(300\text{K})} = \exp(-25.06)$$

$$K_p(800\text{K}) = K_p(300\text{K}) \cdot \exp(-25.06) = 13.12$$

Since the value is greater than unity the reaction is possible at 800 K.

b. The above method assumed there was no temperature dependence of ΔH^0 over the span of 500 K.

Chapter 11

11.13 Sketch the natural log of K_p versus $1/T$ for the following cases:

a. $\Delta H^0 < 0, \Delta S^0 < 0$

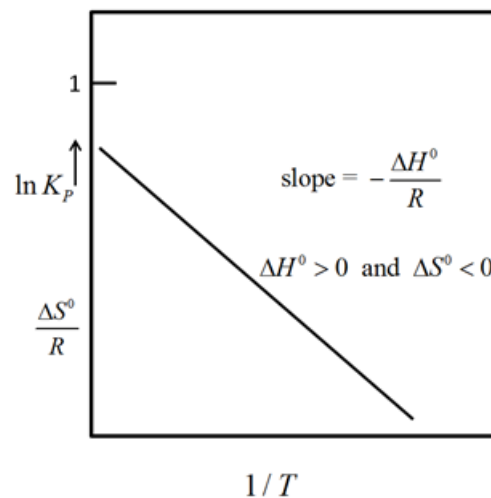
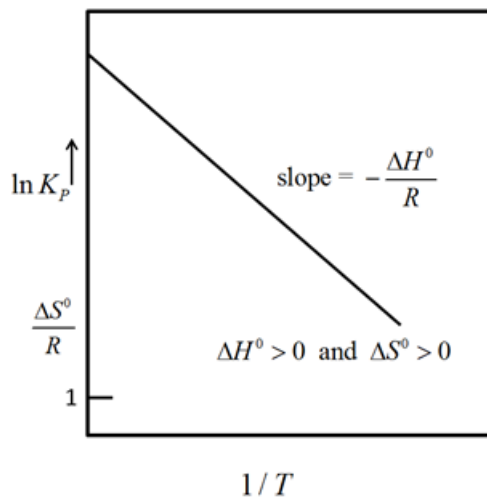
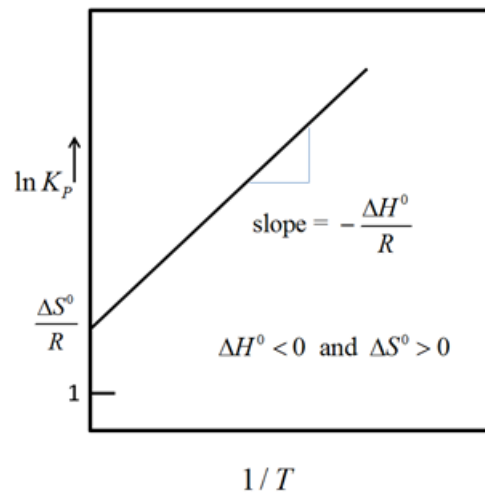
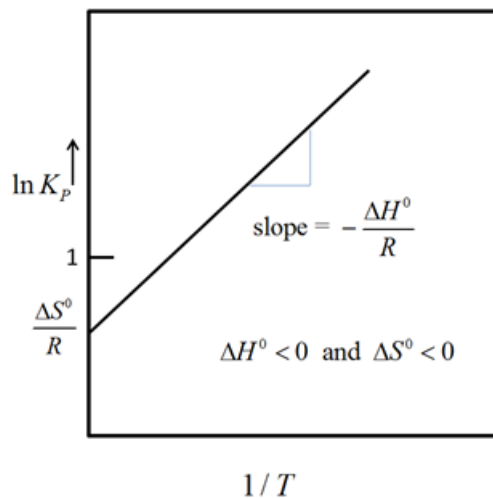
b. $\Delta H^0 < 0, \Delta S^0 > 0$

c. $\Delta H^0 > 0, \Delta S^0 > 0$

d. $\Delta H^0 > 0, \Delta S^0 < 0$

Chapter 11

Solution to 11.13



Thermodynamics of Materials & Phase transformations

The End



Exercise for Chapter 12

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing,
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

Chapter 12

12.1 To what temperature must MgCO_3 be heated in an atmosphere containing a partial pressure of CO_2 of 10^{-2} atm to cause the decomposition of the carbonate?

Chapter 12

Solution to 12.1

$$\Delta G^\circ = -RT \ln K \quad \text{MgO}_{(s)} + \text{CO}_{2(g)} = \text{MgCO}_{3(s)} \quad \begin{array}{l} \text{Textbook P650,} \\ \text{Table A.1} \end{array}$$
$$\Delta G^\circ = -117600 + 170T = RT \ln p_{\text{CO}_2(\text{eq.}T)}$$

So, at $p_{\text{CO}_2(\text{eq})} = 10^{-2} \text{ atm}$,

$$-117600 + 170T = 8.3144 \cdot T \cdot \ln 10^{-2} = -38.29T$$

which gives $T = 565 \text{ K}$.

Chapter 12

12.2 Using the standard Gibbs free energies of formation of NiO from solid Ni and liquid Ni, calculate the melting temperature, molar heat of melting, and the molar entropy of melting of nickel.

Chapter 12

Solution to 12.2

Textbook P650, Table A.1

For $2\text{Ni}_{(s)} + \text{O}_{2(g)} = 2\text{NiO}_{(s)} \quad \Delta G^\circ = -471200 + 172T \text{ J}$

And for $2\text{NiO}_{(s)} = \text{O}_{2(g)} + 2\text{Ni}_{(l)} \quad \Delta G^\circ = 506180 - 192.2T \text{ J}$

Adding gives

$$2\text{Ni}_{(s)} = 2\text{Ni}_{(l)} \quad \Delta G^\circ = 34980 - 20.2T \text{ J} = \Delta H^\circ - T\Delta S^\circ$$

So,

$$T_{m,\text{Ni}} = \frac{34980}{20.2} = 1731 \text{ K}$$

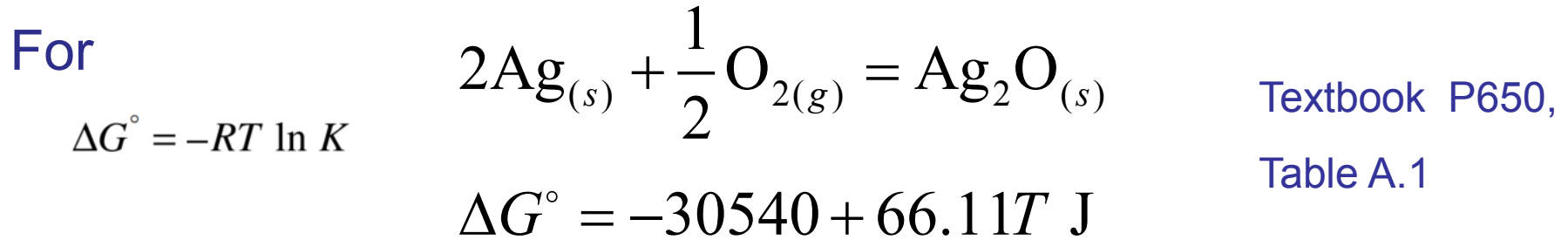
$$\Delta H_{m,\text{Ni}}^\circ = \frac{34980}{2} = 17490 \text{ J} \quad \Delta S_{m,\text{Ni}}^\circ = \frac{20.2}{2} = 10.1 \text{ J/K}$$

Chapter 12

12.3 Calculate the temperature at which pure Ag_2O decomposes to Ag metal and O_2 gas when heated in (a) pure oxygen at 1 atm pressure, and (b) air.

Chapter 12

Solution to 12.3



a. For $p_{\text{O}_2(\text{eq})} = 1 \text{ atm}$,

$$-30540 + 66.11T = RT \ln p_{\text{O}_2(\text{eq})}^{1/2} = 0$$

so $T = 462 \text{ K}$.

b. For $p_{\text{O}_2(\text{eq})} = 0.21 \text{ atm (in air)}$,

$$-30540 + 66.11T = RT \ln p_{\text{O}_2(\text{eq})}^{1/2} = \frac{1}{2} \times 8.3144 \times T \times \ln 0.21$$

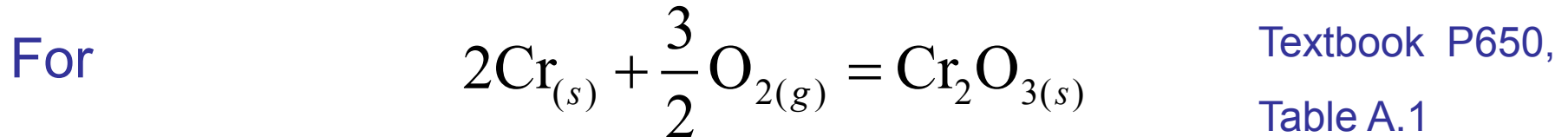
which gives $T = 421 \text{ K}$.

Chapter 12

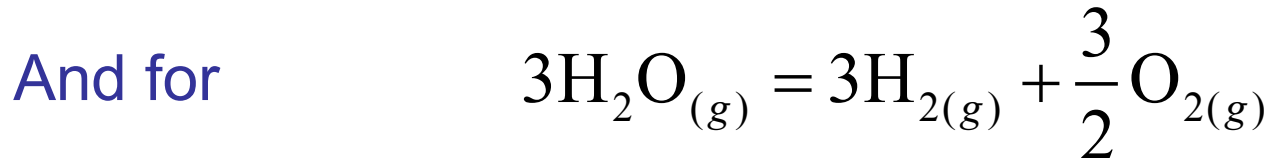
12.4 Determine the maximum pressure of water vapor in wet hydrogen at 1 atm pressure in which chromium can be heated without oxidation occurring at 1500 K. Is the oxidation of Cr by water vapor exothermic or endothermic?

Chapter 12

Solution to 12.4



$$\begin{aligned}\Delta G_{1500\text{ K}}^{\circ} &= -1,110,100 + 247.3T \text{ J} \\ &= -1,110,100 + 247.3 \times 1500 \text{ J} = -739150 \text{ J}\end{aligned}$$

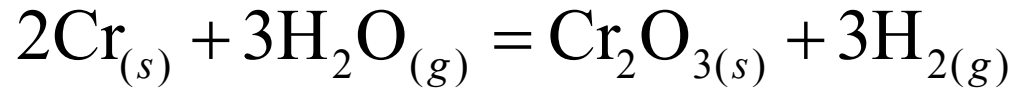


$$\begin{aligned}\Delta G_{1500\text{ K}}^{\circ} &= 3 \times (247,500 - 55.85T) \text{ J} \\ &= 3 \times (247,500 - 55.85 \times 1500) \text{ J} = 491175 \text{ J}\end{aligned}$$

Chapter 12

Solution to 12.4

Adding gives



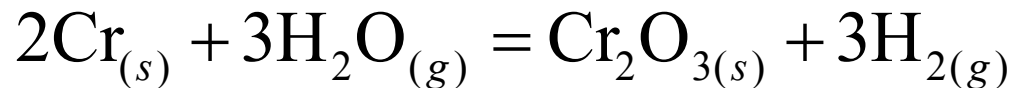
$$\Delta G_{1500\text{ K}}^\circ = -247,975\text{ J} = RT \ln \frac{p_{\text{H}_2\text{O}}^3}{p_{\text{H}_2}^3} = 1500 \times 8.3144 \times 3 \ln \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}$$

So

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 1.32 \times 10^{-3}$$

with $p_{\text{H}_2} = 1\text{ atm}$, $p_{\text{H}_2\text{O}(\text{max})} = 1.32 \times 10^{-3}\text{ atm}$

For



$\Delta H_{1500\text{ K}}^\circ = -371200\text{ J}$, and the reaction is exothermic.

Chapter 12

12.5 A mixture of argon gas and hydrogen gas at 1 atm total pressure is passed through a reaction vessel containing a mixture of liquid Sn and liquid SnCl_2 at 900 K. The composition of the gas leaving the vessel is 50% H_2 , 7% HCl , and 43% Ar. Has equilibrium been attained between the gas phase and the liquid phases in the vessel?

Chapter 12

Solution to 12.5

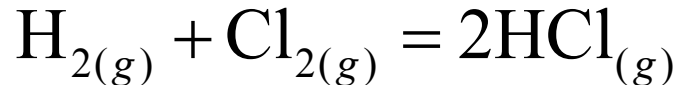
For



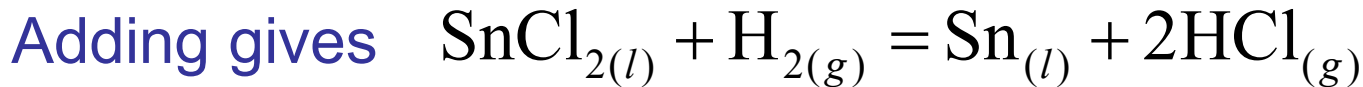
Textbook P650,
Table A.1

$$\Delta G_{900\text{ K}}^{\circ} = -(-333,000 + 118.4T) \text{ J} = 226440 \text{ J}$$

For



$$\Delta G_{900\text{ K}}^{\circ} = -188,200 - 12.80T \text{ J} = -199720 \text{ J}$$



$$\Delta G_{900\text{ K}}^{\circ} = 144,800 - 131.2T \text{ J} = 26720 \text{ J}$$

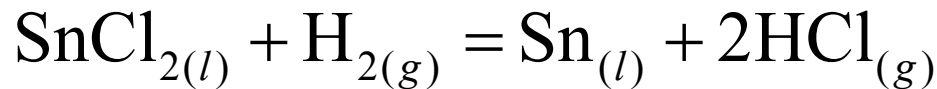
So $K = \exp\left(\frac{-\Delta G_{900\text{ K}}^{\circ}}{RT}\right) = \exp\left(\frac{-26720}{8.3144 \times 900}\right) \text{ J} = 0.028 = \frac{p_{\text{HCl}}^2}{p_{\text{H}_2}}$

Chapter 12

Solution to 12.5

For the final gas, $\frac{p_{\text{HCl}}^2}{p_{\text{H}_2}} = \frac{0.07^2}{0.5} = 0.0098$, which being less than K , indicates that equilibrium has not been attained.

The gas entering contains $0.5_{(\text{leave})} + 0.035_{(\text{reaction})} = 0.535$ mols of H_2 and 0.43 mols of Ar.



before 0.535

after 0.535-x $2x$ $n_{\text{total}} = 0.535 + x + 0.43$
 $= 0.965 + x$ mols

So $K = 0.028 = \frac{(2x)^2}{(0.965 + x)(0.535 - x)}$, which gives $x = 0.0584$.

Chapter 12

Solution to 12.5

For the equilibrium state, the gas composition is:

$$\text{HCl} \quad \frac{2 \times 0.0584}{0.965 + 0.0584} = 11.4\%$$

$$\text{H}_2 \quad \frac{0.535 - 0.0584}{0.965 + 0.0584} = 46.6\%$$

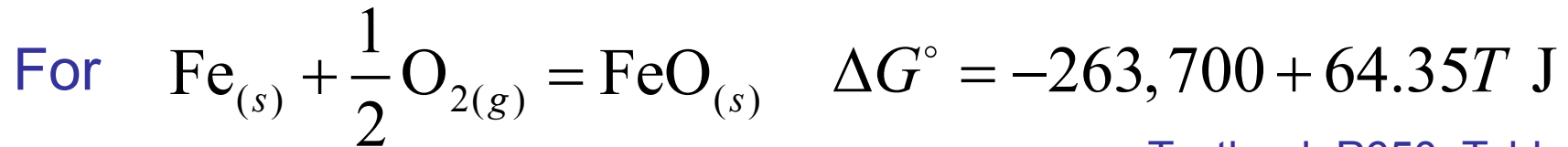
$$\text{Ar} \quad \frac{0.43}{0.965 + 0.0584} = 42\%$$

Chapter 12

12.6 Fe and FeO are in equilibrium with a gas mixture of composition 71.8% CO–28.2% CO₂ at 1273 K. Which of the two solid phases disappears if the composition of the gas is held constant and the temperature of the system is decreased?

Chapter 12

Solution to 12.6



Textbook P650, Table A.1

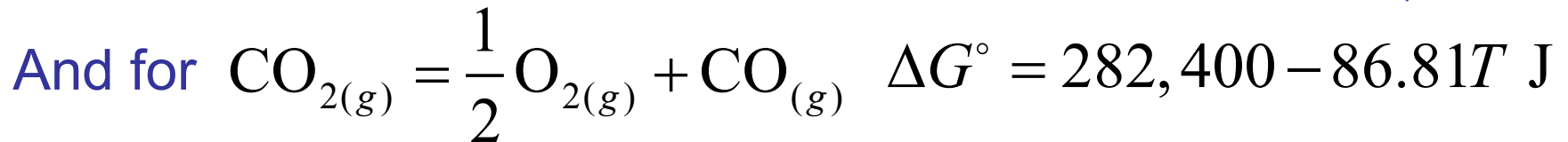


Table A.1 The Standard Gibbs Free Energy Changes for Several Reactions

Reaction	$\Delta G^\circ, \text{ J}$	Range, K
$2\text{Ag}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{Ag}_2\text{O}_{(s)}$	$-30,540 + 66.11 T$	298–463
$\text{Al}_{(l)} = [\text{Al}]_{(1 \text{ wt\% in Fe})}$	$-43,100 - 32.26 T$	
$2\text{Al}_{(l)} + 1.5\text{O}_{2(g)} = \text{Al}_2\text{O}_{3(s)}$	$-1,687,200 + 326.8 T$	993–2327
$\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{CO}_{(g)}$	$-111,700 - 87.65 T$	298–2000
$\text{C}_{(s)} + \text{O}_{2(g)} = \text{CO}_{2(g)}$	$-394,100 - 0.84 T$	298–2000
$\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} + \frac{1}{2}\text{S}_{2(g)} = \text{COS}_{(g)}$	$-202,800 - 9.96 T$	773–2000
$\text{C}_{(gr)} + 2\text{H}_{2(g)} = \text{CH}_{4(g)}$	$-91,040 + 110.7 T$	773–2000
$\text{C}_{(gr)} = [\text{C}]_{(1 \text{ wt\% in Fe})}$	$22,600 - 42.26 T$	

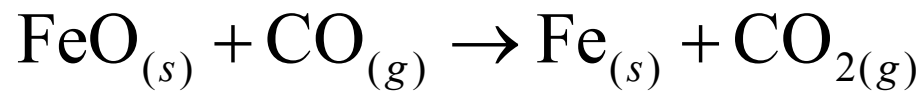
Chapter 12

Solution to 12.6

Adding gives $\text{Fe}_{(s)} + \text{CO}_{2(g)} = \text{FeO}_{(s)} + \text{CO}_{(g)}$

$$\Delta G^\circ = 18,700 - 22.46T \text{ J}$$

As written, the reaction is endothermic. Thus a decrease in temperature causes the equilibrium to shift as:



which decrease $p_{\text{CO}(\text{eq})}$ and increase $p_{\text{CO}_2(\text{eq})}$.

Thus decreasing the temperature at constant p_{CO} and constant p_{CO_2} cause the FeO phase to disappear.

Chapter 12

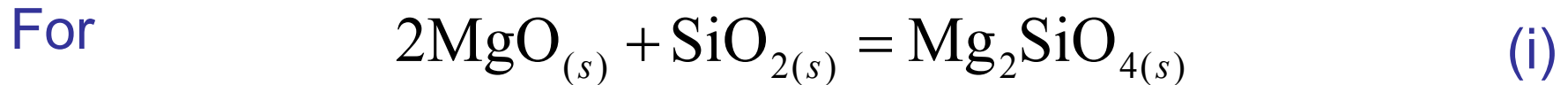
12.7 Calculate the vapor pressure of Mg exerted at 1400° C by the system in which the following reaction equilibrium is established.



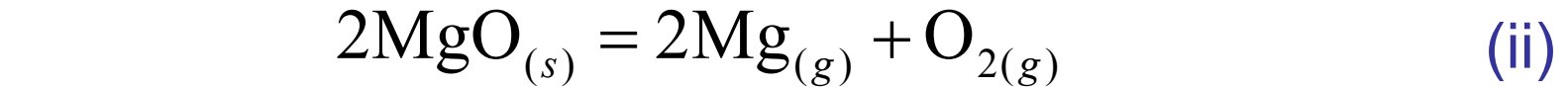
Chapter 12

Solution to 12.7

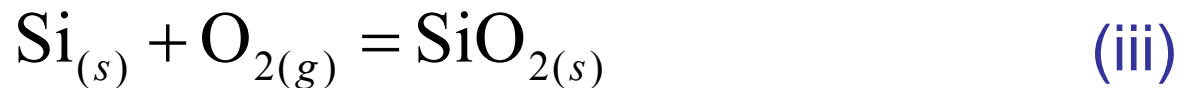
Textbook P650, Table A.1



$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= -67,200 + 4.31T \text{ J} \\ &= -67,200 + 4.31 \times 1673 \text{ J} = -59,989 \text{ J} \end{aligned}$$



$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= 2 \times (729,600 - 204T) \text{ J} \\ &= 2 \times (729,600 - 204 \times 1673) \text{ J} = 776,616 \text{ J} \end{aligned}$$

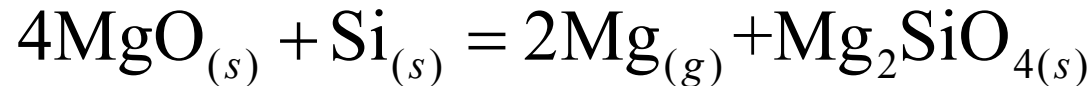


$$\begin{aligned} \Delta G_{1673 \text{ K}}^{\circ} &= -907,100 + 175T \text{ J} \\ &= -907,100 + 175 \times 1673 \text{ J} = -614,325 \text{ J} \end{aligned}$$

Chapter 12

Solution to 12.7

(i)+(ii)+(iii) gives



$$\Delta G_{1673\text{ K}}^{\circ} = 102,302\text{ J} = RT \ln \frac{1}{p_{\text{Mg}(\text{eq})}^2}$$

So

$$\begin{aligned} p_{\text{Mg}(\text{eq})} &= \exp\left(-\frac{\Delta G_{1673\text{ K}}^{\circ}}{2RT}\right) \\ &= \exp\left(-\frac{102,302}{2 \times 8.3144 \times 1673}\right) = 2.53 \times 10^{-2} \text{ atm} \end{aligned}$$

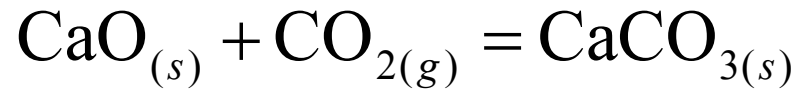
Chapter 12

12.8 One gram of CaCO_3 is placed in an evacuated rigid vessel of volume 1 liter at room temperature, and the system is heated. Calculate (a) the highest temperature at which the CaCO_3 phase is present, (b) the pressure in the vessel at 1000 K, and (c) the pressure in the vessel at 1500 K. The molecular weight of CaCO_3 is 100.

Chapter 12

Solution to 12.8

For



$$\Delta G^\circ = -168,400 + 144T \text{ J}$$

So

$$\ln p_{\text{CO}_2(\text{eq})} = \frac{\Delta G^\circ}{RT} = -\frac{168,400}{RT} + \frac{144}{R} = -\frac{20254}{T} + 17.32 \quad (\text{i})$$

1 g CaCO_3 , $n=1/100=0.01$ moles, so the vessel contains 0.01 moles of CO_2 . If all of the 0.01 moles of CO_2 occur in the gas phase, thus

$$p_{\text{CO}_2} = \frac{nRT}{V} = \frac{0.01 \times 0.082057T}{1} = 8.2057 \times 10^{-4} T \quad (\text{ii})$$

Chapter 12

Solution to 12.8

From (i)

$$p_{\text{CO}_2} = \exp \left[-\frac{20254}{T} + 17.32 \right] \quad (\text{iii})$$

(a) Lines (ii) and (iii) intersect at $p_{\text{CO}_2} = 0.96 \text{ atm}$, $T = 1173 \text{ K}$, which is the highest temperature at which CaCO_3 exists.

(b) At $T = 1000 \text{ K}$, equation (iii) gives $p_{\text{CO}_2(\text{eq})} = 0.055 \text{ atm}$.

(b) At $T = 1500 \text{ K}$, equation (ii) gives $p_{\text{CO}_2(\text{eq})} = 1.23 \text{ atm}$.

Chapter 12

12.9 Calculate the total pressure ($p_{\text{SO}_3} + p_{\text{SO}_2} + p_{\text{O}_2}$) exerted by equilibrated CoO and CoSO₄ at 1223 K.

Chapter 12

Solution to 12.9

For $\text{CoO}_{(s)} + \text{SO}_{3(g)} = \text{CoSO}_{4(s)}$ Textbook P650,
Table A.1

$$\Delta G_{1223\text{ K}}^{\circ} = -227,860 + 165.3T \text{ J} = -25698 \text{ J}$$

$$\Delta G_{1223\text{ K}}^{\circ} = RT \ln p_{\text{SO}_3(\text{eq})} = 8.3144 \times 1223 \ln p_{\text{SO}_3(\text{eq})}$$

Combining the two equations gives

$$p_{\text{SO}_3(\text{eq})} = 7.99 \times 10^{-2} \text{ atm} \quad (\text{i})$$

For $\text{SO}_{3(g)} = \text{SO}_{2(g)} + \frac{1}{2} \text{O}_{2(g)}$

$$\Delta G_{1223\text{ K}}^{\circ} = 94,600 - 89.37T \text{ J} = -14699 \text{ J} = -8.3144 \times 1223 \ln \frac{p_{\text{SO}_2} p_{\text{O}_2}^{1/2}}{p_{\text{SO}_3}}$$

Chapter 12

Solution to 12.9

So

$$\frac{p_{\text{SO}_2} p_{\text{O}_2}^{1/2}}{p_{\text{SO}_3}} = 4.244 \quad \text{and} \quad p_{\text{O}_2} = \frac{1}{2} p_{\text{SO}_2}$$

Substitute (i)

$$4.244 = \frac{p_{\text{SO}_2} \cdot [0.5 p_{\text{SO}_2}]^{1/2}}{7.99 \times 10^{-2}}$$

Which gives

$$p_{\text{SO}_2} = 0.612 \text{ atm}, \quad p_{\text{O}_2} = 0.306 \text{ atm}, \quad p_{\text{SO}_3} = 0.0799 \text{ atm}$$

So

$$p_{\text{total}} = 1.0 \text{ atm}$$

Chapter 12

12.10 A gas mixture initially containing 90% CO, 0.4% COS, and 9.6% inert constituents (by volume) is passed over sponge iron at 1000 K to remove sulfur by the following reaction:

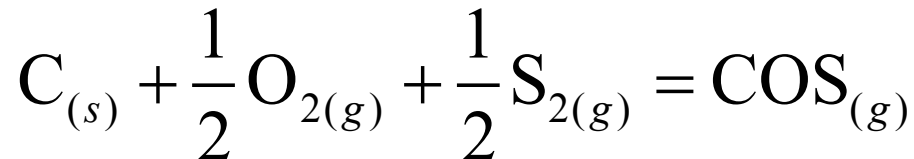


- Assuming that the effluent gas is in equilibrium with Fe and FeS, calculate the percentage of sulfur removed from the gas by reaction with the sponge iron.
- Calculate the partial pressure of S_2 in the effluent gas.

Chapter 12

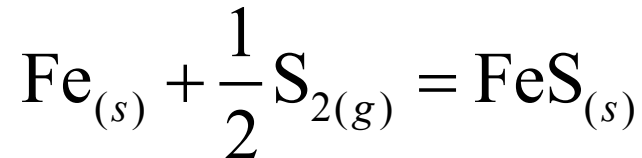
Solution to 12.10

For

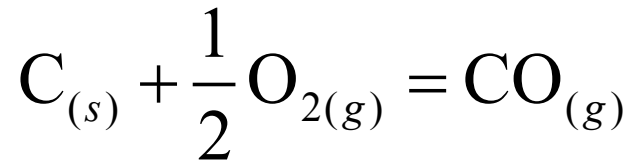


Textbook P650,
Table A.1

$$\Delta G_{1000\text{ K}}^{\circ} = -202,800 - 9.96T \text{ J} = -212,760 \text{ J}$$

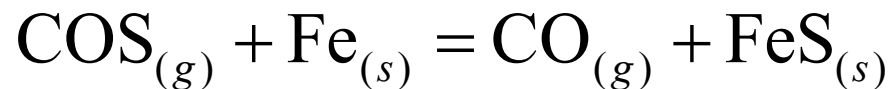


$$\Delta G_{1000\text{ K}}^{\circ} = -150,200 + 52.55T \text{ J} = -97,650 \text{ J}$$



$$\Delta G_{1000\text{ K}}^{\circ} = -111,700 - 87.65T \text{ J} = -199,350 \text{ J}$$

So



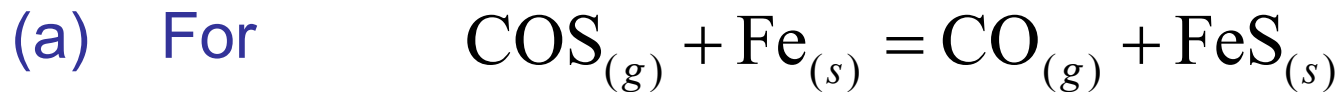
$$\Delta G_{1000\text{ K}}^{\circ} = 212,760 - 97,650 - 199,350 \text{ J} = -84,240 \text{ J}$$

Chapter 12

Solution to 12.10

and

$$K_{1000\text{ K}} = \exp\left[\frac{-\Delta G_{1000\text{ K}}^{\circ}}{RT}\right] = \exp\left[\frac{84,240}{8.4144 \times 1000}\right] = 25130 = \frac{p_{\text{CO}}}{p_{\text{COS}}}$$



before 0.4 90

after 0.4-x 90+x

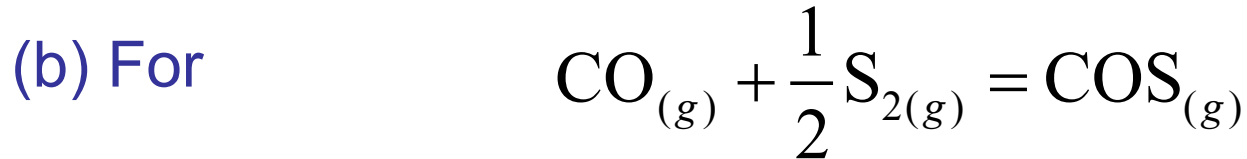
So
$$25130 = \frac{p_{\text{CO}}}{p_{\text{COS}}} = \frac{90+x}{0.4-x}$$

which gives $x=0.3964$.

Thus %S removed = $0.3964/0.4 \times 100\% = 99.1\%$

Chapter 12

Solution to 12.10



$$\Delta G_{1000\text{ K}}^{\circ} = 199,350 - 212,760\text{ J} = -13,410\text{ J}$$

So

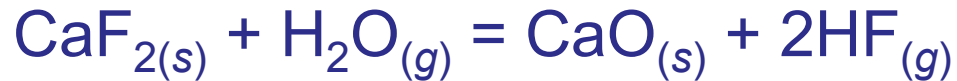
$$K_{1000\text{ K}} = \exp\left[\frac{13410}{8.4144 \times 1000}\right] = 5.017 = \frac{p_{\text{COS}}}{p_{\text{CO}} p_{\text{S}_2}^{1/2}} = \frac{0.0036}{90.3964 p_{\text{S}_2}^{1/2}}$$

Which gives

$$p_{\text{S}_2} = 6.3 \times 10^{-11}\text{ atm}$$

Chapter 12

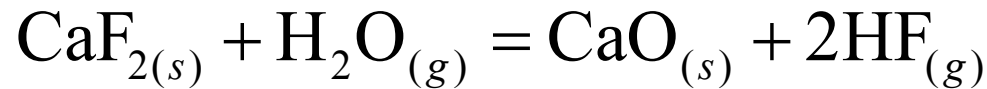
12.11 An Ar–H₂O gas mixture of $p_{\text{H}_2\text{O}} = 0.9 \text{ atm}$ ($P_{\text{total}} = 1 \text{ atm}$) is passed over solid CaF₂, as a result of which CaO forms according to



The reaction proceeds to equilibrium and solid CaO and solid CaF₂ are mutually immiscible. When the gas flow rate (measured at 298 K and 1 atm pressure) over the sample is 1 liter per minute, the measured rates of weight loss of the sample are 2.69×10^{-4} and 8.30×10^{-3} grams per hour at 900 and 1100 K, respectively. Use these data to calculate the variation of ΔG° for the preceding reaction with temperature. The atomic weights are O = 16, F = 19, and Ca = 40.08

Chapter 12

Solution to 12.11



MW	78.08	56.08
----	-------	-------

So a weight loss of 22g produces 2 moles of HF and consumes 1 moles of H₂O, thus a weight loss of 1g products 0.0909 moles of HF and consumes 0.0455 moles of H₂O.

At 900 K for 1 hour,

weight loss of 2.69×10^{-4} g produces 2.44×10^{-5} moles of HF and consumes 1.22×10^{-5} moles of H₂O.

Chapter 12

Solution to 12.11

In 60 L (1 liter per minute) of gas in which $p_{\text{H}_2\text{O}} = 0.9 \text{ atm}$,

$$n_{\text{H}_2\text{O}} = \frac{PV}{RT} = \frac{0.9 \times 60}{0.082057 \times 298} = 2.208 \text{ moles}$$

So consumption of 1.22×10^{-5} moles of H_2O is insignificant.

$$p_{\text{HF}} = \frac{2.44 \times 10^{-5} \times 0.082057 \times 298}{60} = 9.94 \times 10^{-6} \text{ atm}$$

Thus

$$K_{900 \text{ K}} = \frac{p_{\text{HF}}^2}{p_{\text{H}_2\text{O}}} = \frac{(9.94 \times 10^{-6})^2}{0.9} = 1.09 \times 10^{-10}$$

and

$$\Delta G_{900 \text{ K}}^\circ = -8.3144 \times 900 \times \ln(1.09 \times 10^{-10}) \text{ J} = 171,596 \text{ J} \quad (\text{i})$$

Chapter 12

Solution to 12.11

Similarly, at 1100 K for 1 hour,

0.0083 g loss per hour produces 7.545×10^{-4} moles/hour of HF.

So

$$p_{\text{HF}} = \frac{7.545 \times 10^{-4} \times 0.082057 \times 298}{60} = 3.075 \times 10^{-4} \text{ atm}$$

$$K_{1100 \text{ K}} = \frac{(3.075 \times 10^{-4})^2}{0.9} = 1.051 \times 10^{-7}$$

$$\Delta G_{1100 \text{ K}}^{\circ} = -8.3144 \times 1100 \times \ln(1.051 \times 10^{-7}) \text{ J} = 146,962 \text{ J} \quad (\text{ii})$$

Chapter 12

Solution to 12.11

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ} = A + BT$$

Combines (i) and (ii),

$$171596 = A + 900B$$

$$146962 = A + 1100B$$

which gives,

$$B = -123 \text{ J/K}$$

$$A = 282449 \text{ J}$$

So

$$\Delta G^{\circ} = 282449 - 123T$$

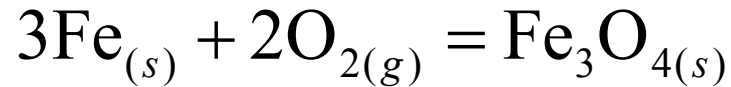
Chapter 12

12.12 Magnetite (Fe_3O_4) is reduced to sponge iron (Fe) in a continuous reactor operating at 800 K using methane gas (CH_4) as the reducing agent. The gaseous reaction product leaving the reactor at a total pressure of 1 atm is a mixture of CO , CO_2 , H_2 , and H_2O with a negligible methane content. The gas is at equilibrium with the Fe – Fe_3O_4 mixture in the reactor. Calculate the consumption of methane as moles of methane used per mole of sponge iron produced.

Chapter 12

Solution to 12.12

For



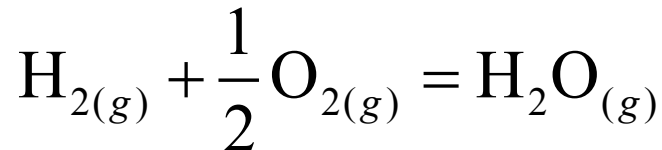
Textbook P650,
Table A.1

$$\begin{aligned}\Delta G_{800\text{ K}}^{\circ} &= -1,102,200 + 307.4T \text{ J} = -856,280 \text{ J} \\ &= RT \ln p_{\text{O}_2(\text{eq})}^2 = 2 \times 8.3144 \times 800 \ln p_{\text{O}_2(\text{eq})}\end{aligned}$$

So in the exit gas

$$p_{\text{O}_2(\text{eq})} = \exp\left[\frac{-856,280}{2 \times 8.3144 \times 800}\right] = 1.11 \times 10^{-28} \text{ atm}$$

For



$$\begin{aligned}\Delta G_{800\text{ K}}^{\circ} &= -247,500 + 55.85T \text{ J} = -202,560 \text{ J} \\ &= -8.3144 \times 800 \ln \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{1/2}}\end{aligned}$$

Chapter 12

Solution to 12.12

So
$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{1/2}} = 1.681 \times 10^{13} \quad \text{and} \quad \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.1771 \quad (\text{i})$$

For
$$2\text{CO}_{(g)} + \text{O}_{2(g)} = 2\text{CO}_{2(g)}$$
$$\Delta G_{800 \text{ K}}^{\circ} = 2(-282,400 + 86.81T) \text{ J} = -425,904 \text{ J}$$
$$= -2 \times 8.3144 \times 800 \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}} p_{\text{O}_2}^{1/2}}$$

So
$$\frac{p_{\text{CO}_2}}{p_{\text{CO}} p_{\text{O}_2}^{1/2}} = 8.02 \times 10^{13} \quad \text{and} \quad \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.845 \quad (\text{ii})$$

Chapter 12

Solution to 12.12

According to the title,

$$p_{\text{H}_2} + p_{\text{H}_2\text{O}} + p_{\text{CO}} + p_{\text{CO}_2} = 1 \text{ atm} \quad (\text{iii})$$

All of the C and H in the gas come from the CH_4 ,

so

$$\frac{n_{\text{C}}}{n_{\text{H}}} = \frac{1}{4} = \frac{p_{\text{CO}} + p_{\text{CO}_2}}{2p_{\text{H}_2} + 2p_{\text{H}_2\text{O}}}$$

$$p_{\text{H}_2} + p_{\text{H}_2\text{O}} = 2p_{\text{CO}} + 2p_{\text{CO}_2} \quad (\text{iv})$$

Solution of (i)-(iv) gives,

$$p_{\text{H}_2} = 0.5662 \text{ atm}, \quad p_{\text{H}_2\text{O}} = 0.1003 \text{ atm},$$

$$p_{\text{CO}} = 0.1806 \text{ atm}, \quad p_{\text{CO}_2} = 0.1528 \text{ atm}.$$

Chapter 12

Solution to 12.12

All of the C comes from CH_4

$$n_{\text{C}} = 0.1806 + 0.1528 = 0.3334 \text{ moles}$$

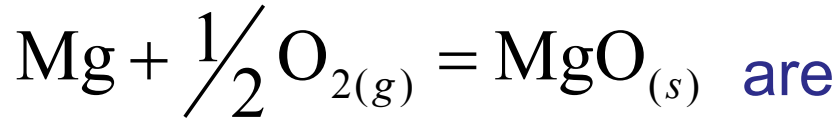
All of the O comes from Fe_3O_4

$$n_{\text{O}} = 0.1003 + 0.1806 + 2 \times 0.1528 = 0.5865 \text{ moles}$$

So 0.3334 moles of CH_4 per $3/4 \times 0.5865 = 0.4398$ moles of Fe,
or 0.76 moles of CH_4 per mole of Fe produced.

Chapter 12

12.13 Three equations for the oxidation of Mg according to



$$\Delta G^\circ = -604,000 - 5.36T \ln T + 142.0T \text{ J} \quad (\text{i})$$

$$\Delta G^\circ = -759,800 - 13.4T \ln T + 317T \text{ J} \quad (\text{ii})$$

$$\Delta G^\circ = -608,100 - 0.44T \ln T + 112.8T \text{ J} \quad (\text{iii})$$

One of these expressions is for the oxidation of solid Mg, one is for the oxidation of liquid Mg, and one is for the oxidation of gaseous Mg. Determine which equation is for which oxidation and calculate the melting and normal boiling temperature of Mg.

Chapter 12

Solution to 12.13

For (I)

$$\frac{\Delta G^\circ}{T} = \frac{-604,000}{T} - 5.36 \ln T + 142.0$$
$$\frac{\partial(\Delta G^\circ/T)}{\partial T} = \frac{604,000}{T^2} - \frac{5.36}{T} = -\frac{\Delta H^\circ}{T^2}$$

so

$$\Delta H_I^\circ = -604,000 + 5.36T$$

Similarly

$$\Delta H_{II}^\circ = -759,800 + 13.4T$$

$$\Delta H_{III}^\circ = -608,100 + 0.44T$$

Chapter 12

Solution to 12.13

Generally,

$$\left| \Delta H_{\text{Mg}(g)}^{\circ} \right| > \left| \Delta H_{\text{Mg}(l)}^{\circ} \right| > \left| \Delta H_{\text{Mg}(s)}^{\circ} \right|$$

So equation (II) is for oxidation of $\text{Mg}_{(g)}$, equation (III) is for oxidation of $\text{Mg}_{(l)}$, equation (I) is for oxidation of $\text{Mg}_{(s)}$.

$$\Delta G_{\text{I}}^{\circ} = \Delta G_{\text{III}}^{\circ} \quad \text{at} \quad T_{m,\text{Mg}} = 930 \text{ K}$$

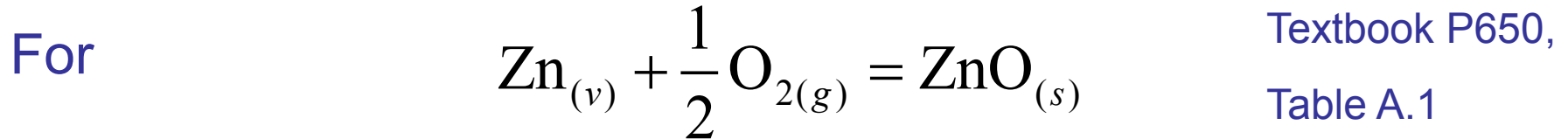
$$\Delta G_{\text{II}}^{\circ} = \Delta G_{\text{III}}^{\circ} \quad \text{at} \quad T_{b,\text{Mg}} = 1372 \text{ K}$$

Chapter 12

12.14 Two hundred grams of liquid zinc are placed in a crucible at 1030 K. Two moles of air are bubbled through the liquid zinc, and the gas comes to equilibrium with the liquid before leaving the system. If the total pressure of the gas remains constant at 0.8 atm throughout the process, how many grams of metallic zinc are left in the crucible? The atomic weights of Zn and O are, respectively, 65.38 and 16.

Chapter 12

Solution to 12.14



$$\Delta G_{1030 \text{ K}}^{\circ} = -460,200 + 198T \text{ J} = -256,260 \text{ J} = -RT \ln K_{1030 \text{ K}}$$

So
$$K_{1030 \text{ K}} = 9.9 \times 10^{12} = \frac{1}{p_{\text{Zn}} p_{\text{O}_2}^{1/2}}$$

$$\ln p_{\text{Zn}}^{\circ} (\text{atm}) = -\frac{15250}{T} - 1.255 \ln T + 21.79$$

Textbook P654,
Table A.4

Thus at 1030 K,
$$p_{\text{Zn}}^{\circ} = 0.1787 \text{ atm}$$

$$p_{\text{O}_2} = \left[\frac{1}{9.9 \times 10^{12} \times 0.1787} \right]^2 = 3.2 \times 10^{-25} \text{ atm}$$

Chapter 12

Solution to 12.14

2 moles air: 0.42 moles O₂ + 1.58 moles N₂

0.42 moles O₂ + 0.84 moles Zn produce 0.84 moles ZnO

and the weight of 0.84 moles ZnO is: 0.84 x 65.38 = 54.92 g

So 54.92 g of Zn are oxidized to ZnO

1.58 moles of N₂ at 1030 K, $p_{\text{total}} = 0.8 \text{ atm}$

So $p_{\text{N}_2} = 0.8 - 0.1787 = 0.621 \text{ atm}$

$$p_{\text{N}_2} V = n_{\text{N}_2} RT$$

Which gives the volume of gas

$$V = \frac{1.58 \times 0.082057 \times 1030}{0.621} = 215 \text{ L}$$

Chapter 12

Solution to 12.14

$$p_{\text{Zn}}V = n_{\text{Zn}}RT$$

So

$$n_{\text{Zn}} = \frac{0.1787 \times 215}{0.082057 \times 1030} = 0.455 \text{ moles}$$

The weight of 0.455 moles Zn is: $0.455 \times 65.38 = 29.78 \text{ g}$

Thus 29.78 g Zn are evaporated, and the remaining liquid Zn is:

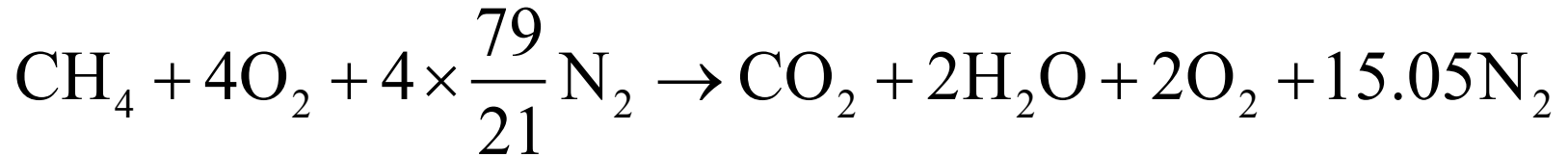
$$200 - 54.9 - 29.8 = 115.3 \text{ g}$$

Chapter 12

12.15 Methane gas is burned with twice the amount of stoichiometric air (i.e., the initial ratio of CH_4 and O_2 in the mixture is 0.25), and the combustion gas produced, in which the concentration of methane is negligible, is used to calcine CaCO_3 in an isothermal furnace. The gas and the solids are at 1080 K and the pressure of the gas is maintained constant at 1 atm. How many moles of CaCO_3 are decomposed per mole of CH_4 burned?

Chapter 12

Solution to 12.15

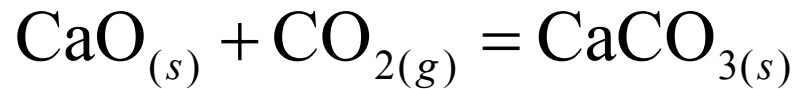


$$n_{\text{total}} = 20.05 \text{ moles}$$

In this gas

$$p_{\text{CO}_2} = \frac{1}{20.05} = 0.0499 \text{ atm}$$

For



Textbook P650,
Table A.1

$$\begin{aligned} \Delta G_{1080 \text{ K}}^{\circ} &= -168,400 + 144T \text{ J} = -13124 \text{ J} \\ &= 8.3144 \times 1080 \ln p_{\text{CO}_2(\text{eq})} \end{aligned}$$

Which gives

$$p_{\text{CO}_2(\text{eq})} = 0.232 \text{ atm}$$

Chapter 12

Solution to 12.15

The decomposition of x moles of CaCO_3 produces x moles of CO_2 and increases p_{CO_2} from 0.0499 atm to 0.232 atm.

That is

$$0.232 = \frac{1 + x}{20.05 + x}$$

So $x = 4.76$

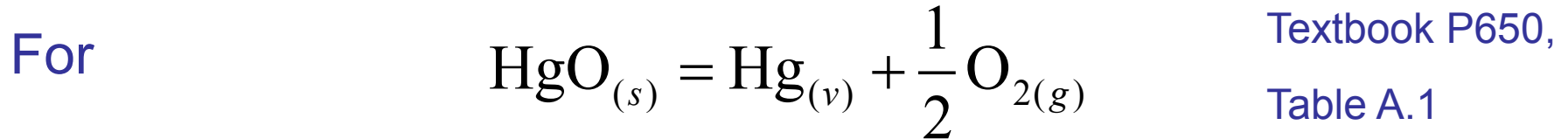
And thus 4.76 moles of CaCO_3 are decomposed per moles of CH_4 burned.

Chapter 12

12.16 Mercuric oxide (HgO) is placed in a vessel which is then evacuated, filled with nitrogen, and heated to 600 K, at which temperature it is observed that the total pressure in the vessel is 2 atm. Calculate the mole fractions of O_2 and Hg vapor in the gas phase.

Chapter 12

Solution to 12.16



$$\Delta G_{600\text{ K}}^{\circ} = 152,200 - 207.2T \text{ J} = 27,880 \text{ J}$$

So $K_{600\text{ K}} = \exp\left[\frac{-\Delta G_{600\text{ K}}^{\circ}}{RT}\right] = 3.74 \times 10^{-3} = p_{\text{Hg}} p_{\text{O}_2}^{1/2}$

Obviously $p_{\text{O}_2} = \frac{1}{2} p_{\text{Hg}}$, so

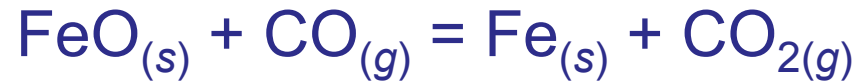
$$3.74 \times 10^{-3} = p_{\text{Hg}} [0.5 p_{\text{Hg}}]^{1/2}$$

Which gives $p_{\text{Hg}} = 0.0304 \text{ atm}$, $p_{\text{O}_2} = 0.0152 \text{ atm}$, $p_{\text{N}_2} = 1.954 \text{ atm}$

Thus $X_{\text{Hg}} = \frac{0.0304}{2} = 0.0152$, $X_{\text{O}_2} = \frac{0.0152}{2} = 0.0071$

Chapter 12

12.17 In Figure 12.17 the line AB, which represents the equilibrium



intersects the carbon deposition line at the point A

($T = 927 \text{ K}$, $p_{\text{CO}_{(s)}} = 0.595 \text{ atm}$, $p_{\text{CO}_2} = 0.405 \text{ atm}$, $P = 1.000 \text{ atm}$).

At what total pressure does the point of intersection of the lines occur at 1000 K, and what are the values of the partial pressures of CO and CO₂ in this state?

Chapter 12

Solution to 12.17

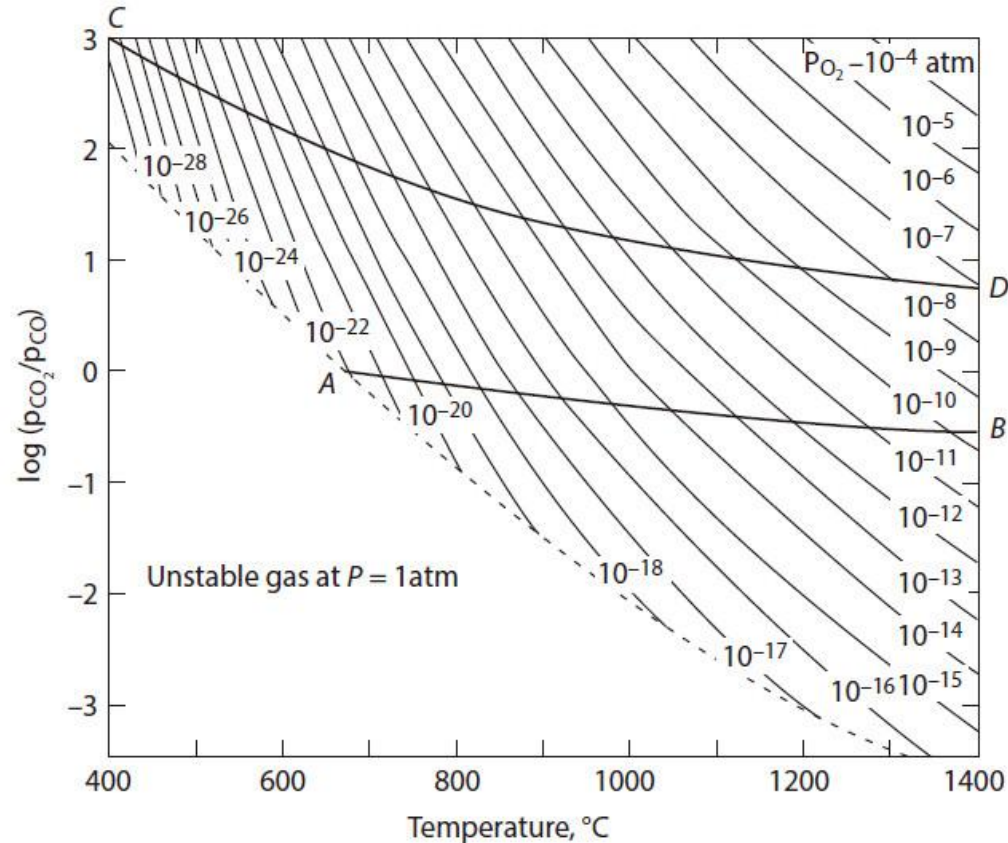
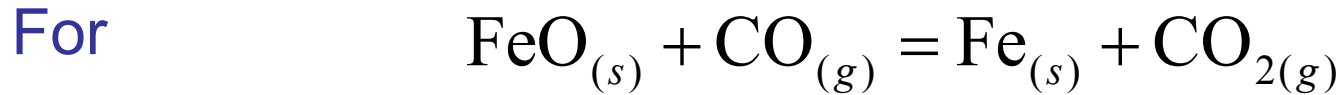


Figure 12.17 The relationship between the partial pressure of oxygen in a CO_2 - CO gas mixture and temperature. The broken line is the variation, with temperature, of the composition of the gas which is in equilibrium with graphite at 1 atm pressure. The lines AB and CD represent, respectively, the equilibria $\text{Fe} + \text{CO}_2 = \text{FeO} + \text{CO}$ and $\text{Co} + \text{CO}_2 = \text{CoO} + \text{CO}$.

Chapter 12

Solution to 12.17



$$\Delta G^\circ = -22,800 + 24.26T \text{ J} = -RT \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}}$$

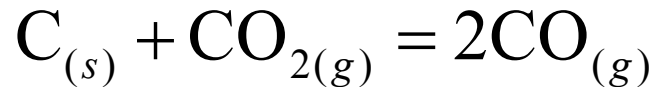
So at the total pressure P_T

$$\frac{2249}{T} - 2.70 = \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = \ln \frac{P_T - p_{\text{CO}}}{p_{\text{CO}}} \quad (\text{i})$$

Chapter 12

Solution to 12.17

For



$$\Delta G^\circ = 170,700 - 174.46T \text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2}}$$

So at the total pressure P_T

$$-\frac{20,530}{T} + 20.99 = \ln \frac{p_{\text{CO}}^2}{P_T - p_{\text{CO}}} \quad (\text{ii})$$

With $T=1000 \text{ K}$, lines (i) and (ii) gives

$$P_T = 1.651 \text{ atm, and } p_{\text{CO}} = 1.009 \text{ atm, } p_{\text{CO}_2} = 0.642 \text{ atm}$$

Thermodynamics of Materials & Phase transformations

The End



Exercise for Chapter 13

Ke Hua (花珂)*

State Key Laboratory of Solidification Processing,
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

<http://teacher.nwpu.edu.cn/kehua>

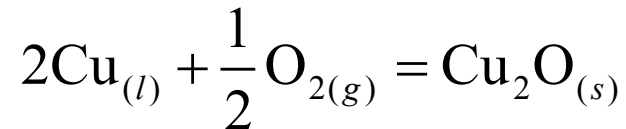
Chapter 13

13.1 Air at atmospheric pressure is blown over a Cu-rich copper–gold liquid solution at 1500 K. If only the copper is oxidized (to form pure solid Cu_2O), calculate the minimum activity of Cu which can be obtained in the solution.

Chapter 13

Solution to 13.1

For



$$\Delta G^\circ = -188,300 + 88.48T \text{ J}$$

$$\Delta G_{1500 \text{ K}}^\circ = -55,580 \text{ J} = RT \ln a_{\text{Cu}}^2 p_{\text{O}_2}^{1/2}$$

So

$$a_{\text{Cu}}^2 (0.21)^{1/2} = 0.0116$$

$$a_{\text{Cu}}^2 = 0.0253$$

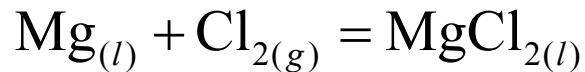
$$a_{\text{Cu}} = 0.159$$

Chapter 13

13.2 Magnesium can be removed from Mg–Al liquid solution by selectively forming the chloride MgCl_2 . Calculate the activity of Mg in the liquid Mg–Al system which can be achieved at 800 °C by reacting the solution with an H_2 –HCl gas mixture containing hydrogen at essentially 1 atm pressure and $p_{\text{HCl}} = 10^{-5}$ atm to form pure liquid MgCl_2 .

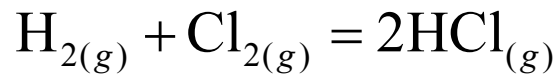
Chapter 13

Solution to 13.2



$$\Delta G^\circ = -603,200 + 121.42T \text{ J}$$

$$\Delta G_{1073 \text{ K}}^\circ = -472,916 \text{ J}$$



$$\Delta G^\circ = -188,200 - 12.80T \text{ J}$$

$$\Delta G_{1073 \text{ K}}^\circ = -201,933 \text{ J}$$

So

$$\text{Mg}_{(l)} + 2\text{HCl}_{(g)} = \text{MgCl}_{2(l)} + \text{H}_{2(g)} \quad \Delta G_{1073 \text{ K}}^\circ = -270,983 \text{ J} = RT \ln \frac{a_{\text{Mg}} \cdot p_{\text{HCl}}^2}{p_{\text{H}_2}}$$

$$\frac{a_{\text{Mg}} \cdot p_{\text{HCl}}^2}{p_{\text{H}_2}} = 6.43 \times 10^{-14} = \frac{a_{\text{Mg}} \cdot (10^{-5})^2}{1}$$

Which gives

$$a_{\text{Mg}} = 6.4 \times 10^{-4}$$

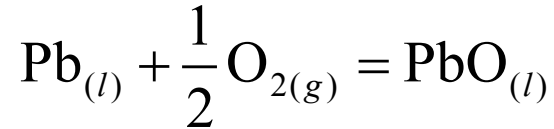
Chapter 13

13.3 The partial pressure of oxygen in equilibrium with pure liquid lead and pure liquid lead oxide at 1200 K is 2.16×10^{-9} atm. When SiO_2 is added to the liquid PbO to form a lead silicate melt, the oxygen pressure in equilibrium with pure liquid lead and the silicate melt is decreased to 5.41×10^{-10} atm. Calculate the activity of PbO in the lead silicate melt.

Chapter 13

Solution to 13.3

For



$$K_{1200 \text{ K}} = \frac{a_{\text{PbO}}}{a_{\text{Pb}} p_{\text{O}_2}^{1/2}} = \frac{1}{1 \times (2.16 \times 10^{-9})^{1/2}} = 21,517$$

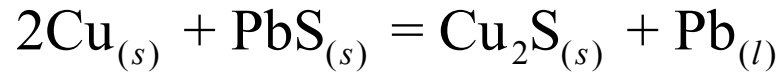
with

$$a_{\text{Pb}} = 1 \text{ atm, and } p_{\text{O}_2(\text{eq})} = 5.41 \times 10^{-10} \text{ atm}$$

$$a_{\text{PbO}} = 21,517 \times (5.41 \times 10^{-10})^{1/2} = 0.5$$

Chapter 13

13.4 Copper, present as an impurity in liquid Pb, can be removed by adding PbS to the Cu–Pb alloy and allowing the exchange reaction



to come to equilibrium.

The solid sulfides are mutually immiscible, Pb is insoluble in solid Cu, and the Cu liquidus, below 850 °C, can be represented by

$$\log X_{\text{Cu}} = -\frac{3500}{T} + 2.261$$

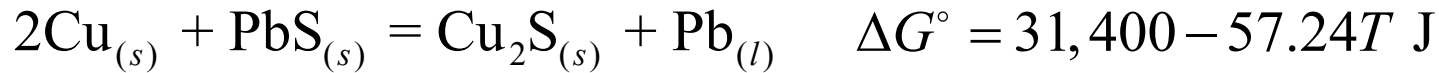
where X_{Cu} is the solubility of Cu in liquid Pb. If Cu obeys Henry's law in liquid Pb, calculate the extent to which Cu can be removed from liquid Pb by this process at 800 °C. Would the extent of purification of the lead be increased by increasing or by decreasing the temperature?

Chapter 13

Solution to 13.4



So for



$$\Delta G_{1073 \text{ K}}^\circ = -30,019 \text{ J} = -RT \ln K_{1073 \text{ K}}$$

Which gives

$$K_{1073 \text{ K}} = 28.93 = \frac{a_{\text{Pb}(l)}}{a_{\text{Cu}(s)}^2}$$

Chapter 13

Solution to 13.4

At 1073 K, $\log X_{\text{Cu}} = -\frac{3500}{T} + 2.261$ $X_{\text{Cu(liquidus)}} = 0.0998$

so $\gamma_{\text{Cu}} = \frac{1}{0.0998} = 10.02$

Thus

$$a_{\text{Cu}} = 10.02 X_{\text{Cu}} \quad \text{and} \quad a_{\text{Pb}} = 1 - X_{\text{Cu}}$$

$$K_{1073 \text{ K}} = 28.93 = \frac{1 - X_{\text{Cu}}}{100.4 X_{\text{Cu}}^2}$$

Which gives $X_{\text{Cu}} = 0.018$

At 850 °C, $X_{\text{Cu}} = 0.024$

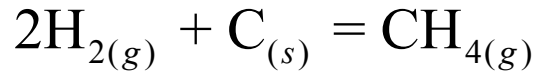
So extent of purification is increased by decreasing the temperature.

Chapter 13

13.5 A $\text{CH}_4\text{--H}_2$ gas mixture at 1 atm total pressure, in which $p_{\text{H}_2} = 0.957$ atm, is equilibrated with an Fe–C alloy at 1000 K. Calculate the activity of C with respect to graphite in the alloy. What would the value of p_{H_2} in the gas mixture (at $P_{\text{total}} = 1$ atm) have to be in order to saturate the Fe with graphite at 1000 K?

Chapter 13

Solution to 13.5



$$\Delta G^\circ = -91,040 + 110.7T \text{ J}$$

$$\Delta G_{1000 \text{ K}}^\circ = 19,660 \text{ J} = -RT \ln K_{1000 \text{ K}}$$

So

$$K_{1000 \text{ K}} = 9.40 \times 10^{-2} = \frac{p_{\text{CH}_4}}{a_{\text{C}} p_{\text{H}_2}^2} = \frac{0.043}{(0.957)^2 a_{\text{C}}}$$

Which gives $a_{\text{C}} = 0.5$

For $a_{\text{C}} = 1$ and $P_{\text{total}} = 1 \text{ atm}$

$$9.40 \times 10^{-2} = \frac{p_{\text{CH}_4}}{(1 - p_{\text{CH}_4})^2}$$

Which gives $p_{\text{H}_2} = 0.92 \text{ atm}$, $p_{\text{CH}_4} = 0.08 \text{ atm}$.

Chapter 13

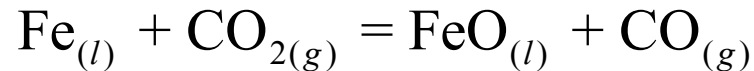
13.6 Calculate the activity of FeO in an FeO–Al₂O₃–SiO₂ melt below which the FeO cannot be reduced to pure liquid iron by a CO–CO₂ mixture of $p_{\text{CO}}/p_{\text{CO}_2} = 10^5$ at 1600 °C.

Chapter 13

Solution to 13.6

	$\Delta G_{1873\text{ K}}^{\circ}, \text{ J}$
FeO(l)	-155,557
CO(g)	-275,868
CO ₂ (g)	-395,673

So for



$$\Delta G_{1873\text{ K}}^{\circ} = -35,753 \text{ J} = -RT \ln K$$

Thus

$$K = 9.93 = \frac{a_{\text{FeO}} \cdot p_{\text{CO}}}{a_{\text{Fe}} \cdot p_{\text{CO}_2}} = a_{\text{FeO}} \times 10^5$$

Which gives $a_{\text{FeO}} = 9.9 \times 10^{-5}$

Chapter 13

- 13.7** A piece of iron is to be heat treated at 1000 K in a CO-CO₂-H₂O-H₂ gas mixture at 1 atm pressure. The gas mixture is produced by mixing CO₂ and H₂ and allowing the equilibrium $\text{CO}_2 + \text{H}_2 = \text{CO} + \text{H}_2\text{O}$ to establish. Calculate
- (a) The minimum H₂/CO₂ ratio in the inlet gas which can be admitted to the furnace without oxidizing the iron,
 - (b) the activity of carbon (with respect to graphite) in the equilibrated gas of this initial minimum H₂/CO₂ ratio,
 - (c) the total pressure to which the equilibrated gas would have to be raised to saturate the iron with graphite at 1000 K, and
 - (d) the effect, on the partial pressure of oxygen in the equilibrated gas, of this increase in total pressure.

Chapter 13

Solution to 13.7

(a)

	$\Delta G_{1000\text{ K}}^{\circ}, \text{ J}$
CO	-199,350
CO ₂	-394,940
H ₂ O(g)	-191,650
FeO	-199,350

So

$$\text{CO}_{(g)} + \text{FeO}_{(s)} = \text{CO}_{2(g)} + \text{Fe}_{(s)} \quad \Delta G_{1000\text{ K}}^{\circ} = 3760 \text{ J} \quad K = \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.636$$

$$\text{H}_{2(g)} + \text{FeO}_{(s)} = \text{H}_2\text{O}_{(g)} + \text{Fe}_{(s)} \quad \Delta G_{1000\text{ K}}^{\circ} = 7700 \text{ J} \quad K = \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.396$$

For



Chapter 13

Solution to 13.7

$$p_{\text{CO}} + p_{\text{CO}_2} + p_{\text{H}_2} + p_{\text{H}_2\text{O}} = 1 \text{ atm}$$

So

$$p_{\text{CO}} + 0.636 p_{\text{CO}} + \frac{p_{\text{H}_2\text{O}}}{0.396} + p_{\text{H}_2\text{O}} = 1 \text{ atm}$$

$$p_{\text{CO}} \left(1 + 0.636 + \frac{1}{0.396} + 1 \right) = 1 \text{ atm}$$

Which gives

$$p_{\text{CO}} = 0.1938 \text{ atm} \quad p_{\text{H}_2\text{O}} = 0.1938 \text{ atm}$$

$$p_{\text{CO}_2} = 0.1233 \text{ atm} \quad p_{\text{H}_2} = 0.4894 \text{ atm}$$

This gas is in equilibrium with Fe and FeO.

Chapter 13

Solution to 13.7

Mix CO_2 and H_2 in the ratio $\frac{\text{CO}_2}{\text{H}_2} = x$

	$\text{CO}_{2(g)}$	$+$	$\text{H}_{2(g)}$	$=$	$\text{CO}_{(g)}$	$+$	$\text{H}_2\text{O}_{(g)}$
before	x		1		0		0
after	x-a		1-a		a		a

Which gives

$$p_{\text{CO}} = \frac{a}{1+x} \quad p_{\text{H}_2\text{O}} = \frac{a}{1+x} \quad p_{\text{CO}_2} = \frac{x-a}{1+x} \quad p_{\text{H}_2} = \frac{1-a}{1+x}$$

For equilibrium with Fe and FeO:

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.636 = \frac{x-a}{a} \quad \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.396 = \frac{a}{1-a}$$

Which gives $a = 0.2837$ $x = 0.4641 = \left. \frac{\text{CO}_2}{\text{H}_2} \right|_{\text{initial}}$

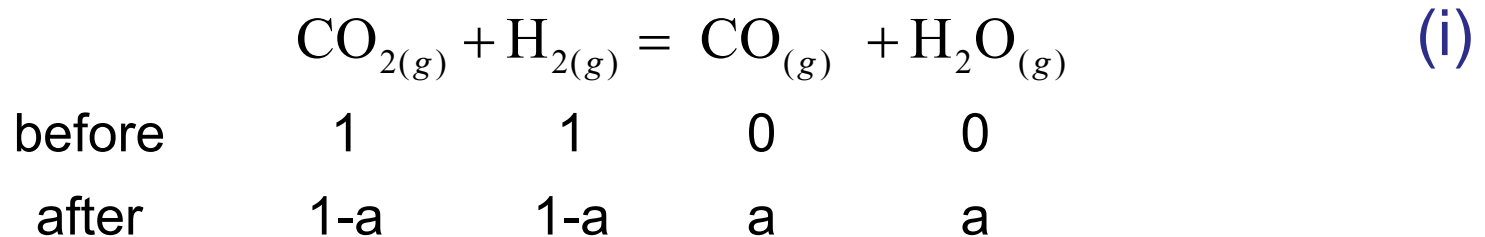
Chapter 13

Solution to 13.7

For $\text{Fe}_{(s)} + \frac{1}{2} \text{O}_{2(g)} = \text{FeO}_{(s)} \quad \Delta G_{1000 \text{ K}}^{\circ} = -199,350 \text{ J} = \frac{1}{2} RT \ln p_{\text{O}_2}$

So $p_{\text{O}_2(\text{Fe/FeO})} = 1.494 \times 10^{-21} \text{ atm}$

Choose a larger value of x, say $x = 1$, then



So $p_{\text{CO}} = \frac{a}{2} = p_{\text{H}_2\text{O}}; \quad p_{\text{CO}_2} = \frac{1-a}{2} = p_{\text{H}_2}$

$$\Delta G_{(i), 1000 \text{ K}}^{\circ} = 3040 \text{ J} = -RT \ln \frac{p_{\text{CO}} \cdot p_{\text{H}_2\text{O}}}{p_{\text{CO}_2} \cdot p_{\text{H}_2}}$$

Thus $0.6226 = \frac{a^2}{(1-a)^2} \quad \text{which gives } a = 0.441$

Chapter 13

Solution to 13.7

So $p_{\text{CO}} = 0.2205 \text{ atm}$ $p_{\text{CO}_2} = 0.2795 \text{ atm}$ $\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 1.268$



$$K_{1000 \text{ K}} = 1.64 \times 10^{-10} = \frac{p_{\text{CO}_2}}{p_{\text{CO}} \cdot p_{\text{O}_2}^{1/2}} = \frac{1.268}{p_{\text{O}_2}^{1/2}}$$

Which gives $p_{\text{O}_2} = 5.93 \times 10^{-21} \text{ atm}$, which is greater than $p_{\text{O}_2(\text{eq, Fe/FeO})}$ at 1000 K.

So 0.4641 is the maximum allowable ratio of $\left. \frac{\text{CO}_2}{\text{H}_2} \right|_{\text{initial}}$

and 2.155 is the minimum allowable ratio of $\left. \frac{\text{H}_2}{\text{CO}_2} \right|_{\text{initial}}$

Chapter 13

Solution to 13.7

(b) $\text{C}_{(s)} + \text{CO}_{2(g)} = 2\text{CO}_{(g)} \quad \Delta G_{1000\text{ K}}^{\circ} = -3760 \text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2} \cdot a_{\text{C}}}$

So $\frac{p_{\text{CO}}^2}{p_{\text{CO}_2} \cdot a_{\text{C}}} = 1.572 = \frac{0.1938^2}{0.1233 \cdot a_{\text{C}}}$, which gives $a_{\text{C}} = 0.194$

(c) For $a_{\text{C}} = 1$ and pressure P ,

$$1.572 = \frac{0.1938^2 P}{0.1233}$$

Which gives $P = 5.16 \text{ atm}$.

(d) The values of the ratios $\frac{p_{\text{CO}_2}}{p_{\text{CO}}}$ and $\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}$ are not functions of the total pressure P .

Chapter 13

13.8 An Fe–Mn solid solution containing $X_{\text{Mn}} = 0.001$ is in equilibrium with an FeO–MnO solid solution and an oxygen-containing gaseous atmosphere at 1000 K. How many degrees of freedom does the equilibrium have? What is the composition of the equilibrium oxide solution, and what is the partial pressure of oxygen in the gas phase? Assume that both solid solutions are Raoultian in their behavior.

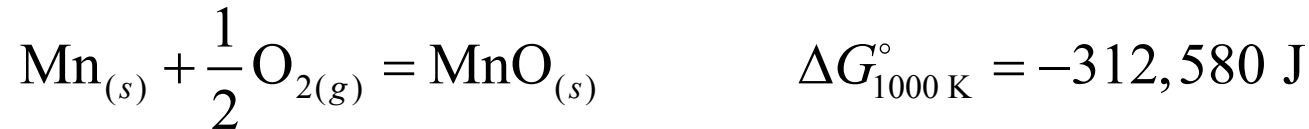
Chapter 13

Solution to 13.8

$C = 3$ (Fe, Mn, O), $P = 3$ (metal, oxide, gas)

So
$$F = C + 2 - P = 3 + 2 - 3 = 2$$

Fixing the temperature at 1000 K leaves 1 degree of freedom



So
$$K_{\text{Mn}} = 2.124 \times 10^{16} = \frac{X_{\text{MnO}}}{X_{\text{Mn}} \cdot p_{\text{O}_2}^{1/2}}$$



So
$$K_{\text{Fe}} = 2.587 \times 10^{10} = \frac{X_{\text{FeO}}}{X_{\text{Fe}} \cdot p_{\text{O}_2}^{1/2}}$$

Chapter 13

Solution to 13.8

$$\frac{K_{\text{Mn}}}{K_{\text{Fe}}} = 821,215 = \frac{X_{\text{MnO}} X_{\text{Fe}}}{X_{\text{Mn}} \cdot X_{\text{FeO}}} = \frac{(1 - X_{\text{FeO}}) \cdot 0.999}{0.001 \cdot X_{\text{FeO}}}$$

Which gives

$$X_{\text{FeO}} = 1.22 \times 10^{-3}$$

Then

$$K_{\text{Fe}} = 2.587 \times 10^{10} = \frac{1.22 \times 10^{-3}}{0.999 \cdot p_{\text{O}_2}^{1/2}}$$

gives

$$p_{\text{O}_2} = 2.23 \times 10^{-27} \text{ atm}$$

Chapter 13

13.9 The elements A and B, which are both solid at 1000 °C, form two stoichiometric compounds A_2B and AB_2 , which are also both solid at 1000 °C. The system A–B does not contain any solid solutions. A has an immeasurably small vapor pressure at 1000 °C, and, for the change of state $B_{(s)} = B_{(v)}$,

$$\Delta G^\circ = 187,200 - 108.80T \text{ J}$$

The vapor pressure exerted by an equilibrated AB_2 – A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{11,242}{T} + 6.53$$

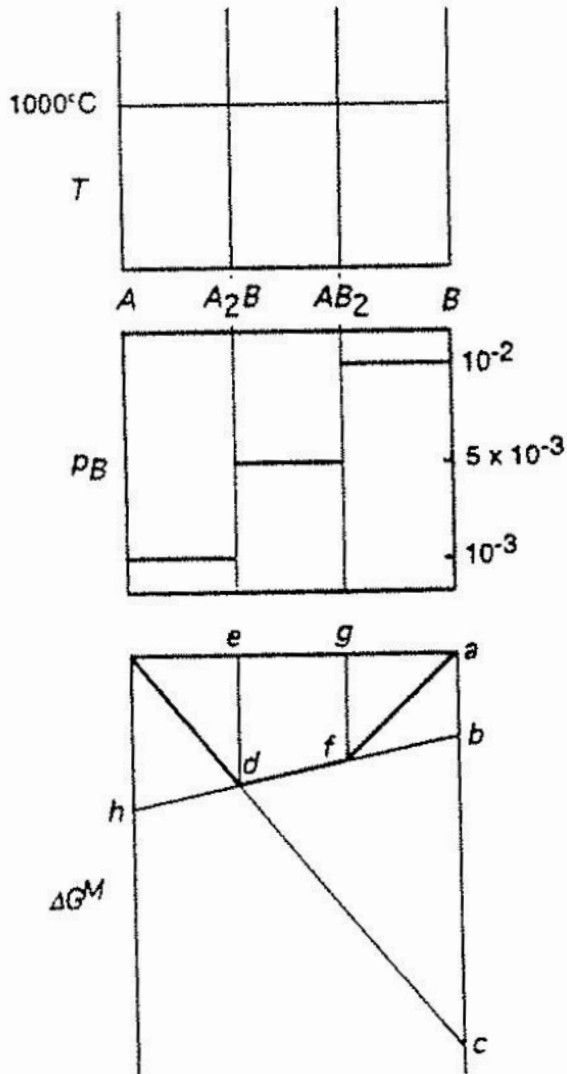
and the vapor pressure exerted by an equilibrated A– A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{12,603}{T} + 6.9$$

From these data, calculate the standard Gibbs free energies of formation of A_2B and AB_2 .

Chapter 13

Solution to 13.9



$$a_B \text{ in } A_2B-AB_2 \text{ mixture} = \frac{5 \times 10^{-3}}{10^{-2}} = 0.5$$

$$\text{So } ab = RT \ln a_B = -7336 \text{ J}$$

$$a_B \text{ in } A-A_2B \text{ mixture} = \frac{10^{-3}}{10^{-2}} = 0.1$$

$$\text{So } ac = RT \ln a_B = -24,371 \text{ J}$$

$$\text{Thus } ed = \frac{-24,371}{3} = -8123 \text{ J}$$

$$\text{For } 2A + B = A_2B \quad \Delta G^\circ = 3 \times ed = -24,370 \text{ J}$$

$$gf = \frac{ed + ab}{2} = -7730 \text{ J}$$

$$\text{For } A + 2B = AB_2 \quad \Delta G^\circ = 3 \times gf = -23,190 \text{ J}$$

Chapter 13

13.10 For the change of standard state $V_{(s)} = V_{(1 \text{ wt\% in Fe})}$

$$\Delta G^\circ = -15,480 - 45.61T \text{ J}$$

Calculate the value of γ_V° at 1600 °C. If a liquid Fe–V solution is equilibrated with pure solid VO and a gas containing $p_{O_2} = 4.72 \times 10^{-10}$ atm, calculate the activity of V in the liquid solution

- (a) with respect to solid V as the standard state,
- (b) with respect to liquid V as the standard state,
- (c) with respect to the Henrian standard state, and
- (d) with respect to the 1 wt% in iron standard state.

Chapter 13

Solution to 13.10

$$V_{(s)} \rightarrow V_{(1 \text{ wt}\%)} \quad \Delta G_{1873 \text{ K}}^{\circ} = RT \ln \frac{\gamma_v^{\circ} 55.85}{100 \times 50.94}$$

So $\gamma_v^{\circ} = 0.14$

$$V_{(s)} + \frac{1}{2} O_2 = VO_{(s)} \quad \Delta G_{1873 \text{ K}}^{\circ} = -274,785 = -RT \ln \frac{1}{a_{V(s)} [4.72 \times 10^{-10}]^{1/2}}$$

So $a_{V(s)} = 10^{-3}$

$$V_{(s)} \rightarrow V_{(l)} \quad \Delta G^{\circ} = 22,840 - \frac{22,840}{2193} 1873 = 3333 \text{ J}$$

So $V_{(l)} + \frac{1}{2} O_2 = VO_{(s)}$

$$\Delta G_{1873 \text{ K}}^{\circ} = -274,785 - 3333 = -278,118 \text{ J} = -RT \ln \frac{1}{a_{V(l)} [4.72 \times 10^{-10}]^{1/2}}$$

Thus $a_{V(l)} = 8.07 \times 10^{-4}$

Chapter 13

Solution to 13.10

$$a_{V(s)} = h_V \cdot \gamma_v^\circ$$

So
$$h_V = \frac{10^{-3}}{0.14} = 7.13 \times 10^{-3}$$

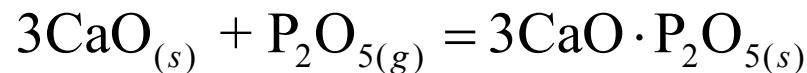
$$\frac{h_{V(1\%)}}{h_V} = \frac{50.95 \times 100}{55.85} = 91.24$$

So

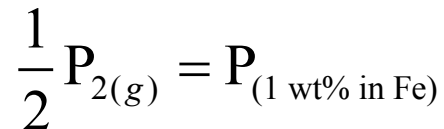
$$h_{V(1\%)} = 91.24 \times 7.13 \times 10^{-3} = 0.65$$

Chapter 13

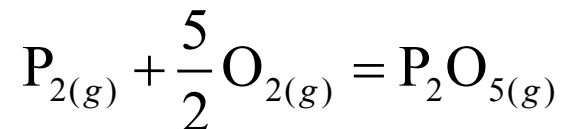
13.11 When an Fe–P liquid solution is equilibrated at 1900 K with solid CaO, solid $3\text{CaO}\cdot\text{P}_2\text{O}_5$, and a gas phase containing $p_{\text{O}_2} = 10^{-10}$ atm, the activity of P in the iron, with respect to the 1 wt% in Fe standard state, is 20. Given that $\Delta G_{1900\text{ K}}^\circ = -564,600$ J for



and $\Delta G^\circ = -122,200 - 19.22T$ J for

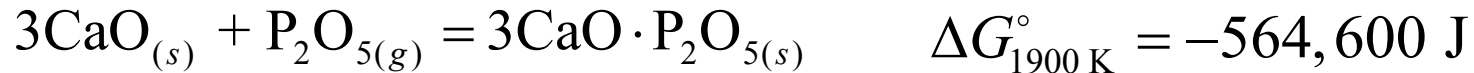


calculate $\Delta G_{1900\text{ K}}^\circ$ for the reaction



Chapter 13

Solution to 13.11



So $K = 3.325 \times 10^{15} = \frac{1}{p_{\text{P}_2\text{O}_5}} \quad p_{\text{P}_2\text{O}_5} = 3 \times 10^{-16} \text{ atm}$

$$\frac{1}{2}\text{P}_{2(g)} = \text{P}_{(1 \text{ wt\% in Fe})} \quad \Delta G^\circ = -122,200 - 19.22T \text{ J} \quad \Delta G_{1900\text{ K}}^\circ = -158,680 \text{ J}$$

So $K = 2.303 \times 10^4 = \frac{h_{\text{P}(1 \text{ wt\%})}}{p_{\text{P}_2}^{1/2}} = \frac{20}{p_{\text{P}_2}^{1/2}} \quad p_{\text{P}_2} = 7.539 \times 10^{-7} \text{ atm}$

Thus for

$$\text{P}_{2(g)} + \frac{5}{2}\text{O}_{2(g)} = \text{P}_2\text{O}_{5(g)}$$
$$K = \frac{3 \times 10^{-16}}{7.539 \times 10^{-7} \times (10^{-10})^{5/2}} = 3.99 \times 10^{15}$$

$$\Delta G_{1900\text{ K}}^\circ = -RT \ln K = -567,500 \text{ J}$$

Chapter 13

13.12 Liquid iron, contained in an Al_2O_3 crucible under a gaseous atmosphere of $p_{\text{O}_2} = 3 \times 10^{-12}$ atm at 1600 °C, contains its equilibrium contents of dissolved oxygen and aluminum. To what value must p_{O_2} be raised in order that solid hercynite ($\text{FeO} \cdot \text{Al}_2\text{O}_3$) appears in equilibrium with the melt and with solid Al_2O_3 ? What is the activity of Al (with respect to the 1 wt% in Fe standard state) in this state? How many degrees of freedom does this equilibrium have at 1600 °C?

Given

$$\frac{1}{2} \text{O}_{2(g)} = \text{O}_{(1 \text{ wt\% in Fe})} \quad \Delta G^\circ = -111,070 - 5.87T \text{ J}$$

$$\text{Al}_{(l)} = \text{Al}_{(1 \text{ wt\% in Fe})} \quad \Delta G^\circ = -43,100 - 32.26T \text{ J}$$

$$\text{FeO} \cdot \text{Al}_2\text{O}_{3(s)} = \text{Fe}_{(l)} + \text{O}_{(1 \text{ wt\% in Fe})} + \text{Al}_2\text{O}_{3(s)} \quad \Delta G^\circ = 146,230 - 54.35T \text{ J}$$

Chapter 13

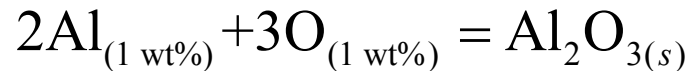
Solution to 13.12

$$\text{Al}_{(l)} = \text{Al}_{(1 \text{ wt\% in Fe})} \quad \Delta G^\circ = -43,100 - 32.26T \text{ J} \quad \Delta G_{1873 \text{ K}}^\circ = 103,523 \text{ J}$$

$$\text{O}_{(1 \text{ wt\% in Fe})} = \frac{1}{2} \text{O}_{2(g)} \quad \Delta G^\circ = 111,070 + 5.87T \text{ J} \quad \Delta G_{1873 \text{ K}}^\circ = 122,064 \text{ J} \quad (\text{i})$$

$$2\text{Al}_{(l)} + \frac{3}{2} \text{O}_{2(g)} = \text{Al}_2\text{O}_{3(s)} \quad \Delta G_{1873 \text{ K}}^\circ = -1,075,103 \text{ J}$$

So



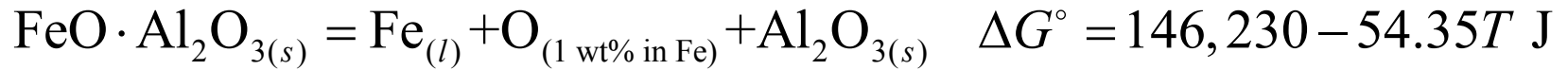
$$\Delta G_{1873 \text{ K}}^\circ = -501,865 \text{ J} = 8.3144 \times 1873 \times 2.303 \log [\text{O}]^3 [\text{Al}]^2$$

For saturation with Al_2O_3

$$2 \log [\text{Al}] + 3 \log [\text{O}] = -14 \quad (\text{ii})$$

Chapter 13

Solution to 13.12



$$\Delta G_{1873 \text{ K}}^\circ = 44,432 \text{ J} = -RT \ln [\text{O}]$$

$$[\text{O}] = 5.77 \times 10^{-2} \text{ wt\%}$$

From (i)

$$122,064 = -RT \ln \frac{p_{\text{O}_2}^{1/2}}{[\text{O}]}$$

So $\frac{p_{\text{O}_2}^{1/2}}{[\text{O}]} = 3.944 \times 10^{-4} \quad p_{\text{O}_2} = 5.17 \times 10^{-10} \text{ atm}$

And from (ii)

$$[\text{Al}] = 7.215 \times 10^{-6} \text{ wt\%}$$

$$C = 3 \text{ (Al, Fe, O); } P = 4 \text{ (metal, Al}_2\text{O}_3, \text{FeO} \cdot \text{Al}_2\text{O}_3, \text{ gas)}$$

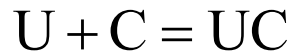
So $F = C + 2 - P = 1$, which is specified by selecting $T = 1600^\circ\text{C}$.

Chapter 13

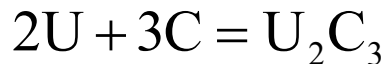
13.13 UC_2 can be equilibrated with UC and C at high temperature and can be equilibrated with U_2C_3 and C at lower temperatures. Calculate the maximum and minimum temperatures at which UC_2 can exist.

Chapter 13

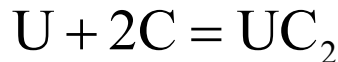
Solution to 13.13



$$\Delta G^\circ = -102,900 + 5.02T \text{ J}$$



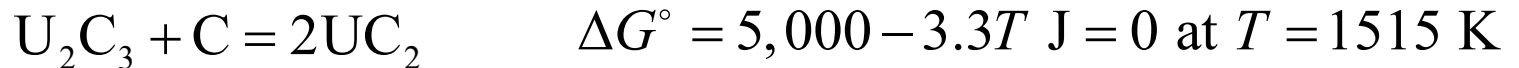
$$\Delta G^\circ = -236,800 + 25.1T \text{ J}$$



$$\Delta G^\circ = -115,900 + 10.9T \text{ J}$$

So $\text{UC} + \text{C} = \text{UC}_2$ $\Delta G^\circ = -13,000 + 5.88T \text{ J} = 0$ at $T = 2211 \text{ K}$

At $T > 2211 \text{ K}$, UC_2 metastable w.r.t $\text{UC} + \text{C}$



Thus at $T < 1515 \text{ K}$, UC_2 metastable w.r.t $\text{U}_2\text{C}_3 + \text{C}$

So $T_{\text{max}} = 2211 \text{ K}$

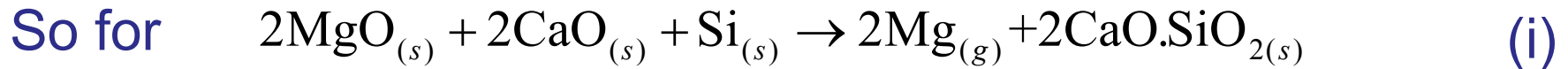
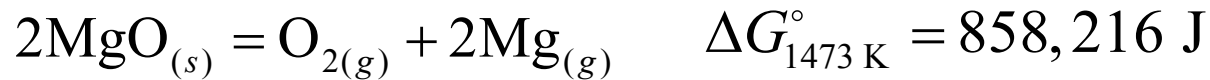
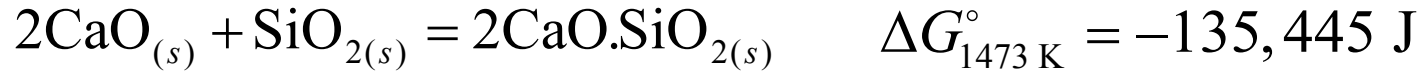
$T_{\text{min}} = 1515 \text{ K}$

Chapter 13

13.14 In the Pigeon process for the production of magnesium, dolomite ($\text{CaO} \cdot \text{MgO}$) is reduced by silicon to form magnesium vapor and $2\text{CaO} \cdot \text{SiO}_2$. Calculate the equilibrium pressure of magnesium vapor produced by this reaction at 1200°C . The Gibbs free energy of formation of dolomite from CaO and MgO is small enough that it can be ignored.

Chapter 13

Solution to 13.14



$$\Delta G_{(i),1473\text{ K}}^{\circ} = 71,973\text{ J} = -RT \ln p_{\text{Mg}}^2 = -8.3144 \times 1473 \ln p_{\text{Mg}}^2$$

Which gives

$$p_{\text{Mg}} = 0.053\text{ atm}$$

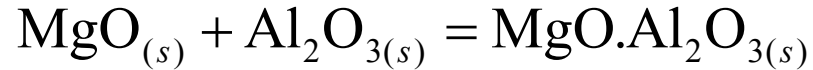
Chapter 13

13.15 What is the minimum value that the activity of MgO can have in $\text{MgO} \cdot \text{Al}_2\text{O}_3$ at 1000 °C?

Chapter 13

Solution to 13.15

For



$$\Delta G_{1273\text{ K}}^{\circ} = -38,220\text{ J} = -RT \ln \frac{a_{\text{MgO}.\text{Al}_2\text{O}_3}}{a_{\text{MgO}} a_{\text{Al}_2\text{O}_3}}$$

The minimum activity of MgO in $\text{MgO}.\text{Al}_2\text{O}_3$ occur when it is saturated with Al_2O_3 at $a_{\text{Al}_2\text{O}_3} = 1$.

So with $a_{\text{MgO}.\text{Al}_2\text{O}_3} = a_{\text{Al}_2\text{O}_3} = 1$

$$a_{\text{MgO}} = 0.027$$

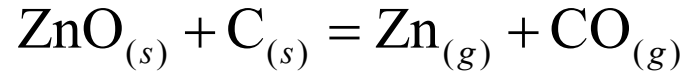
Chapter 13

13.16 A mixture of ZnO and graphite is placed in an evacuated vessel and heated to 1200 K. Calculate the partial pressures of Zn, CO, and CO₂ that are developed.

Chapter 13

Solution to 13.16

For

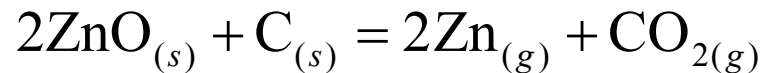


$$\Delta G_{1200\text{ K}}^{\circ} = 5720\text{ J} = -RT \ln p_{\text{Zn}} \cdot p_{\text{CO}}$$

So

$$p_{\text{Zn}} \cdot p_{\text{CO}} = 0.564$$

For



$$\Delta G_{1200\text{ K}}^{\circ} = 50,092\text{ J} = -RT \ln p_{\text{Zn}}^2 \cdot p_{\text{CO}_2}$$

So

$$p_{\text{Zn}}^2 \cdot p_{\text{CO}_2} = 6.60 \times 10^{-3}$$

Chapter 13

Solution to 13.16

In the gas phase

$$\frac{n_{\text{Zn}}}{n_{\text{O}}} = 1 = \frac{p_{\text{Zn}}}{p_{\text{CO}} + 2p_{\text{CO}_2}}$$

So

$$p_{\text{Zn}} = p_{\text{CO}} + 2p_{\text{CO}_2}$$

Thus

$$p_{\text{CO}} (p_{\text{CO}} + 2p_{\text{CO}_2}) = 0.564$$

$$p_{\text{CO}_2} (p_{\text{CO}} + 2p_{\text{CO}_2})^2 = 6.60 \times 10^{-3}$$

Which gives

$$p_{\text{CO}} = 0.740 \text{ atm,}$$

$$p_{\text{CO}_2} = 0.0114 \text{ atm,}$$

$$p_{\text{Zn}} = 0.763 \text{ atm,}$$

Chapter 13

13.17 An assemblage of solid CaO, MgO, $3\text{CaO}\cdot\text{Al}_2\text{O}_3$, and liquid Al exerts an equilibrium vapor pressure of Mg of 0.035 atm at 1300 K. Write the equation for the appropriate reaction equilibrium. Calculate the standard Gibbs free energy of formation of $3\text{CaO}\cdot\text{Al}_2\text{O}_3$ from CaO and Al_2O_3 and the activity of Al_2O_3 in CaO-saturated $3\text{CaO}\cdot\text{Al}_2\text{O}_3$ at 1300 K.

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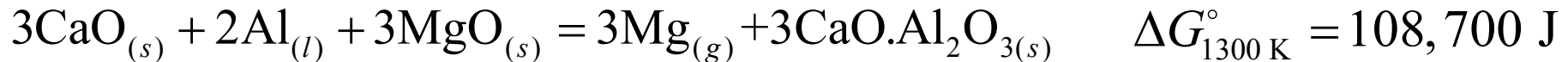
Solution to 13.17

The equilibrium is



for which $K = p_{\text{Mg}}^3 = 0.035^3 = 4.288 \times 10^{-5}$

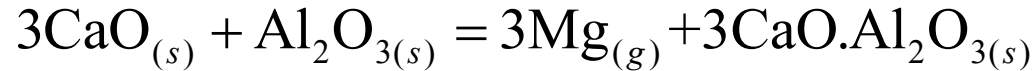
So $\Delta G_{1300\text{ K}}^\circ = -8.3144 \times 1300 \times \ln(4.288 \times 10^{-5}) \text{ J} = 108,700 \text{ J}$



Chapter 13

Solution to 13.17

Thus for



$$\Delta G_{1300\text{ K}}^{\circ} = -22,140\text{ J}$$

with $a_{\text{CaO}.\text{Al}_2\text{O}_3} = a_{\text{CaO}} = 1$

$$-22,140 = -RT \ln \frac{1}{a_{\text{Al}_2\text{O}_3}}$$

Which gives

$$a_{\text{Al}_2\text{O}_3} = 0.129$$

Chapter 13

13.18 An iron–carbon melt containing 0.5 wt% C is prepared in an alumina crucible under an atmosphere of $p_{\text{CO}} = 1$ atm at 1600 °C. Calculate the equilibrium concentrations of O and Al in the melt

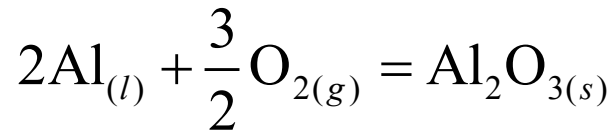
- (a) ignoring all solute–solute interactions and
- (b) considering the solute–solute interactions.

The interaction coefficients are listed in Table 13.1.

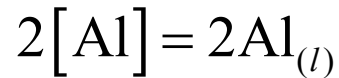
Chapter 13

Solution to 13.18

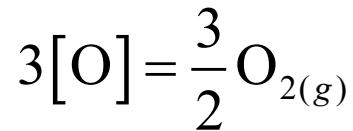
(a)



$$\Delta G_{1873\text{ K}}^{\circ} = -1,075,103\text{ J}$$



$$\Delta G_{1873\text{ K}}^{\circ} = 267,045\text{ J}$$



$$\Delta G_{1873\text{ K}}^{\circ} = 366,193\text{ J}$$

So for

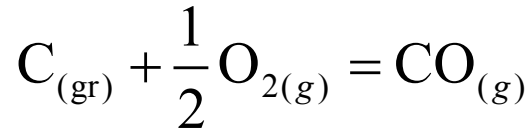
$$2[\text{Al}] + 3[\text{O}] = \text{Al}_2\text{O}_{3(s)} \quad \Delta G_{1873\text{ K}}^{\circ} = -501,864\text{ J} = RT \ln [\%\text{Al}]^2 [\%\text{O}]^3$$

Which gives

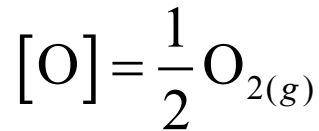
$$2\log [\%\text{Al}] + 3\log [\%\text{O}] = -13.996 \quad (\text{i})$$

Chapter 13

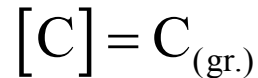
Solution to 13.18



$$\Delta G_{1873 \text{ K}}^{\circ} = -275,868 \text{ J}$$



$$\Delta G_{1873 \text{ K}}^{\circ} = 122,064 \text{ J}$$



$$\Delta G_{1873 \text{ K}}^{\circ} = 56,552 \text{ J}$$

So for

$$[\text{C}] + [\text{O}] = \text{CO}_{(\text{g})} \quad \Delta G_{1873 \text{ K}}^{\circ} = -97,250 \text{ J} = -RT \ln \frac{p_{\text{CO}}}{[\% \text{C}][\% \text{O}]}$$

with $p_{\text{CO}} = 1 \text{ atm}$,

$$\log [\% \text{C}] + \log [\% \text{O}] = -2.716 \quad (\text{ii})$$

$$\text{So } \log [\% \text{O}] = -2.716 - \log 0.5 = -2.414 \quad \text{and} \quad [\% \text{O}] = 0.00385$$

$$\log [\% \text{Al}] = \frac{-13.996 - 3(-2.414)}{2} = -3.377 \quad \text{and} \quad [\% \text{Al}] = 0.00042$$

Chapter 13

Solution to 13.18

(b)

Eq.(i) becomes $2\log[h_{\text{Al}}] + 3\log[h_{\text{O}}] = -13.996$

Eq.(ii) becomes $\log[h_{\text{C}}] + \log[h_{\text{O}}] = -2.716$

$$\log[f_{\text{C}}] = e_{\text{C}}^{\text{C}} [\text{C}] + e_{\text{C}}^{\text{O}} [\text{O}] + e_{\text{C}}^{\text{Al}} [\text{Al}]$$

$$\log[f_{\text{O}}] = e_{\text{O}}^{\text{O}} [\text{O}] + e_{\text{O}}^{\text{C}} [\text{C}] + e_{\text{O}}^{\text{Al}} [\text{Al}]$$

$$\log[f_{\text{Al}}] = e_{\text{Al}}^{\text{Al}} [\text{Al}] + e_{\text{Al}}^{\text{C}} [\text{C}] + e_{\text{Al}}^{\text{O}} [\text{O}]$$

where $e_{\text{C}}^{\text{C}} = 22 \times 10^{-2}$, $e_{\text{O}}^{\text{O}} = -20 \times 10^{-2}$, $e_{\text{Al}}^{\text{Al}} = 4.8 \times 10^{-2}$

$$e_{\text{C}}^{\text{O}} = -9.7 \times 10^{-2}, \quad e_{\text{O}}^{\text{C}} = -13 \times 10^{-2}, \quad e_{\text{Al}}^{\text{C}} = 11 \times 10^{-2}$$

$$e_{\text{C}}^{\text{Al}} = 4.8 \times 10^{-2}, \quad e_{\text{O}}^{\text{Al}} = -94 \times 10^{-2}, \quad e_{\text{Al}}^{\text{O}} = -160 \times 10^{-2}$$

Chapter 13

Solution to 13.18

So

$$2\{\log f_{\text{Al}} + \log [\text{Al}]\} + 3\{\log f_{\text{O}} + \log [\text{O}]\} = -13.996$$

$$\begin{aligned} \text{or } 2\{e_{\text{Al}}^{\text{Al}} [\text{Al}] + e_{\text{Al}}^{\text{C}} [\text{C}] + e_{\text{Al}}^{\text{O}} [\text{O}] + \log [\text{Al}]\} \\ + 3\{e_{\text{O}}^{\text{O}} [\text{O}] + e_{\text{O}}^{\text{C}} [\text{C}] + e_{\text{O}}^{\text{Al}} [\text{Al}] + \log [\text{O}]\} = -13.996 \end{aligned}$$

$$\log f_{\text{C}} + \log [\text{C}] + \log f_{\text{O}} + \log [\text{O}] = -2.716$$

$$\begin{aligned} \text{or } e_{\text{C}}^{\text{C}} [\text{C}] + e_{\text{C}}^{\text{O}} [\text{O}] + e_{\text{C}}^{\text{Al}} [\text{Al}] + \log [\text{C}] \\ + e_{\text{O}}^{\text{O}} [\text{O}] + e_{\text{O}}^{\text{C}} [\text{C}] + e_{\text{O}}^{\text{Al}} [\text{Al}] + \log [\text{O}] = -2.716 \end{aligned}$$

Chapter 13

Solution to 13.18

Thus

$$2\{4.8 \times 10^{-2} [Al] + 11 \times 10^{-2} [C] - 160 \times 10^{-2} [O] + \log [Al]\} \\ + 3\{-20 \times 10^{-2} [O] - 13 \times 10^{-2} [C] - 94 \times 10^{-2} [Al] + \log [O]\} = -13.996$$

and

$$22 \times 10^{-2} [C] - 9.7 \times 10^{-2} [O] + 4.8 \times 10^{-2} [Al] + \log [C] \\ - 20 \times 10^{-2} [O] - 13 \times 10^{-2} [C] - 94 \times 10^{-2} [Al] + \log [O] = -2.716$$

with $[C] = 0.5$

$$\%Al = 0.000541$$

$$\%O = 0.00351$$

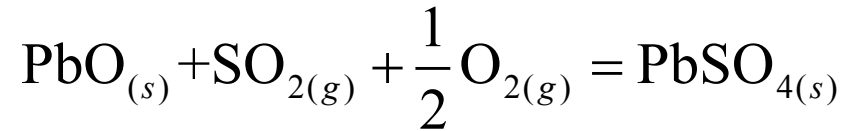
Chapter 13

13.19 It is required that PbO be eliminated from an ore containing PbO, PbS, and PbSO₄ by converting it to PbS or PbSO₄ by reaction with an SO₂–O₂ gas. Although the pressure of O₂ in the gas can vary within wide limits, the partial pressure of SO₂ may not be higher than 0.5 atm. Calculate the maximum temperature at which it can be guaranteed that the PbO phase will be eliminated.

Chapter 13

Solution to 13.19

For

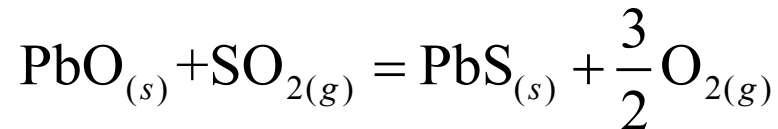


$$\Delta G^\circ = -401,200 + 261.5T \text{ J} = RT \ln p_{\text{SO}_2} \cdot p_{\text{O}_2}^{1/2}$$

So

$$\frac{-48253}{T} + 31.45 = \ln p_{\text{SO}_2} + \frac{1}{2} \ln p_{\text{O}_2} \quad (\text{i})$$

For



$$\Delta G^\circ = 407,200 - 80.6T \text{ J} = -RT \ln \frac{p_{\text{O}_2}^{3/2}}{p_{\text{SO}_2}}$$

So

$$\frac{-48975}{T} + 9.69 = \frac{3}{2} \ln p_{\text{O}_2} - \ln p_{\text{SO}_2} \quad (\text{ii})$$

Chapter 13

Solution to 13.19

With $p_{\text{SO}_2} = 0.5 \text{ atm}$

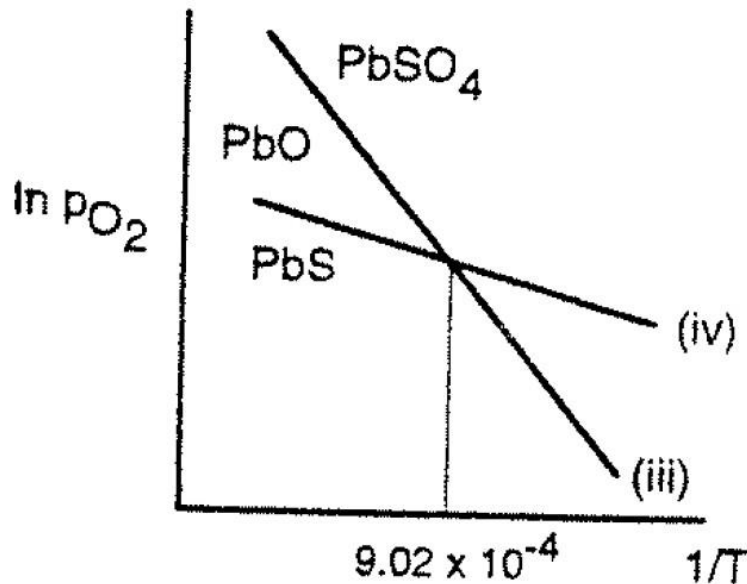
(i) becomes
$$\ln p_{\text{O}_2} = \frac{-96506}{T} + 63.59 \quad (\text{iii})$$

and (ii) becomes
$$\ln p_{\text{O}_2} = \frac{-32650}{T} + 5.998 \quad (\text{iv})$$

(iii) and (iv) intersect at

$$\frac{1}{T} = 9.02 \times 10^{-4}, \quad \ln p_{\text{O}_2} = -23.46$$

So PbO is not stable at $T > 1108 \text{ K}$, which is thus the maximum temperature at which it is eliminated.



Chapter 13

13.20 Cementite, Fe_3C , is metastable with respect to carbon-saturated α -iron and graphite at 950 K and 1 atm pressure. Given that the molar volumes of α -Fe, graphite, and Fe_3C at 950 K are, respectively, 7.32, 5.40, and 23.92 cm^3/mole , calculate the pressure, at 950 K, at which Fe_3C is in equilibrium with carbon-saturated α -Fe and graphite. At what temperature, at 1 atm pressure, is carbon-saturated γ -Fe and graphite in equilibrium with cementite?

Chapter 13

Solution to 13.20



$$V_{\text{Fe}} = 7.32\text{ cm}^3/\text{mole}$$

$$V_{\text{gr}} = 5.40\text{ cm}^3/\text{mole}$$

$$V_{\text{Fe}_3\text{C}} = 23.92\text{ cm}^3/\text{mole}$$

So
$$\Delta V = 23.92 - 5.40 - 3 \times 7.32 = -3.44\text{ cm}^3/\text{mole}$$

Thus equilibrium occurs at the pressure P at which

$$2411 = 3.44(P - 1) \times \frac{8.3144}{82.057}$$

i.e. $P = 6917\text{ atm.}$

Chapter 13

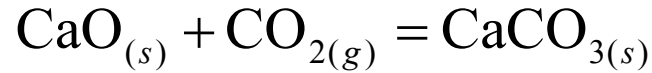
13.21 An experiment is being conducted on an equilibrated mixture of CaO and CaCO_3 contained in a closed vessel at 1200 K. The mixture is contaminated by iron in the form of hematite (Fe_2O_3). The contaminant would not be harmful to the experiment if it occurred as either wustite (FeO) or as cementite (Fe_3C). The necessary changes in the chemical form of the contaminant can be effected by admitting CO gas to the vessel. Calculate the allowable limits of p_{CO} in the vessel for the occurrence of the contaminant as

- (a) wustite and
- (b) cementite.

Chapter 13

Solution to 13.21

For

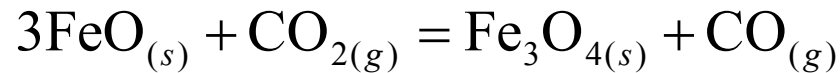


$$\Delta G_{1200\text{ K}}^{\circ} = 3340\text{ J} = RT \ln p_{\text{CO}_2}$$

So p_{CO_2} inside the vessel is 1.398 atm.

(a)

For



$$\Delta G_{1200\text{ K}}^{\circ} = 4348\text{ J} = -RT \ln \frac{p_{\text{CO}}}{p_{\text{CO}_2}}$$

So for equilibrium between FeO and Fe_3O_4 ,

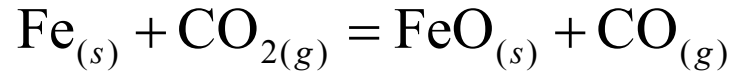
$$\frac{p_{\text{CO}}}{p_{\text{CO}_2}} = 0.647 \quad \text{or} \quad p_{\text{CO}} = 0.904\text{ atm}$$

Thus for FeO to be stable, $p_{\text{CO}} > 0.904\text{ atm}$

Chapter 13

Solution to 13.21

For



$$\Delta G_{1200\text{ K}}^{\circ} = -8252\text{ J} = -RT \ln \frac{p_{\text{CO}}}{p_{\text{CO}_2}}$$

So

$$\frac{p_{\text{CO}}}{p_{\text{CO}_2}} = 2.287 \quad \text{and} \quad p_{\text{CO}} = 3.196\text{ atm}$$

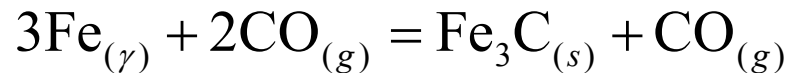
Thus for FeO to be stable, $3.196\text{ atm} > p_{\text{CO}} > 0.904\text{ atm}$

Chapter 13

Solution to 13.21

(b)

For

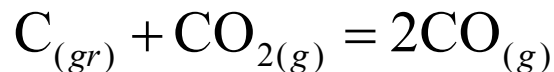


$$\Delta G_{1200\text{ K}}^{\circ} = 36,686\text{ J} = -RT \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}^2}$$

So

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}^2} = 2.53 \times 10^{-2} \quad \text{and} \quad p_{\text{CO}} = 7.43\text{ atm}$$

For



$$\Delta G_{1200\text{ K}}^{\circ} = -38,492\text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2}} \quad (a_{\text{C}} = 1)$$

So

$$\frac{p_{\text{CO}}^2}{p_{\text{CO}_2}} = 47.4 \quad \text{and} \quad p_{\text{CO}} = 8.14\text{ atm}$$

Thus for Fe_3C to be stable, $8.14\text{ atm} > p_{\text{CO}} > 7.43\text{ atm}$

Chapter 13

13.22 A Cu–Au alloy of $X_{\text{Cu}} = 0.5$ is being annealed at 600 °C in deoxidized argon. The argon is deoxidized by being passed over heated pure copper turnings prior to its admission to the annealing furnace. The solid Cu–Au system is virtually regular in its solution behavior, with a molar Gibbs excess free energy of mixing given by

$$G^{\text{XS}} = -28,280 X_{\text{Cu}} X_{\text{Au}} \text{ J}$$

Assuming that equilibrium is attained in the deoxidizing furnace, calculate the maximum temperature at which the deoxidizing furnace can be operated without causing oxidation of the copper in the Cu–Au alloy being annealed.

Chapter 13

Solution to 13.22

For $2\text{Cu}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{Cu}_2\text{O}_{(s)} \quad \Delta G_{873\text{ K}}^\circ = -101,753 \text{ J} = RT \ln p_{\text{O}_2}^{1/2} a_{\text{Cu}}^2$

So $a_{\text{Cu}}^2 \cdot p_{\text{O}_2}^{1/2} = 8.162 \times 10^{-7}$

$$\Omega = -28,280 \quad RT \ln \gamma_{\text{Cu}} = \Omega X_{\text{Cu}}^2$$

Thus
$$\ln \gamma_{\text{Cu}} = \frac{-28,280 \times 0.5^2}{8.3144 \times 873} = -0.974$$

which gives $\gamma_{\text{Cu}} = 0.378$, and $a_{\text{Cu}} = 0.378 \times 0.5 = 0.1888$

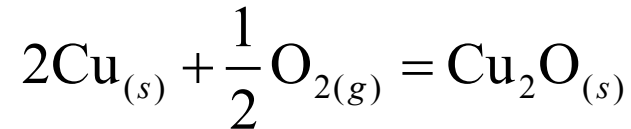
and the oxygen pressure for equilibrium between the Cu-Au alloy and Cu_2O is

$$p_{\text{O}_2} = \left[\frac{8.162 \times 10^{-7}}{0.1888^2} \right]^2 = 5.24 \times 10^{-10} \text{ atm}$$

Chapter 13

Solution to 13.22

For



$$\Delta G^\circ = -162,200 + 69.24T \text{ J} = 8.3144T \ln \left[5.24 \times 10^{-10} \right]^{1/2}$$

which gives 1026 K is the temperature at which 5.24×10^{-10} atm is the oxygen pressure required for equilibrium between Cu and Cu_2O .

The maximum temperature at which the deoxidizing furnace can be operated is thus 1026 K.

Chapter 13

13.23 In a dew-point experiment, a Cu–Zn alloy is placed in one end of an evacuated and closed tube and is heated to 900 °C. When the other end of the tube is cooled to 740 °C, Zn vapor begins to condense. Calculate the activity of Zn in the alloy relative to pure zinc.

Chapter 13

Solution to 13.23

At 740 °C

$$\ln p_{\text{Zn}}^{\circ} = \frac{-15,250}{1013} = -1.255 \ln 1013 + 21.79$$

So

$$p_{\text{Zn}, 1013 \text{ K}}^{\circ} = 0.142 \text{ atm}$$

At 900 °C

$$\ln p_{\text{Zn}}^{\circ} = \frac{-15,250}{1173} = -1.255 \ln 1173 + 21.79$$

So

$$p_{\text{Zn}, 1173 \text{ K}}^{\circ} = 0.923 \text{ atm}$$

Thus

$$a_{\text{Zn}} (\text{in the alloy}) = \frac{0.142}{0.923} = 0.154$$

Chapter 13

13.24 A crucible containing 100.0 g of silver at 1000 °C is placed in the reaction chamber of a Sieverts' apparatus. The chamber is evacuated and filled with 50 cm³ (STP) of argon, which measures the dead volume of the chamber. An external manometer reads the pressure of argon as 0.9 atm. The chamber is reevacuated and filled with oxygen, and it is found that 251.5 cm³ (STP) are required to produce a gas pressure of 0.9 atm in the chamber. Calculate the solubility of O in the Ag (as atom percent) and calculate the value of the Sieverts' law constant at 1000 °C.

Chapter 13

Solution to 13.24

Number of moles of Ar required to fill the chamber:

$$n_{\text{Ar}} = \frac{50}{82.057 \times 273} = 2.232 \times 10^{-3}$$

Number of moles of O₂ required to fill the chamber:

$$n_{\text{O}_2} = \frac{251.5}{82.057 \times 273} = 1.123 \times 10^{-2}$$

So number of moles of O₂ dissolved in liquid Ag is:

$$n_{\text{O}_2} = 1.123 \times 10^{-2} - 2.232 \times 10^{-3} = 8.995 \times 10^{-3}$$

and the weight is $8.995 \times 10^{-3} \times 32 = 0.288$ grams of O₂

Chapter 13

Solution to 13.24

Thus

$$\text{wt}\% \text{ oxygen} = \frac{0.288}{0.288 + 100} \times 100 = 0.287$$

$$\text{mole fraction O, } X_{\text{O}} = \frac{2 \times 8.995 \times 10^{-3}}{2 \times 8.995 \times 10^{-3} + \frac{100}{107.87}} = 0.0190$$

So

$$\text{atom } \% \text{O} = 1.9$$

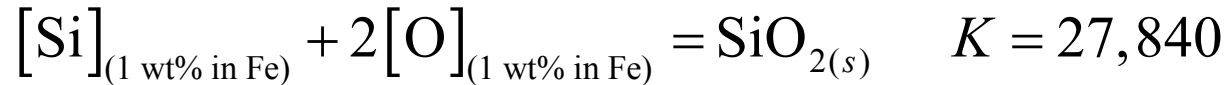
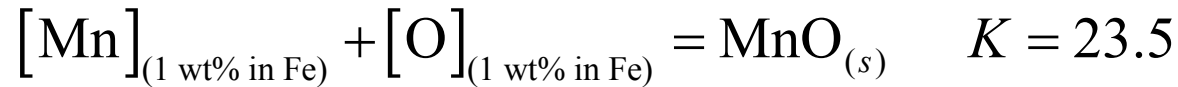
For

$$\frac{1}{2} \text{O}_{2(g)} = [\text{O}]$$

$$k = \frac{[\text{O}]}{p_{\text{O}_2}^{1/2}} = \frac{1.9}{0.9^{1/2}} = 2.0$$

Chapter 13

13.25 Silicon and manganese are commonly used together as deoxidizers for liquid steel. At 1600 °C,



The values of the equilibrium constants show that SiO_2 is considerably more stable than MnO . Why, then, is a mixture of Mn and Si more effective as a deoxidizing agent than Si alone? The activities of MnO and SiO_2 , with respect to solids as the standard states in MnO-SiO_2 melts at 1600 °C, are shown in Figure 13.37.

Chapter 13

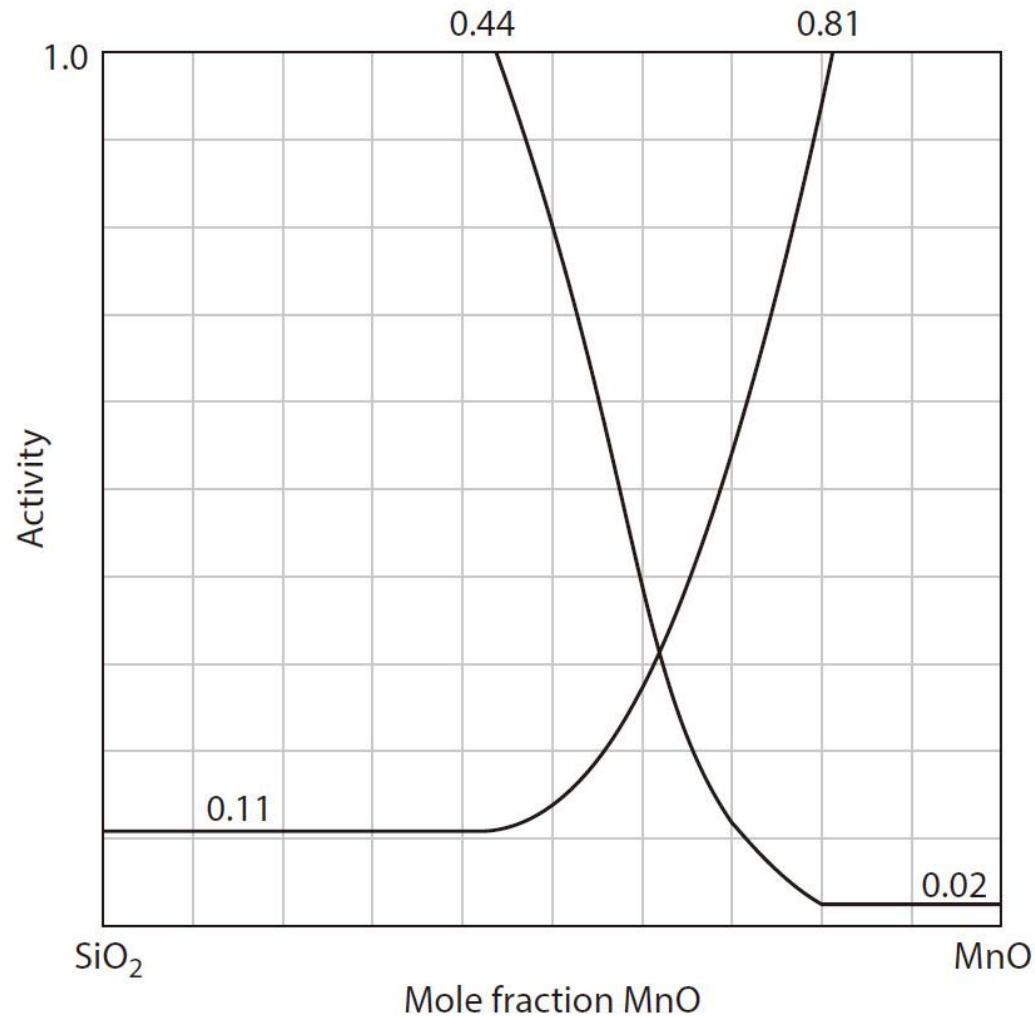


Figure 13.37 Activities in the system MnO–SiO₂ at 1600 °C.

Chapter 13

Solution to 13.25

Using Si alone produces solid SiO_2 $a_{\text{SiO}_2} = 1$ as the deoxidation product. Thus with 0.01 wt% Si, the oxygen content is

$$[\%O] = \left[\frac{a_{\text{SiO}_2}}{k_{\text{Si}} \cdot [\%Si]} \right]^{1/2} = \left[\frac{1}{27,840 \times 0.01} \right]^{1/2} = 0.06$$

Using Si + Mn produces a liquid manganese silicate as the deoxidation product, which, if saturated with MnO gives $a_{\text{SiO}_2} = 0.02$

Thus with 0.01 wt% Si, the oxygen content is

$$[\%O] = \left[\frac{0.02}{27,840 \times 0.01} \right]^{1/2} = 0.0085$$

Chapter 13

Solution to 13.25

The Mn content required to saturate with MnO is calculated as

$$[\% \text{Mn}] = \frac{a_{\text{MnO}}}{k_{\text{Mn}} \cdot [\% \text{O}]} = \frac{1}{23.5 \times 0.0085} = 5.0$$

Thermodynamics of Materials & Phase transformations

The End



MID-Examination 2019

Haifeng Wang (王海丰)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*haifengw81@nwpu.edu.cn

Part I: Regular Questions

1. What is the First Law of thermodynamics?

What are the definitions of work (W) and heat (Q)?

What is the difference between reversible work and irreversible work?

Can you provide one example?

What is the difference between heat and heat energy?

Part I: Regular Questions

Solution:

First law of thermodynamics: A statement of energy conservation

$$\Delta U = U_B - U_A = q - w$$

$$dU = \delta q - \delta w$$

内部力与外界阻力相抵

The work is **reversible** if the forces at the boundary of the system are balanced such that the resisting force at the boundary differs from the force inside the system by only an infinitesimal amount.

内部力与外界存在压差

The work is **irreversible** if there is a finite pressure difference between the system and the surroundings in which the movement is taking place.

Heat is the energy in transfer between a system and the surroundings.

Thermal energy is possessed by the system.

2.8 Reversible isothermal pressure or volume changes of an ideal gas

- ✓ The work done by the reversible isothermal process (which is equal to the area under the curve) exceeds that done by the reversible adiabatic process.
- ✓ This difference is due to the fact that during the isothermal process heat is absorbed by the system in order to maintain the temperature constant, whereas during the adiabatic process no heat is admitted to the system.
- ✓ During the isothermal expansion, the internal energy of the gas remains constant, and during the adiabatic expansion the internal energy decreases by an amount equal to the work done by the gas.

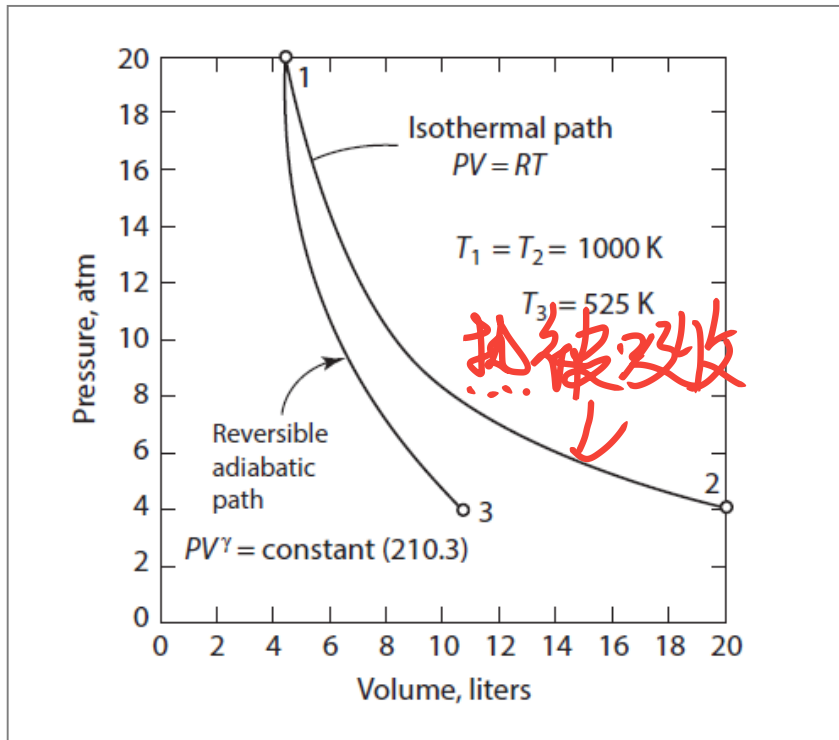


Fig. 2.4. Comparison of the process path taken by a reversible isothermal expansion of one mole of an ideal gas with the process path taken by a reversible adiabatic expansion of an ideal gas between an initial pressure of 20 atm and a final pressure of 4 atm.

3.4 Reversible Processes

- ✓ If the spontaneity is removed, it is apparent that at all times during the process, the system is at equilibrium. Thus **a reversible process** is the one during which the system is never away from equilibrium, and a reversible process which takes the system from the state A to the state B is one in which the process path passes through a continuum of equilibrium states. Such a path is, of course, **imaginary**.
- ✓ But it is possible to conduct an actual process in such a manner that it is virtually reversible. Such an actual process (**a quasi-static process**) is one which proceeds under the influence of an infinitesimally small driving force such that, during the process, the system is never more than an infinitesimal distance from equilibrium. If, at any point along the path, the external influence is removed, then the process ceases, or, if the direction of the external influence is reversed, then the direction of the process is reversed.

3.5 Illustration of irreversible and reversible processes

- ✓ Let us consider the reversible isothermal expansion of 1 mole of a monatomic ideal gas from the state (V_A, T) to the state (V_B, T) , where $V_B > V_A$. The gas is placed in thermal contact with a heat reservoir at the temperature T , and by slowly reducing the weight on the piston by removing one grain of sand at a time, the pressure exerted by the gas is only infinitesimally greater than the instantaneous pressure exerted by the piston on the gas.

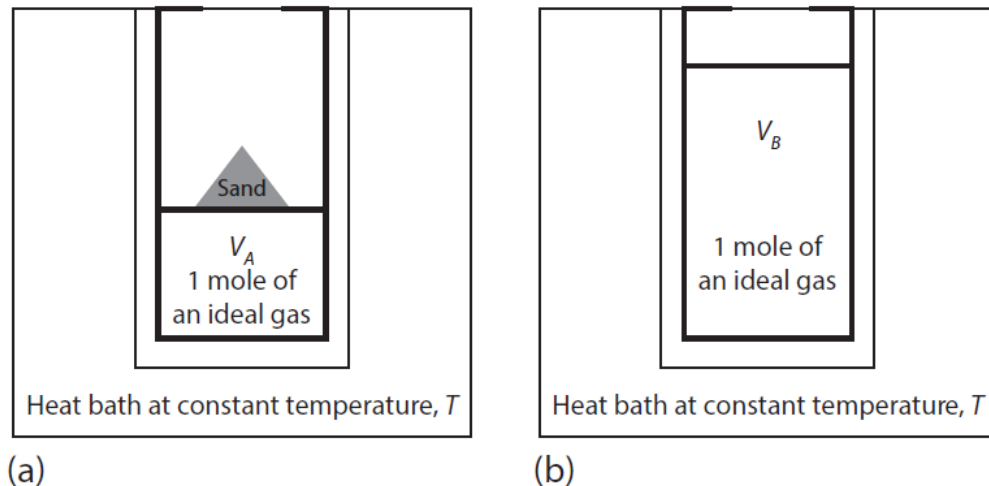


Figure 3.2 (a) One mole of a confined ideal gas being held at V_A by a mass of sand. (b) The mass of sand is removed and the new volume of the gas is V_B .

Part I: Regular Questions

2. What are the two statements of Second Law of thermodynamics from Kelvin and Clausius?

Can you discuss the two statements according to the principles of heat engines, refrigerators and heat pumps?

Part I: Regular Questions

Solution:

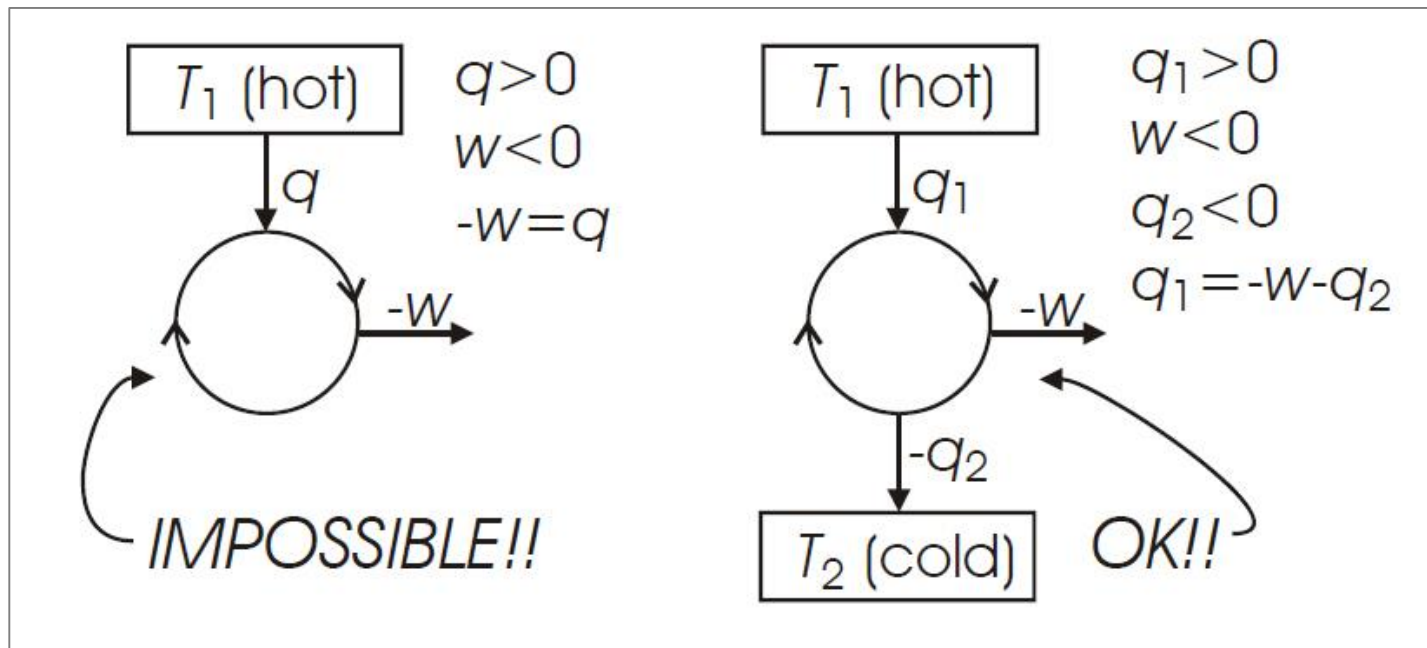
Clausius: It is impossible for any system to operate in a cycle that takes heat from a cold reservoir and transfers it to a hot reservoir without at the same time converting some work into heat. **Refrigerators**

Kelvin: It is impossible for any system to operate in a cycle that takes heat from a hot reservoir and converts it to work in the surroundings without at the same time transferring some heat to a colder reservoir. **heat engines**

Discussion

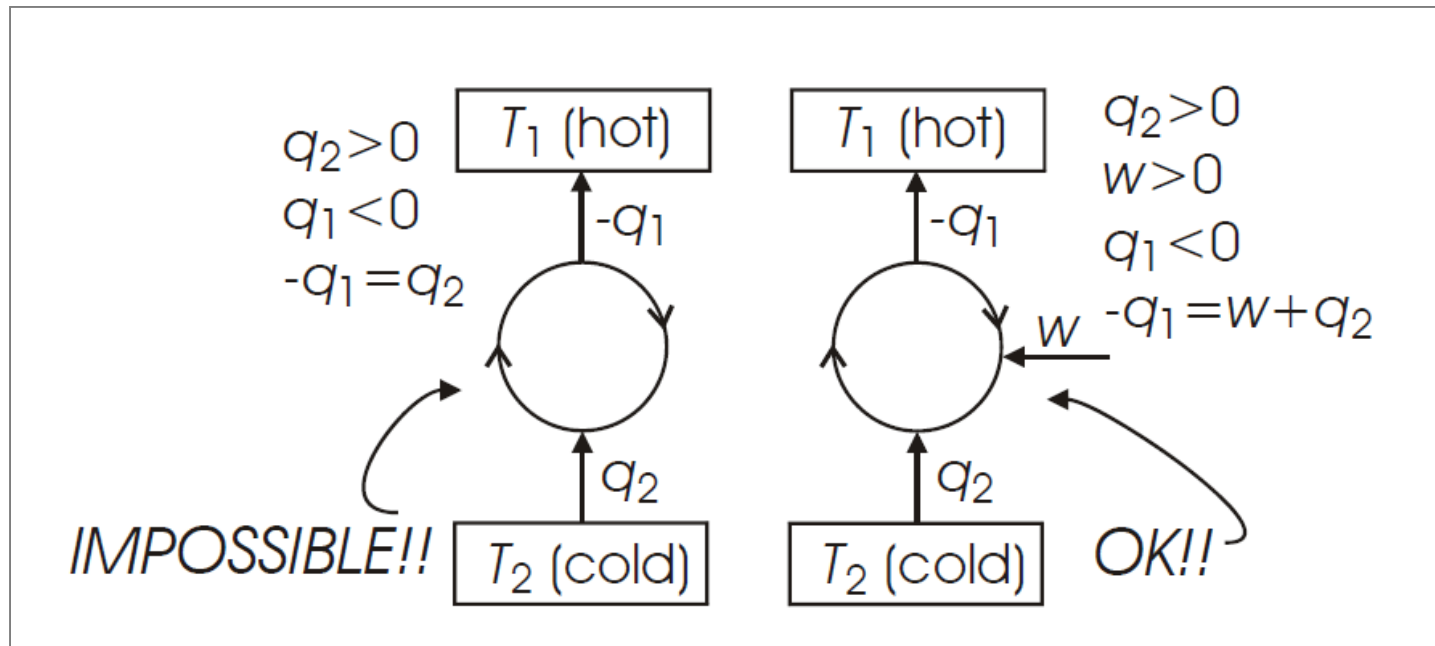
3.10 The properties of heat engines

- ✓ The preceding discussion gives rise to the following two preliminary formulations of the Second Law of Thermodynamics:
- The principle of Kelvin states that it is impossible, by means of a cyclic process, to take heat from a reservoir and convert it to work without, in the same operation, transferring heat to a cold reservoir.



3.10 The properties of heat engines

- ✓ The preceding discussion gives rise to the following two preliminary formulations of the Second Law of Thermodynamics:
- The principle of Clausius states that it is impossible to transfer heat from a cold to a hot reservoir without, in the same process, converting a certain amount of work to heat.



Part I: Regular Questions

3. A reversible heat engine, operating in a cycle, withdraws heat from a high-temperature “reservoir” (the temperature of which consequently decreases), performs work W , and rejects heat into a low-temperature “reservoir” (the temperature of which consequently increases). The two reservoirs are, initially, at the temperatures T_1 and T_2 , and have constant heat capacities C_1 and C_2 , respectively. Calculate the final temperature of the system and the maximum amount of work for the heat engine.

$$\text{热机} \Rightarrow S = 0.$$

)

Part I: Regular Questions

Solution:

The engine will stop producing work when it reaches its equilibrium temperature of T_f . To reach this temperature, the high-temperature bath will expel heat:

$$Q_2 = C_2 (T_2 - T_f)$$

The engine will expel heat to the low temperature bath of

$$Q_1 = C_1 (T_f - T_1)$$

The total work then becomes

$$W = Q_2 - Q_1 = C_2 (T_2 - T_f) - C_1 (T_f - T_1)$$

Part I: Regular Questions

In this reversible engine, the total entropy change (reservoirs plus engine) must be zero.

The engine operates in a cycle and thus must have no entropy change. Assuming constant heat capacities, the entropy change from the reservoirs is:

$$\Delta S = \int_{T_1}^{T_f} \frac{C_1}{T} dT + \int_{T_2}^{T_f} \frac{C_2}{T} dT = 0 = C_1 \ln \frac{T_f}{T_1} + C_2 \ln \frac{T_f}{T_2}$$

$$T_f = \exp \left(\frac{C_1 \ln T_1 + C_2 \ln T_2}{C_1 + C_2} \right)$$

Part I: Regular Questions

The maximum amount of work is

$$W = Q_2 - Q_1 = C_2 \left[T_2 - \exp \left(\frac{C_1 \ln T_1 + C_2 \ln T_2}{C_1 + C_2} \right) \right] - C_1 \left[\exp \left(\frac{C_1 \ln T_1 + C_2 \ln T_2}{C_1 + C_2} \right) - T_1 \right]$$

$$T_f = \exp \left(\frac{C_1 \ln T_1 + C_2 \ln T_2}{C_1 + C_2} \right)$$

$$W = Q_2 - Q_1 = C_2 (T_2 - T_f) - C_1 (T_f - T_1)$$

Part I: Regular Questions

4. Please prove for a real gas

$$C_P - C_V = \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

Solution:

$$H = U + PV$$

$$c_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH}{dT} \right)_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P$$

$$c_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V$$

Part I: Regular Questions

$$U = U(V, T)$$

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V$$

$$= \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

 $= U$

Part I: Regular Questions

5. In contrast to the internal energy (U), entropy (S) is not conserved in the natural processes. How to prove this unusual and sometimes confusing characteristic of entropy function?

(Hint: one could consider a case that body I and body II are at two different temperatures and are connected by a thermal conductor, such as a bar of copper, i.e., Fig. 1).

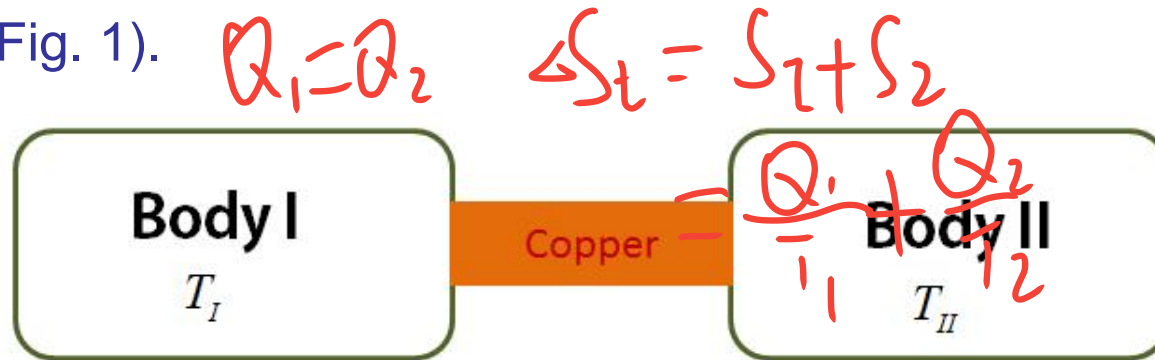


Fig. 1 Heat conduction from body I to body II with different temperatures.

Part I: Regular Questions

$$\Delta U_{I+II} = \Delta U_I + \Delta U_{II} = Q_I + Q_{II} = 0$$

One can see that the change in the internal energy is zero, i.e. energy is conserved.

The heat transfer at the boundary of reservoirs I and II is considered to be **reversible**. That is the temperature of the copper bar immediately adjacent to a reservoir is essentially at the temperature of the reservoir.

$$\Delta S_{I+II} = \Delta S_I + \Delta S_{II}$$

Part I: Regular Questions

$$\Delta S_{I+II} = \Delta S_I + \Delta S_{II}$$

$$\Delta S_I = \frac{Q_I}{T_I} \quad \downarrow \quad \Delta S_{II} = \frac{Q_{II}}{T_{II}}$$

$$\Delta S_{I+II} = \frac{Q_I}{T_I} + \frac{Q_{II}}{T_{II}}$$

$$Q_I = Q_{II} \quad \downarrow$$

$$\Delta S_{I+II} = \frac{Q_I}{T_I} + \frac{Q_{II}}{T_{II}} = Q_{II} \left(\frac{T_I - T_{II}}{T_I T_{II}} \right)$$

Part I: Regular Questions

$$\Delta S_{I+II} = \frac{Q_I}{T_I} + \frac{Q_{II}}{T_{II}} = Q_{II} \left(\frac{T_I - T_{II}}{T_I T_{II}} \right)$$

Entropy is not conserved in this isolated system. The heat flow was irreversible because the two temperatures differed by more than an infinitesimal amount.

Because the heat flows spontaneously only from higher to lower temperature, it should be apparent that the entropy of this isolated system can never decrease; it can only increase or remain constant.

Part I: Regular Questions

6. Please calculate the change of entropy for the six mixing processes as follows. Noting that both A and B are ideal gases.

$$(1) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_B \ T \ V \ P) \\ \hline \end{array} \Rightarrow (n_A + n_B \ T \ 2V \ P)$$

$$(5) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V) & (2n_A \ T \ V) \\ \hline \end{array} \Rightarrow (3n_A \ T \ 2V)$$

$$(2) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_B \ T \ V \ P) \\ \hline \end{array} \Rightarrow (n_A + n_B \ T \ V)$$

$$(6) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V) & (2n_B \ T \ V) \\ \hline \end{array} \Rightarrow (n_A + 2n_B \ T \ 2V)$$

$$(3) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_A \ T \ V \ P) \\ \hline \end{array} \Rightarrow (2n_A \ T \ 2V \ P)$$

$$(4) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_A \ T \ V \ P) \\ \hline \end{array} \Rightarrow (2n_A \ T \ V)$$

Isothermal reversible process of ideal gas

$$\Delta S = \frac{Q_R}{T} = \frac{-W}{T} = nR \ln \frac{V_2}{V_1} = nR \ln \frac{p_1}{p_2}$$

Part I: Regular Questions

Solution:

$$(1) \quad \Delta S = n_A R \ln \frac{2V}{V} + n_B R \ln \frac{2V}{V} = (n_A + n_B) R \ln 2$$

$$(1) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_B \ T \ V \ P) \\ \hline \end{array} \Rightarrow (n_A + n_B \ T \ 2V \ P)$$

$$(5) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V) & (2n_A \ T \ V) \\ \hline \end{array} \Rightarrow (3n_A \ T \ 2V)$$

$$(2) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_B \ T \ V \ P) \\ \hline \end{array} \Rightarrow (n_A + n_B \ T \ V)$$

$$(6) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V) & (2n_B \ T \ V) \\ \hline \end{array} \Rightarrow (n_A + 2n_B \ T \ 2V)$$

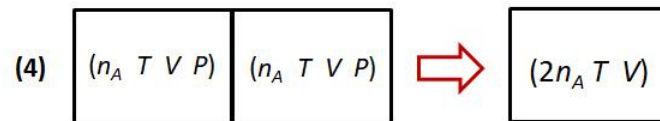
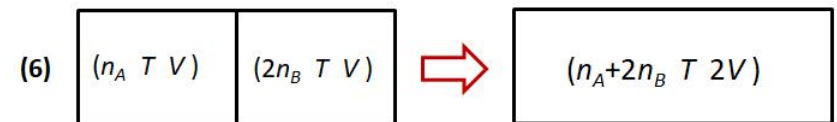
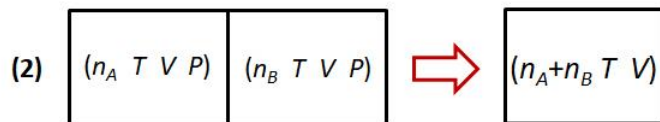
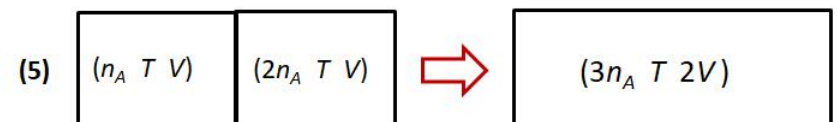
$$(3) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_A \ T \ V \ P) \\ \hline \end{array} \Rightarrow (2n_A \ T \ 2V \ P)$$

$$(4) \quad \begin{array}{|c|c|} \hline (n_A \ T \ V \ P) & (n_A \ T \ V \ P) \\ \hline \end{array} \Rightarrow (2n_A \ T \ V)$$

Part I: Regular Questions

(2)

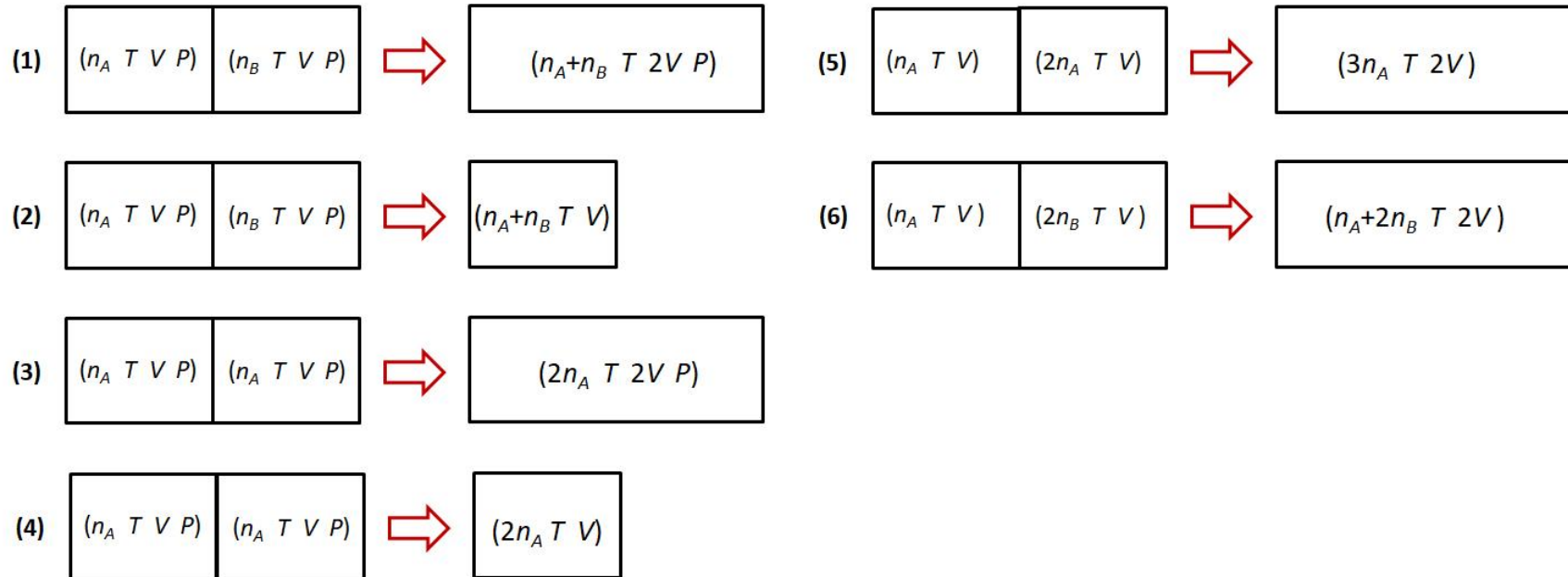
$$\Delta S = n_A R \ln \frac{V}{V} + n_B R \ln \frac{V}{V} = 0$$



Part I: Regular Questions

(3)

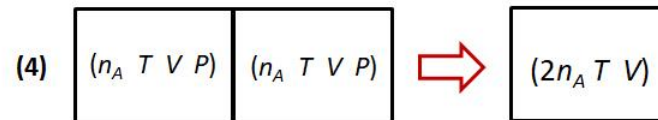
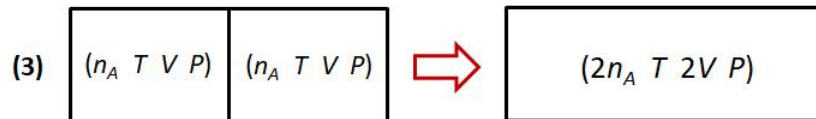
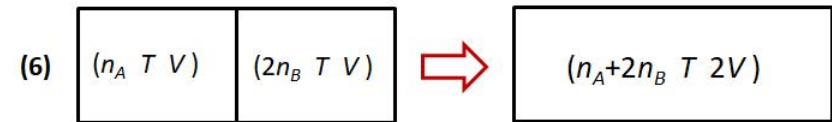
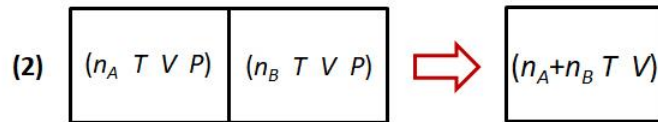
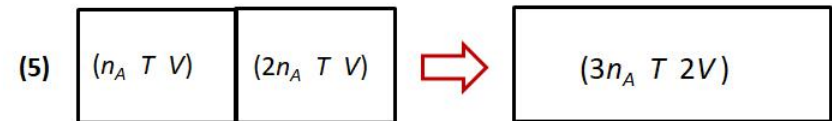
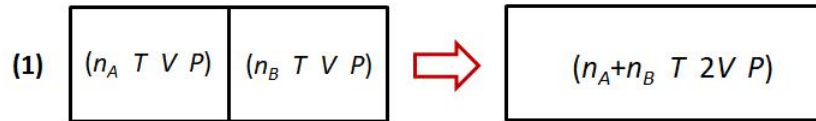
$$\Delta S = 0$$



Part I: Regular Questions

(4)

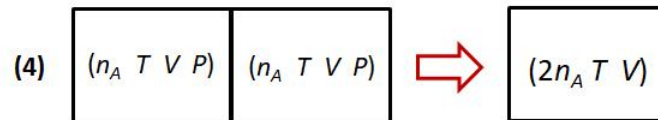
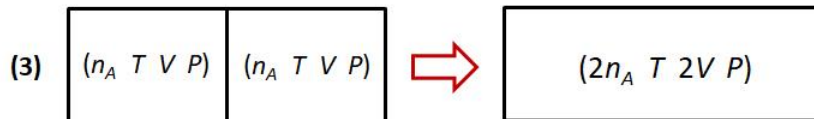
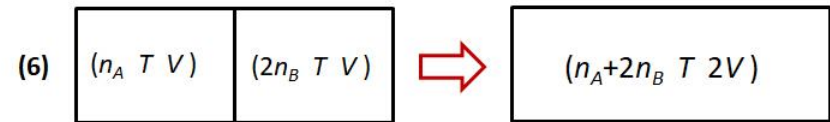
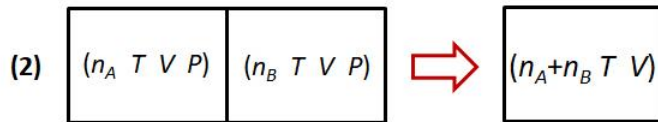
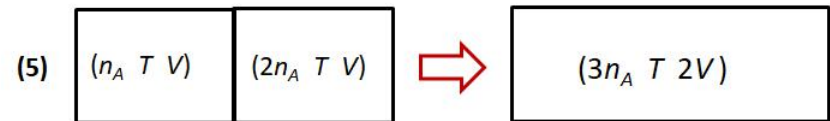
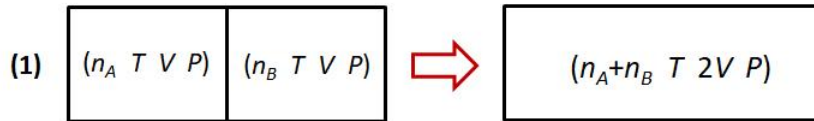
$$\Delta S = 0 + 2n_A R \ln \frac{V}{2V} = -n_A R \ln 4$$



Part I: Regular Questions

(5)

$$\Delta S = \Delta S_{\text{left}} + \Delta S_{\text{right}} = n_A R \ln \frac{\frac{2}{3}V}{V} + 2n_A R \ln \frac{\frac{4}{3}V}{V} = n_A R \ln \frac{2}{3} + 2n_A R \ln \frac{4}{3}$$



Part I: Regular Questions

$$(6) \Delta S = \Delta S_{\text{left}} + \Delta S_{\text{right}} = n_A R \ln \frac{2V}{V} + 2n_B R \ln \frac{2V}{V} = (n_A + 2n_B) R \ln 2$$

(1)

$(n_A \ T \ V \ P)$	$(n_B \ T \ V \ P)$
---------------------	---------------------

 $\Rightarrow$

$(n_A+n_B \ T \ 2V \ P)$

(5)

$(n_A \ T \ V)$	$(2n_A \ T \ V)$
-----------------	------------------

 $\Rightarrow$

$(3n_A \ T \ 2V)$

(2)

$(n_A \ T \ V \ P)$	$(n_B \ T \ V \ P)$
---------------------	---------------------

 $\Rightarrow$

$(n_A+n_B \ T \ V)$

(6)

$(n_A \ T \ V)$	$(2n_B \ T \ V)$
-----------------	------------------

 $\Rightarrow$

$(n_A+2n_B \ T \ 2V)$

(3)

$(n_A \ T \ V \ P)$	$(n_A \ T \ V \ P)$
---------------------	---------------------

 $\Rightarrow$

$(2n_A \ T \ 2V \ P)$

(4)

$(n_A \ T \ V \ P)$	$(n_A \ T \ V \ P)$
---------------------	---------------------

 $\Rightarrow$

$(2n_A \ T \ V)$

Part I: Regular Questions

7. Calculate ΔH_{1600} and ΔS_{1600} for the reaction $\text{Zr}(\beta) + \text{O}_2 = \text{ZrO}_2(\beta)$.

Given the thermochemical data:

$$\text{ZrO}_2 : \Delta H_{298}^0 = -1100800 \text{ J}, S_{298}^0 = 50.4 \text{ J K}^{-1}$$

$$\text{Zr} : S_{298}^0 = 39.0 \text{ J K}^{-1}$$

$$\text{O}_2 : S_{298}^0 = 205.1 \text{ J K}^{-1}$$

$$\text{Zr}(\alpha \rightarrow \beta) \Delta H_{\text{trans}} = 3900 \text{ J at } 1136 \text{ K}$$

$$\text{ZrO}_2(\alpha \rightarrow \beta) \Delta H_{\text{trans}} = 5900 \text{ J at } 1478 \text{ K}$$

$$c_p : \text{For O}_2 : 29.96 + 4.18 \times 10^{-3} T - 1.67 \times 10^{-5} T^2$$

$$\text{For Zr}(\alpha) : 21.97 + 11.63 \times 10^{-3} T \quad 298 \text{ K} - 1136 \text{ K}$$

$$\text{For Zr}(\beta) : 23.22 + 4.64 \times 10^{-3} T \quad 1136 \text{ K} - 2128 \text{ K}$$

$$\text{For ZrO}_2(\alpha) : 69.62 + 7.53 \times 10^{-3} T - 14.06 \times 10^{-5} T^2 \quad 298 \text{ K} - 1478 \text{ K}$$

$$\text{For ZrO}_2(\beta) : 74.48 \quad 1478 \text{ K} - 2950 \text{ K}$$

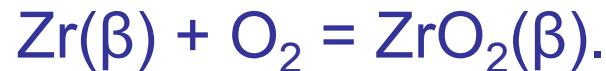
Part I: Regular Questions

Solution:

Process (1)

$$\begin{aligned}\Delta H_{Zr}^1 &= \int_{298}^{1136} 21.97 + 11.63 \times 10^{-3} T dT + \Delta H_{trans}(\alpha \rightarrow \beta) + \int_{1136}^{1600} 23.22 + 4.64 \times 10^{-3} T dT \\ &= \left(21.97T + 5.815 \times 10^{-3} T^2 \right)_{298}^{1136} + 3900 + \left(23.22T + 2.32 \times 10^{-3} T^2 \right)_{1136}^{1600} \\ &= 25398.69898 + 3900 + 13719.32928 = 43018 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{Zr}^1 &= \int_{298}^{1136} \frac{21.97 + 11.63 \times 10^{-3} T}{T} dT + \frac{\Delta H_{trans}(\alpha \rightarrow \beta)}{T} + \int_{1136}^{1600} \frac{23.22 + 4.64 \times 10^{-3} T}{T} dT \\ &= \left(21.97 \ln T + 11.63 \times 10^{-3} T \right)_{298}^{1136} + \frac{3900}{1136} + \left(23.22 \ln T + 4.64 \times 10^{-3} T \right)_{1136}^{1600} \\ &= 39.14 + 3.433 + 10.10 = 52.673 \text{ J/K}\end{aligned}$$

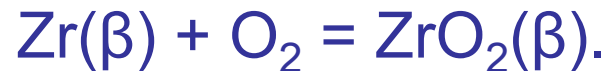


Part I: Regular Questions

Process (2)

$$\begin{aligned}\Delta H_{O_2}^2 &= \int_{298}^{1600} 29.96 + 4.18 \times 10^{-3} T - 1.67 \times 10^5 T^{-2} dT \\ &= \left(29.96T + 2.09 \times 10^{-3} T^2 + 1.67 \times 10^5 T^{-1} \right)_{298}^{1600} \\ &= 43716.69 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{O_2}^2 &= \int_{298}^{1600} \frac{29.96 + 4.18 \times 10^{-3} T - 1.67 \times 10^5 T^{-2}}{T} dT \\ &= \left(29.96 \ln T + 4.18 \times 10^{-3} T + 0.835 \times 10^5 T^{-2} \right)_{298}^{1600} \\ &= 54.89 \text{ J/K}\end{aligned}$$

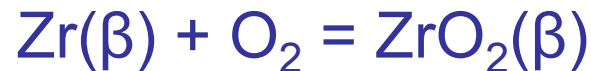


Part I: Regular Questions

Process (3)

$$\begin{aligned}\Delta H_{\text{ZrO}_2}^3 &= \int_{298}^{1478} 69.62 + 7.53 \times 10^{-3} T - 14.06 \times 10^5 T^{-2} dT + \Delta H_{\text{trans}}(\alpha \rightarrow \beta) + \int_{1478}^{1600} 74.48 dT \\ &= \left(69.62T + 3.765 \times 10^{-3} T^2 + 14.06 \times 10^5 T^{-1} \right)_{298}^{1478} + 5900 + (74.48T)_{1478}^{1600} \\ &= 86274.999 + 5900 + 9086.56 = 101261 \text{ J}\end{aligned}$$

$$\begin{aligned}\Delta S_{\text{ZrO}_2}^3 &= \int_{298}^{1478} \frac{69.62 + 7.53 \times 10^{-3} T - 14.06 \times 10^5 T^{-2}}{T} dT + \frac{\Delta H_{\text{trans}}(\alpha \rightarrow \beta)}{T} + \int_{1478}^{1600} \frac{74.48}{T} dT \\ &= \left(69.62 \ln T + 7.53 \times 10^{-3} T + 7.03 \times 10^5 T^{-2} \right)_{298}^{1478} + \frac{5900}{1478} + (74.48 \ln T)_{1478}^{1600} \\ &= 112.777 + 3.991 + 5.907 = 122.675 \text{ J/K}\end{aligned}$$



Part I: Regular Questions

Process (4)

$$\Delta H_{298} = -1100800 \text{ J}$$

$$\Delta S_{298} = 50.4 - 39 - 205.1 = -193.7 \text{ J/K}$$

According to

$$\Delta H_{1600} = \Delta H_{ZrO_2}^3 + \Delta H_{298} - \Delta H_{Zr}^1 - \Delta H_{O_2}^2$$

$$\Delta S_{1600} = \Delta S_{ZrO_2}^3 + \Delta S_{298} - \Delta S_{Zr}^1 - \Delta S_{O_2}^2$$

We have:

$$\Delta H_{1600} = -1086273 \text{ J}$$

$$\Delta S_{1600} = -178.588 \text{ J/K}$$

Kirchhoff's law

Part I: Regular Questions

8. Carbon has the following three allotropes: graphite, diamond, and a metallic form called solid III. Graphite is the stable form of 298 K and 1 atm pressure, and increasing the pressure on graphite at temperatures less than 1440 K induces the transformation from graphite to diamond. Calculate the pressure which, when applied to one mole graphite at 298 K, induces the transformation from graphite to diamond. (Assuming the same isothermal compressibilities for graphite and diamond)

Given the thermochemical data:

$$H_{298\text{ K, (graphite)}} - H_{298\text{ K, (diamond)}} = -1900\text{ J.}$$

$$S_{298\text{ K, (graphite)}} = 5.74\text{ J K}^{-1}.$$

$$S_{298\text{ K, (diamond)}} = 2.37\text{ J K}^{-1}.$$

1 mol graphite/diamond is 12 g.

The density of graphite at 298 K is 2.22 g cm^{-3} .

The density of diamond at 298 K is 3.515 g cm^{-3} .

$$1\text{ cm}^3 \cdot \text{atm} = 0.1013\text{ J}$$

Part I: Regular Questions

Solution:

For the transformation graphite $\rightarrow$ diamond at 298K:

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= 1900 - 298(2.37 - 5.74) = 2904 \text{ J}\end{aligned}$$

For the transformation of graphite to diamond at any temperature

$$\left(\frac{\partial \Delta G_{\text{graphite} \rightarrow \text{diamond}}}{\partial P} \right)_T = \Delta V_{\text{graphite} \rightarrow \text{diamond}} \quad V_{\text{graphite}} = \frac{12}{2.22} = 5.405 \text{ cm}^3 / \text{mole}$$

$$\Delta V = -1.99 \text{ cm}^3 / \text{mole} \quad V_{\text{diamond}} = \frac{12}{3.512} = 3.415 \text{ cm}^3 / \text{mole}$$

Part I: Regular Questions

Equilibrium between graphite and diamond at 298 K requires that

$$\Delta G_{\text{graphite} \rightarrow \text{diamond}} = 0$$

Since

$$\left(\frac{\partial \Delta G}{\partial P} \right)_T = \Delta V$$

Then

$$\Delta G(P, T = 298) = \Delta G(P = 1, T = 298) + \int_1^P \Delta V dP$$

If the difference between the isothermal compressibilities of the two phases is negligibly small (i.e., if the influence of pressure on ΔV can be ignored), then

$$1 \text{ cm}^3 \cdot \text{atm} = \frac{8.3144}{82.057} = 0.1013 \text{ J}$$

$$\Delta G(P, T = 298) = 2904 + (-1.99 \times 0.1013)(P - 1)$$

Part I: Regular Questions

$$1 \text{ cm}^3 \cdot \text{atm} = \frac{8.3144}{82.057} = 0.1013 \text{ J}$$

$$\Delta G(P, T = 298) = 2904 + (-1.99 \times 0.1013)(P - 1)$$

$$(P - 1) = P = \frac{2904}{1.99 \times 0.1013} = 14400 \text{ atm}$$

Transformation of graphite to diamond at 298 K requires the application of a pressure greater than 14,400 atm.

Part II: Bonus Question

9. A block of rubber weighing 100 g is to be isothermally compressed from a pressure of 1 atm to a pressure of 1001 atm. How much heat must be added (or removed) from the rubber to maintain it at a constant temperature of 298 K?

For rubber: Volumetric thermal expansion coefficient is $100 \times 10^{-6} \text{ K}^{-1}$.

The density of rubber at 298 K is 1.0 g cm^{-3} .

Note: one can assume that the volume of the rubber does not change much during the compression process.

Part II: Bonus Question

Solution:

The density of rubber at 298 K is 1.0 g/cm³. So the volume of the rubber is:

$$V = 100 \times 1.0 = 100 \text{ cm}^3 = 0.0001 \text{ m}^3$$

The Volumetric thermal expansion coefficient is $100 \times 10^{-6} \text{ K}^{-1}$. We have:

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P = 100 \times 10^{-6} \Rightarrow \left(\frac{\partial V}{\partial T} \right)_P = 100 \times 10^{-6} \times V \approx 10^{-8}$$

According to the MAXWELL' S RELATIONS,

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P = -10^{-8}$$

Part II: Bonus Question

So the change of the entropy is

$$\Delta S = \int_{1\text{atm}}^{1001\text{atm}} \frac{\partial S}{\partial P} dP = -10^{-8} \times (1000 \times 101325) = -1.01325 \text{ Pa}$$

$$\Delta S = \frac{q}{T} \Rightarrow q = T \Delta S = -301.9485 \text{ J}$$

So 301.9485J heat must be removed from the rubber to maintain it at a constant temperature of 298 K

Final-Examination 2019

Haifeng Wang (王海丰)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*haifengw81@nwpu.edu.cn

Basic Concepts

1. Please discuss the irreversibility of a process according to the following example.

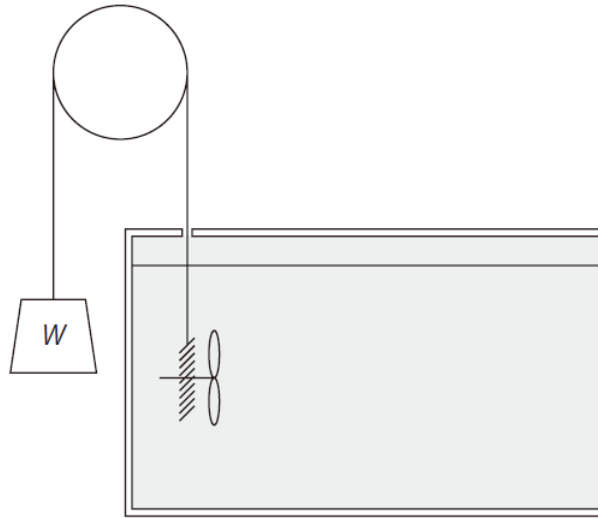


Fig. 1. A weight–pulley–heat reservoir arrangement in which the work done by the falling weight is degraded to thermal energy, which appears in the heat reservoir.

Basic Concepts

Solution:

This system consists of a weight-pulley arrangement which is coupled to a constant temperature heat reservoir, and the system is at equilibrium when an upward force acting on the weight exactly balances the downward force, W , of the weight. If the upward force is removed, then the equilibrium is upset and the weight spontaneously falls, thus performing work, which is converted, by means of a suitable system of paddle wheels, to heat which enters the constant-temperature heat reservoir. Equilibrium is re-attained when the upward force acting on the weight is replaced, and the net effect of this process is that mechanical energy has been converted to thermal energy.

Basic Concepts

- (1) The heat reservoir in the weight-heat reservoir system is at the temperature T_2 . The weight is allowed to fall, performing work, w , and the heat produced, q , enters the heat reservoir.
- (2) The heat reservoir at the temperature T_2 is placed in thermal contact with a heat reservoir at a lower temperature T_1 , and the same heat q is allowed to flow from the reservoir at T_2 to the reservoir at T_1 .
- (3) The heat reservoir in the weight-heat reservoir system is at the temperature T_1 . The weight is allowed to fall, performing work, w , and the heat produced, q enters the reservoir.

Basic Concepts

Each of these process is spontaneous and hence irreversible, and degradation occurs in each of them. However, as process (3) is the sum of processes (1) and (2), the degradation occurring in process (3) must be greater than the degradation occurring in each of the processes (1) and (2). Thus it can be said that process (3) is more irreversible than either process (1) or process (2). Examination of the three processes indicates that both the amount of heat produced, q , and the temperatures between which this heat flows are important in defining a quantitative scale of irreversibility. In the case of comparison between process (1) and process (3), the quantity q/T_2 is smaller than the quantity q/T_1 , with is in agreement with the conclusion that process (1) is less irreversible than process (3)

Proof and derivation questions

3. Please prove for one mole of a substance:

$$TdS = c_V \left(\frac{\partial T}{\partial P} \right)_V dP + c_P \left(\frac{\partial T}{\partial V} \right)_P dV$$

Proof and derivation questions

Solution:

$$S = S(P, V) \quad \Rightarrow \quad dS = \left(\frac{\partial S}{\partial P} \right)_V dP + \left(\frac{\partial S}{\partial V} \right)_P dV$$

$$c_V = \left(\frac{\partial U}{\partial T} \right)_V = T \left(\frac{\partial S}{\partial T} \right)_V$$

$$c_P = \left(\frac{\partial H}{\partial T} \right)_P = T \left(\frac{\partial S}{\partial T} \right)_P$$

$$c_V \left(\frac{\partial T}{\partial P} \right)_V = T \left(\frac{\partial S}{\partial P} \right)_V$$

$$c_P \left(\frac{\partial T}{\partial V} \right)_P = T \left(\frac{\partial S}{\partial V} \right)_P$$

$$c_V \left(\frac{\partial T}{\partial P} \right)_V dP + c_P \left(\frac{\partial T}{\partial V} \right)_P dV = T \left(\frac{\partial S}{\partial P} \right)_V dP + T \left(\frac{\partial S}{\partial V} \right)_P dV = T dS$$

Proof and derivation questions

4. Please derive and discuss the maximum work done for a process change from state A to state B. (*Hint*: By combining the first law with the second law)

Proof and derivation questions

Solution:

For a irreversible process, from the second law, we have:

$$dS_{system} = \frac{\delta q}{T} + dS_{irr} \quad (1)$$

From the first law, we have:

$$\delta q = dU_{system} + \delta w \quad (2)$$

Combing Eq. (1) with Eq. (2), we have:

$$dS_{system} = \frac{dU_{system} + \delta w}{T} + dS_{irr} \Rightarrow \delta w = TdS_{system} - dU_{system} - TdS_{irr}$$

Proof and derivation questions

$$\delta w = TdS_{\text{system}} - dU_{\text{system}} - TdS_{\text{irr}}$$

$$dS_{\text{irr}} \geq 0$$



$$\delta w \leq TdS_{\text{system}} - dU_{\text{system}}$$



$$w_{\text{max}} = q_{\text{rev}} = T(S_B - S_A) - (U_B - U_A)$$

Calculation and Analysis questions

5. A block of rubber weighing 100 g is to be compressed from a pressure of 1 atm to a pressure of 1001 atm.

(1) What is the volume of the block of rubber?

(2) The block of rubber is compressed isothermally at 298 K. What is its change of entropy?

(3) How much heat must be added (or removed) from the rubber to maintain it at a constant temperature of 298 K?

Specify whether heat is added or removed.

For rubber: Volumetric thermal expansion coefficient is $100 \times 10^{-6} \text{ K}^{-1}$.

The density of rubber at 298 K is 1.0 g cm^{-3} .

Note: one can assume that the volume of the rubber does not change during compression.

Thermodynamics of Materials & Phase transformations

The End

